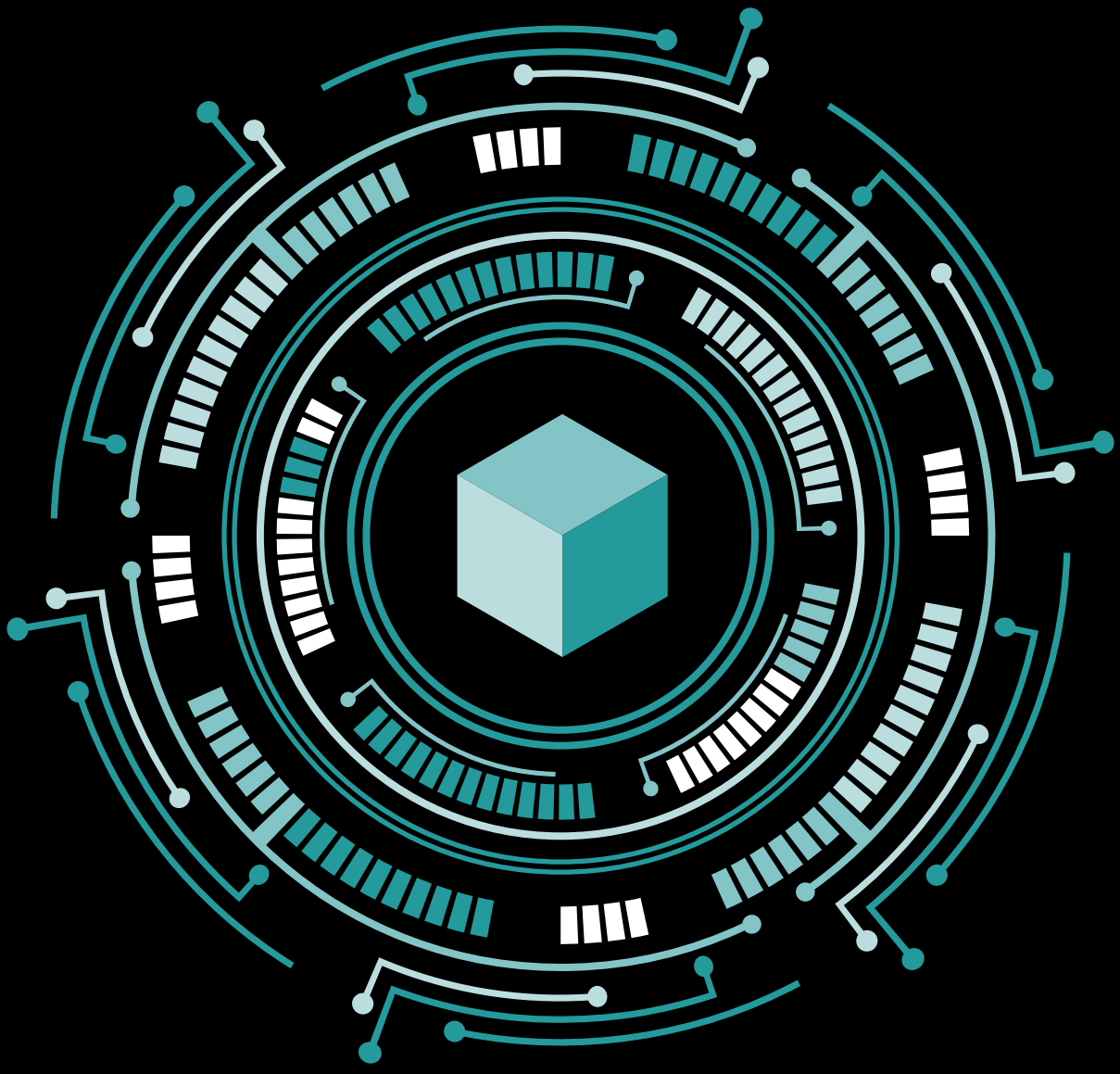
FUTURISTIC OUTLOOK

to Product Management

Revamped Edition 2024-25



Prepared By

THE RODMAN CLUB OF IIM AHMEDABAD

ACKNOWLEDGEMENT

There has been a steep increase in the number of firms offering product management roles in the recent past. This is not only maneuvered by an increase in the number of firms visiting the graduate students, but also by expansion in their scope to associate product manager roles for undergrads. In spite of this, there exists a gap in terms of comprehensive preparation material suited for the Indian audience. Filling this gap was an uphill task, which couldn't have been possible without the stakeholders who were a part of the journey at different points in time.

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The ProdMan Club of IIM Ahmedabad

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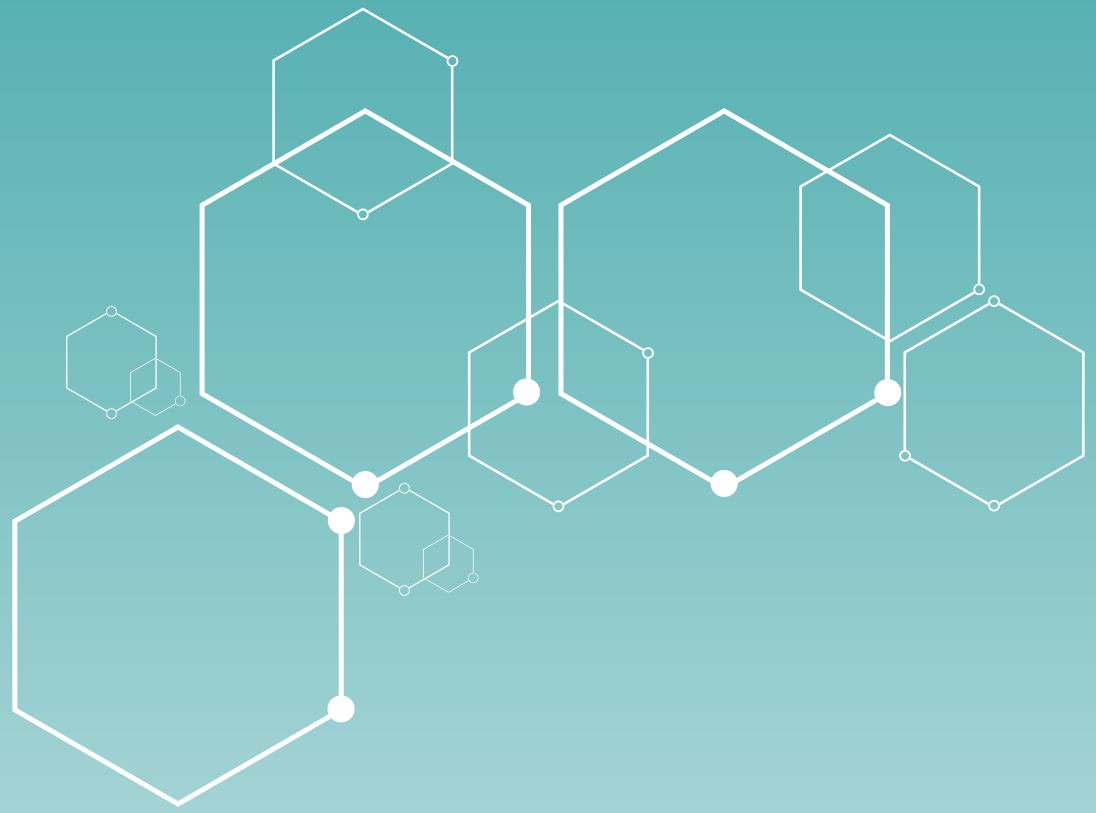
MEET THE TEAM

SECTION 1

INTRODUCTION



WHAT IS PRODUCT MANAGEMENT?

PRODUCT MANAGEMENT: IT HAS ALWAYS BEEN THERE

It is rare to spot tech companies today without a product manager. It is common in the contemporary business ecosystem to associate the definition of product management with the role of a product manager in the software field. However, if we look at it in hindsight, traditionally, every business has been creating products since the inception of it, long before the advent of technology, computers, and software. The history of business is full of examples of people who have conceptualized ideas to solve a problem. They have built these ideas, tested them, and delivered them to customers for bringing a significant change in their lives. A lot of times, they have failed too. But they reiterated, they pivoted, keeping a single goal in mind: creating something to solve a problem and making sure the solution is built and shipped in the right way. And this is precisely what a product manager does in organizations today.

So, what is Product Management? The most common explanations look like:

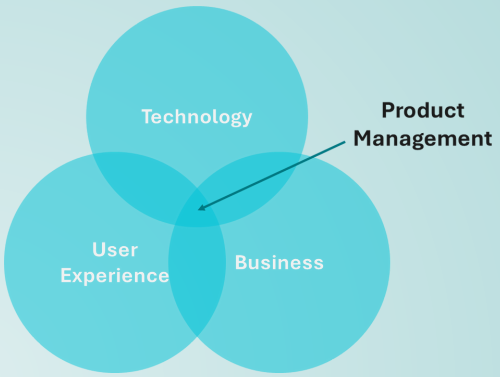


Figure 1.1 What, exactly, is a Product Manager ?

It is like when someone asks, “Where is the Statue of Liberty?” and gets a reply, “It’s between Manhattan, Jersey City, and Governor’s Island.” This kind of information is orienting but incomplete. So, what does a product manager actually do? Do not worry, no one knows!

There are numerous definitions of Product Management around the world, and the internet is full of them. The most popular one is perhaps that " A product manager is the CEO of a product." This makes sense in a way because a product manager indeed possesses almost the entire responsibility of the product. But it may not entirely be true in a lot of cases.

However, what every product manager does in common is to "guide a team in discovering, developing, and shipping the right product to users and bridge the gaps among different stakeholders of that product."

In contrast to what the name suggests, a product manager may actually not be a manager of anybody. A PM is a communicator that puts all the pieces together by getting feedback from all the stakeholders involved. The PM is also a prioritizer, a researcher, a presenter and the most responsible for the ultimate success of a product.

THE "JOB" OF A PRODUCT MANAGER

The words "job" and "product" may make us think of Steve Jobs, one of the most significant product visionaries of all time. In the subsequent sections, we will try to draw some learnings from his works. Once he said, "A lot of times, people don't know what they want until you show it to them." This quote became legendary as one of the most popular opinions of a highly opinionated man. Let us look at a few.

To Listen or Not to Listen

The first, foremost, and perhaps the most essential trait of a Product Manager is to listen to your customers. To solve a customer problem, you need to understand the customer's needs, both stated and unstated. It might sound cliché and very obvious, but that's indeed the hardest thing to do, following the unstated needs. To understand the customers' unstated needs, you need to "really listen." When you do not do that, you end up building solutions to problems, which no one wants.

Going back to Steve Jobs's quote, he indeed went ahead and showed the world what they wanted. So does it mean that you are better off by not listening to your customers?

Absolutely not! You should listen to your customers to understand the problem, but not when you are building the solution. Your job is essentially to manage the solution, your way. The quality of the solution, regardless of its acceptability, is what differentiates great product managers from average product managers.

Solutions Prioritization and Decisions

Coming up with multiple ideas/solutions is just the starting step. Prioritizing them comes next. While prioritizing work, PM needs to look at both internal as well as external factors. Prioritization of work will enable the team to work on the most critical features first, and these features/work will be delivered to the customer as soon as possible.

Here customers' needs play an essential role. Based on the objective of the deliverable, a Product Manager can use the below metrics to prioritize the work. Based on the impact and feasibility of the work, PM can decide which one to pick from the list. There are other metrics as well, such as based on revenue generation, the effort required, etc. to prioritize work and make informed decisions.

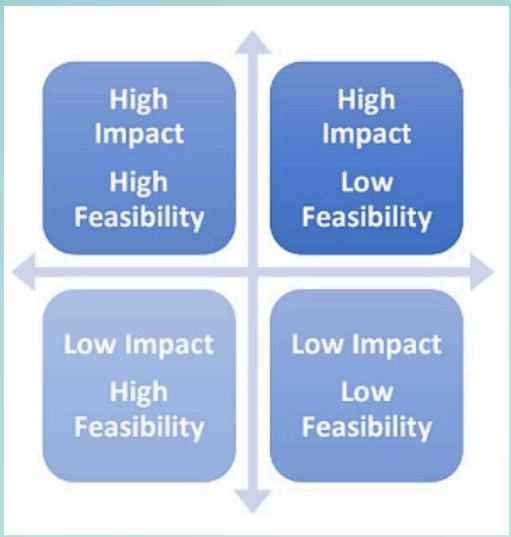


Figure 1.2 Impact vs Feasibility Matrix

FORMAL ROLES & RESPONSIBILITIES OF A PRODUCT MANAGER

As already discussed, the roles of product managers and their exact responsibilities change across different industries and different companies. It also depends on the product type, size of the product, life stage of the product, and organizational structure of the company. However, the essence of it remains the same. We will talk about the general duties of the product manager.

Market Research: As a PM, you need to understand the customer's requirements and the latest industry trends. Extensive market research will help you to get relevant information that can be used to understand the needs and make decisions. Further, Porter's five forces framework can be used to understand the industry's dynamics.

Vision and Strategy: Product vision is simply the crux of the product - why are we building this product, for whom, and benefit of it. It should have alignment with your company's vision. PM defines the vision and crafts a strategy to achieve it, and translates the strategy into short term and long term product roadmap. This product roadmap should be made known to all the stakeholders involved in the product as well as to the executive management team.

Product Ideation: Once the customer's needs have been identified, the next step for PM is to come up with new and innovative ideas to address the needs. Multiple brainstorming sessions with the team and iterative revision of ideas lead to the final selection.

Product Design & Execution: PM aims to provide a unique customer experience, so the design of the product is customer-centric. PM design the product in an innovative, simple, yet easy to understand way using wireframe or other mock-up tools. Further, PM writes user stories and communicate this information to the engineering team. The product manager continually supports the engineering team throughout the development and testing process.

Marketing Strategy: PM collaborates with the marketing team to create a marketing strategy for the product. PM shares all the required information about the product, customer's needs, and vision to the marketing team to create a unique marketing campaign. PM plans the marketing activities in the roadmap of the product. Based on the product's launch timeline, marketing team starts the campaign and reach out to target customers.

Product Launch: Finally, the day of launching the products to customers arrives. PM plans and coordinates with different stakeholders to launch the product.

PM needs to ensure that there is a customer service team to address and resolve all customer queries. The success of the launch is measured using specific metrics such as customer usage, customer satisfaction, customer retention, and other metrics.

TRAITS AND CHALLENGES OF BEING A PRODUCT MANAGER

A PM should be capable of creating a vision for his/her product(s). A PM needs to have a structured thinking capability regarding the vision, and the ability to lead the collaboration between people involved in the process, guide the execution and courage to take accountability for the product. Five qualities that make a great Product Manager are:

Strategic Thinking: A strategic mind-set is required to create a vision and build a roadmap for the product.

Decisiveness: It is the ability to make a decision. It is a desirable skill to enable drastic prioritization.

Empathy for the Customer: It is to understand the unstated needs of the customer, and their willingness to pay calls for the ability to empathize with the customer.

Lead through Influence: In most cases, no one reports to a Product Manager except for other product managers. The development team in general reports to a Project Manager or a Programme Manager. It is the ability to lead through the influence, which helps to get the work done and delivered by the team.

Excellent Communication Skills: Explaining the same thing to the CEO, to the marketing team, to the Project Manager, to the design team and the development team, calls for different sets of vocabulary and requires the Product Manager to have excellent communication skills directed at the specific requirements of the person.

No matter how exciting that may sound, like all roles, a PM role comes with its own set of challenges. Three such situations that a PM can face are:

1.Multiple, never-ending and different sets of demands and opinions from different teams disrupting the working plan of a PM.

2.The temptation to get too technical or too business-oriented, which results in losing the required balance.

3.The feeling of a lack of control that an unruly and unorganized team brings along can dismantle the composure of a PM.

KEY PERFORMANCE INDICATORS (KPIS) FOR A PRODUCT MANAGER

You can say it loud that your product is successful, or you have exceeded expectations. But no one is going to believe it until you prove it using data. So, the performance of PM can be measured using activities associated with product success, customers, and the overall process of product design and launch. KPIs help the PM to track the progress and identify risks. The following are examples of some of the metrics used to measure performance.

These KPIs will vary according to different product types and business requirements.

- Monthly Revenue – Month-wise revenue generated by the product
- Delivery-On-Time - Delivery of product or new feature on time
- Product Quality - Number of deviation from standard quality practices
- Customer Counts - Increase in customer over a specified duration
- Stickiness Ratio - Daily active user/ daily monthly user (DAU/DMU)
- Customer Retention - Number of customers stay with the company over a specified period
- Customer Satisfaction Score - Measures overall satisfaction of customers with the product or new features

PRODUCT MANAGEMENT VS PROJECT MANAGEMENT

Product Managers are responsible for the success of a product. Here, success is defined by KPI or metrics. The method by which they reach their goal is not defined. It is up to them or their team. On the other hand, Project Managers are responsible for accomplishing a project, not a goal. A project usually has a timeline and a budget as constraints. A Product Manager needs to have the skill of project management also, in order to manage the execution of the product.

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WHAT DIFFERENT COMPANIES EXPECT FROM A PRODUCT MANAGER

In this chapter let's look at the different kinds of PM's there are and what different companies expect from their PM's.

According to an article from Deepak Singh, there are mainly 5 different types of PM.

The Tech Product Manager
These PM's mainly solve the engineering problems, eventually they can end up as engineering managers rather than product managers. In this role one should focus on WHY you are building and WHAT you are building and leave the HOW to engineers.

Some advantages of this role are it's easy to gain trust and respect of engineers and PMs work well with the engineers.

Google and Amazon hire for these roles. Some of the examples for this type are Elon Musk and Marissa Mayer.

What should you focus on: Build good business sense and user empathy to see the big picture and what product to build.

The Designer Product Manager
PM's in these roles should strongly sense the future needs of consumers and focus on design and aesthetics.

Some advantages of this role are PMs empathize well with customers which helps them build much needed and beautifully designed products.PMs can understand the difference between stated and unstated goals design wise. Some challenges that PMs can face are as the design PM would not have much idea about coding it might create troubles when working with the engineering team because they have different styles of working.

Instagram, Facebook, Apple and Tesla hire for these roles. Some of the examples for this type are Steve Jobs, Brian Chesky and Joe Gebbia.

What should you focus on: prioritizing things in this role is really important so developing business sense helps. One should get really good at recognising flows in a product. Prioritisation is the key.

The Business Product Manager

These PM's are well versed with terminologies in business, finance and operations. They communicate really well and understand people around them.

Some advantages are they are able to think through and paint the bigger picture and lead teams. Some challenges can be because the understanding on topics like technology is limited it might create trouble when working with engineers, makes it hard for PMs to build credibility and respect with engineers. Ken Norton is an example for this category.

What should you focus on: One should start by focusing on the technology. Have an eye for detail and rigor.

The Data Product Manager

This is one of the emerging roles with the advancement of AI, ML. They mainly focus on personalisation, recommendation products etc., They make all their decisions based on data, they will gain influence for sound decisions they make over time.

Some advantages are being well versed with data and good analytical skills will help understand business and product, but relying on data so much sometimes is harmful because one loses to see the bigger picture.

Netflix, Amazon and Google would love PMs in this category. Sebastian Thrun is one of the examples in the category.

What should you focus on: PMs should concentrate on WHY, combining WHY with data will work wonders. Concentrating on design aspects along with data will help.

The Growth Product Manager

Growth for different organisations has different meanings. Some may find it hard to generate demand while some may find it hard to supply when they have demand, PMs in this category come up with ideas to solve these problems.

PMs in these roles are strong with data and communication. Some advantages of this role are by solving the problem they make a real impact in the company, clear metrics identification would help them move in the right direction. Some challenges would be though there are a lot of ideas to solve lack of quick experiments and right prioritisation will lead to low impact work. Chamath Palihapitiya falls in this category of growth product managers.

What should you focus on: Prioritization is important. User research and psychology will help you understand what to build and
why.

COMPANIES & WHAT THEY EXPECT

Amazon

Amazon is a American multinational technology company founded in 1994. They work on a wide variety of products. Amazon expects a PM to be excellent planners, have ability to develop long term plans, good analytical skills and good writing and communication skills. Amazon focuses a lot on good customer service, so empathy and understanding what customers need is a must. Amazon has 14 principles that drive their work and lead them to success, aligning your goals to them is really important for a PM.

In summary this is what Amazon wants:

- Understanding and utilising analytics
- Launch preparedness
- Customer centricity
- Communicating and collaboration

Microsoft

Microsoft is a American multinational technology company founded in 1975. It is best known for its software products. Microsoft expects its PM to be analytically capable, and understand technology. They are mainly interested in the big picture and how it solves customer problems. They want the PM to be able to think clearly, understand business and prioritise the important features to have maximum impact.

Paytm

Paytm is an Indian e-commerce payment system and financial technology company founded in 2010. At PayTM, they mainly focus on product development strategy. Product managers are expected to take control of every minute detail and be obsessed with technology and product innovation.They expect people to be good at communicating not just with the insider team but external customers as well.

Flipkart

Flipkart, founded in 2007, is an eCommerce platform that started by selling books before expanding to various other product categories. It has acquired companies like Myntra, PhonePe over the years. In 2018, US retail giant Walmart acquired a major stake in Flipkart for $15 billion.

5 competencies that Flipkart looks at in a PM candidate:

- Product thinking
- Problem-solving
- Leadership skills
- Domain and Tech
- Execution Excellence

Sprinklr

Sprinklr, founded in 2009, is a Software as a Service company that develops a Customer Experience Management (CXM) platform. It utilizes artificial intelligence, and combines different applications for social media marketing, social advertising, content management, collaboration, employee advocacy, customer care, social media research, and social media monitoring. Sprinklr PMs are expected to be client focused.

Media.net

Media.Net, founded in 2010, is a leading global advertising technology company. It got acquired by Beijing Miteno Communication Co. in 2016 for $900M. They connect advertisers to the websites that want to host the advertisements. They aggregate the ad space supply from different websites and offer it to the potential advertisers.

Media.net expects PMs to be analytically sound. Knowledge of basic algorithms and the ability to build analytical models is usually tested during the interviews. PMs should know how the advertising industry works and where exactly Media.net fits in the value chain.

Adobe

Adobe, founded in 1982, is best known for its media products like Photoshop, Illustrator, etc. They have a wide range of products both in the B2B and B2C segment. PMs in Adobe are expected to be highly creative and have a great story-telling ability.

They look for PMs who care deeply about the customers, mostly in creative fields, and work towards improving their experience. PMs should understand the fundamentals of business, ability to coordinate with multiple functions while working quantitative data.

In general, a Product Manager:

- should have a good understanding of the industry they are working in. They should be able to identify trends within the industry and use these insights to develop a roadmap for the product.
- act as a bridge between various functions (engineering, design, marketing, etc.) and therefore communication becomes vital skill in aligning these teams.
- is the voice of the customer. User Empathy, in terms of understanding the user, their needs and pain points, is essential for PMs to succeed. This helps PMs better understand the motivations behind users and 'why'/'how' they use the product.

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PM CONCEPTS

USER PERSONAS

For any product to hit it off in the market, it is necessary to achieve product-market fit. A vital element of achieving this is through understanding who exactly are the users of the product and what are their characteristics. This is where User Personas help product managers in understanding more about the product users. User Personas is a profile of a product's typical customer. These profiles help PMs understand the key traits, goals, challenges, responsibilities, and needs of the user. User Personas are usually created after user research tools like in-depth interviews and FGDs to emphasize better customers needs, goals, and behavior

A typical user persona includes the following (but not limited to):-

Age, education level, geographic location, or demographic status
How exactly do the users currently go about using the product
Motivation and goals for the usage of the product
Challenges, frustration, and pain points of the customer in product usage
Generally, a product targets a wide range of audiences, and hence most of the time, it is

It is recommended to come up with at least 3-4 user personas. Anything more than that might lead to lack of clear goal and focus. Here's an example of a user persona for a music streaming product like "Spotify" or "Ganaa". The customer segment chosen here are people who use the product during their workout session.

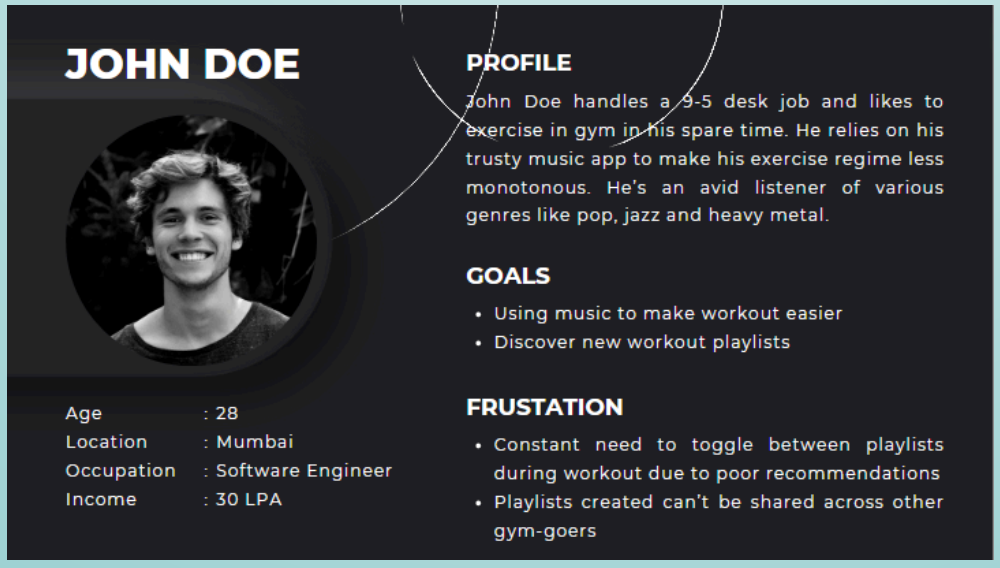

Figure 2.1
User Persona example

USER STORIES

As a product manager, being customer-centric is one of the vital qualities, and it's necessary to define product features from the lens of a customer. User stories are the smallest unit of product development and are written from the perspective of customers. They are primarily used to explain the product features easily to all stakeholders involved in it. Converting product features from a software perspective to a customer perspective helps the stakeholder focus on the end-user, motivating them to deliver the goals sooner.

As [a user persona], I want [to perform this action] so that [I can accomplish this goal]

As a fitness enthusiast, I want to be able to track my diet meticulously so that I can change my workout schedule based on that

As an office goer, I want to be able to turn on my water heater on the way home so that I can enjoy a refreshing bath immediately

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Communicating the requirements clearly without any loss is one of the critical jobs of any product manager. Product Requirement Document is an artifact to communicate to all the stakeholders about the product features. Any product development process includes stakeholders like engineering, design, marketing, sales, service, etc. Few of these stakeholders can be internal to the organization and others external. There will be multiple discussions with different subsets of stakeholders regarding various topics, but PRD is like a bible that glues all of them together and ensures every individual works towards a common goal.

PRD is beneficial in both waterfall and agile methodologies of development. PRD doesn't have any standard format and varies across organizations, but there are few common aspects. The product manager takes care of creating and making any updates to the PRD in the future.

Figure 2.2 PRD Outline

The basic outline of PRD contains:

Scope and Assumptions: This section gives an overview of the document and states the assumptions clearly so that they are no different interpretations.

Business Objectives and Goals: It states how the product or features fit the overall vision and what the product accomplishes for different stakeholders.

Use Cases and Features: We can start with use cases and derive features from them or vice versa. Each feature follows a detailed description, use cases, metrics for the relevant goal, and any exceptions specific to the feature.

User Interactions: Generally, after the finalization of PRD, UI/UX work starts with developing wireframes, designs, and relevant assets where all exceptions and failure cases are covered in detail. However, it is good to have high-level user interactions at this stage, which helps communicate better.

System Requirements: It contains details about the external environment required like operating systems, stacks or external libraries, etc., in the case of software products and Microprocessor, memory, etc., in the case of physical products.

Apart from the above major aspects, there are hygiene things like document name, version, release date, and change history. These might seem insignificant but play a major role in agile development where requirements are bound to change and to ensure every stakeholder refers to the latest version.

A/B TESTING

A/B testing is a process where two versions of a particular variable (color, size, etc.) are shown to different users to understand which version leaves a maximum impact and drives the metrics. It is commonly used by advertising and marketing professionals to understand which version of the ad or marketing mail resonated with most users. Product Managers started using it to build better products.

A/B testing starts with splitting the entire user base into two different groups randomly. If you have different customer segments, then there are chances that a group is dominated by one segment, leading to wrong insights. In that case, it is advisable to do randomization at each segment level. Later these two groups receive different versions of the same variable. In A/B testing, only one variable is changed, keeping the rest of the variables the same. However, there are advanced versions like Multivariate testing in which multiple variables are changed. The data from the target groups can be a click on a button, responding to a campaign, etc. After receiving enough responses from customers,managers analyze the data and decide which version of the variable should be used.

Today's streaming experience of Netflix is a result of numerous A/B testing. How many titles should be displayed in a row, what should be the cover image, etc., results from constantly running experiments. For the same series Money Heist, one day its professor, and the other day it's Tokyo on the preview.

Generally, large organizations use different software like Google Optimize, Adobe Target, VWO, etc., to carry out A/B testing. But individuals or small organizations who are not very particular about specific customer segments started using platforms like Linkedin and Instagram to decide on the color, the layout of a page, etc.

RICE: A PRIORITIZATION TOOL FOR PRODUCT MANAGERS

RICE(Reach Impact Confidence Effort) is the feature or idea prioritization tool co-developed by Sean McBride while working as a PM with Intercom. During the development of MVP or finalization of the ideas, prioritization is required as it helps decide what to work on first. But the dominating individual, emotional or gut feelings can add biases to the prioritization process. RICE helps in avoiding these biases to a significant extent. RICE stands for Reach Impact Confidence Effort, which quantifies the estimated value of features that allows sorting when deciding on the order.

The RICE Score is calculated using the below formula.
Detailing of what each letter stands for is as follows.

Reach: Reach estimates how many new or existing users the feature or idea can affect within a given period. For example, it can be 100 customers per quarter.

Impact: Impact tries to answer by how much the feature will affect the users. However, it isn't easy to understand the impact of a single feature on a customer's product purchase decision. Hence, having a clear goal like customer acquisition, engagement, etc., will help to understand impact better. To standardize, below impact values are used.
3 = massive impact
2 = high impact
1 = medium impact
0.5 = low
0.25 = minimal

Confidence: Confidence tries to offset the biases that went into different scores. It tries to answer how confident we are about the reach, impact, and effort scores. Below scores are used as a standard.
100% - High Confidence
80% - Medium Confidence
50% - Low Confidence

Effort: Effort tries to factor in the amount of time different stakeholders need to spend on the development. The unit of measurement is person-months or person-weeks. If five persons require four weeks each, then the Effort value is 20 person-weeks or 5 person-months.

HOOK MODEL

In today's world, adding more users to your platform isn't enough. Companies need to actively instill habits to increase user engagement with the platform to reap economic benefits.
Some of the most successful products like YouTube, Facebook have thrived by creating a solid habit-forming product making their users spend countless hours on their platform. Hook model by Nir Eyal in his book "Hooked: How to Build Habit-Forming Products" explains in detail how such a habit-forming product can be created.

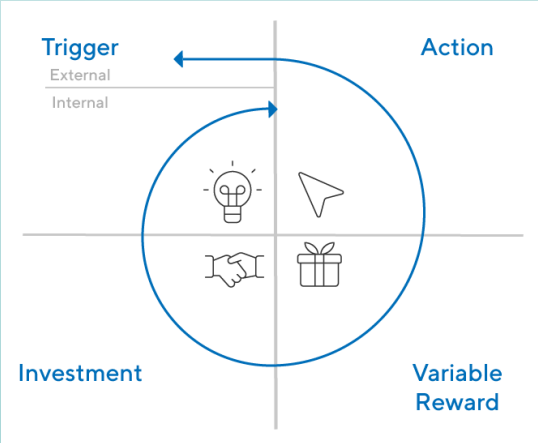

Figure 2.3 Hook Model

Trigger:- To form a habit-forming product, you need a trigger to remind/create desire among users. The triggers could be either external or internal. External triggers could be email or app notifications reminding the users to use the product. Internal triggers could be anything in your mind which subconsciously makes you relate to the product. Prolonged exposure to these triggers is the first step in creating habits among users

Action:- Once the trigger has been created, it is vital to maximizing the chances of the intended action being taken. Behavior designers constantly work on this to motivate users to follow the intended action. Google famously tested around 41 shades of blue color to identify which color maximizes the search link click rate.

Variable Reward:- Variable or Unpredictable positive outcomes have a chance of creating better habit-forming products than predictable outcomes. This is one of the reasons why people engage a lot with slot machines/loot-box microtransactions due to their unpredictable nature. Similarly, apps like YouTube, Instagram offer randomized but related contents to hook you to the product

Investment:- Investment involves two phases, where the first phase involves making the user go through the loop again. The second phase involves extracting some returns from the users. These returns could be the user spending money/inviting other people to the platform/providing data etc

WEIGHTED PRIORITIZATION MODEL

Usually, not all features identified are implemented in a stretch. Organizations are usually restricted by the time involved in feature development, employee resource, or the cost element attached to developing a feature. Hence a product manager needs to prioritize and select the right features out of the product roadmap. The weighted prioritization model involves choosing a list of metrics that includes but is not limited to Cost, Effort, Revenue, Impact, Customer Satisfaction, Risk, etc. The weights are attached to each criterion, and a score is computed for each feature. Based on the score, the feature is ranking in descending order, and top n features are taken for the sprint

Features	Complexity	Time to Market	Customer Value	Development Efforts	Total	Rank
Weightages	30%	20%	30%	20%		
Fitbit enabled tracking	8	9	6	9	7.8	1
Computer Vision to identify postioning	6	7	8	7	7	2
Training using Google glasses	5	6	9	7	6.8	3

Figure 2.4 Weighted Prioritization model example

HEART FRAMEWORK

HEART, by Google, solves the problem of measuring user experience success. It is a scorecard focussed on five factors. It can be applied to whole products as well as individual features.

Happiness – How do users feel about your product, usually measured via surveys, ratings, and reviews.

Engagement – How often are people using the product? It can be measured by looking at a key action (uploading a photo for Instagram) or other metrics like session length or visits per month.

Adoption – How many new customers do your product have that complete a certain action?

Retention – What percentage of users are leaving your product – this is called churn. The opposite of churn is retention.

Task Success – How easy or difficult is it for the user to complete a task in your product? It can be measured along with efficiency (how long), effectiveness (percentage completed), and error rate.

However, not all factors are applicable to all products. For example, engagement is usually not a concern for enterprise apps since usage is dictated by the business. So, to start using the framework, you need to define the goals of your product. A set of three goals is a good start and lets you focus on specifics. For a new social networking app, happiness, adoption, and engagement might be good goals to have. Then you need to define signals within your product that show that the goal has been met or not. For Netflix, searches in which the user does not click anything might indicate irrelevant results.

Once signals are identified, you can create normalized metrics (in terms of percentages or ratios) that measure the success of your goal. For example, under happiness, you may want to measure NPS (net promoter score) and Appstore ratings over time.

MOSCOW

MoSCoW is a prioritization framework it helps bucket features into four categories – must have, should have, could have, and will not have. Before applying the framework, key stakeholders and the product team align on objectives, factors used for prioritization as well as the items to prioritize.

Must-Have – Non-negotiable and mandatory
Will the product/release work without it?
Is there another way to accomplish the same thing?
Should-Have – Important, but not vital and add significant value
What value is missed out when I don't ship this?
Can it be a part of a future release?
Could-Have – Nice to have, will have a small impact if left out
Smaller impact than should have if left out
Will-Not-Have – Not a priority for the current time frame
Will not be a part of this release

AARRR

AARRR or pirate metrics is a set of metrics that measure user behavior (or user funnel) that is especially useful for product-based high-growth businesses. It was created by Dave McClure, a Silicon Valley investor.

- Acquisition – where do your customers come from? (SEO, SEM, Newsletters, Campaigns, etc.)
- Activation – what percentage have a happy first experience? (Signup flow, first action, etc.)
- Retention – do customers come back and use your product? (Churn rate)
- Referral – do customers like your product enough to tell others? (Campaigns, contests, etc.)
- Revenue – can you monetize this behavior? (Subscription, ads, etc.)

Based on the conversion rate at each stage and the revenue model, each user can be assigned a monetary value.

Category	User Status	Conv %	Estimated Value
Acquisiton	Vists site/installs app	100%	$0.02
Acquisiton	Stays 2 mins or does 5 clicks	50%	$0.05
Activation	Happy first visit (views 4 page, stays 4 mins and 10 clicks)	30%	$0.25
Activation	Signs up for free trial	10%	$2
Retention	Repeat visitor (5 visits in first 30 days)	5%	$7
Referral	Users refer others	2%	$5
Revenue	Referred users activate their accounts	20%	$3
Revenue	User generate minimum revenue	1%	$20

Figure 2.5 AARRR example

MVP

Minimum Viable Product is a product with just enough features to entice customers, is quick to build, and enables insights. It is a vital part of agile development where MVPs are constructed quickly and iterated based on user feedback.

MVPs can be done for the following reasons:

- Release a product as quickly as possible
- Test an idea on real users with a small budget
- Test acceptance of new ideas on an existing target market

Features to be included on an MVP are a result of the following

- User research
- Competitor research
- Cost vs Impact trade-off

- MVPs do not need to be full-fledged products as well. They can be high or low fidelity. Fake door MVP is just a signup page for a product that does not exist yet.
- Mock-up or 3D prototypes like high fidelity wireframes or mock-ups that can be shown to customers for their feedback.
- Wizard of Oz MVP is when a product looks like a real one but is powered by humans
- Piecemeal/Concierge MVP is similar to Wizard of Oz, but the customer is aware that it is powered manually

Before embarking on an MVP, you need to consider potential risks (reputational, leak of the idea, etc.), a time window for the MVP, and objectives.

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- https://www.atlassian.com/agile/product-management/requirements
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- https://clevertap.com/blog/google-heart-framework/
- https://www.productplan.com/glossary/aarrr-framework/
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- https://www.productplan.com/glossary/minimum-viable-product/

SECTION 2

TECH CONCEPTS

AGILE METHODOLOGY

Waterfall model – Software Development Life Cycle

Waterfall Model is one of the earliest Software Development Lifecycle approaches. It represents the linear sequential flow of the software development process. In this model any phase in the development process begins only after the previous phase is complete and the phases do not overlap. This type of model is more suited for small and less complex projects where goals are clear and minimum changes are expected during the course of the project. Shown below are the phases that a typical waterfall software development life cycle follows.

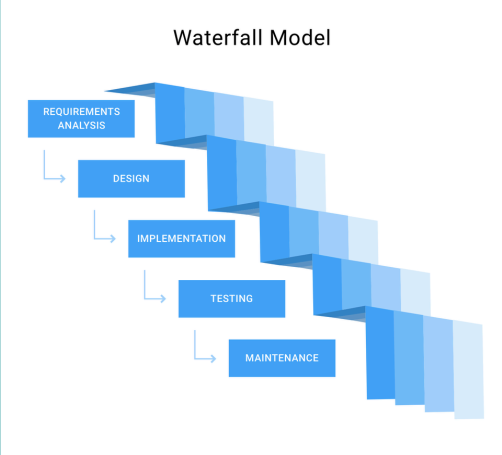

Figure 3.1 Illustration of the waterfall model

The waterfall model poses a number of drawbacks, and is not suitable for projects with unexpected and frequent revision. Few of the shortcomings of this model are mentioned below:

Not adaptable to change
It lacks adaptability at all the stages of development. A design flaw found in the testing phase can lead to a big fall back to the earlier stages of the process and can even lead to serious flaw to the legitimacy of the entire system. The rework and delay due to sequential processes can lead to late delivery of the product and can affect the competitive edge of the client over its competitors.

Risk of defects in the end phase
The testing is done at the end of the process and might result in delays, in case defects are found. The risk that a project might fail is concentrated towards the end phase of the cycle.

Absence of Client feedback mechanism
It does not have a mechanism to continuously collaborate with the client and incorporate the feedback of the client in between the development phases.

AGILE

Agile is the ability to respond to change. It is an iterative and time boxed approach to software delivery that builds software in increments from the start to the end of the project,

In Agile methodology, the collaboration between self- organizing cross-functional teams leads to the evolution of requirements and solutions. Agile development helps a cross-functional team to deliver value faster. It also helps in responding quickly to the customer needs. It enhances the quality and predictability of the work done.

Any development process that closely follows the concepts put in the Agile Manifesto is known as Agile development. The Agile Manifesto came into place when a group of fourteen software enthusiasts came together to devise a new way of working.

Agile Manifesto
We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

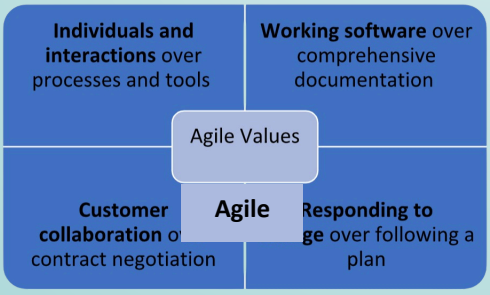

Figure 3.2 Agile Manifestto

Individuals and interactions over processes and tools
Processes and tools are time consuming and less responsive to change, which is why Agile gives greater importance to people over processes and tools.

Working software over comprehensive documentation
While required documentation is necessary, in the agile mindset, working software is the prime importance.

Customer collaboration over contract negotiation
Agile promotes customer collaboration and feedback throughout the life cycle.

Responding to change over following a plan
Agile works in sprints that are adaptable to change.

Agile acknowledges the value in the items on the right but it seeks to value the items on the left more while working on any project.

AGILE PRINCIPLES

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
The customer expects fast delivery of the product, and each delivery should add some value for the customer.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage. The market and businesses are continuously changing and it's almost impossible to predict exact requirements. Agile embraces the changes, even late in the cycle. The latest product gives a competitive edge to the customers over its competitors.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
It promotes incremental delivery of workable software. Frequent delivery leads to frequent feedbacks from the customers and less defects in the delivered software.

Businesspeople and developers must work together daily throughout the project.
This agile principle encourages organizations to increase the daily collaboration between the developers and the business, increasing the mutual understanding and respect. Communication barriers should be removed and teams should be motivated to interact with business people.

Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done. The motivated developers should be enabled with correct coaching, environment, and tools to do their work professionally and contribute to the team.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
This principle promotes that team members should have more of face- to-face communication rather than emails, messages and calls.

Working software is the primary measure of progress.
Working software at the end of each cycle is the primary measure of progress and if the working software is not delivered then no progress is considered.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.
This principle talks about identifying the velocity of the team and making everyone work collectively with a constant pace.

Continuous attention to technical excellence and good design enhances agility.
We should avoid creating technical debts and look for technical excellence and efficient designing of the deliverables.

Simplicity–the art of maximizing the amount of work not done–is essential.
This principle helps us identify the work which is not very important for the end user. Agile says that the work which doesn't add much value to the customers can be postponed or avoided.

The best architectures, requirements, and designs emerge from self-organizing teams.
Agile places a lot of importance in the role of self-organizing teams. It gives power to these teams to make decisions and allows management to trust the teams with the outcomes. It believes that the self-organizing teams are capable of coming up with the best architectures, designs and ideas.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.
A self-organized team tunes to itself and does the regular retrospective of its processes, behaviors to adapt and change according to the shifting realities. Retrospection helps in course correction.

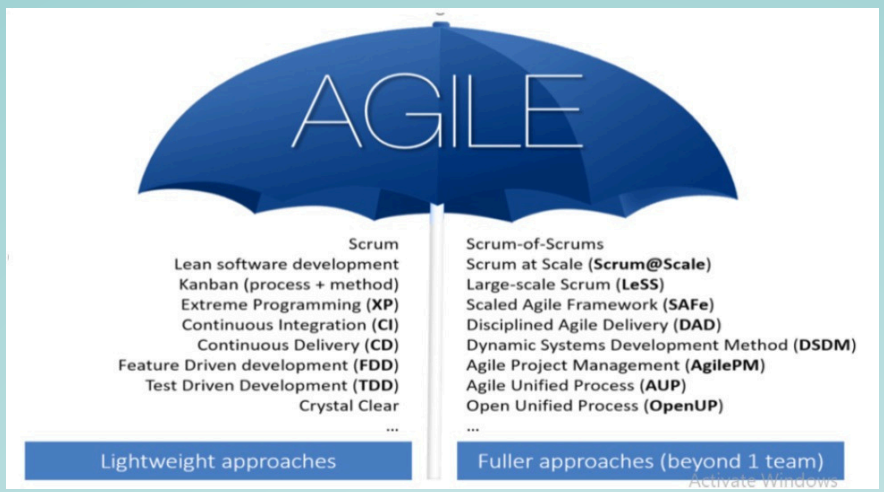

Figure 3.3 Illustration of some popular agile frameworks

Agile Frameworks

Agile is a development methodology and it requires a framework to implement its principles into a project. Here we shall discuss some of the most frequently used frameworks.

Scrum Framework

It is an agile framework used for developing and delivering complex projects. Scrum is lightweight, easy to understand, but difficult to master. Scrum is driven by five values that include Courage, Focus, Commitment, Respect, and Openness. Roles of the scrum team includes product owner, the development team, and a scrum master. Pillars of the scrum framework are transparency, adaptation and inspection.

Kanban

In the Kanban framework, the tasks are visually represented on a board called the Kanban Board. The board carries different states such as “to do”, “in progress”, “completed”, etc. to depict the status of each task. The board helps in avoiding bottlenecks and track the progress of all tasks in a single frame.

Figure 3.5 Illustration of the kanban framework

XP – Extreme Programming - It is one of the most sophisticated agile frameworks. It is used when the requirements are frequently changing. It comes with a set of engineering practices that should be followed in order to fully experience the capabilities of XP.

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USER EXPERIENCE DESIGN FOR DIGITAL PRODUCTS

DESIGN THINKING PROCESS

As defined by the Interaction Design Foundation, User Experience Design is the process of fabricating products that provide purposeful and relevant experiences to users. To achieve this, designers follow the process of design thinking. Design thinking is a problem-solving methodology through an iterative process, in which designers seek to step into the users' shoes, to understand their needs, goals, and frustrations, redefine the problem, and identify alternative approaches to solve the problem. Figure 4.1 illustrates the process of design thinking. The six stages involved will now be discussed in detail.

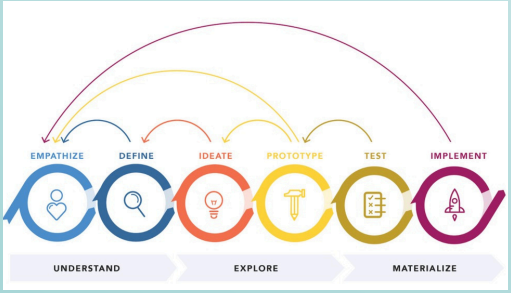

Figure 4.1 The design thinking process

Empathize

In this stage, one should try to get a better understanding of the problem. This will begin with secondary research through databases, books, journals, etc to understand the context and the current state of the art. At this stage, one can also perform a competitive analysis to identify the market gaps. This is followed by primary research. Through primary research, one should try to gather the information that can be used to fulfill the gaps identified in secondary research. Some of the tools that can be used to perform primary research are as follows:

- Ethnography research
- Focus groups
- Diary study
- Cultural probe
- Surveys, interviews and digital questionnaires
- Interviews of subject matter experts

Define

After completing the research, one can use the process of affinity diagramming (also called KJ method)

to understand the research data, and to capture insights, gaps, and identify pain points. It is a participatory process, and the whole product team takes part in it. The qualitative data from the research is broken into smaller chunks and is written on cards. The team members collaboratively organize the cards into groups of related information. The process results in identifying the opportunities for future studies, identifying the range of problems, and uncovering similarities among problems from multiple stakeholders.

Then, one moves forward to segment their users and develop user personas. As defined by Alan Cooper, "Personas are archetypal users that represent the needs of larger groups of users, in terms of their goals and personal characteristics." Personas communicate the product's goals and users' behavior patterns to the product team. One starts with identifying behavioral variables like activities, attitudes, skills, aptitudes, and motivations of research subjects. Then, one maps the subjects to these behavioral variables and identifies significant behavior patterns. For each persona, synthesize relevant goals and check for completeness and redundancy. Lastly, expand the description of attributes and behaviors for ease in communication among the product team.

A product team can prepare the following types of personas:

- **Primary:** Identifies the primary target user for the design.
- **Secondary:** They are mostly satisfied with the primary persona's design, but have specific additional needs that can be integrated without upsetting the product's ability to serve the primary persona.
- **Supplemental:** Their needs are completely represented by a combination of primary and secondary persona.
- **Customer:** They address the needs of customers, who can be different from end-users.
- **Served:** Though they are not users of the product, they are directly effected by the use of the product.
- **Negative:** They are used to communicate to stakeholders and product team members that these are the type of users that the product is not being developed to serve.

After defining the users, the product team moves forward to define a problem statement. This helps in defining the purpose of the design initiative, for both the personas and for the business. Following is an example of a problem statement.

"Company X's customer satisfaction ratings are low and market share has diminished by 10% over the past year because users don't have adequate tools to perform A, B, and C tasks that would help them meet their goal of G."

Implement

After prototyping and testing, developers take over the prototype for development. During this stage, it is essential to collaborate with developers to ensure that the workflow of the developed product is as planned, and that the aesthetics of the developed product matches with that of the prototype prepared earlier. A basic knowledge of CSS and Javascript is a plus for this type of collaboration.

Heuristics for User Interface Design

Nielsen and Morich developed 10 general principles for interaction design which can be used as broad rules to test the usability of an online product:

Visibility of system status

The users should be kept informed about what is going on using feedback within appropriate time. Eg: loading indicators.

Match between system and the real world

The phrases used in the system should be familiar to the user and fit their mental model. Eg: using appropriate icons as metaphors and using language which matches the real-world conventions.

User control and freedom

If users get into an unwanted scenario by mistake, they should have an emergency exit option. Eg: undo and redo.

Consistency and standards

A consistent terminology and paradigm should be used throughout the system. Eg: use either 'Add to basket' or 'Add to cart', but they should not be used interchangeably.

Error prevention

The system's flow and design should be such that it eliminates error prone conditions and confirms their inputs before they perform the action. Eg: checking if the password protocols are met before the user presses register.

Recognition rather than recall

The system should decrease the users' cognitive load by making actions and alternatives available. Eg: autocomplete in search.

Flexibility and efficiency of use

The system should cater to the needs of both, novice and experienced users. Eg: use of shortcuts by expert users.

Aesthetic and minimalist design

The system should not be cluttered and show only relevant information upfront. Eg: keeping details hidden under 'view more' hyperlink.

Help users recognize, diagnose, and recover from errors

The error messages should be useful. They should convey the reason for why error has occurred and how to overcome the issue.

Help and documentation

The system should provide documentation in the interface. The documentation section should be easy to search, crisp and focussed on the user's tasks.

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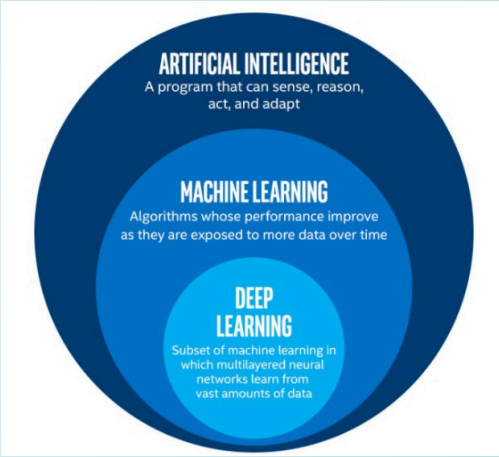
MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

INTRODUCTION

In the past decade, technology and computing have evolved significantly. Machine Learning & Artificial Intelligence have disrupted almost every industry. Traditional businesses were forced to adapt, change their business model, and develop technological expertise. In 2019, 7 out of the 10 most valuable companies, based on market capitalization were technology firms. They have replaced big oil and natural gas companies, which dominated the market for the last 50 years. Companies like Facebook, Alibaba, WhatsApp, Instagram did not even exist 20 years ago. With an exponential increase in the data being generated every second, utilizing it properly gives any company a competitive advantage.

Artificial intelligence is a field of computer science developed to give machines the ability to think like humans. It is a broader field that includes areas like Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Robotics, etc.



WHAT IS MACHINE LEARNING?

Compared to rule-based systems followed earlier, Machine Learning has the advantage that with more data, the system improves itself and gets rid of underlying biases. For example, in a bank, to sell a deposit product, they decide on potential buyers based on their savings account balance. Thus, if the balance is greater than threshold, then send the advertisement to the buyer. In the ML approach, we can look at the characteristics of people who bought the product in the past and target similar customer profiles to reduce the cost of advertisements and improve conversion rates.

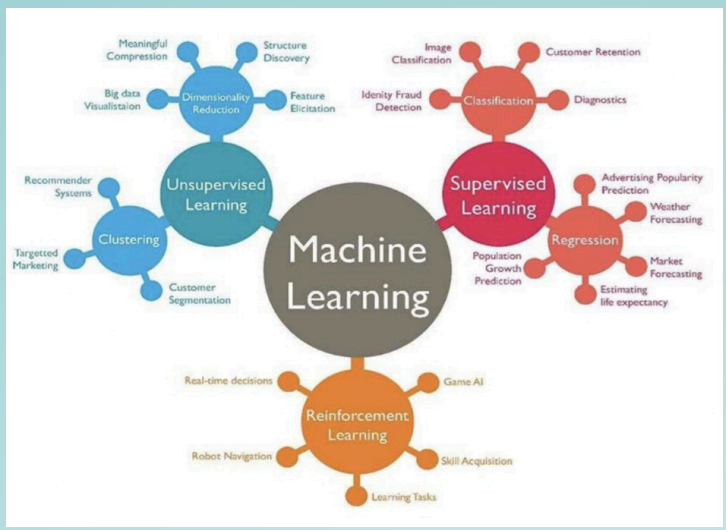


Figure 5.2 Machine learning and related areas

Machine Learning can be divided into three major areas:

Supervised Learning

Making a prediction based on a set where the output is labeled.

- Regression: To predict a continuous variable. For example, weather forecasting, indicating the price of a stock, etc.
- Classification: Classifying input into a set of predefined classes. For example, fraud detection, image classification, spam detection, etc.

Unsupervised Learning

Organizing/describing the structure of unlabeled data.

- Clustering: Finding similar groups between data. For example, customer segmentation, recommender systems, etc.
- Dimensionality Reduction: Reducing the number of features in the dataset.

Reinforcement Learning

To take action to maximize the notion of cumulative reward. For example, game AI is trained for several generations until it learns from its mistakes how it ought to take each step, so that it would lead to victory or survival.

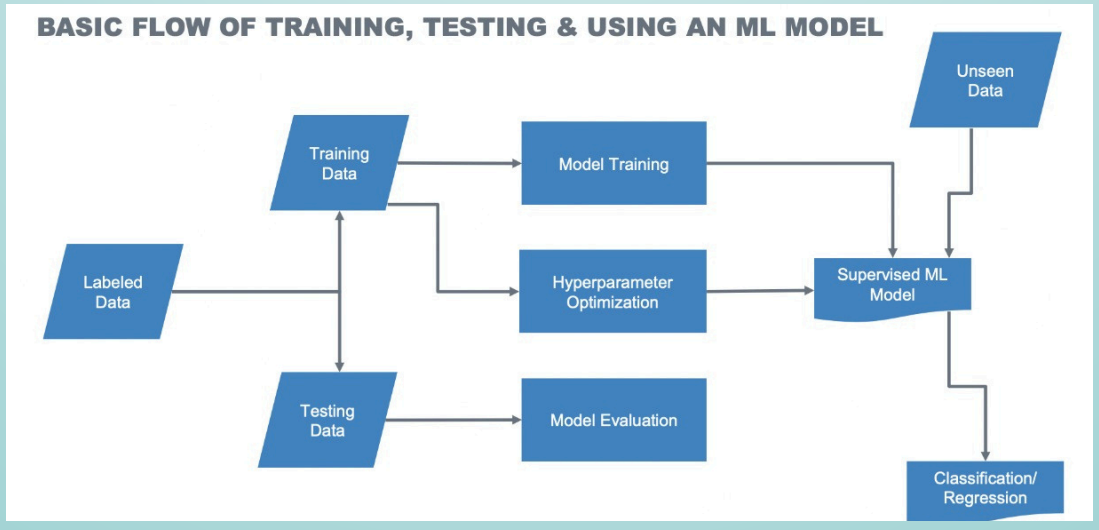


Figure 5.3 Training testing flow of an ML model

In the previous diagram, the critical step is to decide which model to use based on each algorithm's advantages and limitations. Below are few of the standard algorithms with their pros and cons.

Algorithm	Used for	Pros	Cons
Linear Regression	Regression	Easy to use and interpret.	Assumption of Linearity.
Logistic Regression	Classification	Convenient probability score for obs., interpretable.	Doesn't perform well if feature space is large.
Decision Tree	Classification	Intuitive Decision rules, handle non-linear features.	Can overfit the model.
Random Forest	Classification, Regression	Reduces overfitting, Robust technique.	Computationally expensive.
SVM	Classification, Regression	Identify Non-Linear patterns in the data, and can handle large feature space.	Difficult to fit the kernel and requires high computational time, resources.
K-Means	Clustering	Simple and efficient and easy to explain the results.	The number of clusters is to be selected in the beginning. The different initial partitions will lead to different clusters.

Another very important class of ML algorithms is recommender systems. Companies like Netflix, Amazon, Facebook, Google, YouTube, Spotify rely on recommender systems for their core product. As Steve Jobs said, “People don't know what they want until you show it to them.” Recommender systems help people identify their needs, which they are not aware of. It can be determined by looking at relationships characterised by ideas like, “people bought X products also bought Y products.”

TYPES OF RECOMMENDER SYSTEMS

Content-based: This approach identifies attributes/descriptive keywords from different items and recommends on the basis of similarity between two items. The drawback of this approach is that if there is no common attribute then the users won’t be recommended a particular item.

Knowledge-based: This approach is used in the initial stages where a user is explicitly asked for his/her preferences and creates a model to recommend the next items. Example: Netflix asks users to select three preferred movie/tv series of their liking.

Collaborative filtering: This approach utilizes both item attributes and user preferences, including the information obtained from other users’ ratings. Hence it is known as collaborative filtering.

We’ll now look as example of the above recommender systems:

Content Based

Figure 5.4 Example of content-based recommender systems

The table contains a list of movies and their attributes. It shows that Inside Out is an animated, non- marvel movie without any supervillain, which failed the bechdel test, and includes people from “Parks and Recreation”. Based on this information, if somebody had watched Inside Out, we want to know which among the three – Ant-Man, Avengers and Minions we should recommend. We take the similarity between the features of each movie and Inside Out. The highest similarity is with Minions, and hence Minions should be recommended. The drawback is that this approach doesn't consider user preferences and relies solely on the movie's attributes.

Knowledge Based

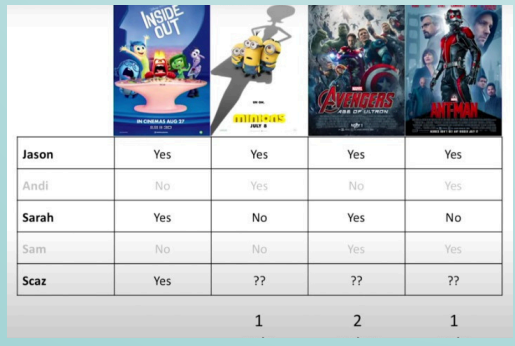


Figure 5.5 Example of knowledge-based recommender systems

The above example contains a list of users and information as to whether they liked a particular movie or not. If Scaz liked the movie Inside Out, we want to know which movie we should recommend that he watch next. We take a weighted average of the opinions of people who have similar reviews as Scaz does for Inside Out, and choose the movie with maximum votes. In this case, Ant-man has the highest votes. This approach's drawback is that it doesn't consider the fact that we have no information about the genre preference. We are not utilizing the movie-attribute information in this scenario.

Collaborative Filtering

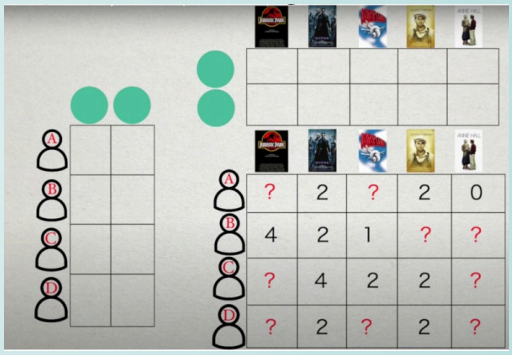


Figure 5.6: Example of collaborative filtering recommender systems- step 1

The matrix contains half-filled information with reviews that users have filled for different movies. The missing values correspond to movies that haven’t yet been watched by those users. The aim is to find the missing values. We will try to decompose the matrix into two matrices, known as latent factors.

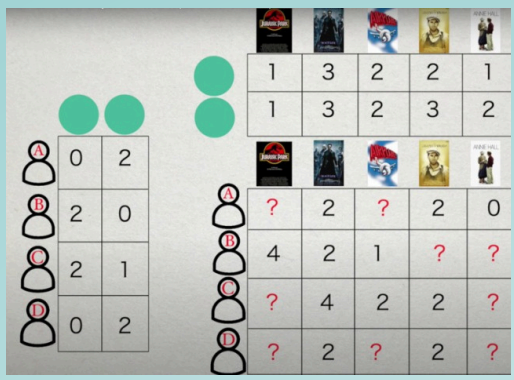


Figure 5.7 Example of collaborative filtering recommender systems - step 2

We start with random values in the two matrices and multiply them to get the information matrix. We calculate the difference between actual and predicted values. Finally, we find the optimal values for both matrices by using gradient descent, which minimizes the error between predicted and given information matrix. The algorithm used here is known as 'Alternating Least Squares'.

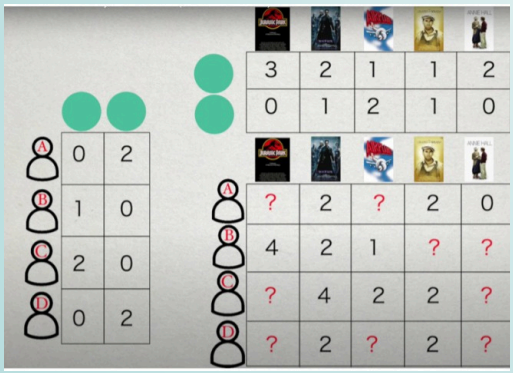


Figure 5.8 Example of collaborative filtering recommender systems - step 3

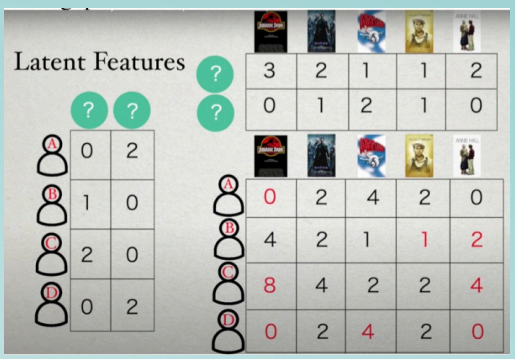


Figure 5.9 Example of collaborative filtering recommender systems - step 4

Using the two matrices, multiply and get the missing values. The final predicted values are the recommendation scores. This method takes care of user preferences and object attributes, and is a very powerful and simple recommendation algorithm.

NEURAL NETWORKS

All the algorithms described earlier worked differently from how humans learn. The goal of machine learning and artificial intelligence was to mimic our way of learning in computers. Neural Networks were developed in the 1960s but gained popularity in the current era, with increased computational capabilities of existing systems. Below is a diagram that compares the working mechanism of a biological neuron and an artificial neuron.

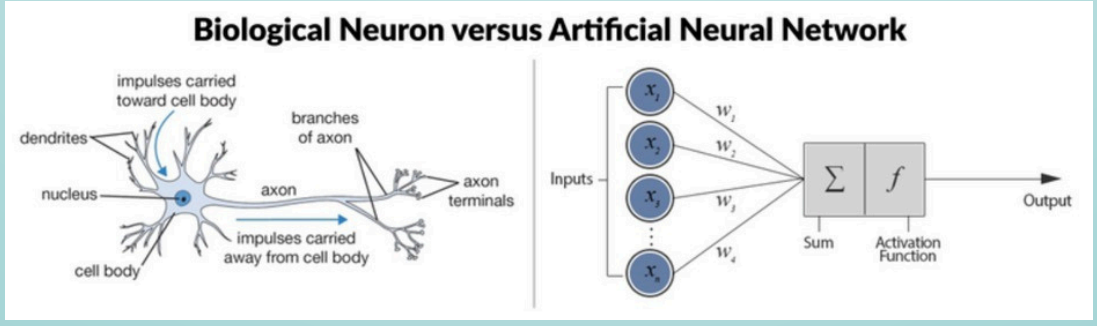


Figure 5.10 Comparison of biological neuron vs artificial neural network

If one puts their hand on a hot stove, the immediate action would be to pull their hand away from the stove. In this process, information that the stove is hot, is taken as input from dendrites to the nucleus, where it decides if it is hotter than some threshold; if yes, the neuron is fired. The axon carries the impulse or signal to take the hand away, till axon terminals. Similarly, in an artificial neural network, inputs are taken from all the input nodes, and weights are assigned to minimize error. Based on the value of the weighted sum and activation function, the output is predicted. An activation function is simply a function such as y = tan(x), that is intended to transform the input.

Each neural network has a different architecture, and is used for different purposes. However, neurons, artificial networks, and hidden layers are common, which extract hidden features.

The more the number of layers, the more capable the ANN in extracting features of a higher dimension.

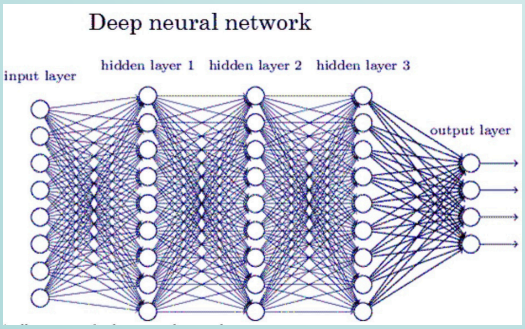


Figure 5.11 Illustration of a deep neural network

The deep neural network has more number of hidden layers, which help in feature extraction. Compared to classical ML models, where users have to derive or specify features from the dataset. In Deep Learning, the model itself selects and creates features. We shall see a specific type of Deep Learning Neural Network, a state of the art network, known for its future potential.

GENERATIVE ADVERSARIAL NETWORKS

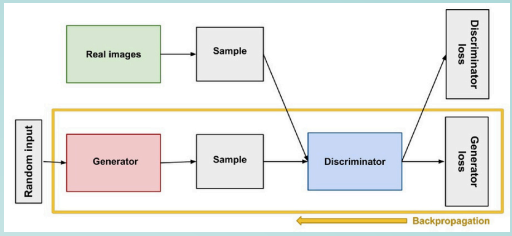


*Figure 5.12 I*llustrative diagram of a generative adversarial

It is a special type of neural network where we have two optimizing functions instead of one, and both of them compete with one another. We have a generator network and a discriminator network, with their loss functions, and the task is to reduce the loss. Initially, the generator is provided with random input, and hence the discriminator can easily distinguish between real and fake images. The generator loss will be too high. After several iterations, generator loss is minimized, and it creates a more and more realistic image. In a nutshell, both neural networks have achieved state of the art by letting one machine compete with the other. The final output is two networks, one to generate fake images and the other to detect. Both of them can be used independently.

Conclusively, the product manager should understand what different tools they have available, and how to improve the product by applying ML and AI. The following framework provides an idea as to whether you should apply ML to the existing product or not:

- Does it fit the product goal?
- How does the product revolve around the ML?
- How should the product start using ML?
- Benchmark to compare it with?
- How quickly should the product change?
- What interactions, actions, and control do users have?

Several innovative solutions like Tesla self-driving cars, sentiment analysis, and object detection in robotics and the futuristic store – Amazon GO, wouldn't have been possible without an understanding of advanced algorithms and where these tools could be used. The expectation is not to code, but know what requirements to ask of the technical team members.

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- https://www.youtube.com/watch?v=njKP3FqW3Sk&list=PLtBw6njQRU-rwp5__7C0olVt26ZgjG9NI&index=1&ab_channel=AlexanderAmini
- https://towardsdatascience.com/introduction-to-recommender-systems-6c66cf15ada
- https://towardsdatascience.com/understanding-confusion-matrix-a9ad42dcfd62
- https://www.youtube.com/watch?v=ImpV70uLxyw&ab_channel=Luis Serrano

AUGMENTED REALITY & VIRTUAL REALITY

INTRODUCTION

Augmented Reality and Virtual Reality are two particularly popular buzzwords. They have been a part of popular culture since at least 2012- 13, which is quite a long time in the tech-world. However, clarity about their functioning and use cases is still pretty limited and niche. Frequently, these two terms are used interchangeably as well. However, there is a significant difference between both the technologies.

Augmented Reality (AR) refers to observing the real-life environment with a digital augmentation overlaid on it. The fundamental idea here is to enhance real-life experiences with relevant content, information etc. ("What is Augmented Reality", 2019)

In Augmented Reality, we are using a computer to decrypt, or understand the live video stream from camera. By understanding, we mean identifying & recognizing the objects in the video and using the contents of the video to create a context. The computer then renders the relevant 3D graphics and superimposes it on the live camera feed to enhance a user's experience of the real world.

In **VR**, we are creating a parallel reality itself. We are creating an entire simulated environment for a complete, immersive experience.

A user is placed inside a virtual environment, which in turn tries to trigger and engage (i.e. simulate) as many senses as possible. The ultimate motive is to create an all- round experience with complete self-immersion. The only limits to virtual reality (VR) are the availability of content and cost effective computation power. As is apparent from the definition itself, detailed graphics are an important part of a powerful and convincing virtual reality (VR) experience, and hence the requirement for high computation power (Bardi, J. 2019).

POPULAR PRODUCTS & BRANDS

AR has a wide range of applications, and hence its product set is also diverse. Examples of AR include multiple app based software products like "Pokémon Go": an AR based smartphone game that became a viral sensation in 2016-17. Similarly, the "IKEA Places" app by IKEA- a furniture retailer has an AR feature

which allows customers to visualise the selected IKEA product in their homes to confirm product fit. Amongst hardware-based AR products, "smart glasses" is the most popular category, which is led by brands like Meta (with their Oculus series), Toshiba ("dynaEdge") etc. The "smart glasses" category is highly fragmented with no clear winner.

VR products can also be segmented into Hardware and Software. VR headsets, popularly called "Head Mounted Display" (HMD) constitute the most popular hardware product category in VR. HMDs are further classified into three types:

- **Category 1:** VR headsets to be used in combination with Smartphones.
- **Category 2:** VR headsets to be used in combination with PCs or Consoles.
- **Category 3:** Standalone VR headsets.

Each category has a different source of computation power. For example, in category 1, smartphones provide the computation power, while in Category 2 PCs or consoles provide the same. In category 3, the onus of computation is embedded in the headset itself. Generally, category 2 and category 3 HMDs are also accompanied with "hand controllers", which improve motion tracking and therefore, the virtual device experience. The major brands operating in the VR headset space are Sony PlayStation, HTC Vive and Facebook's Oculus.

PRICING AND PRODUCT FEATURES

AR devices, having varied applications, are meant for long duration use, to the tune of an entire day at a time. As a result, the make or break features include the device being light in weight and having long-lasting batteries. Google Glass, for example weighs around 46 gms and has a battery which is sufficient for a typical day's use. However, it costs the customer around $1500.

VR devices on the other hand are typically used for short durations, in the range of 2-3 hours of continuous use. They weigh more, in the range of 400-600gms and have less battery life, to the tune of 3-4 hours. With lower battery life and heavier weight, however, comes the benefit of a lower price tag compared to AR devices. Pricing starts from $120 and touches $600-700. The inefficiency in product design (battery life and weight) are the major bottlenecks in VR and provide immense scope for improvement.

HOW AR WORKS

Computer Vision is the most fundamental building block of AR technology. In this section, we will first introduce Computer vision and then further explore the working of AR technology.

Computer Vision Basics: The goal of Computer Vision is "to use observed image data to infer something about the world" (Browlee, J. 2019). Given an image (or video stream – which is again a set of image frames), what can we understand about the scene being portrayed in that image? Does it have an object, what kind of object, how many such objects are there etc. These are the key questions that computer vision focuses on. It is an extremely advanced research field and can be further broken down into sub fields like object

classification, object verification etc. We'll delve a little deeper into object classification, to develop a clearer idea of how computer vision works.

We will try to understand the underlying theory of how object classification works through the character recognition problem.

The problem statement for us is: given a set of images containing only two characters: "c" and "e", how can we develop an automated algorithm to classify each image as either the character "c", or character "e"?

The solution here is broken into two parts: the training phase and the testing phase. The two possible outcomes ("c" or "e") are called classes. Any given image will belong to either class "c", or class "e".

In the training phase, a labelled data set is passed through the "feature extraction algorithm". A labelled data set is a set of images, where each image is labelled as either class "c" or class "e".

In the testing phase, we use an unlabelled dataset. The only information that we have about the unlabelled data set is that each image in the data set will either be of class "c" or of class "e". Unlike the labelled data set, we don't know beforehand whether a given image is of class "c" or class "e". However, we now have a classifier in place. We will pass the unlabelled data through the same feature extraction algorithm used earlier, thus deriving a feature vector Y. We then compare feature vector Y with Xc and Xe, to identify which one it is closer to. If it is closer to Xc, our algorithm will classify the image as Class "c". If it is closer to Xe, our algorithm classifies it as Xe. In this example, since only two classes are used, not belonging to class "c" implies belonging to class "e". But, in case of 3 or more classes, the same would not be possible.

Note: This is a very simplistic model. Advanced algorithms which are used for classification are much more sophisticated.

AR technology works through the following steps:

- First step is Computer vision, where a computer understands the content of the video streaming from camera feed and generates a basic context for the world around the user.
- Second step is Rendering. Once the computer has understood what the camera (and hence, the user) is looking at, it searches for the relevant digital content and then super impose it on the camera feed, so that it looks like a part of real world and enhances the experience.

AR is a real time application & hence; both the above steps have to be performed on every frame that the camera captures. iPhone 11 has a camera which captures 4k resolution frames at 60fps, which gives us

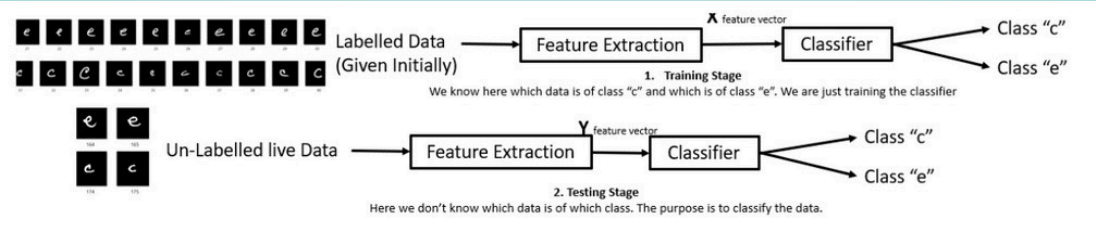

Figure 6.1 Basic character recognition algorithm

around 16-17 millisecond per frame to complete both the above step for a 4k frame. Considering the computational speed of modern processors, 16-17 ms duration gives developers enough bandwidth to design sophisticated AR algorithms.

HOW VR WORKS

In the case of VR, it is a three-step process:

- Step 1: Video is sent from PC (via HDMI cable), or smartphone slotted into headset, or from the graphics controller in headset itself, for a standalone VR headset.
- Step 2: VR headsets either send 2 feeds on a single display, or have two LCD displays, each for one eye.
- Step 3: VR headsets have a lens assembly between the user's eyes and the display, which reshapes the picture appropriately for each eye and creates a 3D image.

We are just trying to mimic the way our eyes work. There is a minimum requirement of 60fps from the VR headset for a convincing experience. The modern devices have capabilities which far exceed this limit. Oculus gives 90fps level performance, while Sony's PlayStation VR gives 120fps level performance. Another spec. to focus on for VR headset will be field of view. Ideally, VR headset should provide a 360-degree display to mimic the real life, but that will be too expensive. Most of the VR headsets have 100 to 11o degree field of view which is enough for human capability.

Tracking is an area where VR headsets try to differentiate themselves from each other, in order to justify their premium pricing. VR headsets have sophisticated tracking sensors, which are essential to create a quality experience. Tracking of the user's head is done via internal components like gyroscopes, magnetometers and accelerometers. Many headsets have a set of external sensors like LED lights and cameras to further improve the tracking accuracy. Image data from the headset camera is used to create a 3D model of the room and detect external landmarks like stairs, corners in the room etc. The earlier mentioned computer vision techniques are used here. Rotational and linear acceleration data from sensors in the hand controllers is used to track the user's hand movements.

Latency is an important performance parameter in VR. Head-tracking tech needs low latency to be effective - 50 milliseconds or less or we will detect the lag between our head movement and the VR environment changing. The Oculus Rift has lag of around 30ms.

FOVE VR headset has implemented eye tracking as well, using an internal camera and infrared sensor that monitors the user's eye. Even keeping one's head still, while moving only one's eyeball, will lead to the VR environment changing.

INDUSTRIAL APPLICATION

AR and VR headsets are expanding their market by creating enterprise use cases. There are multiple companies using AR smart glasses in their supply chain to improve their productivity. Ford designers use Oculus Rift to visualise and experience the interior and exterior design of their cars. Lockheed

Martin's engineers use AR glasses to get real time visuals of their aircrafts and overlay images with instructions in a real working environment. The company has realised an annual savings of $10 million in its product line (Source: Statista).

Vital Enterprise has created an AR software which makes instructions available in a hands free and voice-controlled format. Assembly and maintenance engineers can access instructions using AR glasses instead of reading long pdfs of instructions.

"Focus is shifting from talking about technology benefits **to showing real and measurable business outcomes**, including productivity and efficiency gains, knowledge transfer, employee's safety, and more engaging customer experience" – **IDC report**

INDUSTRY STATUS AND PRODUCT MANAGER ROLES

Figure 2 shows the current industry size and the forecasts till 2031. The market size is expected to increase significantly, from $23.73 billion in 2024 to $214.82 billion by 2031, at a CAGR of 31.70% (source: www.verifiedmarketresearch.com).

Overall, the industry is moving away from hardware based technological improvements towards software, with a focus on creating enterprise level use cases, while creating more content on play stores. Customer enterprises are also in the process of exploring how to use AR-VR for productivity related improvements in their value chains, such as production supply chains. PM roles in customer enterprises are extremely generic and exploratory, while in AR-VR manufacturing enterprises, PM roles are more software focused.

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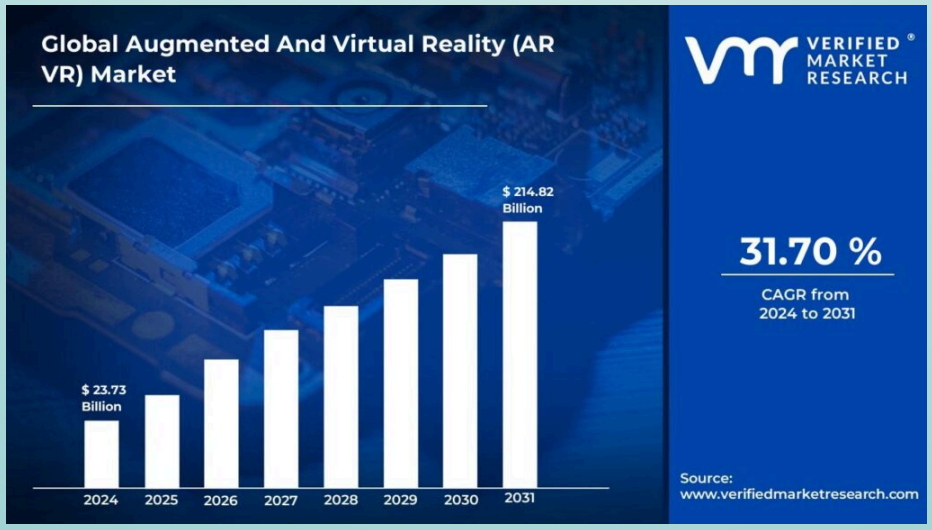

Figure 6.2 AR-VR market size and 2020 forecasts

BIG DATA

WHAT IS BIG DATA

Data is the most essential component in decision-making and serves as the raw material for accountability. Big Data refers to data which is so large, generated at such a pace and is so complex that it is difficult to process and use it with traditional methods. Major improvements in processing power, storage density and internet access with cloud have led to a rapid evolution in the field, making Big Data accessible to any company.

Big Data was characterised in the form of three V's by Doug Laney, an industry analyst back in the 2000s:

- **Volume** refers to the large size of the datasets from many sources ranging from transactions, IoT sensors, audio-visual data, social media mining and more.
- **Velocity** refers to the high speed of data generation. With adoption of IoT, data comes at an unprecedented speed and is handled in real time. Eg. RFID tags, sensors and smart meters.
- **Variety** refers to the diversity of data generation sources and formats. Source can be structured, numeric data to unstructured text documents, emails, videos and financial transactions.

Big Data can be classified based on the source that it is coming from, or based on the form in which it is coming.

Internal Data
Data generated within the operations of a business. For example, transactions, inventory, customer data etc.

External Data
Data sourced from external sources.For example, traffic, census, industry data etc.

Structured Data
Data which is highly specific and is stored in a predefined format. For example, geo-locations, employee details, marketing campaigns etc.

Unstructured Data
Data stored in varied forms, which cannot be analysed in a fixed form. For example, social media, blog comments, video/image etc.

Nowadays, the concept of horizontal scaling (more machines working parallelly) is used for intense data storage or processing. However, this strategy incurs an additional penalty of complexity. Most cloud applications are on distributed systems (network that stores and shares data on more than a

single node at the same time). The CAP theorem states that while designing a management system we can only choose 2 out of the following 3 characteristics – Consistency (all clients see the same data across nodes at the same time), Availability (any client making a request gets a valid response all the time) and Partition Tolerance (cluster continues to work despite any number of communication breakdowns between the nodes in a system).

Big Data Architecture

The architecture components of a big data system typically consist of four layers:

- **Sources Layer** – This is the layer where we source the data from. This is different from where data is originated, as the Sources layer refers to data which has already been recorded and stored in the form of data warehouses, Database management systems (SQL or NoSQL) etc.
- **Storage Layer** – Here, the data is taken from the source layer and stored in a suitable format to proceed with analysis. The format is decided based on the requirements of the data analytics tool.
- **Analysis Layer** – In this layer, there are various analytics and business intelligence tools which extract insights from the stored data.
- **Consumption Layer** – This is the final layer of the architecture which interfaces with the end user. The output of the analysis later is presented in visual or dashboard type of formats.

Big Data Technologies

The platform of Big Data is like an ecosystem. There are three broad components associated – Data Management, Data Analysis and the Infrastructure side.

Data Management – Within data management, there are ingestion platforms like Flume and Sqoop, through which data is transported from various sources to a storage medium. Then there are storage technologies like Hadoop and NoSQL. They help in storing the data in the form of clusters, with a node assigned, that can be accessed from any part of the system. There are also cluster administration technologies like Ambari and Chukwa, which help in managing and monitoring clusters. Finally, we have data governance technologies like Eagle and Falcon,

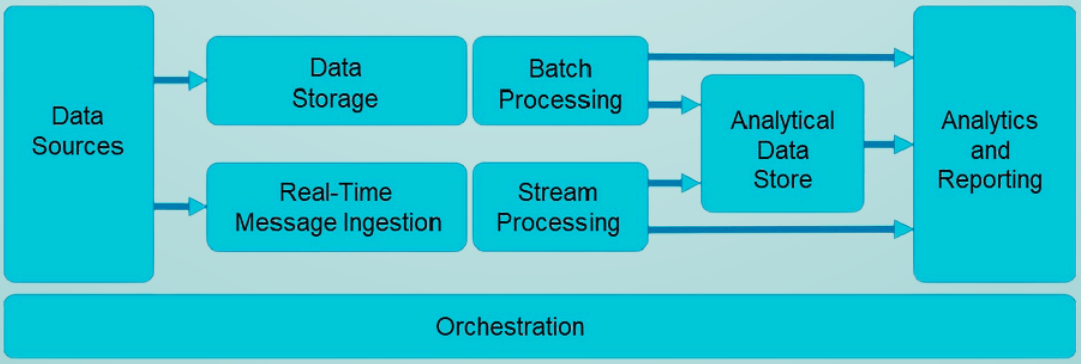

Figure 7.1 Big Data Architecture

which help in controlling the quality and security aspects of data.

Data Analysis – For data analysis, there are query technologies like Hive, Impala and Drill, which provide interactive query support for the stored data. There are batch processing engines like ETL, Pig and Oozie, which help in scaling out computations, in order to handle a large volume of data. Unlike real- time processing, however, batch processing is expected to have latencies (the time between data ingestion and computing a result). There are also data stream engines, which are an improvement over batch processing. They process data over rolling windows or most recent records. Examples are Kafka and Spark. Finally, there are advanced analytical technologies like Mahout & Spark which apply ML to Big Data to perform advanced computations.

Infrastructure – Infrastructure refers to the medium through which these technologies are stored, provided and accessed. These are mainly cloud services like AWS, Azure, GCP, etc. Figure 2 shows Microsoft's Big Data ecosystem.

B2C Applications

Major companies using Big Data in their operations can be classified based on their industries.

- BFS – Banking and financial service sector includes insurance providers, banks, credit card providers, etc. Big Data is used in insurance and trading operations, identifying and detecting fraud, analysing portfolio and valuations. Data security and governance is critical in this industry.
- Retail – This includes sales of goods and services and use cases across supply chain, vendor management, analysing shopping baskets, and other service based categories.
- Healthcare & Life Sciences – This includes biotech companies, research institutes, health insurance providers, who use big data for pharma R&D, disease management, drug efficacy etc.

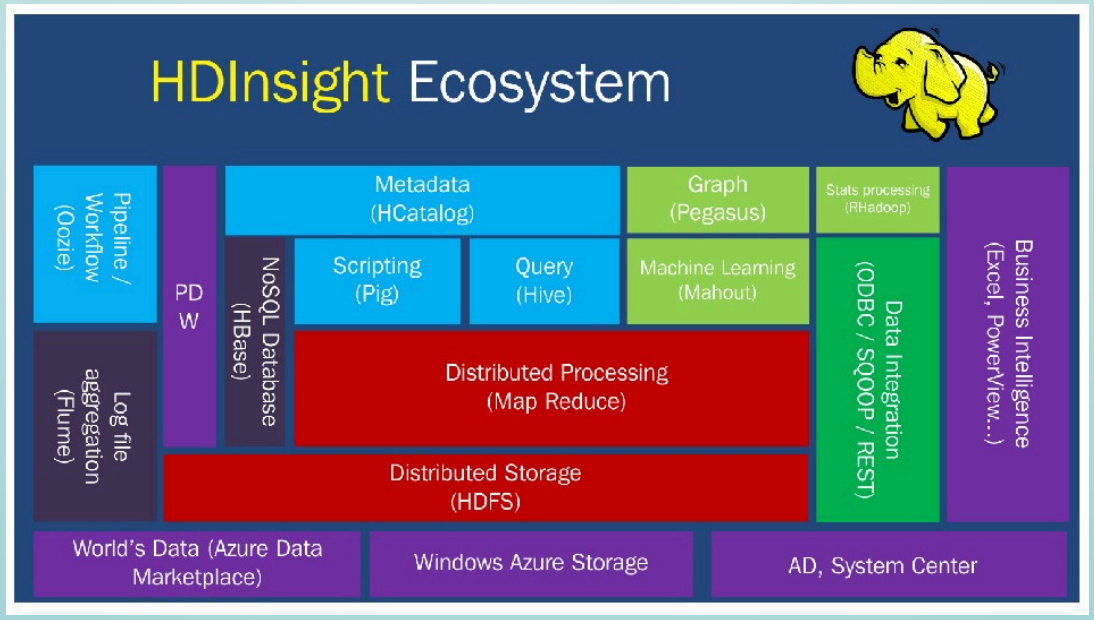

Figure 7.2 Microsoft HDInsight ecosystem

ICT – This includes companies that work on software and information systems, and other technology based companies. Use cases include data analysis.
Media & Entertainment – This includes companies working in print media, television, radio, gaming, and online advertising. Most commonly big data is used in delivering content and optimizing workflows.
Telecommunications – Telecom service providers and operators use big data to improve customer service, churn management and for optimizing their network.
Government – This includes all government affiliated entities such as intelligence agencies, defence & military. Some of the most common use cases are Cybersecurity, defence, legal intelligence, disaster preparedness, & law enforcement.

Big data analytics can address and improve many issues in B2B industries – operational, strategic, financial, etc. Consider the example of the power sector. The industry infrastructure is becoming smart, with built-in processing, connectivity, and sensing capabilities. Technology systems like electric vehicles (EVs), internet of things (IoT), smart homes, grid management systems, and others are likely to interface with utilities and provide them with potentially valuable data. Currently, there are few players in the world who have actively adopted big data analytics.

Big Data analytics help **power companies** avert turbine failures – By utilizing predictive models, companies can study various factors causing equipment failure and then perform maintenance activities in time, leading to cost savings. Some examples include:

- Schneider Electric's PRiSM, a big data software solution, diagnoses equipment issues in advance and provides early warnings. Southern Company in the U.S currently uses the system.
- GE's Predix platform uses advanced predictive analytics to identify patterns from datasets.
- Enel, an Italian MNC, active in electricity generation and distribution sectors, uses Predix platform to achieve the following 4 goals – increased performance, increased reliability, optimized output, and reduced costs.

Big data plays a large role in **utilities consumption management.** Utility companies use a demand response strategy to spread the energy consumption from the peak hours of the day to other parts of the day when demand is lower.
The national grid in the U.K has partnered with Open Energi to explore the use of automated demand response in stabilizing the grid. The focus was on frequency control which requires fast response from the system.

Big data can be used to optimize **electric vehicle** charging. EVs generate datasets of various parameters and from various sources which can be collected via vehicle- to-grid (V2G) charging systems. This data can then be analysed to develop new effective charging station policies, develop smart charging algorithms, solve the issues of energy efficiency, and even calculate the required capacity of electrical distribution systems to manage the future load.

PM & BIG DATA

Organisations usually have data scientists leading a team of data analysts for all the big data operations. By creating a separate team, businesses sometimes create yet another 'silo', which is isolated from the rest of the functional groups. For a product manager, big data can be an important tool in a typical product development cycle. We explain it through the Design Thinking method:

Empathize – This phase is all about understanding the customer - who they are, their pains, problems, attitudes, and what their emotional goals are. PMs can look at website or app analytics, observe the user journeys, understand the heat maps and where users end up, to gain insights.

Define – In this phase, a PM defines the problem that they are going to solve. Data visualization tools allow PMs to isolate pain points and attach root causes to them. It is also important to chart out different user personas, where the product positions itself vis-à-vis competitors to segment users.

Ideate – Here, PMs come up with a bunch of different ways to solve the problem. It is necessary to evaluate different ideas. Large amounts of historical data (internal + external), along with the data from the empathize phase can be used to assign weights to different parameters and shortlist the most relevant ideas.

Prototype – The ideas are turned into an MVP in this stage. As they build these MVPs, Product analytics can inform PMs about the features that are working and those which are not. Analytics help in creating a product roadmap, which can tell where the product is currently, where they want it to go and how to get there.

Test – Prototypes are tested in front of users at this stage. Usually products are released into apps or websites for A/B testing, hence app/web Analytics data gives a broad user feedback. Nowadays, social media mining helps perform sentiment analysis, and gauge the perception of users as well.

THINGS TO LOOK OUT FOR

Personas – Understand the end user of the product. Different groups will be at different levels of data literacy. So, possible groupings could be: data producers, data consumers, data maintainers etc. These could be further divided into different functional roles and use case scenarios. For example, data producers could be divided into engineers, executives (web based use or phone based), etc.

Analytics – This involves the application part of the data and extracting insights from it. One must be cognizant of all the steps involved between storing data and retrieving it to perform analysis. One must consider how the data will be retrieved and processed, and consequently what the requirements of it are. There are multiple companies offering ready to use solutions, and some of these can be outsourced as well.

Visualization – To make best use of the data, present it in an intuitive and user-friendly way. A PM must understand user personas and demographics, their mindset when using data and the accompanying form factor.

Storage – While a PM does not define the technical requirements, he/she is responsible for defining the use cases and business model. The data type used will impact the kind of database to use. For example, for a structured database a relational database like an SQL Server would work, but for unstructured data, one would use a schema less database such as MongoDB or Hadoop. A PM's job is to define clearly the use cases and then convey those to the engineering team. Similarly, on deciding where to store data, on cloud or on-premise, a PM must analyse the ROI for each approach.

Multiple Access Points – There are multiple products that different people prefer, and it is difficult to make a one-stop tool. Hence include import and export capabilities and other migration features that allow the user to transfer data between systems through various methods, and not restricted just through the user interfaces (UIs). Creating an open API is one way to do so.

Security – Security is a big concern for various stakeholders, since we are dealing with private and sensitive data. As a PM one must ensure that security is kept a priority since the beginning of the development process and is incorporated in all features of the product, across the full stack, right from the database to the UI and APIs. Correct quality control measures should be in place and bugs removed in frequent tests.

PRACTICE CASE STUDY

Sample – B2B social media data analytics process in a firm like Sprinklr

The steps involved in this process would be – Identifying data sources, Capturing data, Processing data, Presenting findings, and finally Generating Insights.

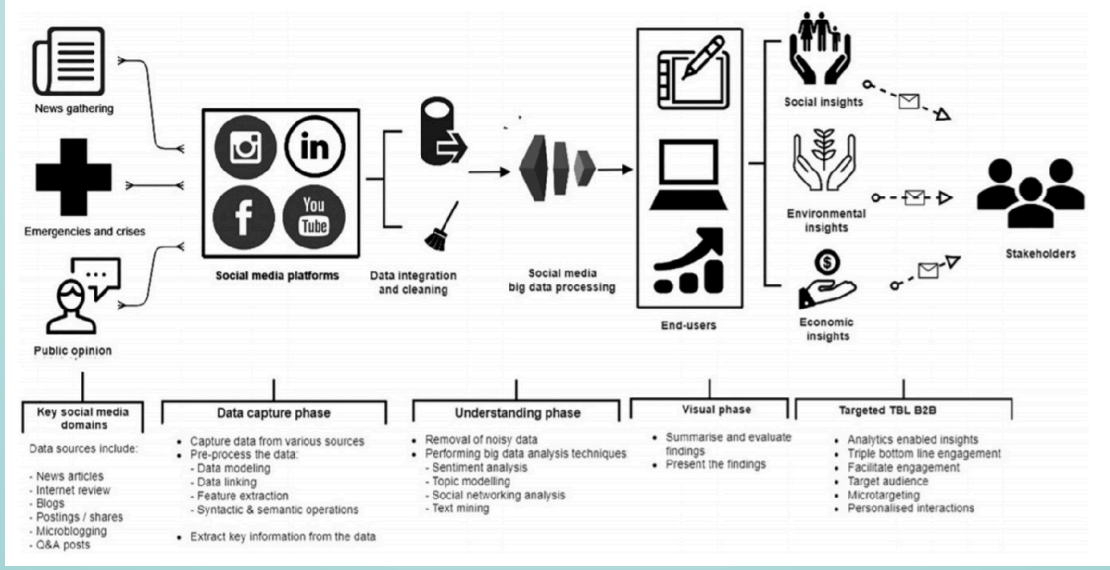

Figure 7.3 Social Media Data Analytics

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INTERNET OF THINGS

WHAT IS IOT

The Internet of Things is essentially a bunch of devices connected together. IoT doesn't require that the network connecting the devices be the internet. However, given that the Internet is the network connecting most systems, the terminology holds.

IoT is widely regarded as the fourth stage of the industrial revolution and is touted to be as impactful as the steam engine, assembly line, or personal computer. The rapid development in artificial intelligence, big data analytics, and internet connectivity, are the key drivers in the adoption of Internet of Things based technologies.

THE IOT TECH STACK

The IoT Tech stack consists of 5 layers. These are the device hardware, device software, communication, data & analytics, and the application layer.

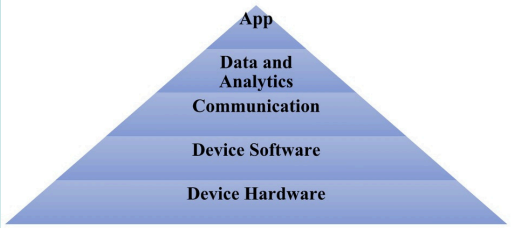

Figure 8.1 The IoT stack

Device Hardware

The device hardware forms the bottom-most layer of the IoT Tech stack. The two primary components of the device hardware are the sensors and the processors.

The sensors collect raw data from the surroundings, which is processed further to generate an output that can influence some other system. A typical IoT system can have anywhere between tens and millions of sensors depending on its complexity.

The microprocessors form an integral part of the device hardware, and a lot of them put together, form the basis for crunching the high-velocity data that are generated by the sensors and use it for processing in other stages.

Device Software

The device software is primarily the operating system and the applications built on top of it, to connect to the upper layers in the tech stack.

The operating system to be used is determined on the basis of the complexity of the system, along dimensions such as whether it has to be real-time, and the amount of data that needs to be processed. Another critical factor in deciding on the OS is the kind of interface available in the layers above. Linux is one operating system that is widely used in IoT design.

Communication Layer

The communication layer forms the backbone of any IoT system and connects the different devices that comprise the system.

There are five broad types of connectivity: bluetooth, radio, wi-fi, cellular, and satellite. These differ mainly in their operating frequency and the range (distance) that they support. The trade-off between the type of connectivity is often in terms of cost and the kind of hardware and software that has been used in the lower layers.

There are also communication protocols that define how these devices interact with each other. Some examples include IPV4/V6, JSON -Ld. and WebSocket.

Data and Analytics

There is a lot of data that is generated in an IoT system. The data can be either in a structured or an unstructured format (which needs to be interpreted before being processed for further use).

The most important thing to consider when designing an IoT system is to understand the nature and volume of data you will receive, and see which of it is relevant and needs to be extracted. One can then design the system to have the ability to process such data.

There are a lot of cloud APIs that can help generate insights using sophisticated data analytics techniques and can help make sense of the data collected and processed so far.

Applications

The last layer in the tech stack is the applications layer. These can take different forms like web-based or app-based depending on the requirement. The applications layer is typically the layer that the end- user interacts with and sees.

There are multiple examples of areas in both B2B and B2C segments of IoT applications. Some examples in the B2C space are Pawscout (A dog tracking app), or Amazon go.

One Example in the B2B space is South East Water, which made an app to improve its clients' service delivery and CRM scores. Another company CSG has made IoT solutions specifically for the telecommunications sector.

Figure 8.2 Images of pawscout and amazon go

GROWTH POTENTIAL

The Internet of Things is taking over mainstream business use cases, and provides opportunities to create disruptive solutions for different use cases across industries. The number of IoT connected devices is estimated to increase from 16.7 billion in 2024 to 38.8 billion in 2030. As per the growing demand IoT market will grow globally at a 18% CAGR, from $1 trillion in 2024 to $3.3 trillion by 2030 (source: Statista). The adoption of the cloud platform, digital transformation, 5G, and reduction in the cost of sensors are other attributes that have contributed to the fast growth of the Internet of Things.

IMPORTANT USE CASES

Amazon is reinventing warehousing and uses "armies" of wi-fi connected robots to locate and dispatch products at its warehouses.

Power and Robotics Major ABB has used connected sensors to monitor the performance of its robots and maintain them.

Airbus is building the factory of the future to improve operational efficiencies and have incorporated features like the smart glasses to minimize errors.

These are just glimpses of a few use cases of the way IoT has seen adoption.

Smart Cities - One of the most important and promising use cases of IoT is in developing smart cities. IoT systems can sync up different traffic lights across cities as per real-time data, which will help redirect and synchronize traffic as and when needed. The sensors in buildings like CO2 sensors, smoke detectors, cameras etc. can optimize buildings' energy usage. Power energy grids can be synced further to predict energy surges and then redistribute energy as per requirement. Numerous other functions can be integrated using IoT devices to make cities more efficient.

Healthcare - Health trackers and wearables are already used heavily in urban areas. However, IoT provides a much bigger opportunity to revolutionize medicine completely. IoT devices are being used to collect critical patient information and send it to some other location. They are also used to continuously monitor heart rate, blood pressure, and glucose for patients like diabetics who need regular attention. Apart from remote monitoring, caregivers can use IoT devices to monitor and administer medicine remotely.

Industry 4.0 - IoT has only accelerated all use cases related to Industry 4.0. Real time data are collected through sensors, RFID tags, and various electronics and then integrated with industrial machines and various systems. From asset management in real time and tracking to predictive maintenance, all have the useful application of IoT.

Autonomous and Connected Vehicles - Cameras, LIDAR, and other onboard sensors can get data about road conditions and inform appropriate driving actions to be prevented. IoT is helping in making "vehicle-to-everything" or "V2X" technology a reality.

IOT PRODUCT MANAGEMENT

An IoT product combines hardware(s) and software(s), measures real-world data, and connects to the internet to create value for the user.

An IoT product manager would be different from others because of the complexity of multiple layers in the IoT stack, and the need to keep all layers and various interfaces in mind while designing. One does not need to be an expert in all layers, but it is vital to have a fundamental understanding of them. It's essential to adopt a system thinking approach for the success of the IoT product.

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CLOUD COMPUTING

WHAT IS CLOUD COMPUTING?

Cloud computing can be defined as the collection of computing services that are delivered such as servers, databases, networking, storage, analytics and even intelligence over the internet, aka “the cloud”. This helps in achieving not just faster innovation but flexibility as well and one of the reasons for very fast adoption had been lowering overall operating costs, no or low management of infrastructure and option of quick scaling up as and when needed.

MARKET SIZE & KEY CHARACTERISTICS

1. In 2023, the global cloud computing market was estimated at 602.3 billion USD and is expected to grow at a CAGR of 21.2% from 2024 to 2030. (Source: Grand View Research).
2. India cloud computing market is expected to grow significantly at a CAGR of 26.7% from 2024 to 2030.

MAJOR PLAYERS

- Amazon Web Services (AWS)
- Microsoft (Azure)
- Google Cloud (GCP)
- Alibaba Cloud
- IBM Cloud
- Oracle Cloud Infrastructure
- SAP Cloud

TYPES OF CLOUD

Public Cloud: Amazon Web Service or Microsoft Azure are example of third party cloud vendors who deliver their computing resources like storage and server over the internet. All Software, hardware and other supporting infrastructure management is done by the cloud vendors. The resource model can be shared and dedicated, hence helps to leverage economies of scale.

Private Cloud: It can be defined as a cloud computing offering that can be described as a proprietary environment of resources used dedicatedly by a single organization or many a times the cloud environment given by public vendor itself but the environment is completely isolated. This option results in less cost saving but ensures high security. The resource model is dedicated, hence it is capital intensive.

Hybrid Cloud: Hybrid clouds combines flavor of both public and private clouds. Hybrid Cloud allows both application and data to be

shared between. This easy sharing model allows business to have a greater flexibility as well as many more deployment options which helps in optimizing existing infrastructure, compliance and security.

TYPES OF CLOUD SERVICES

IaaS [Infrastructure as a Service]: Iaas is the basic offering of cloud computing services with no frills. With IaaS, IT infrastructures like virtual machines (VMs), servers, storage, networks, operating systems can be rented. It is a pay-as-you-go based on the usage. Users manage their own infrastructure. In simple terms it can be understood as an on- premise machine that you buy on cloud at a very low cost. For example, AWS EC2, Rackspace, Google Computer Engine.

PaaS [Platform as a Service]: Very similar to IaaS, PaaS provides a framework around the offerings. Here the barebone infrastructure is hidden from the end users. It provides the developer with various tools and services. PaaS is more focused on providing an environment for developers to start working instantly. For example, AWS Elastic Beanstalk, Windows Azure.

SaaS [Software as a Service]: The most opted offering where applications is provided over the Internet to the end-user on demand and typically on a subscription basis. For example, Google Workspace, Dropbox, Salesforce.

Serverless Computing: Intersecting with PaaS offering, serverless computing intrinsically focuses on building app functionality and scaling up-down. The developer does not have fine-grain control in the environment, as they would have had on PaaS. Here the cloud provider handles the setup, capacity planning and server management for you. The use case can be for developing any event-driven programming.

FaaS [Function as a service]: Function as a service (FaaS) is a very new model which is becoming increasingly famous . The design of Faas is based on the premise of serverless architecture. It provides environment to develop, run and deploy the code unit on the fly and gets executed in response to any event. For example, Lambda functions from AWS, Microsoft Azure functions, Oracle Cloud Fn etc.

Figure 9.1 Types of Cloud Services

BENEFITS

It is estimated that in the USA, 69% of businesses are using cloud in one or the other form, and 18% are considering adopting it. So let's look at what makes the cloud so appealing.

Cost Savings – Adopting Cloud Services can help to drastically reduce the huge capex incurred while setting up computing infrastructures like machines or even large-scale data centers. Most cloud services give you the flexibility to choose what you want to do and pay only for how much you use. For example, for storage, the pay-as-you-

go model ensures that the organization/individual is paid exactly for the space one is currently using. Taken together, such factors lead to huge cost savings, and organizations need not worry about scalability at all. In one word it reduces the total cost of ownership (TCO).

Mobility and collaboration - As resources are over the internet, cloud ensures that resources can be accessed from anywhere, anytime 24X7, and on any device. This makes collaboration a simple process by empowering employees to connect and share information seamlessly and securely.

Reliability – Backup and disaster recovery is an intrinsic feature. Even the backups are geographically distributed, to add a layer to reliability. For example, loss prevention of data that would not have been possible in case of hardware failure in a local computer.

Security - Cloud host's full-time job to carefully monitor security, which is significantly more efficient than a conventional in-house system. Cloud storage providers implement baseline protections for their platforms and the data they process, such authentication, access control, and encryption. Many enterprises supplement these protections with added security measures of their own to bolster cloud data protection and tighten access to sensitive information in the cloud.

REVENUE MODELS

The revenue model for cloud resources are under the following buckets:

Freemium – In this model, the company offers their product for free, with limited features based on users, bandwidth, and other criteria. When you cross that threshold, you might need to upgrade to a paid account.

Tiered pricing - Some services have volume pricing tiers across certain usage dimensions that lowers price rates when usage increases. With consolidated billing, the system determines which volume pricing tiers to apply, giving a lower overall price whenever possible.

Pay-as-you-go/Usage based - PAYG cloud computing is a payment method that charges the user exactly on the amount of usage. For example: The electricity bill is charged on the entity based on the usage.

Reserved capacity – For certain resources flat monthly or yearly rates are charged.

CHALLENGES IN CLOUD COMPUTING

- Data security and privacy concerns
- Compliance with regulations and standards
- Downtime and service availability risks
- Vendor lock-in and interoperability issues
- Managing multi-cloud environments
- Data migration complexity
- Performance and latency issues
- Cost management and unexpected expenses
- Scalability and resource management
- Limited control and flexibility over infrastructure
- Dependency on internet connectivity
- Integration with legacy systems

Privacy of data – This is the biggest blocker for non-adoption of cloud in many organizations and specially in data sensitive domains like healthcare, military and government. There have been reports around data breaches, insecure access of data etc. Another reason is the secondary usage of data by cloud companies resulting in loss of privacy , targeted advertisement.

Service levels – Although most cloud vendors guarantee 99.x% SLA's (service level agreements), for mission critical application this is still a no go.

Cloud migration and integration – Although creating a new application in the cloud is very easy, migrating existing applications to the cloud is not so straightforward. It might even require an architecture change. Many times, the application might have a dependency on other applications, or tools that in turn might be on-premise. Such issues lead to a lot of interoperability issues.

Compliance - Transferring data on the cloud can involve compliance clearings. For multinational companies, it can be very difficult to remain in compliance with regulations of all countries. GDPR, Canada's privacy act or health laws and the U.S patriot act are trying to force stop exposure of data to third parties. Many a times companies need to do due diligence or consult data security professional to ensure they comply with the law.

USE CASES

Cloud strategy - This is used by companies for the business transformation strategy. It helps the organization to understand the best way to use cloud services. For example, which apps are right for the cloud, what are the prerequisites for cloud application, migration , cost benefit analysis, ROI etc.

Cloud first application development - E2E services – Projects to build progressive, scalable, value driven cloud applications to match the business needs.

Cloud migration - Designing (or re-designing) your software to fully leverage cloud deployment and to gain access to scalable infrastructure, as business needs require it.

High Potential in SME's - With rise of growth stage Startups as well as existing SMEs moving towards digital transformation ,the cloud service market forecasts a huge growth. Major reason being cost saving and flexibility.

Design Considerations - To design cloud applications following consideration can be useful:

1.Scalability
2.Resilience
3.Loose coupling between interacting modules
4.Take advantage of multi tenancy to optimize resources and cost.
5.Performance Monitoring

To conclude : Till recently IT was considered as a cost center, but now with advent of cloud, IT is considered as a business enabler, helping tremendously in organizational transformations.

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BLOCKCHAIN

INTRODUCTION

A blockchain is a form of a mathematical data structure and is programmed such that it is very difficult or even impossible to manipulate it. A blockchain consists of a list of records called blocks that keep on growing as more information is added. These records (blocks) are linked with one another using cryptographic functions. There are three elements in each block-

- A cryptographic hash function pointing to the previous block in
- the chain.
- Transaction data of the current block + Timestamp, corresponding
- to the current block.
- A cryptographic hash of the
- current block.

Some of the properties that make blockchain very interesting are listed below-

Distributed: All network participants possess a copy of the ledger, and any addition to the records will reflect for participants in real-time.

Immutable: Any record once validated and added to the ledger cannot be changed and is irreversible. In order to change
any block, every block preceding it must be changed, which is
computationally impossible.

Time-stamped: Each transaction/block has an immutable timestamp associated with it. Having a timestamp for each entity not only helps in its unique identification but also ensures easy tracing of the chronology of events at any subsequent stage.

Programmable: Blockchain transactions could be automated using programmed codes called smart contracts. Smart contracts are stored on the blockchain as lines of codes and give blockchains the ability to automatically complete transactions when certain predefined terms and conditions written on the contract are met.

TYPES OF BLOCKCHAIN

Traditional enterprise applications are built on a client-server-database architecture, where a centralized server services requests from clients. The database acts as an information storage device that communicates with the server. In the blockchain environment, there is no centralized server or database; each node in the network owns their copy of the shared ledger. The following diagram gives a brief note on possible architectures in a blockchain environment, based on different

nodes' ability to access and edit the shared ledger.

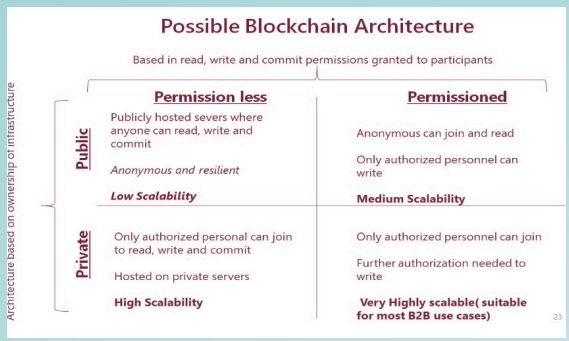

Figure 10.1 Types of blockchain

TECHNICAL DETAILS

A blockchain is technically defined as a collection of blocks that contain data or information linked together in an immutable and decentralized structure. The following two components are useful to understand how blockchains function at its core.

Cryptographic Hash Function - A special type of function named the cryptographic hash function is pivotal to the functioning of the blockchain protocol. It can be imagined as a BlackBox, that takes some random input (images, text, etc.) and will convert it into a unique output of a fixed size. There is no mapping possible between the input and the output of the hash functions, and the only way to inverse this function is by brute force checking.

Digital Signatures – Digital signatures are pivotal to allow secure messaging in any network. Each identity consists of two unique components – public key and private key. These keys are generated in pairs by a mathematical function. The public key of a user is visible to everyone in the network, whereas the private key is only visible to the user. Each message that is sent over this network is converted into a signature by using a private key. The receiver of the message and signature will use the sender's public key to verify the message's authenticity. This allows secure and anonymous messaging within a network.

Broadly, each block in a blockchain consists of three major parts - the hash of the previous block, the contract specific data along with a timestamp, and also the hash of the current block. Since each block refers to the previous block, the chain essentially is an immutable structure. If someone tampers with an intermediate block's data, they will also have to change every subsequent block to make the blockchain consistent again.

Example: Transactions Ledger - This section will cover the working of a sample transaction ledger that records every transaction on a decentralized and immutable database. The network consists of five nodes, all of which have a copy of the blockchain with records of every transaction since the inception of the chain. This record of transactions is depicted into a simplified ledger, which stores the account balance of each node. In this network, every node also acts as a verifying node. Consider a case where the Node "A" wishes to transfer two units to node "B." In this scenario, "A" will transmit the message "A pays 2 units to B" in a protocol that is fixed by the blockchain. This message's validity is established as soon as it reaches the rest of these nodes using a digital signature. Once the message sender is verified, each node will then separately validate the transaction in two significant steps. In its first step, every node would check if "A" has the

required amount of balance to complete the transaction (greater than 2 units in this case). Blockchains have underlying protocols that determine how transactions get verified. One common protocol is termed as “Proof of Work,” which is a consensus mechanism that crudely states that the node in the network that has the highest available computing power will validate a transaction, create an updated ledger and send the updated ledger across the network. The process of finding the node with the highest computing power differs across each blockchain, but all have the same goal – to validate a transaction. The node that achieves this gets a reward, essentially the transaction fees. Finally, the transaction is completed as all the network players receive an updated ledger from the verifying node.

BLOCKCHAIN BUSINESS VALUE

The discussed properties of blockchain infrastructure make it suitable for a variety of business sectors due to the following advantages-

Greater Transparency - Transparency due to the distributed ledger technology. It increases accountability for each stakeholder and motivates them to act in the best interest of the firm.

Increased Efficiency - Peer-to-peer transactions remove the need for a middleman even while doing cross-border transactions. Programmable contracts or smart contracts automate many manual processes based on some predefined conditions further improving efficiency.

Improved Traceability - On a blockchain ledger, each transaction/exchange of goods is recorded in a timely manner. This time sorted information helps in tracking the movement of the goods as a single point of the proof trail is present. This helps in improving security and prevent counterfeit related frauds. This has become especially important in the current pandemic context where consumers need and demand the ability to track their product's origin.

Better security - Blockchain is highly secure as each transaction is encrypted and linked to the network's previous transaction. This makes tampering with any past transaction practically impossible. This immutable and incorruptible nature of Blockchain makes it safe from falsified information and hacks.

BUSINESS APPLICATIONS OF BLOCKCHAIN

The uses of Blockchain can be categorized into the following functions:

Record Keeping - The feature of immutability and easy accessibility makes Blockchain a powerful candidate that could be used for storing static information, which has potential uses in keeping records for land titles, patents, government databases like IDs and social security numbers. Storing of voter information on blockchain could potentially be the first step towards secure and trusted digital voting.

Dynamic transactions Registry - As the transactions of a blockchain network are time stamped and programmable (via smart contracts), they could be used for storing and

real-time updating of tradable information. Widespread use of this is the Bitcoin network, where transaction information is dynamically stored and acts as a single point of truth for everyone. Other use cases include monitoring supply chains in real-time, information of drug trials, cross- border peer-to-peer transactions, insurance, and medical claims, etc.

The following segment highlights a few industry use-cases where blockchain can be used to drive business value

Blockchain Augmented Supply-Chain – As businesses grow complex, the network that allows firms to deliver manufactured products to market also grows complex. In this scenario, a trust-based method to transfer information is crucial to ensure operational efficiency. Along with operational efficiency, such a mechanism would allow players in a supply chain to trust the nodal firm. Currently, only limited information about products is recorded, with none of the players having complete visibility or access to a product's origin or path. Providing a centralized database to track products in a supply chain would not be appreciated by other players, as trust is not reinforced. A private permissioned blockchain-based solution can be used in this scenario, where only supply chain partners can access and edit product quality and transaction data. This has multiple use-cases such as flagging counterfeit products/raw materials, tracking provenance of damaged goods, etc. A blockchain based supply chain solution will thus add business value by allowing businesses to avoid risk and ensure better planning.

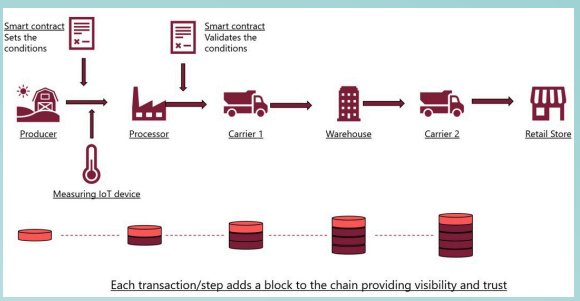

Figure 10.2 Blockchain augmented Supply-Chain

Government/Civic Tech Applications - As governments push their digital initiatives, ensuring that the government's record-keeping facilities are tamper-free and transparent is essential. Blockchain's trust-based record-keeping and transaction mechanisms can be used to solve this issue, especially by enabling services such as Digital Voting. Since blockchains are immutable, secure and anonymous, they can be used to create digital voting alternatives to existing systems.

Banking and Finance – The banking and finance sector has been one of the earliest adopters of blockchain technology. Bitcoins and other cryptocurrencies function on the infrastructure of Blockchain. The primary need for blockchain technology arises from the need for peer-to-peer transactions to facilitate cross-border payments and trade. Currently, many intermediary banks and financial institutions that facilitate cross border transactions. Nevertheless, due to the presence of intermediaries, the process is both time-consuming and costly. Utilizing the framework used for Bitcoins, blockchain infrastructure could eliminate the need for intermediaries for any transaction. The blockchain protocols also eliminate the problem of double-spending, which is prevalent in other digital currencies.

Healthcare – Healthcare industry needs the functionality of immutability and a distributed database of blockchains. The primary need arises due to the need for accessibility of medical records to accelerate clinical and biomedical research. Currently, centralized data stored with medical institutes, which is shared according to their will, and generally, time-consuming processes are involved in obtaining these records. Using blockchain technology and the ability of distributed storage with correct access permissions, medical data could be democratized. This will allow medical institutions other than hospitals to leverage this data for further development.

BLOCKCHAIN AND PRODUCT MANAGEMENT

Deciding whether to adopt a blockchain infrastructure for your organization could be a highly challenging task, as there are trade- offs between speed and security/reliability. While the traditional database system would work better in specific industries, blockchain would provide unparalleled advantages in others. The below framework could be really helpful in choosing between various options of data storage infrastructures.

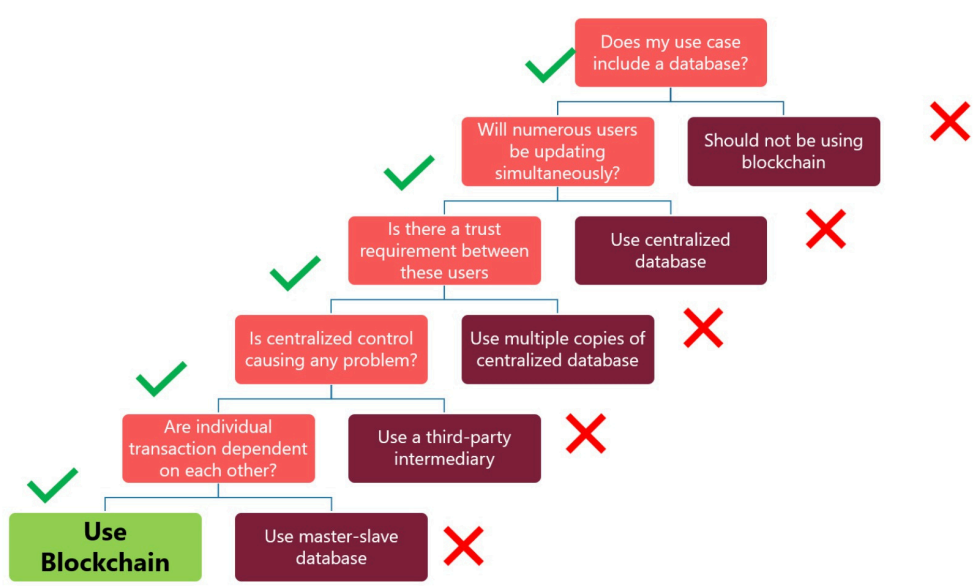

Figure 10.3 Assessing need for blockchain solution

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ECONOMICS OF PLATFORMS

Understanding the microeconomics of an industry indeed lends one an excellent theoretical understanding of the business, even as a complete outsider. Microeconomics forms the backbone of management education and is most critical to understand various subjects. Naturally, it also lends insights to product managers, and it is fitting to start the book by looking at the technology industry through the lens of microeconomics.

This chapter will help PMs and aspiring PMs to understand what type of features are essential, how to compete with companies, and to align personal goals with the strategy of the organization to move into a strategy role eventually.

The chapter has been structured into three parts. The first part introduces a framework to understand various elements to understand platforms. The second part has exercises which help drive the concepts further. Finally, we have three cases (Google, Amazon & Uber) where we apply the above framework and draw insights about the strategies that the company should choose.

INTRODUCTION TO THE FRAMEWORK

The definition of platforms for this chapter is going to be very broad. Platform businesses are essentially any business which creates value by facilitating an exchange between one or more "sides" or "segments of users," e.g. in the case of Uber, drivers and riders are the two sides of the platform. Various examples of platform businesses include Uber, Google, Amazon, Facebook and even Windows OS.

There are the following three factors comprising the framework:

Economies of Scale: When looking at economies of scale, we try to ask the question “how much does it cost the company to produce one extra unit/to service one extra user/to produce one extra minute of streaming etc.?” We will only look at economies of scale from the point of view of costs.

Network Effects: Network effects are about customer value. Network effects are said to exist when the value of the product/platform increases by increasing one extra

customer or user on the platform. The question that we ask here is "What type of network effects does the company have if it does have network effects?"

Differentiability: Differentiability helps us understand competitive evolution. Differentiation is created when there is a unique space in the customer's mind about the company's product. It is about the perception of the customer and is not internal.

ECONOMIES OF SCALE

Economies of scale are said to exist when the cost of producing one extra unit or servicing one extra user reduces. Usually, when there are fixed costs like a factory which can produce multiple units or serve multiple users, the costs get distributed. Therefore, once a company has invested in a factory with a capacity of 100 users and current demand is 50, the extra cost incurred to produce one extra unit, i.e. the 51st unit is minimal.

The extra cost associated with producing one unit is called the **marginal cost**. On the right side, we see the standard Demand-Supply curve. The y-axis represents the price, while the x-axis represents the number of units. The demand curve represents the customers, i.e. what the customer is willing to buy at that price. The supply curve represents what the suppliers are willing to serve when they are getting the price. **Therefore, the supply curve is nothing but the marginal cost curve of the suppliers,** because the supplier would be willing to supply a particular unit only when he gets a price higher than its marginal cost.

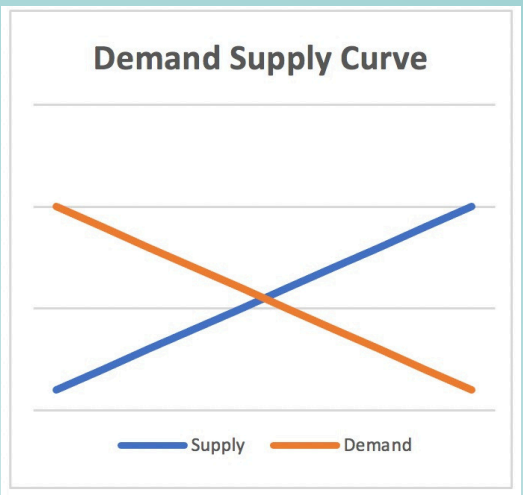

Figure 11.1 Demand Supply Curve

That begs the question if the supply curve is just the marginal costs curve, why is the supply curve (marginal cost curve) increasing? What happened to economies of scale? Shouldn't the cost be reducing?

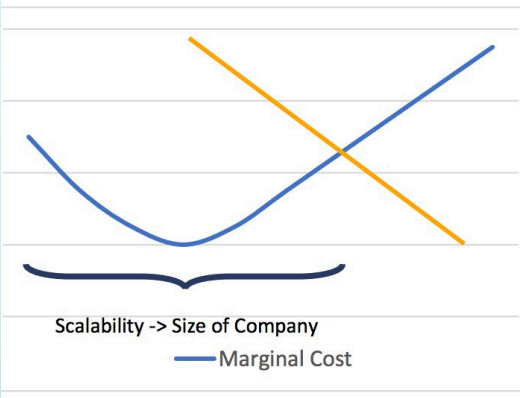

Figure 11.2 Marginal Cost

Well, economies of scale do exist. The actual cost curve (supply curve) looks like the one in the figure below. So, the economies of scale do exist for the first few units, but then the marginal cost curve goes upward, and diseconomies of scale set in. This starts to happen because after a certain point several costs begin to increase, e.g. managerial costs, costs of raw materials, cost of capital etc. In the above example of a factory, to produce the 101st unit, the cost

incurred will be to set-up an entirely new factory which is going to be much higher than the marginal costs for the previous units.

We can also see that the size of the company is limited by the upward curving supply curve, which leaves space for competitors to carve out the remaining customers and compete with the company. We can also see from here that highly scalable companies delay the upward curving of the marginal cost curve, which automatically results in a bigger sized company.

Hence, there are also **natural monopolies** which exist, whose marginal cost curve is like that shown in the figure on the left. The curve suggests that the companies are infinitely scalable, which means that there is no space left in the market for another competitor – hence the name natural monopolies. Some examples of natural monopolies are railways, pipelines etc.

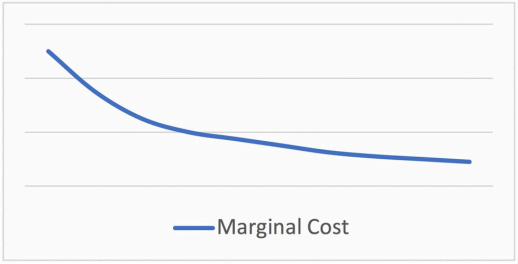

Figure 11.3 Marginal Cost

The cost curves of most technology companies are similar to that of a natural monopoly, i.e. highly scalable.

Network Effects

The second element of the framework is Network effects. Network effects are said to exist when the value of the product or platform increases when one additional user is using the product. The simplest example of network effects is messaging services. The service holds zero value to a person when they are the only person on the platform. Add 1,000 more people, and you have probably got some people who know each other. Keep adding more people on the platform, and there will be a point where the platform reaches a critical mass where the value addition for every customer snowballs. Eventually, there is a point reached where the value that is created by adding more people to the platform diminishes and the network effects begin vanishing.

There are two different categories of network effects:

- Side Network Effects: Cross side network effects vs Same side network effects.
- Scale Network Effects: Global vs Local Network Effects.

Side Network Effects

For side network effects, we can have either cross-side or same-side network effects.

Cross Side Network Effects

When the value of the platform increases when users get added to the other side.In case of Uber, if there are more drivers on the platform, it is more attractive for the riders because the waiting times will be lower because of higher availability and lower surge prices. Similarly, the more the number of riders, the more attractive it becomes for drivers to join the platform, and take advantage of the high capacity utilization and higher prices.

Same Side Network Effects

When the value of the platform increases when more users get added to the same side. Messaging service that we discussed above, has same- side network effects. The more the number of people using the service, the higher value it holds for everyone.

Same side network effects are more potent than cross-side network effects because of the existence of a flywheel. More people keep increasing the value for customers because of which even more people join because of which value increases even further. It is no coincidence that the apps and companies with the most massive scale have same-side network effects (WhatsApp, Google, Facebook).

In cross-side network effects, however, there is always an equilibrium which is reached. In the case of Uber, adding more drivers makes more customers join in, and the cycle ends there creating an equilibrium. One side needs to be consistently incentivized to keep joining the platform to make the other side join. There is no flywheel effect in place, and hence it is more challenging to grow in this case.

Scale Network Effects

For scale network effects, again, we have two types, i.e. local and global network effects.

Local Network Effects

When the network effects are restricted to local geographical regions or a particular segment of customer, the network effects are called local network effects.In the case of Uber, the network effects which exist are local. Having an additional driver in Mumbai does not help a rider in Delhi.

Global Network Effects

When the network effects are spread across global geographical regions or several segments of customers, the network effects are global. In the case of Airbnb, where having an additional room in New York provides value to a customer travelling from Mumbai.

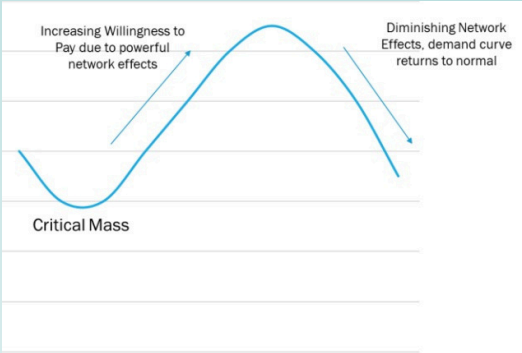

Figure 11.4 Demand Curve

Global network effects are more powerful because it helps companies scale over a broader market much faster, while for companies which have local network effects it requires much more investment to reach a similar scale. Since network effects are a demand-side effect, let us examine how the shape of the demand curve changes due to the existence of network effects. The demand curve is the cumulative willingness to pay of all the customers, and we can see the resulting curve on the image on the right.

Initially, the value added by the network effects is minimal, and hence the demand curve keeps falling. However, when critical mass is reached, the value increases due to powerful network effects. At some point, the marginal value provided by network effects goes to zero, after

which the demand curve returns to normal.

The demand curves of most technology companies are characterized by the above curve where the network effects are strong and pervasive.

DIFFERENTIABILITY

The third element of the framework is differentiability. Differentiability is when one creates a unique space in the customer's mind about the product. It is strictly external and not related to internal processes. If Uber creates a new algorithm which increases cab availability for customers, but the customers do not perceive any significant difference over Ola, then Uber has not differentiated itself. A difference in the internal processes of the company should change into a tangible, customer value to be considered differentiation.

Cost of Convergence: Differentiation requires a unique value proposition. A unique value proposition in the context of technology platforms is provided through unique features. However, maintaining the uniqueness of features in technology products is not easy. As soon as company A launches a new feature, product managers in company B and company C immediately start reverse- engineering the feature to add it to their products. This is because the cost of them copying the feature is quite low, i.e. the cost of convergence is quite low.

Customer Switching Costs: The second aspect we should take a look at are customer switching costs. If the customer switching costs are high, then the first mover in the market has a significant advantage. Having high customer switching costs can lead to differentiation since the platforms do not have an incentive to converge features.

PUTTING IT ALL TOGETHER

Initially, let us assume that there is no competition and consider the effect of economies of scale & network effects on the demand curve and the supply curve in conjunction. Before we do that, let us look at the company and customer surplus in case of a standard demand-supply curve and the natural monopoly.

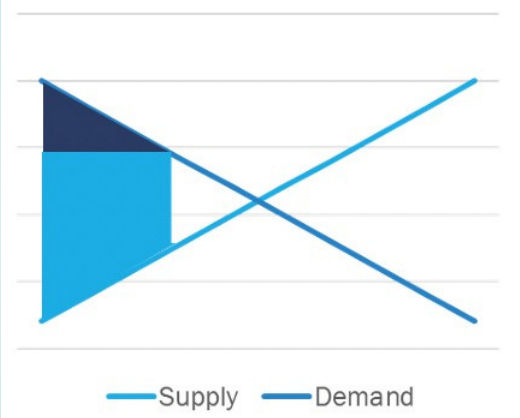

Figure 11.5 Supply Demand Curve

For a standard demand-supply curve shown on the right, the darker area represents the consumer surplus, while the entire light blue area represents the surplus of a monopoly firm. For a technology platform, we juxtapose the demand curve and the supply curve we discussed previously as is shown in the figure above on the left. Since technology platforms can also price discriminate, the entire surplus is captured by the company. Even a very preliminary comparison reveals that technology companies can capture a vast amount of supernormal profits.Now let us try and introduce competition and understand how the market evolves.

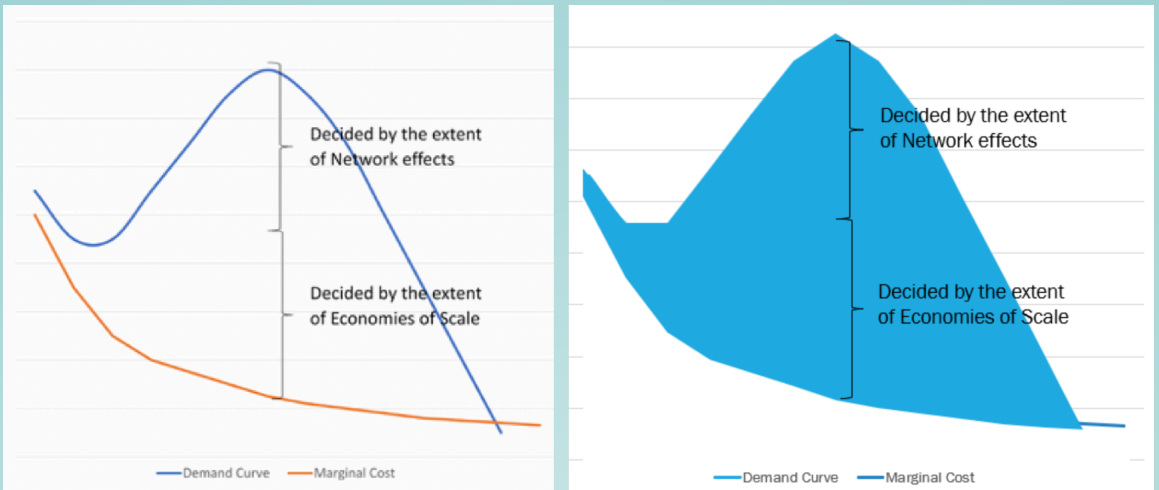

Figure 11.6 Marginal Cost

If there are powerful economies of scale, powerful network effects and low differentiability (low customer switching costs and high convergence costs), the industry will undoubtedly end up as a monopoly.

Because there are scale economies, the bigger player can price lower because of which the bigger player grows faster. If there are strong network effects, the bigger player adds customers faster, because of which the bigger player grows faster. The bigger player would also find it attractive to acquire smaller players and increase its scale. Hence, whichever company (typically the first mover) reaches the critical mass of users faster will become the dominant player. Infinite economies of scale leave no space for competitors, and robust network effects mean the company grows to occupy the space faster than any other company.

As a late mover, the only way to compete is by creating differentiation. To create the differentiation, the company needs features which can increase the cost of convergence or increase the customer switching costs.

As a result, we can draw three important implications from the entire discussion about the framework that was introduced.

1.New companies burn cash to reach the critical mass point. The companies ramp up fast because they are likely to get eliminated from the market if they do not
reach the point first.

2.A company needs to ask itself whether it inherently has vast economies of scale. If it has 100 users today, it needs to understand whether it can handle 5,000 users tomorrow and a million users in a year, if it does business the same way it has been
doing until now.

3.As a smaller player in the market, the only way to compete is by differentiating. Increasing switching costs and introducing features that make it difficult for
the bigger player to copy, will
create differentiation.

EXERCISES

Which company amongst Uber and Airbnb would find it easier to expand overseas?

AirBnB has global network effects, whereas Uber primarily has local

network effects. Hence, AirBnB would find it much easier to expand overseas, while for Uber to expand, it will need cabs operating in several local regions.

You are examining a start-up company you want to join. In its chosen market, a bigger player already exists. The switching costs are low, and convergence costs are high. Do you think the start-up will do well?

Since the customer switching costs are low, there is scope for the company to attract a segment of the customers of the bigger company with some new features. The bigger company cannot also copy the features to win back their original customers.

You are examining a start-up company you can invest in. In its chosen market, the switching costs are low, and convergence costs are low. There are no other players in the market, will you invest?

Since the customer switching costs are low, this company might quickly lose customers to new players. However, since the convergence costs are low, the start-up can easily copy any new features the other player comes up with.

When should you go for a cloud service provider and when should you own your infrastructure?

Having your own infrastructure ensures vast economies of scale and scalability. However, having your own infrastructure also undermines flexibility. Having a cloud service provider significantly increases flexibility. However, since the marginal costs increase with scale, it undermines scalability. Hence, for smaller companies or for very early-stage companies, where flexibility is essential, cloud service providers would be best. However, when the scale increases, and the company starts getting bigger, owning infrastructure is more critical.

You are a PM in Flipkart. Amazon has approached you with a potential collaboration. They want to partner to create a tool which lets suppliers manage their inventory and pages through a common portal for all e-commerce platforms. Should you accept?

Flipkart is the first mover in India while Amazon is the second mover. Both companies have similar scale. This would result in reducing the customer switching costs by way of which differentiability would be reduced. Since you need to maintain your own space in the market, it is best not to go ahead with the proposal.

CASE STUDIES

Google & the Search Market

Economies of Scale: Google has considerable economies of scale. Google's file system is highly scalable. In fact, there is an interesting story associated with Google and Yahoo. Even though Yahoo was the first mover, Google won the search market. Why? This is because Google built its file system to be highly scalable while Yahoo's story is captured in the following sentence extracted from a Techcrunch article. Yahoo was "unable to keep up with increasing demand due to complex & inefficient infrastructure and rising vendor costs."

Network Effects: Google's search algorithm improves with every new search. Google also has global network effects. Google has been further strengthening its global network effects by expanding the languages it supports. Google primarily has same-side network effects. Cross side network effects exist for businesses that want to advertise on Google Search. More the search market share, more businesses are interested in Google.

Differentiability: This is a classic example of a market where a smaller company needs differentiability to create space for itself. This can be done either by decreasing switching costs or by increasing convergence costs. Google has so much of our data stored with it, including passwords and search history, which makes browsing the internet much more comfortable, hence the switching costs are relatively high. Hence, a feasible way to increase differentiability is by increasing convergence costs. DuckDuckGo exploits precisely that characteristic of the industry. Since Google will find it extremely costly to provide privacy to people, DuckDuckGo positioned itself as a company which provides privacy.

Hence, this way, we can draw insights about how Google came to dominate the search market, overtook Yahoo and how some niche players are still competing with the giant.

Amazon & the E-Commerce Market

Economies of Scale: Amazon does not necessarily have high economies of scale. To ensure delivery, it needs to build and maintain a long supply chain. However, the alternative, i.e. physical retail stores have much higher overheads than Amazon. Hence, the economies of scale of Amazon are more significant than standard retail chains.

Network Effects: Amazon has countrywide network effects as its international supply chains are still not very well developed. Amazon also has strong cross-side network effects which means that one side needs to be continuously incentivized.

Differentiability: In terms of differentiability, the convergence costs are low, but the customer switching costs of its customers are very high.

Anyone who has taken a look at Amazon's business model flywheel will realize that Amazon understood the economics of its business well. Amazon understood that the scale of its retail business is limited and knew that competitors would soon be able to create space for themselves. Hence, they created high customer switching costs in the form of Amazon Prime. Amazon did not focus on features because it knew that the convergence costs are low. Instead, once they created a good website, they have hardly made any significant changes to it over the past few years and moved on to newer, more attractive businesses like cloud service.

Amazon can improve by creating more network effects as its current ones are underwhelming. Hence, Amazon has taken a few measures recently, which, when put in the context of this, makes more sense. Amazon has introduced vernacular languages to make the countrywide network effects more powerful. It introduced a social network for

customers to talk about what they shopped. It is also introducing advertisements for sellers and third parties to strengthen the cross side effects further.

Uber and the Ride-hailing market

Economies of Scale: Uber's software developed anywhere can be deployed across the world. Uber also owns some of the cars in some markets; hence higher capacity utilization drives its economies of scale.

Network Effects: Uber has strong cross-side network effects but is restricted locally to cities. One side needs to be continuously incentivized to join; hence it undermines the economies of scale.

Differentiability: Uber's model has been copied across the globe because it could not expand fast enough owing to its local network effects. The ride-hailing market currently also has very low switching costs, and the customer goes to the player with lower costs and higher availability.

Though the market has not completely evolved yet, we can predict that the market will end up in local monopolies or duopolies. Since the economies of scale are not pervasive, and the network effects are local, other companies can create space for themselves.

What can Uber do today to improve value? One, since it has economies of scale, it should look for mergers and acquisitions in all key markets. Two, Uber should take measures to increase customer switching costs by instituting highly attractive loyalty programs. Three, the company can instantly create value by getting into the transportation business and not just a ride-hailing business because that creates global network effects.

EDGE COMPUTING

WHAT IS EDGE COMPUTING?

Think of the cloud ecosystem that AWS etc. have created. Data of Indian clients is stored across the world in data centres across various geographies. What if there was a better way to store data and perform processing and at the same time potentially save some money? That is exactly the advantage that Edge Computing provides over the cloud. Edge Computing is a distributed Information Technology architecture in which the client's data is stored as close to the client's origin location as possible and the computing machines are also as close to the client as possible.

WHY EDGE COMPUTING?

The following improvements are offered by Edge Computing over the traditional cloud:

Huge Latency Improvements: As the data centers are closer to the client, the data has to travel lesser physical distances through optic cables and hence, latency improves drastically

Lower & better Bandwidth Usage: The amount of data being processed everyday by organizations is increasing at a very high pace and hence, keeping the processing units and storage as close to the client location reduces the bandwidth usage in transmitting the data across the globe. This even makes it easier to avoid internet outages across the globe. Places where there are perennial bandwidth constraints due to geo location etc., can benefit massively as the computing becomes on-site for these locations.

Cybersecurity: Distributed systems make it easier to prevent physical cyber security attacks on the servers.

Legislation (Data Sovereignty): Recent legislative directives from many countries including India have made it mandatory to store Indian data in Indian data servers. This is especially true for financial transaction data. This has been necessitated to protect against the threat of other countries

DISADVANTAGES OF EDGE COMPUTING

Limited Scalability: Part of the allure of Cloud computing stems from the fact that its scale can be increased on-demand. Users can be added to CRM systems etc. and computing capacity can be scaled very easily. But, with edge computing, the scope and scale need to be defined early and hence, it is not as easy to scale-up or down as in the cloud.

- **Connectivity**: Even with the low bandwidth requirement, edge computing does still have the basic connectivity requirements. Provisions need to be made for poor and erratic connectivity in case some failure occurs. At the same time, provisions for what happens to the data/on-going computing at the time of failure need to be made.
- **Data maintenance and backup**: With the limited scalability, it becomes important to understand which data to store for long and which one to remove. It would be immensely critical to make such provisions in the edge computing framework. Also, cloud servers generally have multiple data backups at various locations to safeguard against data loss due to various reasons like server failure etc. It would become imperative to make similar adjustments for edge servers and needs to be incorporated while thinking about the scale.
- **Security and server lifecycle control**: With the distributed network, a lot of the software updates and maintenance work needs to be done online and may prove to be difficult to do over a short span of time. Also, with the distributed systems, it also becomes imperative to have more secure local communications which doesn't exist at this scale for edge computing. It needs to be scaled down from the cloud infrastructure, especially for the IoT environment.

TECHNOLOGIES AND APPLICATIONS ENABLING EDGE COMPUTING

With the ever improving tech stack, it becomes important to know what would drive the growth of edge computing. The following are expected to contribute the most:

- 5G and Wi-Fi 6: Faster, low latency communication enabled by 5G and Wi-Fi 6 would enable faster and real-time data transfer required for various IoT and connected environment concepts. The biggest example is the connected car concept arriving with 5G services in various parts of the world.
- IoT enabled devices/appliances: Many industries are moving towards the concept of smart devices for electricity usage and many other applications like health monitoring, home appliance improvement etc. With the improvements provided by edge computing, it would become easier for such devices to transfer and process data with significant improvements in power consumption and bandwidth requirements.
- AI: AI-enabled edge computing services will help make smart decisions on how to handle a particular processing task like sending it to the cloud for faster processing or having an overall better network utilization for all the tasks running etc. It would even help handle bandwidth, latency and storage issues.
- Nano scale devices: With servers occupying significant amount if physical space and having high power and cooling requirements, it becomes difficult to setup decentralized edge computing locations. With the advent of nano devices, all these problems can be solved. More on nano devices is available here: link

MARKET SIZE AND GROWTH

The industry is expected to grow to **~USD 11bn** by 2028 with a CAGR of **24%** over the next few years.

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LLM

WHAT IS LLM?

LLM refers to a special type of AI model called a Large Language Model. LLMs, as the same suggests, are trained with large text data and with an input of text, it can respond with probabilistically possible next set of words or text based on the statistical patterns it had learnt about the natural language of humans from the training data.

WORKINGS OF LLMS?

ChatGPT is currently a well-known example of LLMs which is based on the GPT-3.5 LLM model. LLMs represent words with numbers called word vectors. These word vectors are multi-dimensional and capture the essence of words in a mathematical space.

Words with similar meanings or contexts are placed closer to each other in this multi-dimensional space. For example, "cat" might be close to "dog," "kitten," and "pet." That a word vector learns the relationship between one word and every other word. It learns in the process its linguistic features. Pronouns and words that can be replaced easily with each other will have similar-looking word vectors and will be clustered if we plot them.

Word vectors themselves are highly complex. The most potent version of GPT-3 employs word vectors with 12,288 dimensions, capturing extensive information and context about words. The challenge is to determine how to place words in this word high-dimensional space so that their relationships and meanings are accurately represented. In order to that, a machine learning technique called transformers helped the cause.

Transformers changed language processing by enabling computers to comprehend entire sequences of text at once, enhancing their ability to capture context and patterns. This accelerated training and scaling of Large Language Models (LLMs), resulting in more accurate text generation and efficient processing of complex linguistic structures. This ability is called self-attention

Self-attention empowers LLMs to consider context beyond sentence boundaries. It can understood how the same word can have different meanings in different contexts. It can also understand which word is referring to what ie "Dog chewed bone because it was hungry". In this, the transformer can understand that “it” relates to dog. If hungry is replaced with delicious, it can understand that “it” refers to bone.

CHATGPT A SPECIAL LLM?

At its core, ChatGPT's main objective is to produce a "reasonable continuation" of the text it has processed so far. To achieve this, it assesses billions of web pages and digital books to determine the likelihood of each word following a given text. This process involves considering words that match in meaning rather than relying on literal text. This allows ChatGPT to rank words and calculate their probabilities for the next token in the text. The intriguing aspect is how ChatGPT selects words. While one might expect it to choose the highest-ranked word based on its probability, ChatGPT introduces randomness to promote creativity and diversity in its responses. Picking the highest-ranked word consistently results in monotonous and repetitive text. However, by occasionally choosing lower-ranked words, ChatGPT produces more interesting and creative content. This randomness arises from a parameter known as "temperature," where a value around 0.8 yields the most engaging essays or responses. The concept of temperature, however, lacks a theoretical foundation and is determined based on empirical findings rather than a scientific principle.

Salient Points about LLMs:

Automated AI detection systems are ineffective. Their susceptibility to manipulation through subtle text alterations is problematic. Additionally, they exhibit substantial false positives, often wrongly attributing AI usage, particularly among non-native English-speaking students. Proving innocence is challenging, as OpenAI acknowledges the their own detection models failed.

AI also acts as a skill leveller. Those who initially performed poorly will exhibit the most significant performance improvement, while top-performing people benefit, albeit to a lesser degree. AI's capabilities keep improving every day. Initial versions could not solve crossword puzzles, but a recent study of GPT-4 found improvement on that front. Updates to chatGPTs are also leading users to adjust their input sentences or prompts to get optimal results.

IMAGE GENERATORS

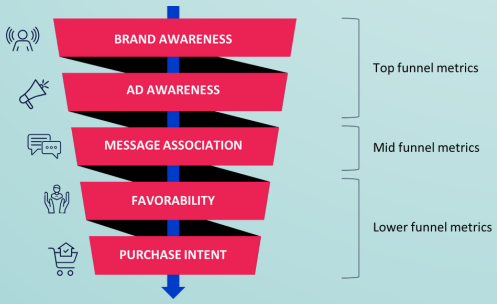
In simple terms, images consist of pixels, which be represented as numbers on RBG dimensions i.e. every color in a pixel can be defined as a combination of darkness of red, green and blue colors. When images are converted to numbers, similar operations can be performed on them.

CUSTOMER FUNNEL & BRAND LIFT

As a product manager, understanding the impact of your marketing efforts on a feature improvement or new product development is crucial. A brand lift study is a powerful tool to measure how well your campaign has changed people's awareness and perception of your product.

A brand lift study helps you measure the effectiveness of your marketing campaigns in improving how people perceive and are aware of your new feature or product. It's like taking a before and after snapshot to see how your campaign has influenced people's opinions and behaviour.

Brand Lift Studies probe brand parameters amongst control and exposed audience sets, to measure the impact (delta) of any feature improvement.



The primary goal is to understand whether your marketing efforts for a new feature or product are effective.

- Awareness: Are more people aware of the new feature or product after the campaign?
- Perception: Do people have a better opinion of the new feature or product?
- Intent: Are more people interested in using or buying the new feature or product?
- Engagement: Are people more engaged with the new feature or product (e.g., increased usage, positive feedback)?

BENEFITS OF A BRAND LIFT STUDY

- Understand Impact: Clearly see how well your marketing efforts for the new feature or product are working.
- Make Data-Driven Decisions: Use the insights to improve future campaigns and product strategies.
- Optimize Budget: Allocate resources more effectively by knowing what works best.
- Prove ROI: Demonstrate the return on investment (ROI) of your marketing spend to stakeholders.

HOW IT WORKS

1. **Setting Up the Study:** Determine what you want to measure, such as awareness, favorability, or usage intent for the new feature or product. Develop questions to ask people about their awareness and perceptions of the feature or product before and after the campaign.
2. **Collecting Data:** Split your audience into two groups. One group sees the marketing campaign (exposed group), and the other doesn't (control group). Ask both groups about their awareness and perception of the feature or product before the campaign starts.
3. **Running the Campaign:** Promote the new feature or product as planned.
4. **Post-Campaign Survey:** After the campaign ends, ask the same questions to both groups again to see if there are any changes in their awareness and perception.
5. **Analyzing Results:** Look at the differences in responses between the control and exposed groups before and after the campaign. Calculate the "lift," which is the improvement in awareness, perception, or intent caused by the campaign. For example, if 20% of people in the control group knew about the feature before the campaign and 40% in the exposed group knew about it after, the brand lift in awareness would be 20%.

Section 3

APP CRITIQUES

ECOMMERCE

UNDERSTANDING E-COMMERCE

E-commerce (electronic commerce) is the activity of electronically buying or selling of goods and services over the internet. The industry can be classified into following categories based on the business application:

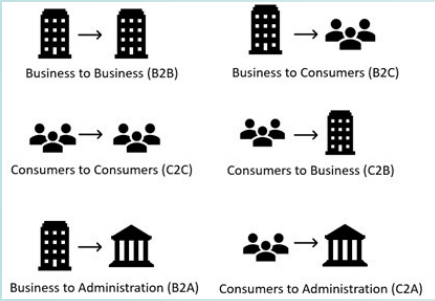


Figure 12.1 Industry categories

USER PERSONAS

The users of an e-commerce platform are the customers, sellers, advertisers and delivery executives. The customers can be classified basis their behaviour as:

Customer

1.People who are aware of the exact product they want to purchase.
2.People who have a purpose in mind but not a final product.
3.People who are casually shopping without a specific purpose.

Customers can also be classified as:

1. The **Value Seeker** who is searching for the best deals.
2. The **Researcher** is looking to compare different products.
3. The **Brand Devotee** usually purchases from the same brand.
4. The **Social Butterfly** who recommends products to others.
5. The **Replenisher** who brings in repeat sales.
6. The **Gifter** is looking to buy for friends and family.

In an interview, you can decide on customer personas based on any factor you choose. Describing 3-4 personas is usually sufficient.

You should select the best persona for the customer journey with the interviewer's consent.

CUSTOMER JOURNEY

From the user's perspective an E-commerce platform may have two typical customer journeys:

First is if the user launches the app directly, they will be taken to the home screen, where they will have to research/search the product and proceed from there. (Assuming user is already signed in)

Figure 12.2 Customer Journey 1

The second is when a user searches for a product on a search engine and clicks on a link that directs them to the PDP (product description page) of the product on the e-commerce platform. Then, the research part took place off the e-commerce platform.

Figure 12.3 Customer Journey 2

From the perspective of the e-commerce app, the important steps in the journey are:

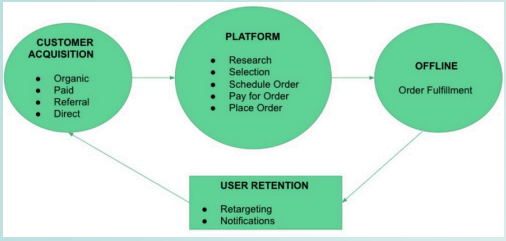


Figure 12.4 App's perspective

APP CRITIQUE: AMAZON

After logging in, the app can be divided into three sections based on the user's purpose:

Section I

This comprises the research and selection phase and has the following features.

- Banners for upcoming or existing sales, offers, product releases, prime membership promos, and so on the homepage. These banners lacked carousel indicators, which would have made it very clear to the user that they could swipe left or right on the banners to navigate through them.
- Deal of the day area which includes a live clock that displays the time when the deal for a certain product expires, which is an useful feature.

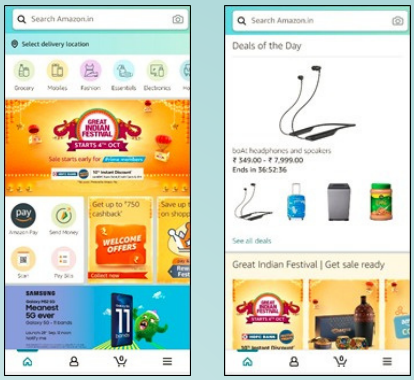


Figure 12.5 Banners & Deal of the Day

- Product Categories and personalized recommendations are available on the home page after scrolling down. Unlike Myntra, which has a well-organized categories approach, Amazon's categories list is not very user friendly.

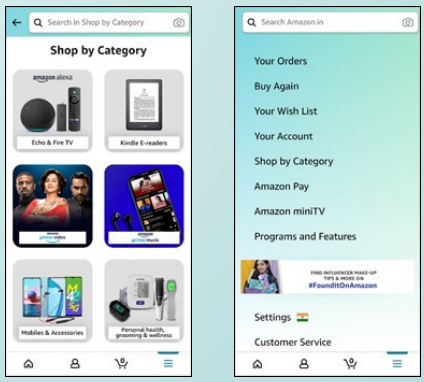


Figure 12.6 Product Categories

- Product Listing Page shows all products retrieved by the search. The user has the option to search by text or use the camera function to search for products by image.

The page includes a variety of filters and sorting options to make the user's research and decision- making process easier.

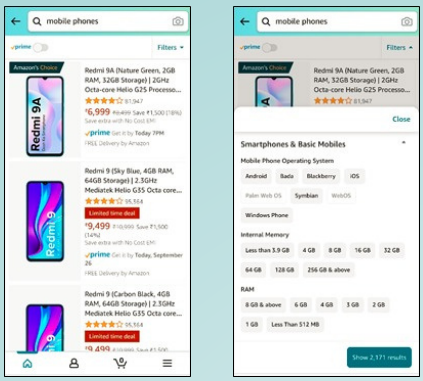


Figure 12.7 Product Listing Page

- The Product Description Page (PDP) contains all of the product's details. The pricing, media (photos/videos), user ratings and reviews and similar product recommendations. As the user scrolls down the page to learn more, amazon provides a navigation bar at the top so that the user can easily browse to different areas. This is a good feature for easy navigation within the page.

Section II

This comprises the cart page and order confirmation sections of the app.

- Cart page lists all of the products in the cart, along with their quantities. Amazon allows users to change the quantity of their order and add gifting options.
- Confirm delivery schedule screen allows the user to choose the delivery option as well as supply their contact information.

Payment screen takes care of payment methods available to the customer. Rewards and discount options are also provided.

Section III

This section houses the post order workflow for users.

- The order review screen allows users to provide feedback. Once an order has been placed, an e-commerce platform needs to know the quality of the goods and the vendor.
- The order tracking page displays a list of all prior orders, together with the invoice and status information. It also shows the order(s) that have yet to be fulfilled, as well as the projected delivery date and current status.

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FINTECH

FINTECH SPACE IN INDIA

Fintech is one of the fastest growing sectors in India, and India is amongst the fastest growing fintech markets globally. Within fintech, the payments space has been most active in India, with 19% of India's 2200+ start-ups working in payments. The sector grew tremendously after both Demonetization and the Covid-19 pandemic. The value of monthly transactions has more than tripled since Jan 2020.

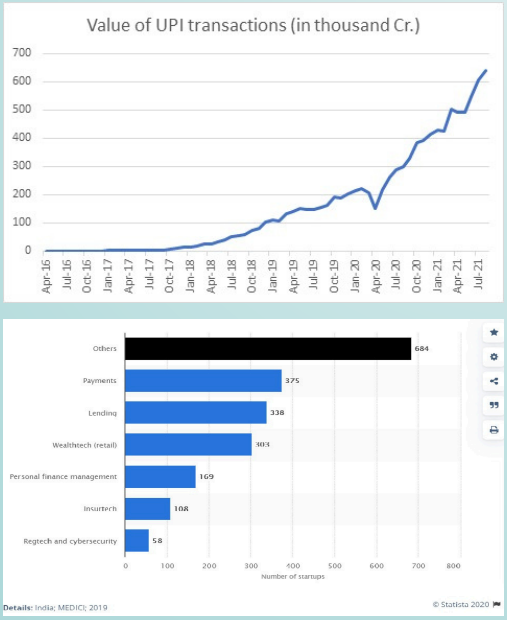

Figure 13.1 Payments statistics and Number of startups by fintech model

FINTECH BUSINESS MODELS

The popular business models of fintech are:

- Payment Gateways
- Digital Wallets
- Digital Insurance
- Digital lending
- PoS
- Neo banking
- Wealthtech

Payment Gateways and Wallets

Ever since demonetization happened in India, the emergence of payments has been phenomenal. With 375 payment start-ups in the country, the digital payments market in India is expected to grow exponentially till 2023. A proactive government, and deep mobile penetration are the driving forces for this industry. Some emerging companies in this space are PayTM, MobiKwik and FreeCharge and RazorPay.

Digital Lending

Digital lending FinTechs are mainly targeting the unmet demand of MSMEs as well as consumers, for credit. The emergence of technology, easy access to data has led to the popularity of digital lending based on credit scores for loans, mortgages,

and small businesses. Some promising start-ups in this space are Paytm, PhonePay, etc.

Point of Sale (PoS)

The point of sale (POS) or point of purchase (POP) is the time and place where a retail transaction is completed. A digital retail point of sale system typically includes a POS terminal that can read debit/credit cards and enable payments through a swipe. Service providers that provide both POS hardware and software solutions are called POS companies.

Some of the notable POS terminal providers here are Ezetap, Mswipe, Pine Labs, Innoviti, Mosambee, Payswiff (earlier known as Paynear), among others.

NEO Banking

Neobanking is a relatively new concept in India. Being a digital-only platform, neobanks are fast, efficient, straightforward, adaptable and highly cost-effective. Different neobanks have different focuses – some help with managing online bank accounts, and with budgeting and saving tools. SBI Yono, Open, Kotak 811 are some of the examples of neobanks in India.

Wealth Management

WealthTech describes a new class of financial technology companies which are creating digital solutions to transform the investment management process. This includes both end investors and firms that service them. Some of the profiting companies in this space are Zerodha, PayTM Money and Groww.

GOOGLE PAY

Initially called Google Tez, the app was launched in India in August 2017. Initially the app gained huge traction by offering Rs. 51 to a user every time they invited a friend. The app also rewarded users with lucky draw scratch cards for completing transactions. The user journey and features needed on the app are shown below:

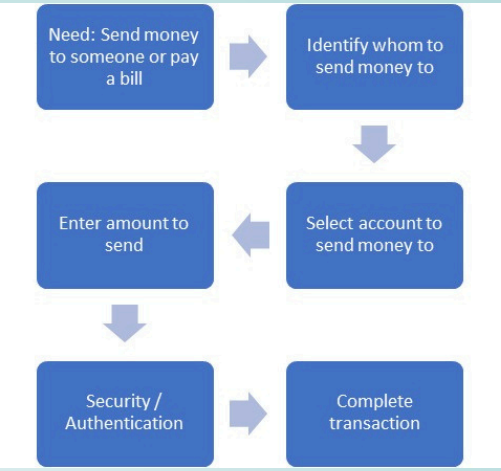

Figure 13.2 User Journey

Minimum Features Needed
• Send and receive Money
• Find other users and businesses
• Link Google Pay to bank account / cards
• View transaction history
• View bank balance

Figure 13.3 User Journey

Interviewers often ask what your favorite app is and why it is so. It is recommended to prepare in advance to answer this question. The analysis here will give you one possible answer as to why Google Pay is a great application.

GOOGLE PAY'S INITIAL SUCCESS

In payment apps, the challenge is to acquire users and get them familiarized with the app / platform. Once the initial hurdle is crossed, users will be more likely to stick to the app.

To overcome this challenge Google used the ₹51 offer as an incentive for inviting new users. The amount was awarded after the referred user makes their first transaction. This meant that every new user acquired was also using the app at least once and getting familiarized with it.

Lucky draw scratch cards provided incentive for users to come back and use the app again. In later years Google also used gamification with Diwali stickers increase usage of the app. The stickers were awarded for completing different tasks within the app, thus showing users that the app can also be used for bill payments, mobile recharge, etc.

The internet was filled with advertisements like the one shown below which made the app go viral.

Figure 13.4 Google Pay Advertisments

GOOGLE PAY'S KILLER FEATURE

In day-to-day life repeat transactions for rent, groceries, bill split among friends are extremely common. People's usage behavior indicates that they often send money to the same few people.

Upon opening the app, the very first screen shows a list of contacts that the user has sent money to in the past. Therefore, Google Pay's UI is extremely helpful for sending money to people you have sent money to before.

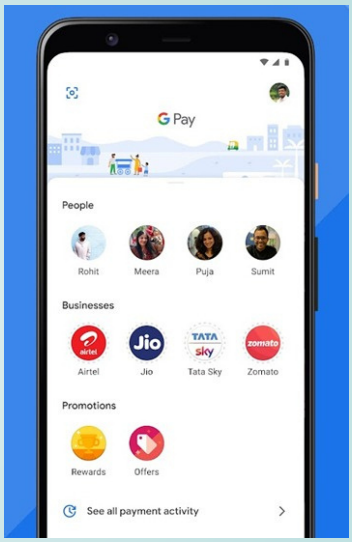

Figure 13.5 Google Pay Home Screen

While other apps also show a history of recently paid accounts, this is usually kept away from the home page. Google pay is unique and intuitive as it puts the most used feature right on top.

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HEALTH TECH

INDUSTRY OVERVIEW

Healthcare Technology, popularly known as Healthtech, alludes to the usage of technologies created to improve all aspects of the medical care system. From telehealth to robotic-assisted medical procedures, from sports apps to calories counters, all domains of health assistance leveraging technology can be termed as healthtech.

Drivers of HealthTech opportunity for the future

1. A growing landscape - Increase in health tech firms.
2. Growing market opportunity for Indian Health Tech providers - Increase in market size.
3. Acceleration in funding - Birth of new health tech-focused startups.
4. Increase in consumer awareness - Proliferation of B2C segments.

HEALTHCARE APPS

The healthtech industry is growing at a very fast rate, and is attracting a lot of VC funding, which has bolstered startups building apps to create specific solutions in the space. The global health market is expected to reach $111.8 billion by 2025, growing at a CAGR of 44.2%. As can be seen from the above figure, healthtech is a very broad segment with several different categories of apps - from building up habit and productivity, to healthcare checkups, from meditation and fitness to medicine deliveries and reminders. In this chapter, we have taken up the Indian startup - Practo to dive into details from a Product Management perspective.

Figure 14.1 Healthtech apps in different spaces

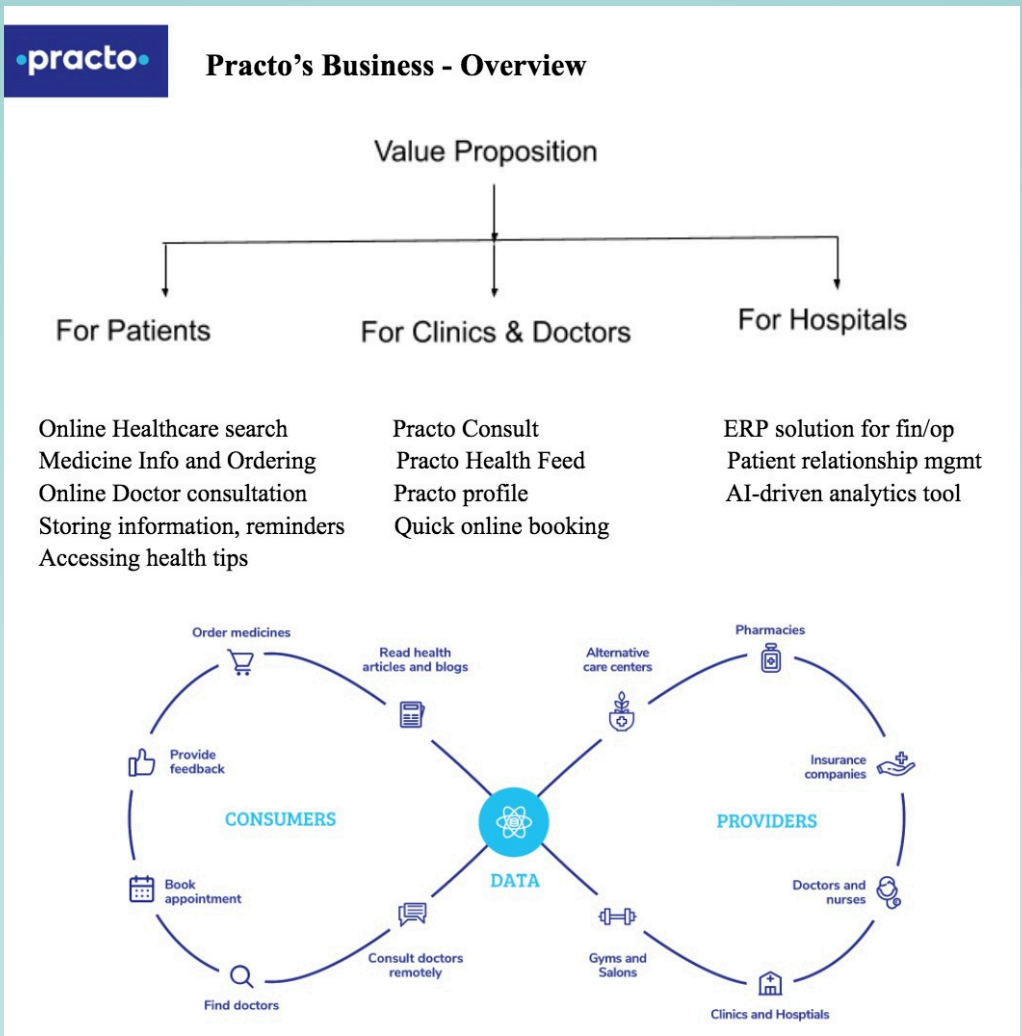

Figure 14.2 Practo data flow

- Key activities: Software development, Networking with multiple stakeholders, Promotions.
- Consumer/Partner segments: Patients, Clinics, Hospitals, Pathology Labs, Pharmacies, Insurance companies, Doctors, Delivery partners.
- Key resources: Website, app, Practo Manpower skills, IT.

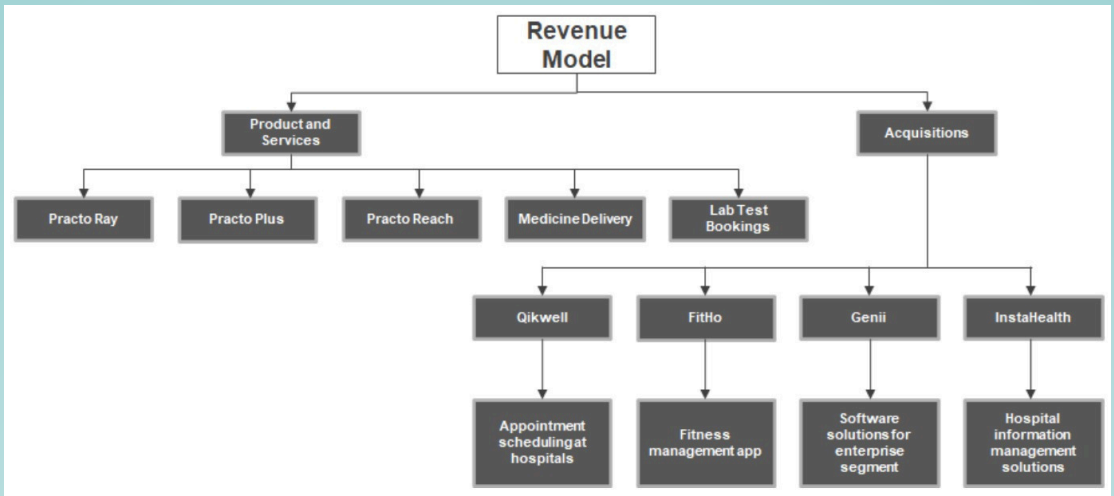

Figure 14.3 Practo revenue model

HOW PRACTO CROSSED THE 'CHASM'

Crossing the chasm is a theory by Geoffrey Moore built around the technology lifecycle for tech products. As we can see in the graph below, there is a chasm between the early market and mainstream market, crossing which can only help make any company leverage gains and a sizable market chunk for the product. To cross the chasm, companies need to provide products that add value to consumers and are easy enough to use.

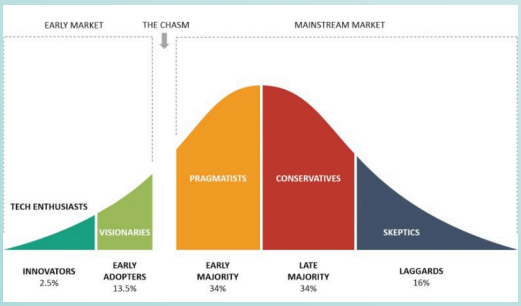

Figure 14.4 Crossing the chasm

What problem did they start with: To make clinic visits cashless, paperless, and painless.

MVP - Building the multi-sided platform: Practo was, in a way designed to help the patient find and book doctors in a seamless manner, avoiding unnecessary existing hassle. This made Practo a multi-side platform with several other stakeholders like clinics/hospitals, even for the MVP.

Most successful multi-sided platforms first need to set up the supply side, and only then can they expect the demand side. This meant doctor onboarding was a very important step. The pain point for doctors, clinics and hospitals at the time was maintaining patient information, records and maintaining billing details. So, Practo furnished them with a complete end-to-end practice management electronic tool that managed their consultations, prescriptions, bills, and medical records. Within 2 years, had onboarded 500 doctors and 500,000 patients.

Crossing the chasm: Now they needed to onboard large and popular doctors, as only that could pull in more patients, and thereby create network effects to pull in further more doctors and more patients. They came up with several features for the doctors, to assist them and had a 3-tier pricing plan and a free

version with limited features. Free service for patients and minimal monthly charge for listing doctors, once the number started growing. In this next year, the number of doctors jumped from 500 to 8000 doctors and 3 million patients.

For marketing, they went ahead with the 'Fleet on street model'. This same model has been used by Byju's, to grow. This model was to bring doctors onboard as we already discussed, this was the key in growing the app. They also initiated developing for B2C features – discovering the doctors and booking the appointments comprehensively. 'Practo Ray' got doctors on board and 'Practo Search' got patients onboard. They began a spree to map every doctor and clinic to ensure genuine doctors. By 2013, they had 10,000+ registered doctors and 7.5mn+ unique visitors. Once doctors were on board, they became habituated, so even though the cost of acquisition was high, LTV was higher. It was the two-sided marketplace which began showing network effects and helped them cross the chasm.

Entering the Tornado: This is the phase right after crossing the chasm, surviving which, the company becomes able to reap the benefits of the mainstream market. They introduced several features including Practo software and ads. They started with multiple acquisitions in 2015. They also expanded their product portfolio - medicine delivery, chat with doctors and even undertook geographic expansion.

User Profiling and Jobs To Be Done The most important step in designing any product or solution is to understand the WHY and WHOM. It begins by trying to understand the user and create a user profile, which is more than just demographic information. A good user profile should illustrate the behavior, tendencies and pain points of the user. From the JTBD primary statement, we create a job map. This is a step by step, what a customer is trying to do. It gives a 'needs view', rather than a 'solution view'.

User Profile	Riya 45, married, chronic thyroid issue, working woman, on the lookout for solutions that make her healthcare easy.	Bhushan, 28, married. Tech savvy, stays away from parents and wants to take the best care of them. Is always worried for them.	Shobhna, 22, single. Works in Bangalore, is new to the town. Wants to keep her minor health issues away without losing on time.
JTBD Statement	**"My Thyroid drains me physically and psychologically. I want the pain of managing my health reduced".**	**"My parents are getting old and I am always scared about their health".**	**"I need an easy way to look out for the most trustable doctors nearby".**

Pain Points	- Medicines prescribed by doctors are not easily available. - Forgets to take medicines sometimes due to schedule. - Not too many good doctors in the city. - Managing a huge health file. - Long waiting lines in hospitals.	- Forgets to remind parents. - Remote consulting not possible. - Constant fear of parents falling ill. - Finding new and better doctors is cumbersome. - Managing parent's profile.	- Limited knowledge of locality. - Don't know how to judge if a doctor is good or not. - Long waiting lines in clinics. - No reliable source for great health tips.
Jobs to be done	- Give me an easy way to manage health records. - Help me in planning doctor and lab visits. - Provide an easy way to order medicines. - Keep me updated with knowledge on thyroid patients' life.	- Give me easy accessibility to info on parents' well-being. - Give me ways to reduce the burden from my parents. - Provide an easy way to consult a doctor remotely. - Help remind parents to take medicines.	- Help me find a relevant and good doctor easily. - Give me a hassle-free and fast booking experience Trusted source of information on health. - Provide an easy way to manage bills. - Give me important healthcare tips.
Solution by Practo	Profile, doctor booking, medicine orders, personalized articles.	Practo Plus, reminders, medical history, profile, online consultation.	Test bookings, appointments, Health readings, Online payments.

APP CRITIQUE - CUSTOMER JOURNEY SNAPSHOT

Steps in general:
1.Identification of symptom
2.Quick research and self-treatment
3.Doctor selection
4.Appointment
5.Post appointment

How is it facilitated on the app - A Critique:

- Identification of symptom - Symptom to issue mapping is still relatively weak, and doesn't come out in the UI, though there is a provision for the same.
- Quick research and self-treatment: The app provides articles on issues and free expert answers. However, it does not seem to promote this idea, as the home screen should have had a search button to really make this a part of their offering.

Doctor selection - The process of selecting doctors is seamless and even personalized, as the doctors previously consulted, show up on the home screen. This has been the core offering of the app for a long time. Also provides reviews of doctors. Appointment - Seamless appointment booking process, where after selecting the doctor, you can easily select the required date and time from the simple UI. Post appointment - This is a new segment that they have entered and have been adding feature after feature with flavor of personalization. Be it ordering medicines, or lab tests and checkups, or reminders - the UI/UX is seamless.

A user's journey begins with the signup and onboarding. Then they can directly book an appointment or add more profile details. The app has a lot of other features, like setting reminders, adding medical history, viewing personalized articles, ordering medicines etc.

Here, we will only look at the snapshots of only some processes:

Home page:

Pros: To begin with, has a very good looking, familiar UI. Major offerings in the home screen and other offerings can be easily accessed from the menu bar in the home. Has a very effective nudge to the "Practo Plus" feature, which is a paid subscription. Has major features listed on the top of the home screen. Also, has a personalized home screen with info on doctors consulted.

Cons: It's all well and good until 'Not feeling too well', which provides a simple way to choose between options. At this stage usage starts getting overwhelming. A very lengthy home page (mostly seem to be ads). The app should have followed the maximum 3-scroll screen rule. People will just scroll through without reading. Has to be some bifurcation. Additionally, the plus feature only supports debit and credit card payments.

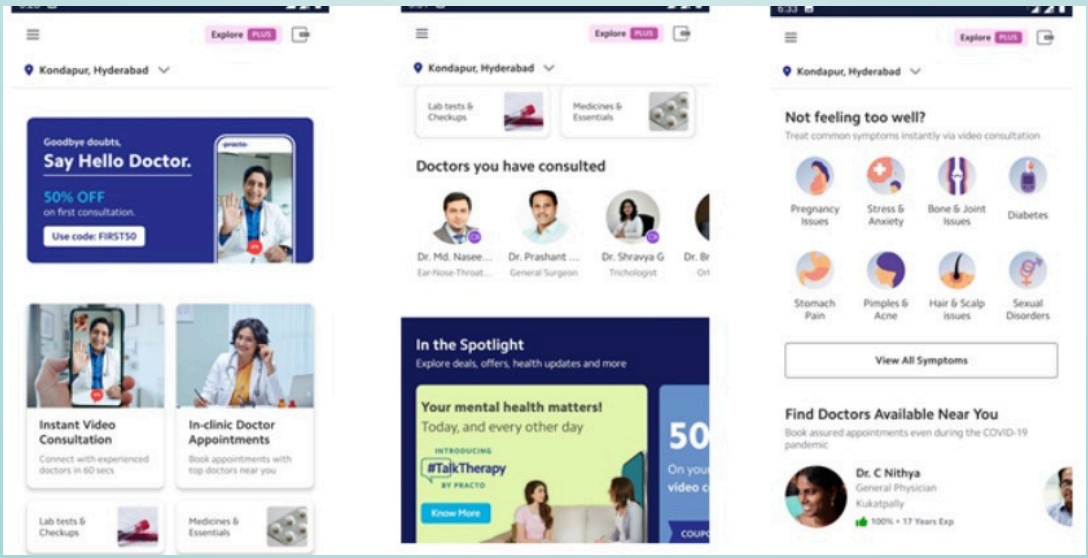

Figure 14.5 Practo home page

Menu bar

Pros: A very well made menu bar with standard icons that are intuitive, and with feature names that are easy to understand. Option to add medical records, add reminders. Very personalized. Also, the features are very simple and easy to use. The design is intuitive.

Cons: Your profile option is comprehensive which is a good thing. But people may not want to fill in so much detail manually without any incentive. Either the process could be made more seamless or it could be incentivized.

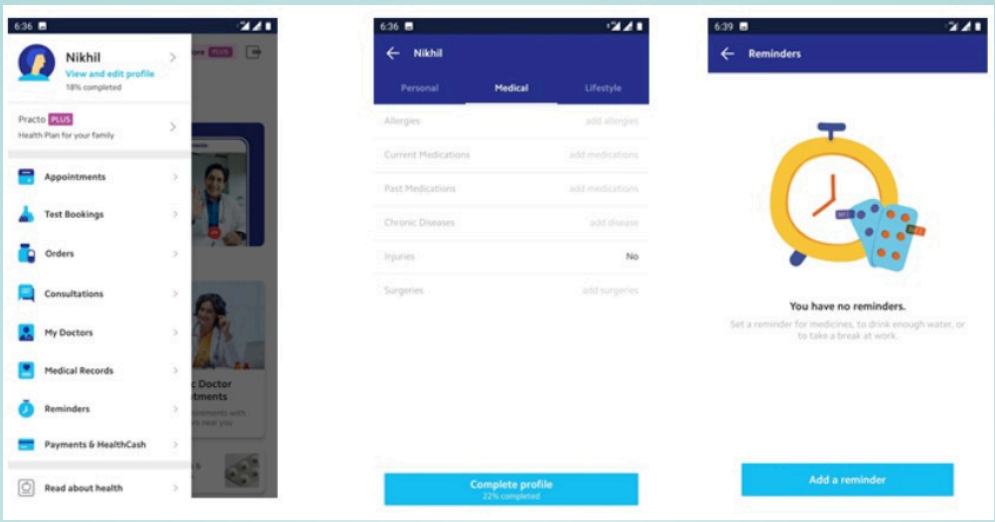

Figure 14.6 Practo menu bar

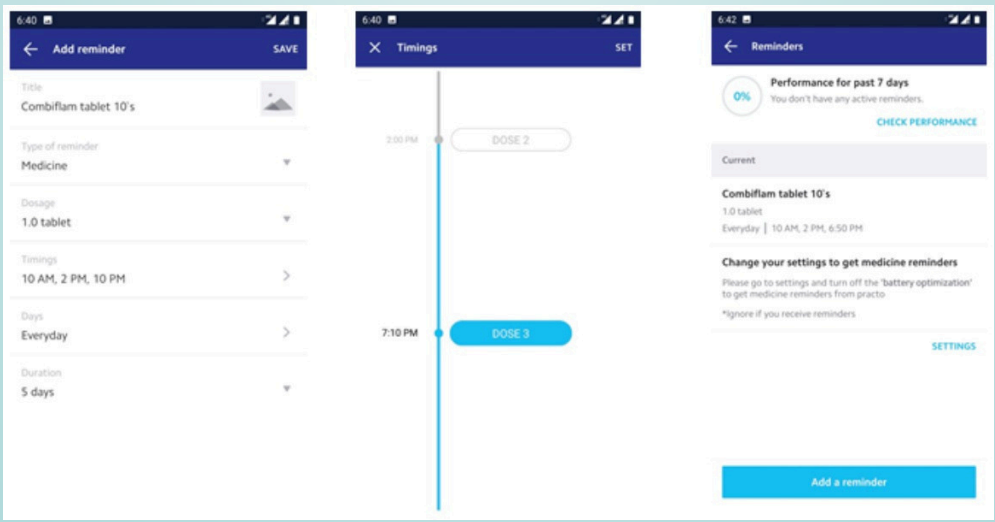

Figure 14.7 Practo reminder feature

APP CRITIQUE - PARAMETRIC REVIEW

The app has over 5 million downloads on the PlayStore and a rating of 4.2. However, the rating in the iOS version is 4.6, clearly showing the upside potential of improvement in Android. We will analyse the app based on several standard parameters of app critique:

Parameter	Comment	Verdict
Onboarding	The 20 sec onboarding video on playstore is too small to educate consumers. Features are easy enough though for self-onboarding for tech-savvy users .	Neutral
UI Design	Great colors, visible text. Easy access to features through the menu and easy to understand. Home screen is too lengthy.	Positive
Inclusivity	Despite being a healthcare app, it does not really cater to the specially abled users. No voice support, or easy motor assistance or haptic touch, or easy zoom in.	Negative
Ease of Use	For the literate English speaking tech savvy user, using the app is a cakewalk and extremely intuitive and familiar.	Positive
Diversity	Despite targeting an industry as huge as healthcare in a diverse country like India, no multi-language support could be easily found on the app while sign or even later on.	Negative
Innovation	Makes full use of technology today with added personalization and online consultation.	Positive
Addressing user pain points	As we had seen in the user profiling and JTBD, a lot of the user pain points are resolved by the app.	Positive
Compliance with Fitts's laws	The UI touch points on the app are large enough and have sufficient space between them for an ordinary user.	Positive
Nielsen Heuristic	Medicine order tracking, booking status visible. Gives control to users for canceling orders, changing locations. Has a help section. Design is minimalist but not throughout all screens.	Positive-Neutral

PRODUCT MANAGEMENT AND HEALTHTECH

Future Trends

Forefront versatile healthcare innovations are going to be founded on the following layers:

Open Marketplace - A great multi- sided platform that seamlessly integrates the entire healthcare ecosystem.

- Mobile Application - Has to be mobile, always on always connected.
- Blockchain Back-end - Blockchain is revolutionizing security and supply chain. Healthtech already has data privacy concerns and a growing number of stakeholders.
- AI & Big Data Analysis - Personalization, optimized recommendations, self diagnosis, the entire doctor, clinic and hospital ecosystem will be available on the phone.

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- Landscape overview - 2018 http://analytics.dkv.global/data/pdf/Health-Tech-Mobile-Apps-Analytical-Report.pdf

RIDE HAILING TECH

INDUSTRY OVERVIEW

The taxi-booking market was highly fragmented, and there was no transparency in the entire process (from booking to payment). Earlier, there were some regional players, but they were not established players, and the value proposition they offered were not significantly different from the traditional taxi- booking services.

Uber and Ola brought transparency to the entire process and have also significantly improved cab services' reliability. Lower search cost is also one of the additional value-adds that a customer/user enjoys through these services. The ride-hailing market in India is estimated to be ₹20000- 25000 cr. (Philip, 2020).

BUSINESS MODEL CANVAS

The following must be kept in mind when you address the question. It gives us an exhaustive list of the business dependencies.

Value Proposition - What is the problem/need addressed by the product to the different customer segments?

Value proposition also talks about the competitive advantage of the product.

Customer Segment - Understand the user who will interact with the product and how they will use it.

Channels - The channel through which the product reaches the customer.

Customer Relationships - Relationships that the customer expects from the company, and the cost required to provide that.

Key Activities - Activities required to provide the value proposition, and to maintain the channel, customer relationships and revenue source.

Key Resources - Resources required to provide the value proposition and to maintain the channel, customer relationships and revenue source.

Key Partners - Mention the partners and suppliers, and keep in mind the resources that they provide.

The business model canvas in the ride-railing business is as follows:

Figure 15.1 Business model canvas for ride hailing

When analysing platform apps, it is imperative to consult the interviewer regarding the side to be explored and the metric to be focused on. For example, in Uber and Ola, they will have different apps for the passenger and the driver. Also, keep in mind the 10 heuristics of user interface design that were developed by Nielsen and Morich and are mentioned in the chapter, "User Experience Design for Digital Products". The product can be assessed from the following point of view:

- UI/UX
- Current features
- Metrics used
- Near future improvements
- Long term product roadmap

The metrics that can be focused on are:

- Awareness
- Acquisition
- Activation
- Engagement
- Retention
- Monetisation
- Referral

CUSTOMER JOURNEY

Discussing an application using a customer journey ensures that you don't miss any critical parts. It also shows that you can think through any problem in a structured manner. To start with let's divide the user journey in three easy to understand parts – pre-ride, during-ride and post-ride

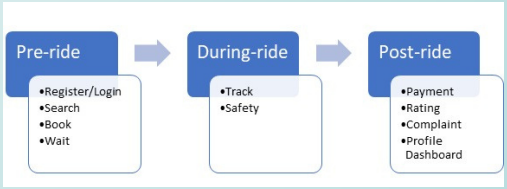

Figure 15.2 User Journey

Section I : Pre-Ride

This part of the app is most critical to business as it enables conversion of user into a rider and generate revenue. It can further be broken down into following four parts:

- **Register/Login**: A minimal details (usually mobile no.) based login is first step towards a lightweight experience.

Search: The current location is auto-picked, and the user needs to put in the destination. Apps also display the route on a map, which aids in selecting the destination location. Various options, like changing the starting location and adding stops, are included to account for alternate scenarios.

Book: Platform businesses add value by reducing the search cost and building trust between two different sides of the platform. The two sides here are drivers and riders. Based on the search, the user is shown various details like cab type and estimated cost before booking. For value-conscious customers, a sharing scenario has also been added.

Wait: After booking the cab, the user will receive the details of the cab. The ability to track the movement of the cab to the pickup location adds transparency and optimises travel time for the customer. The feature to cancel the cab at this stage boosts user confidence. Communication channels like calls and chat enable a smooth pick-up experience.

Section II: During the ride

Track: After the user boards the cab, the ability to track the movement on the map further builds consumer confidence.

Safety: It is difficult to ensure that only good drivers are onboarded and to control their actions. The app should provide features that can reduce untoward incidents during the journey. A few of such features are a boldly visible red emergency button and sharing ride details (and tracking ability) with others.

Section III: Post-Ride

Payment: Auto-debit facilities make user experience lightweight. Multiple payment options are provided to cater to everyone's needs.

Rating: Feedback loop enables forward movement of any system. In platforms, it also encourages its users (both riders and drivers) to perform their tasks well. A simple star-based rating input with default fill-in scenarios like driver was rude, fast driving, etc. ensures users can fill this easily.

Complaint: The ability to register complaints and get them resolved further builds user confidence. Companies often make the complaint feature less apparent to avoid misuse. Pre-filled common scenarios like charged more than shown earlier, missed belongings etc. adds to the simplified user experience and modularizing the system.

Profile Dashboard: A dashboard or a menu where user can check past trips, change default payment options, rating etc. also caters to various use cases. For instance, a user may want to share the bills of past rides for reimbursement from the employer.

DRIVER JOURNEY

Similar to customer's driver's journey can discussed.

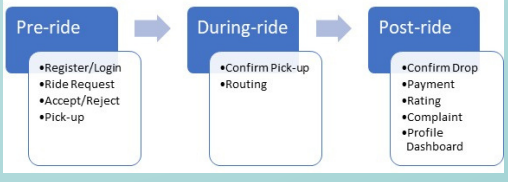

Figure 15.3 Driver Journey

HOW WOULD YOU IMPROVE A RIDE HAILING APP

Often app critique interviews turn into suggesting improvements in the product. A product manager is expected to be practical and as such one need to empathetically discuss the features with respect to both consumer and development. When asked for improvement, candidate can continue a structured approach. CIRCLES (Lewis, 2016) is a good framework which can be applied here.

1. Clarify your goals

In app critique, the goal will arise out of the part you are criticizing. Let's say we want to improve 'wait' part in the user journey discussed above.

2. Identify User Personas

Following are common users:

Student	Working Professional	Business Traveler
These are young individuals who mostly use ride hailing apps when public transport is not convenient and are conscious about prices.	These are young professionals; many of them don't own four-wheelers. Often use it for travelling to office and for travelling outside with family. Reasonably conscious about prices.	These are individual travelers who use ride hailing apps for business travels. Their expenses are reimbursed by the employers.

3. Report Use Case

The use cases for the above described personas are:

Student	Working Professional	Business Traveler
• I would like to reduce wait time and save cost by traveling to a nearby convenient pick-up location.	• I try to optimize travel time by simultaneously comparing wait time on multiple apps. • When travelling individually and in hurry, I won't mind travelling to a nearby convenient pick-up location.	• I try to optimize travel time by simultaneously comparing wait time on multiple apps. • I even cancel the ride when wait time is more to try and re-book a cab with lower wait time.

4. Cut through use cases

While the use case of comparing wait time across multiple ride hailing is a common scenario, building such feature looks out of scope. And the focus should be on improving the app to the extent that user don't feel like checking competitor app. With this let's discuss further about use cases of convenient alternate pick-up location and alternate ride finder with less wait time.

5. List Solutions

To solve above use cases we can consider below solutions:

Alternate pick-up suggester	Alternate cab finder with less wait time
• In case, the user can avoid large wait time by walking to a nearby location (<100 m) for pick up, the feature will suggest this to the user. Many times, although cab is close to user location but for reaching pick up spot user needs to travel around. In such cases, this feature can be of great use.	• After the cab is booked, this feature will keep searching for alternate cab which has less wait time. • It will be a user triggered feature and a nominal fee will be charged for it in case alternate cab is found.

6. Evaluate trade-offs

The trade-offs and metrics for evaluating success are:

Criteria	Alternate pick-up suggester	Alternate cab finder with less wait time
Target Market	Student (Major), Working Professionals	Business Traveler, Working Professionals
Frequency of Use	Medium	Medium
Development Complexity	Medium	High
Revenue Source	Increased app usage on account of increased user satisfaction	Increased user satisfaction + fees charged for switching to a less wait time cab
Test Metric	1) % Of users who use it when alternate pick-up location is shown 2) Difference in rating given by user after the feature is launched; further observe such difference for sub-groups – who use it and who don't use it.	1. % Of users who use it when feature is shown 2. Difference in rating given by user after the feature is launched; further observe such difference for sub-groups – who use it and who don't use it (to assess if there is any negative impact) 3. To assess the impact on driver, compare (before and after) following metrics at driver level: # cab rejects, time spent by driver on the app

7. Summarize

To improve the wait time experience of the users, we started by discussing common user personas – student, working professional and business traveller. Two solvable use cases were suggesting alternate nearby pickup locations and finding alternate cabs with less wait time. While the first feature is mostly intended for youngsters, the second is geared towards working professionals and business travellers. Further, the second feature is more complex to develop, and it may hurt drivers' morale. To assess the same, ratings given by users and time spent by drivers on the app should be tracked.

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TRAVEL TECH

The COVID-19 pandemic has very severely affected the travel industry and will probably continue to be the reason for its transformation. The sector was already transforming before the lockdown, with tech companies enabling users to travel better. In 2019, the travel industry peaked, with only leisure travel spending of $4.7 trillion (Statista, 2020). A user journey for travel currently comprises mainly discovery (where to go, what to do), booking (for transport, lodging, activities) and experience (supporting the during travel needs).

There are a variety of players in each step of the user journey and many players that cut across all three. Given a very long customer journey and technology advancements in both the digital and physical world, the strategy of a company is determined by the choices it makes on its business model.

On the basis of a known umbrella business model, we list nine different categories which are shaping the modern day travel experience of the customers (Skift, 2016).

Figure 16.1 List of choices across business model elements for travel tech Companies

Figure 16.2 Examples of companies in different business categories

Convenience and ability to compare prices have pushed the adoption of Online Travel Agencies (OTA) in India. The leaders now compete for a larger share of the market, and to do the same, continually focus on offering better services and improving customer experience. The website and mobile app experience become a big part of the overall user interaction with an OTA.

To develop or evaluate UX in a product we must look at it by keeping in mind five basic principles of UX Design (Barua, 2019)

1. Design for your users, not for yourself: Build for an identified user persona and not for yourself. Make it easy and delightful for these users to use the product.
2. Feedback: Give users a little visual confirmation/response for their actions.
3. Digestibility: Show users only limited content that they can comprehend.
4. Clarity: Design to minimise confusion by ensuring elements work as expected.
5. Familiarity: Take design cues from real words or acceptable digital standards.

We compare the UX of two popular OTA apps in India: MakeMyTrip & Yatra.

Use Case #1: Experience of Booking Flight Tickets

Stage	MakeMyTrip	Yatra
Login Page*	- Clean UI with friendly priming on Best Deals to increase user excitement	- Lot of login Alternatives - User not asked to log in till personal information needed for booking
Home Page*	- Highly Overloaded with options and banners causing confusion	- Highly Overloaded with options including some that aren't a user's priority
Flights Discovery Page	- Easy Flight Search & Date Picker with Pricing information - Complex sorting menu requiring an extra click	- Easy Flight Search & Date Picker with Pricing information - Easy sorting menu with a mostly familiar process
Flights Booking Page	- Lot of information affecting user digestibility and causing user fatigue - Auto apply of best offers and picking of saved traveller information add to ease of use - Cross-sell of cab services to user needs well timed and beneficial to both parties	- Lot of information affecting user digestibility and causing user fatigue - Auto apply of best offers and picking of saved traveller information add to ease of use - Additional pop up for the purchase of insurance before allowing a confirmation might seem pushy
Payment Page*	- Multiple payments options - Recommended and Saved options help save the user time	- Multiple payments options - Recommended and Saved options help save the user time

Use Case #2: Experience of Booking Hotels

Stage	MakeMyTrip	Yatra
Hotels Discovery Page	- Easy Hotel Search with the ability to input Location, Dates, Guests & Room requirements - "Trip Type" filter might enable personalization for users	- Easy Hotel Search with ability to input Location, Dates, Guests & Room and Check-In/Out timing requirements
Hotels Listing Page	- Recommended deals to help the user discover special offers - Map view helps user select as per their location preference - Sorting is complex, and the filters are not aligned with user needs and rather are set as per MMT offers	- Easy and multiple sorting options help users find their hotel faster - Search option allowing users to look for a specific hotel saves time and effort - Filters based on amenities help in better shortlisting - 3rd party ratings giving more confidence during selection rather than app native ratings
Hotels Booking Page	- Video + photo gallery helped built user confidence - Easy checkout page with all essential details mentioned again before confirmation	- Photo gallery for all hotels helped built user confidence - Auto best offer applies to save the user some time on the checkout

**Login Page, Home Page and Payment Page are same across both experiences*

RECOMMENDING A NEW FEATURE USING "CIRCLES" FRAMEWORK

The CIRCLES framework (Lewis, 2016) is used by product managers for identifying new features & products as it provides an exhaustive & structured framework for the same.

Stage 1: Clarify Your Goals

At this stage, one must aim to clarify doubts in order to provide suitable recommendations.

Some possible questions at this stage could be:

- What is the end goal of this feature?
    - Increase user engagement
    - Improve UI/UX
    - Increase revenue/sales
    - Additional In-App
    - Purchases
    - Increased bookings
    - Increased advertising revenues
    - Increase market share of product

Does the feature have to be in any specific domain of the current app structure, e.g. Hotels, Flights, etc.?

Yes, it has to be cohesive with the existing structure

No, it has to cover a new domain

We have clarified that the new feature needs to increase revenues through one of the three revenue streams, and it has to be cohesive with the existing structure.

Stage 2: Identify User Personas Knowing and understanding customers and their needs are crucial. Let us consider the customers we had identified using the Business Model developed earlier in this chapter. We will attempt to create a background of their general behaviours, demographics, app journey and possible needs to achieve clarity.

Family	Solo Traveler	Business Traveler
- Families are groups of individuals who are likely travelling with kids. - This customer segment would like procedures that are family-friendly, i.e. suitable to their kids' needs as well as their own.	- Solo Travelers are individuals who are commuting by themselves without any colleagues, family or friends for accompaniment. - Since they are alone, they would prefer systems which they can navigate by themselves as well. For example, they might like vending machines over takeaway food counters since they don't have to worry about keeping a check of their baggage.	-A business traveller is an individual who travels between destinations typically alone or with their colleagues for pure business purposes. - They prefer services that are good quality and professional and want to avoid all forms of delay. In general, they look for very low interface and hassle-free procedures

Stage 3: Report Use Case

In this stage, we try to connect the aforementioned user personas with their needs.

Family	Solo Traveler	Business Traveler
- I want to be able to compare the entertainment options at different hotels for my kids - I want convenient seating areas at the airport with my kids with food, toilets etc.	- I want a place to keep my luggage securely when I visit the washroom or food stalls at the airport - I want someone to accompany me for activities & tours that can't be done alone	- I don't want the hassle of checking into hotels for short term trips. I want a comfortable rest stop with few disturbances, good quality WiFi and charging ports where I can rejuvenate. - I want easy web check-in to avoid lines - I want to be able to compare hotels based on their amenities like business centres, WiFi speeds etc.

Stage 4: Cut-Through Use Cases

In this step, we cut through different use cases listed above and choose a few that stand out based on our criteria or goals, like revenues (in this case) or customer benefits. Based on the above use cases & our end goal of increasing revenue, we shortlist our top cases:

1. Quality Airport Seating
2. Compare / Filter Hotels

These are chosen as they would be more likely to have a higher frequency of use and be relatively easier to monetize as compared to the other options.

Stage 5: List Solutions

In this step, we list a couple of solutions to solve the above-shortlisted use cases. The most feasible solutions, meeting users' needs post this step, will be considered for evaluation.

Use Cases	Quality Airport Seating	Compare/Filter Hotels
Solutions	**- Lounge Discovery & Booking** The mechanism to book seats at an airport lounge any time between booking & journey dates. Check for compatible card offers, view the number of empty seats etc. - Internal Airport Maps Contain details like current footfall, eating and charging options. This option might be infeasible due to airport security restrictions	- New Filters Provide additional filters which suit user needs. However, a large number of filters can reduce user-friendliness. **- Hotel Comparator** Add multiple hotels to your consideration set & compare them on a customizable set of attributes (similar to mobile phone specs comparison)

Stage 6: Evaluate Tradeoffs

In this step, we evaluate the solutions on various metrics to identify which solution will be most beneficial for the end-user while simultaneously meeting our goals and not overburdening our development resources.

Metric	Lounge Discovery & Booking	Hotel Comparator
Frequency of Use	Medium to High	High
Dev. Complexity	Medium	High
Target Market	Business Travelers	Families & Solo Travelers
Revenue Source	Increases revenue through in-app purchases	Increases revenue by increasing customer engagement, leading to higher advertising revenues
KPI Metrics	- % of flight ticket customers who use this feature - % growth in lounge bookings - % growth in revenues from flight lounge bookings	- % increase in customer activity time - %increase in customer visit frequency - % increase in advertising revenues

Step 7: Summarize

In the final step, we discuss our identified solution and the rationale behind the same: "For our identified goal of increasing revenue by developing a new feature within the constraints of the existing infrastructure, I recommend developing a Lounge Discovery and Booking Feature. This feature will primarily target Business users with families being secondary targets. It has been chosen as business travellers typically have a high frequency of travel, and since their bills are reimbursed through company accounts, they are less likely to be price elastic. We can promote this feature to be an alternative to booking hotel rooms for shorter trips. This feature is also less complex to develop than the alternatives and the KPIs are easier to measure for an increase in revenue as they share a direct link as compared to advertising revenues"

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EDTECH

INDUSTRY OVERVIEW

'EdTech' refers to the use of technology in the education sector, starting from smart devices in classrooms, test prep platforms and online certifications for skill development. India has over 1,100 million wireless internet subscribers (Statista, 2019) with a high rate of adoption. The EdTech sector stood fourth amongst all sectors, with 8.5% of total startup investments in 2018 (Statista, 2018).

The startups involved in K12 and online certifications have accounted for a significant share (~88%) of $1.6 billion worth of investments, in the education sector between 2014 and 2019 (inc42, 2020). This skewed funding is in part due to the grades- first approach of the Indian market and the need for skilled labour in the fast-growing services sector. Moreover, Indians have a higher willingness to pay for education, and consumer income inelasticity in this market is greater than even healthcare in some cases.

CUSTOMER NEEDS

Currently, the apps in the market work as a supplement to the classrooms, like the tuition classes where the student's learning is self- paced and tailored to their learning speed.

A school student may start using such an app at a very young age, and continue using it through to grade 12, and for associated competitive exams.Hence, their journey becomes all the more critical. Therefore, the experience should not be homogenous for everyone, as is the case in industries like ride-hailing, where the service is the same irrespective of the characteristics of the user. It needs to be tailored to individual ability and interests.

STUDENT JOURNEY

The typical customer journey of a school student on an EduTech platform looks like:

Figure 17.1 Student Journey

PARENT JOURNEY

Parents play an essential role in all the major decisions of their kids, and education is no different. As the amount involved is significant with typical yearly packages in the tune of 10000 and above, the sale is only possible when both the students and their parents see value in the platform.

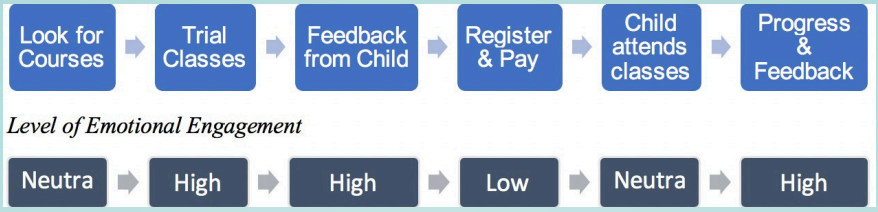

Figure 17.2 Parent Journey

Byju's

Founded in 2011, Byju's is the highest valued EdTech company in India with a valuation of $11 billion as of September 2020 (inc42, 2020). It engages in online tutoring with pre-recorded course videos, which can be viewed either through a mobile application (iOS/ Android) or on the Byju's tablet.

Byju's has a range of apps, which targets students from kindergarten right up to grade 12, as well as for competitive exams, which significantly increases the customer lifetime value.

Figure 17.3 Byju's range of apps

Vedantu

Founded in 2011, it operates as a marketplace for teachers and students where the students can enrol in classes of the tutor of their choice. It uses a proprietary virtual learning environment, WAVE (Whiteboard Audio Video Environment) to deliver live classes through its website and mobile applications.

Vedantu caters to students of grades 6 to 12, primarily from the CBSE and ICSE boards. They provide summarised notes at the end of each class.

BUSINESS MODEL CANVAS

The points in red are specific to Byju's & those in yellow are specific to Vedantu, while the remaining points apply to both the companies.

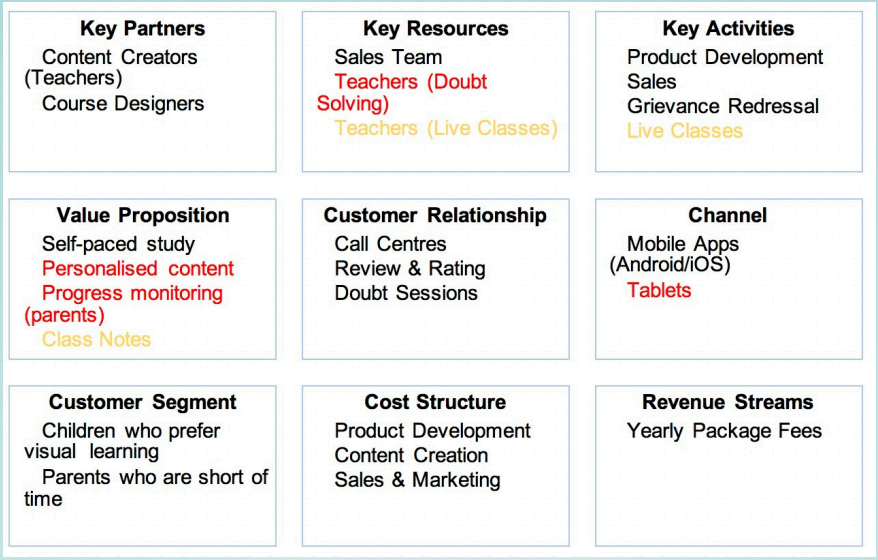

Figure 17.4 Business model canvas

EVALUATION OF THE PRODUCT

The companies operating on the subscription model have two primary tasks at hand. Firstly, signing up the customer, and secondly, reducing the churn through higher engagement and value delivery. The UI/UX and features of the apps have been evaluated based on the following three stages:

- Trial & On-boarding
- Nurturing & Expansion (Nielsen's Heuristics)
- Renewal & Loyalty

Trial & On-Boarding

Critical Factors	Byju's	Vedantu
Pricing	Yearly subscription fees. Actual prices vary based on location sales agents.	Freemium model. Upgrade to a yearly subscription.
Trial Classes	Minimum trial period pre-buying.	Trial live classes & notes.
Awareness/ Promotion	Free classes for one year on account of COVID-19. Advertisements, Celebrity Endorsements. Watch on any device, any time.	Free courses for one year on account of COVID-19. TV Ads, Social Media. Watch on any device, any time.
Ease of Use	Mostly single click. Design based on user: Cartoons for kids, simple vanilla for parents. Recommendations to start.	Option to explore courses & webinars on the landing page. Mostly single click. Minimalist design.
Options of Subscription	In-app purchases not allowed, need to contact service agents. Only yearly subscriptions.	Only yearly subscriptions.

Nielsen's Design Heuristics

Critical Factors	Byju's	Vedantu
Visibility of System Status	Messages like *'Ready, set, go'*, *'Let's keep exploring'* appear when loading content.	Loading status not visible while changing tabs.
Match between System & the Real World	Animated lessons with familiar Disney characters for kids up to grade 3. Test feedback and progress monitoring for parents.	Live classes that mimic the classroom sitting with students asking questions. Students get to take back notes for revision.
User Control & Freedom	Easy navigation through categories and sub-categories.	Mismatch in the redirection while using the on-screen and smartphone's back button.
Consistency & Standards	Consistent themes & graphics. Seamless integration of the various apps with the Parent Connect app.	Consistency of colour of graphics. All live classes happen in the standard WAVE environment.

Error Prevention	Minimizes the videos into a thumbnail, if back button pressed by mistake.	A countdown timer for the upcoming classes. Options on the navigation drawer inadvertently lead to an error page.
Recognition rather than Recall	Recommended list of lessons. Progress monitoring for parents to guide the child. Dedicated search bar absent.	Timer for upcoming classes. Alert to view any missed lessons. Dedicated search bar absent.
Flexibility & Efficiency of Use	Handholds the student for most of the duration.	Provides notes for easier revision of the lessons.

Aesthetic & Minimalist Design	Landscape Mode. Background: Dark (Disney Early Learn), Light (Parent Connect). High-resolution graphics. The kids' app at times looks cluttered due to a lot of dark colours and characters.	Portrait mode, changeable to landscape for videos. Light background. Minimal yet sufficient data during ongoing classes. Landing page, on reopening the app shows promotions of extracurricular classes/events.
Recognize & Recover from Errors	Quick prompt on errors in user input and option to correct.	Quick prompt on errors. Redirects to the home page on most occasions.
Help & Documentation	No documented help section.	No documented help section. Option to mail a query.

Renewal & Loyalty

Critical Factors	Byju's	Vedantu
Service Quality	Freemium Model: Pay to unlock features. Schedule sessions for working on doubts & queries. Seamless integration of Parent Connect app with others for monitoring progress.	Freemium Model: Pay to unlock features. Continuous resolution of doubts during live classes.
Renewal	Yearly subscription renewed on moving to next grade.	Yearly subscription renewed on moving to next grade.
Value Added Products	No upselling/cross-selling on the app. However, the same is prevalent when agents are selling.	The landing page shows extracurricular classes/events.

RECOMMENDATIONS & PRODUCT IMPROVEMENT

- **Introduction of a Search Bar:** It would allow students to directly look for the lesson without wasting time remembering the path and navigating to the same.
- **Inclusion of a Help Document**: A guide to help resolve issues on the go would improve the user experience, as the current methods of calling the service executive or waiting for a reply in the mail are too cumbersome .
- **In-app Purchases**: Allow upgradation of the content by paying from within the app, without calling up the service centre, with transparent pricing.

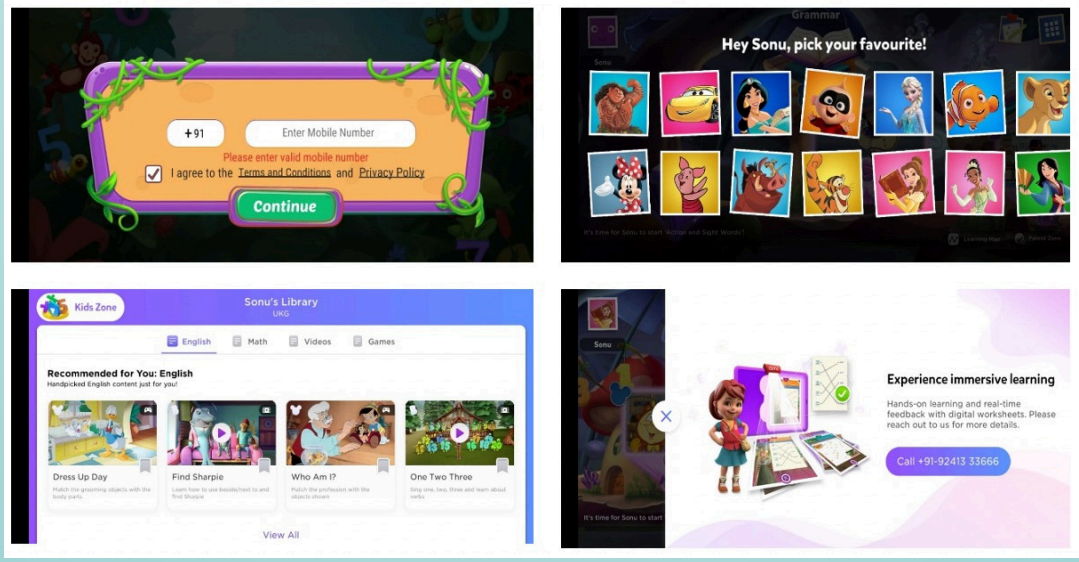

Figure 17.5 Disney Byju's early learn app (v 2.8.0; 21st sep 2020)

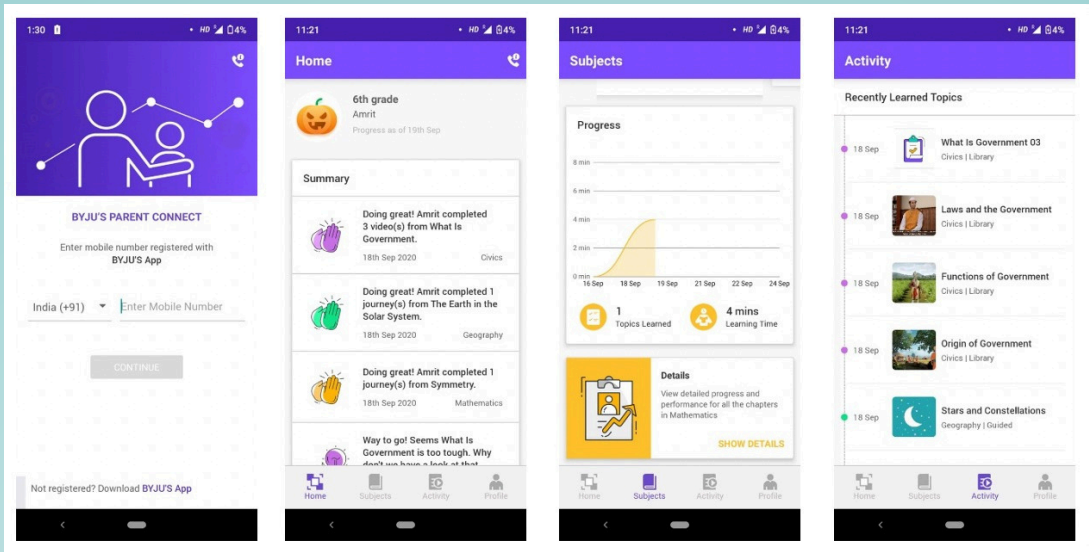

Figure 17.6 Byju's parent connect app (v 3.3.0; 21st sep 2020)

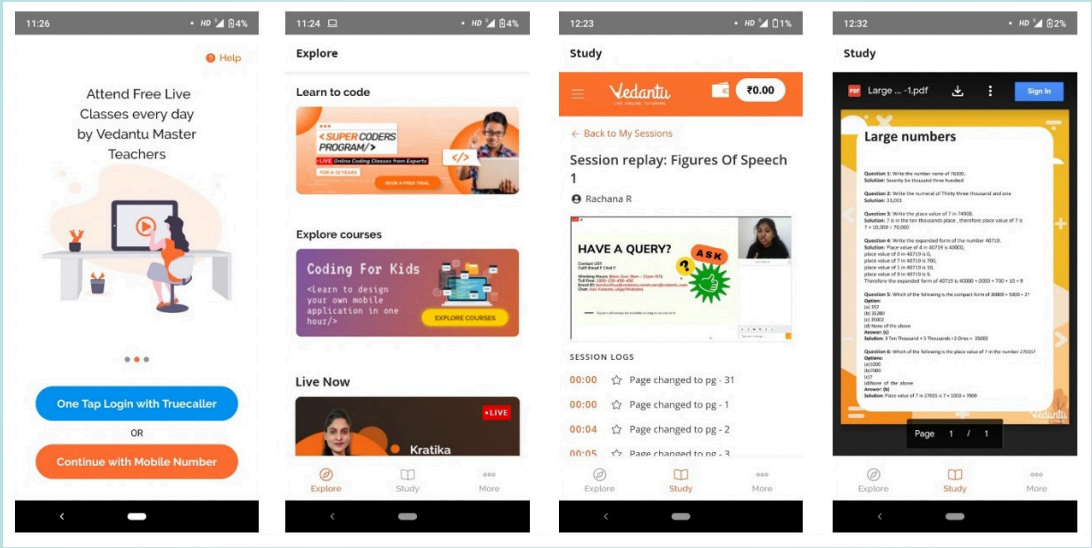

Figure 17.7 Vedantu app (v 1.7.3; 21st sep 2020)

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FOODTECH

INDUSTRY OVERVIEW

Foodtech applications have seen tremendous growth in recent years. The contribution of online orders to the overall food delivery market has grown from 2% in 2016 to 11% in 2020. Zomato has shown an increase of 4x in the number of orders in the first quarter of 2020 as compared to 2019, showing the increasing popularity of online food delivery applications. The industry comprises a multitude of business models. Some of the prominent business models include:

- **Restaurant Aggregator:** These platforms host restaurants and allow users to search for restaurants. For example, Zomato, Swiggy.
- **On-Demand Food:** These restaurants have their centralised kitchens and take orders through their application/website. For example, Rebel Foods and Freshmenu.
- **Ready-to-heat food:** The platforms that provide ready-to-heat meals depend on the demand or the subscription. For example, Freshly.
- **Home Food to the Office:** These companies enable users to place delivery orders and offer pickup facilities for the food prepared by home chefs. For example, Ootabox.
- **Recipe Discovery:** These platforms enable users to search, store, and upload their recipes. For example, Tastemade.
- **Personalized nutrition:** These companies offer personalized nutrition plans and diets to the users. For example, DayTwo.

Other business models include, but are not limited to dine out apps, recommendation apps, innovative restaurants, catering platforms, chef marketplaces, and novel foods.For further discussion, the restaurant aggregator business model is chosen, and in particular, Zomato and Swiggy are the focus applications.

BUSINESS MODEL ANALYSIS

Swiggy: Swiggy is limited to helping users to find restaurants/dishes. Once the user places an order, Swiggy's delivery partners will visit the restaurant to pick the order and will deliver to the customers.

Figure 18.1 Swiggy business model canvas

Zomato: In addition to what Swiggy offers, Zomato also provides a platform to explore the detailed reviews of the restaurant along with a pickup option. They also provide a 'hyperpure' option to restaurants, aimed at providing high quality fresh raw material to the restaurants. The elements of business canvas which differ from Swiggy are colored in red.

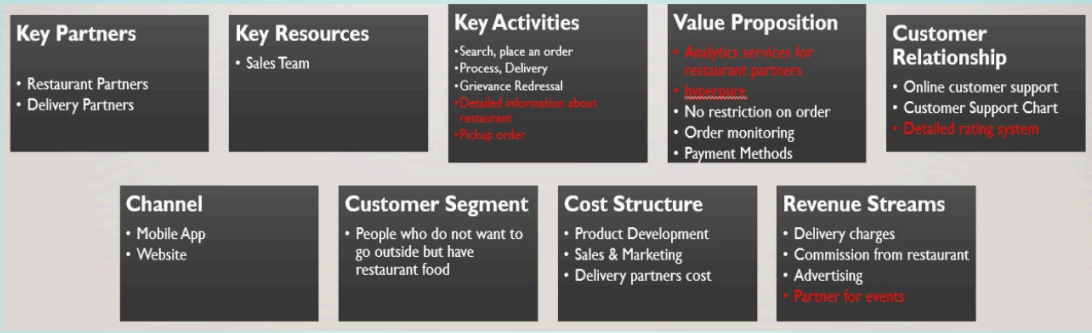

Figure 18.2 Zomato business model canvas

USER PERSONA

Three user personas can be defined based on the frequency of orders and the affinity towards dishes.

1. A user who is new to the platform wants to explore various restaurants or has no bias towards a particular restaurant or dish.
2. A repeat user who wants to explore the dishes of a particular cuisine/restaurant only.
3. A repeat user who frequently orders previously ordered items from favourite restaurants.

TYPICAL USER JOURNEY

Swiggy: The typical user journey in Swiggy is as follows:

- **Login/Signup**: The user logs in using a mobile number/email or social media account. The user can skip the log in process and explore the restaurant menu.
- **Restaurant Search**: The user searches for the restaurant and selects a restaurant from which they want to order.
- **Placing the Order**: The user selects the dishes they want from a restaurant and then places an order.
- **Tracking**: After the payment is done, the user tracks the order's progress.

Zomato: The typical user journey in Zomato is as follows:

- Login/Signup: The user logs in either using a mobile number/email or social media account. The user can skip the log in process and explore the restaurant menu.
- Restaurant Search: The user searches for the restaurant and selects a restaurant from which they want to order.

Two choices are available to Zomato users: similar to Swiggy users. They can place an order or explore more about the restaurant through detailed reviews and photos.

- Tracking: If the user goes to place an online order through the application, he/she can track the order's progress.

APP CRITIQUE AND COMPARISON

Disclaimers: The critique is done on the basis of the apps' design as on 1st Oct, 2020.

Sign-in Option: Lazy sign-in is an important feature, considering a hungry user will be looking for food quickly and hence forced sign in at the start can increase the bounce rate. Both the applications give an option of lazy signing in and allow users to explore the dishes and restaurants without mandating them to log in. The sign in can be done when the user is placing the order. But, the option to skip the login is more intuitive in the case of Zomato with the button title as 'Skip'. In the case of Swiggy, the option of lazy sign-in is available with button tile 'Swiggy' making it less intuitive than Zomato.

Restaurant Explorer Page: This is the page where the users will explore the different restaurants available.

Swiggy: Swiggy's restaurant explorer page looks cluttered, with offers floating everywhere. They could instead have only one row for offers, rather than bombarding the users with offers. They do provide 'Top Picks for You' at the beginning, accounting for the user preferences or the past orders of a user. In the restaurant display panel, they have time for delivery, the average cost per person, and the rating of the restaurant. But they do not have any filters on the page. Also, the extracted number of people who have rated the restaurant is unknown, impacting the trust of the user. Towards the bottom of the page, the user has search, cart, and account options. The cart option seems unwarranted when there are no items in the option. The option could have been made dynamic.

Zomato: Zomato's restaurant explorer page contains a list of restaurants, similar to the Swiggy. But the information related to the number of orders since the lockdown has been imposed is given so as to gain the trust of the user. There is also an option to bookmark a particular restaurant. Also, the filtering options including cuisine, rating, cost per person along with sorting options by delivery time, cost, and popularity are available, easing out the process of finding the best suitable restaurant for the user. The app also offers top brands in the spotlight, what's new on Zomato, and other options to help out explorers to try new items or the items liked by the community. Towards the bottom of the screen, there are options for history, videos, and

account management. The history option is particularly useful for the third persona, who might be interested in repeating the past orders frequently. Also, the video option gives access to Zomato originals, cook with pros, easy to cook recipes increasing the user engagement in the application.

Restaurant Landing Page: When the user finalizes which restaurant to order from, they land on the restaurant's landing page, where they can select dishes, from the restaurant's menu.

Swiggy: The restaurant landing page in Swiggy shows a rating, delivery time, and cost per person on the page. The number of people who have rated the restaurant is not made public and is shown as 20+ or null on the page. Swiggy could show the number of reviewers explicitly, to gain more customer trust and add credibility to the rating system. Also, it provides an option to mark a restaurant as a favorite. As compared to Zomato's bookmark feature, mentioned in the restaurant explorer page, the favorite restaurant feature in Swiggy is well placed, as the user will mark a restaurant favorite only after looking at the menu. The page also marks some of the dishes as best sellers on the menu, but they are randomly placed on the menu. Swiggy could have placed the best seller items at the top. to provide the user convenience of finding the popular dishes.

Zomato: Zomato's landing page has detailed reviews of the restaurant, along with time for delivery. The review highlights are also provided, to get the gist of all the reviews making the process of going through the review easy and comfortable for the user. The app also provides reviews for all the dishes in the restaurant for gaining the trust of the user. This, in my opinion, is one of the top differentiating features in Zomato, as not all the dishes in the restaurants are equally good. It also provides best seller dishes, and unlike Swiggy, best seller dishes are displayed at the top, making the ordering process convenient for the users. Zomato also tries to increase the order value through suggestions such as popular add-ons, which get displayed when the user is in the process of completing the payment.

Covid Related Initiatives: In times like Covid, users care about safety the most. In any food delivery platform, food transfers from restaurant personnel to delivery partners to customers. As multiple people are involved in the end-to- end delivery of food, gaining the trust of the users, providing them with confidence around the safety measures taken by the platform is of paramount importance to any food delivery platform.

Swiggy: Swiggy mentions 'delivery partner safety' on its restaurant explorer page. But it does not mention the safety precautions followed by a restaurant. Also, the delivery partner safety banner is not at the top of the page. Hence, the user at first glance, would not see the initiatives taken by Swiggy to deliver the food safely.

Zomato: Zomato mentions how food delivery is safe and how customers' safety is a priority for them. They have also mentioned the number of orders delivered since the lockdown started. They also mention the number of orders from each restaurant, giving a sense of safety to the customers. On the restaurant's landing page,

Zomato explicitly mentions the safety measures followed by a particular restaurant in an attempt to reinforce the trust in the customer.

In a nutshell, zomato appears to be doing better in terms of customer convenience, easing out the process of food ordering, bringing in more transparency and customer engagement. Order tracking on the apps is not covered in this chapter, and is left to the reader to explore.

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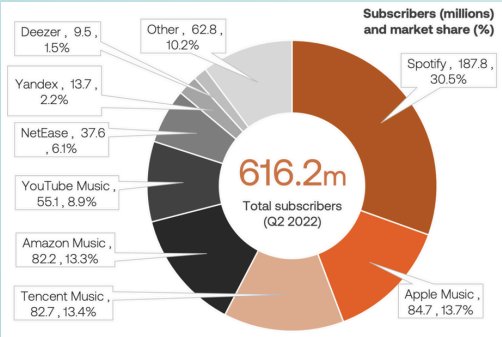
APP CRITIQUE: SPOTIFY

THE MUSIC STREAMING BUSINESS:

In today's digital age, music streaming has rapidly ascended as the predominant mode of music consumption for a vast majority of global listeners. At its core, the music streaming sector can be analogized to a commodity business – where the intrinsic business model yields scant differentiation among competitors. The true competitive advantage, thus, lies not in the infrastructure but in the user interface and experience. Over recent years, Spotify has carved a significant market share, distinguishing itself not by the mere provision of songs but through designing an unparalleled user experience.



USER PERSONA:

To better grasp Spotify's target user base, we can define it using user personas, categorized by their listening habits:

1. The Passive

- **Demographics:** 32, Female, Account Manager
- **Insight:** Relies on familiar sources for music, seeking simplicity in her choices.
- **Needs:** Effortless streaming; mood-aligned selections.
- **Barriers:** Overwhelmed by vast choices; unsure about discovering new tracks.

2. The Curator

- **Demographics:** 20, Male, Student
- **Insight:** Crafts thematic playlists; values shared musical experiences.
- **Needs:** Tools for collaborative playlists; expansive music catalogue access.
- **Barriers:** Misses out-of-app suggestions; often gets misaligned recommendations

3. The Nostalgic

- **Demographics:** 40, Male, Photographer
- **Insight:** Treasures past music; desires updates on favoured artists.
- **Needs:** Direct access to cherished artist catalogues.
- **Barriers:** Unaware of favourite band updates; inundated with unfamiliar tracks.

The customer journeys arise from the personas driven by their respective primary needs.

APP CRITIQUE: SPOTIFY

Upon entry, Spotify delineates three core navigational paths:

1. Home
2. Search
3. Your Library

1. The Home Screen

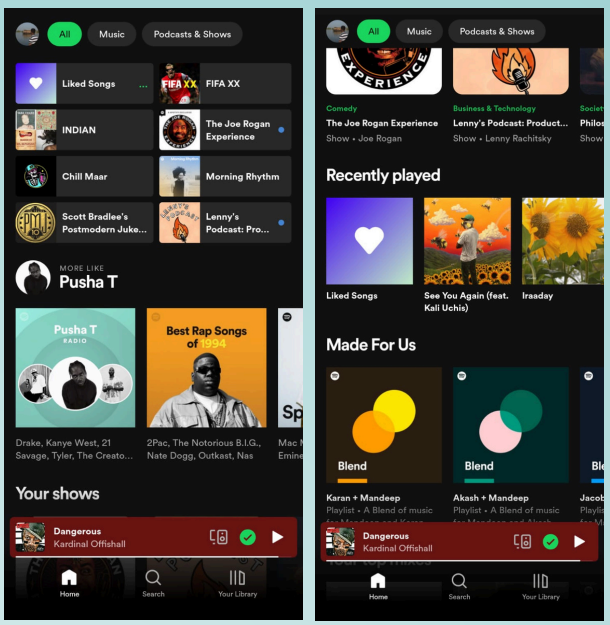
At the top, three chips segregate the experience into ‘All’, 'Music' and 'Podcast & Shows'. This initial categorization allows users to tailor their homepage journey by intent.

Below it, the focus is on continuity. The recently played content is prominently displayed, facilitating a seamless continuation of prior sessions.

Scrolling down, driven by both time-of-day and historical engagement, the algorithm curates a selection of suggestions. The visual appeal is enhanced through dynamic playlist covers and animated GIFs, catalyzing user engagement.

Below it, algorithm-driven recommendations are shown to let users select what they want to listen to. The recommendations are heavily driven by the time of the day and previous listening activity.
There is visual variety in the way recommendations for playlists are shown. They utilize playlist covers and GIFs to drive a CTA for listening.

A key User Delight experience is the “Made For You” playlists, which also include social aspects like Mixed playlists with friends. They capture both personalization and sociability.



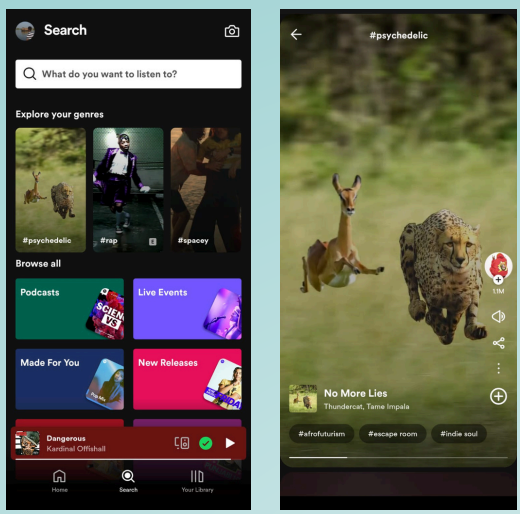
The Home Screen

2. The Search Screen

The Search Screen is designed for users who approach the platform with a clear listening objective in mind. Dominating the upper section is a prominent Search Bar, accommodating not just song titles but even lyrics. Users can also input genres/moods to find playlists, reflecting a commitment to a frictionless search journey.

Further down, there are short-form videos under the tile “Explore your Genres”. This is an experimental feature that aids users in discovery by the TikTok format of short-form videos. The videos, however, are very short GIFs, and the user is unlikely to be engaged for long.

Scrolling along, there are cards to browse all genres and categories of playlists that exist. This helps with users who have high intent on a genre/category.

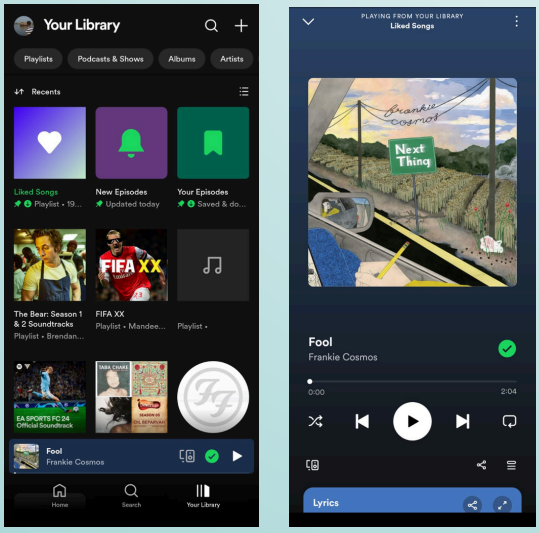


Search Screen & “Explore your Genres” Screen

3. The Your Library Screen

This screen caters to users wanting to listen to their curated content via saved playlists, podcasts and albums. It‘s a relatively simple screen and allows users to quickly play what they really like.

The chips at the top allow users to search by category for quick action.



The Your Library & Now Playing Screen

4. The Now Playing Screen

The Now Playing Screen is fairly standard. Users can add songs to their Liked Songs playlist. A unique feature is to have a “Magic Shuffle“, which adds recommendations sporadically to the already playing queue.

Live Lyrics below Playing Now add user delight and add something new to the experience.

Scrolling down, there is more information about artists and also tour dates catering to live music fans.

In The Pipeline - Spotify AI DJ

The DJ is a personalized AI guide that knows you and your music taste so well that it can choose what to play for you. This feature, first rolling out in beta, will deliver a curated lineup of music alongside commentary around the tracks and artists Spotify thinks you’ll like in a realistic voice.

Strengths

Spotify's core competencies lie in its unerring commitment to personalization and its distilled simplicity. The platform ensures user-centric navigation, facilitating prompt content discovery and initiating the listening experience.

The Home Screen has been meticulously designed to cater to the 'Passive User' persona. They quickly find something fitting the mood and start listening.

Conversely, the Search Screen resonates adeptly with the preferences of the 'Curator', and the 'Your Library' section appears tailored to the needs of the 'Nostalgic' persona.

A pivotal element in Spotify's value proposition is its robust emphasis on social connectivity. Consider the 'Spotify Wrapped'–a curated, annual audit of a user's listening habits. It transcends mere analytics by fostering shared musical experiences and reinforcing interpersonal bonds through music. The platform's continuous innovation in social-centric features underscores its understanding that music is deeply intertwined with individual identity. For users, the discovery and sharing of music isn't just a feature–it's an intrinsic part of their journey on the platform.

Weaknesses

1. **Discoverability of Features:** Many features like Lyrics, Magic Shuffle, and Collaborative Playlists are not easily discoverable. Many users are unaware of these features as very specific actions are required to find these features.
2. **Degrading Free-Tier experience:** Spotify deploys a Freemium model where the Free-Tier allows for limited features. This served as a great way to show users the simplicity and personalization. However, the features available have heavily reduced over time and might be unable to drive the point of a seamless experience that leads to conversion.
1. **Unclear following payoff:** Spotify allows users to follow artists and friends. However, the payoff that comes out of it is not very clear. No place is dedicated to following artists.
2. **Lack of a social corner:** Spotify does not have a dedicated place for Social features and playlists.
3. **Short-form videos:** While this tool can be very helpful for discovery, Spotify is using short GIFs that play in the playing now Screen on the TikTok-styled content as well. It is unlikely to engage the user. Users will need something more entertaining.

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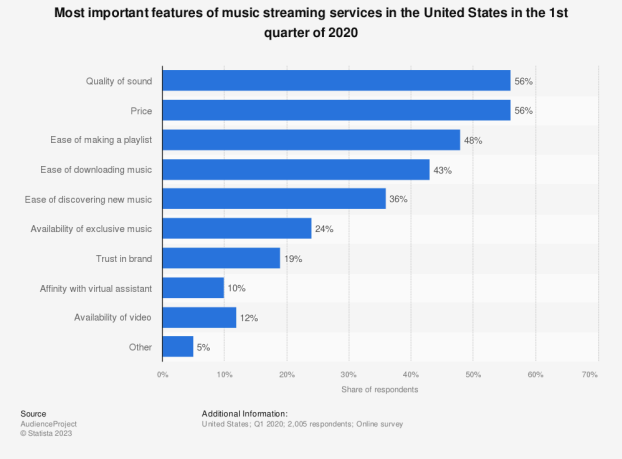


APP CRITIQUE: INSTAGRAM

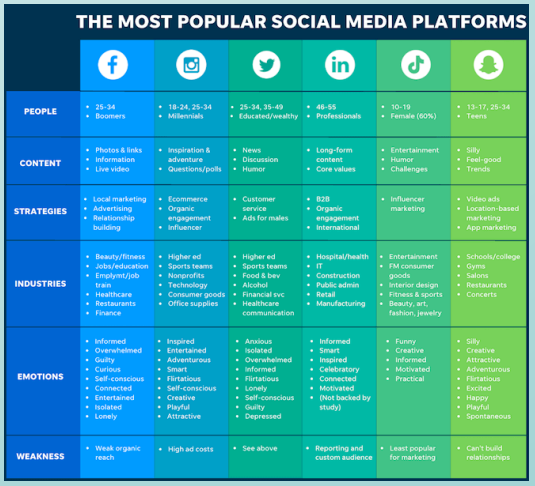
SOCIAL MEDIA INDUSTRY

The social media industry in India is a battlefield where both global giants and local champs vie for user attention.

Among the heavyweights, WhatsApp, Instagram, and Facebook lead the pack with user base of half a billion users each offering platforms for messaging, photo sharing, and community building. On the other hand, TikTok, with its short video content, has created ripples in the digital domain. Local platforms like MX TakaTak and ShareChat have swooped in to fill the void left by TikTok's ban, providing a stage for short video content creators and catering to regional language speakers. Each of these platforms brings distinct flavours to the social media milieu.

USER PERSONAS OF SOCIAL MEDIA USERS

- **The Casual Browser**: found on Facebook and Instagram predominantly, casually browsing through updates, engaging with content that catches their eye.
- **The Content Creator**: Thrives on platforms like Instagram, TikTok, and Youtube where they can showcase their creativity and build a following. Instagram, with its extensive user base and brand collaboration opportunities, is a major player for this persona.
- **The Communicator**: Prefers straightforward communication platforms like WhatsApp and Telegram for personal and professional conversations. WhatsApp, with its simplicity and widespread adoption, dominates this space.
- **The Business Professional**: Utilises LinkedIn and Facebook for professional networking, brand promotion, and customer engagement. LinkedIn, with its professional-centric platform, has a significant foothold on this persona.
- **The Information Seeker**: Shares and consumes news and information on Twitter and Facebook. Twitter, with its real-time update feature, holds a prominent position for this persona.



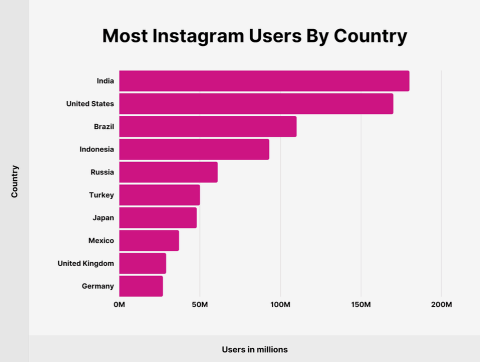
USER JOURNEY

The overarching user journey of a social media user typically unfolds in the following manner:

1. **Discovery & Onboarding**: Users come across a platform through referrals, advertisements, or organic searches, followed by a straightforward sign-up process.
2. **Profile Setup**: Post-signup, they personalize their profiles, adding details and preferences that help in tailoring the platform experience.
3. **Connection Building**: They begin connecting with friends, family, or professional contacts, and start following brands, influencers, or groups aligned with their interests.
4. **Content Exploration**: Users engage with the content, be it through browsing, liking, commenting, or sharing posts.
5. **Content Creation**: As they become comfortable, they may transition to creating and sharing their own content, seeking engagement from their network.
6. **Advanced Engagement**: Over time, they may explore advanced features such as shopping, using business tools, or participating in community groups.
7. **Feedback & Adaptation**: They provide feedback through various channels, and adapt their usage based on platform updates, evolving interests, and experiences.

APP CRITIQUE: INSTAGRAM

Instagram is a premier photo and video sharing app. It has become a digital haven for over a billion users globally. Renowned for its simple, visually-centric interface, it enables users to capture, edit, and share moments, while also discovering a world of content through a scrollable, engaging feed, embodying a modern social media experience. Interestingly, it derives its largest userbase from India.

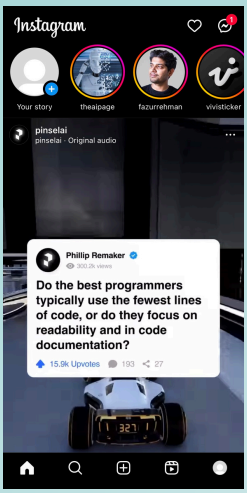


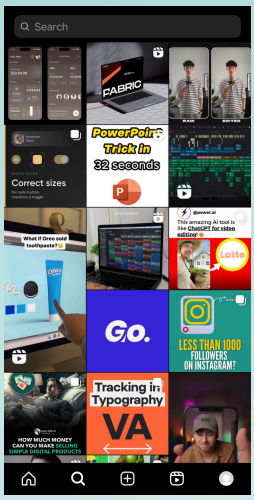
Instagram is particularly advantageous for businesses aiming to provide a visual representation of their brand or products, as visual content tends to engage audiences more effectively compared to text-only posts.

It provides a variety of features to enhance user experience and engagement. These include direct messaging, filters, stickers, GIFs, profile customization, diverse video content options including live-streaming, IGTV, and Stories. Additionally, the geotagging feature and feature-rich business accounts provide further value, especially for businesses aiming to reach a local audience or showcase their offerings in a structured manner.

Analysing Instagram

Analysing Instagram's UI can provide a clear understanding of its design principles and usability. Let's break down the UI into the specified sections:





1. **Home Page**:

- **Feed**: The core of the home page is the feed, displaying photos and videos from accounts the user follows. Each post has icons for liking, commenting, sharing, and viewing comments below the post.
- **Stories Bar**: At the top, there's a horizontal scroll bar for Stories from people the user follows.
- **Navigation Bar**: A navigation bar at the bottom allows quick access to different sections of the app like the Search, Reels, Shop, and Profile tabs.
- **Post Button**: In the centre of the navigation bar is the Post button for creating new posts.

2. **Search Page**:

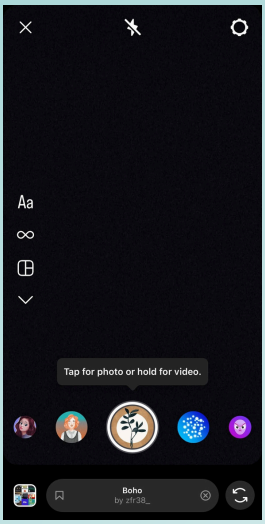
- **Search Bar**: At the top, there's a search bar for finding specific users, tags, or places.
- **Category Filters**: Below the search bar, category filters help narrow down the search to Top, Accounts, Tags, or Places.
- **Content Display**: The bulk of this page displays either a grid of photos/videos or a list of accounts depending on the search or selected category.
- **Reels**: A separate section accessed from the bottom navigation bar where users can scroll through a feed of short video clips, with a dedicated interface for engaging with and discovering Reel content. This seems to be sort of a redundant section given that users can always go to reels through the search section, but for some reason Instagram has it as a separate tab as well.

Instagram's search page ranks content based on their algorithm. The algorithmic nature of the feed can disrupt the chronological display of posts, which might affect content visibility and user engagement. Furthermore, the influx of sponsored posts and ads can overwhelm users, possibly detracting from the organic social interaction and content discovery experience on the platform.

3. **Profile Page:**

- **Profile Header**: Contains user's profile picture, follower/following counts, bio, and a button for editing the profile.
- **Post Tabs**: Below the header are tabs for Posts, Reels, IGTV, and tagged photos.
- **Content Display**: Displays a grid of the user's posts, or a feed of their Reels or IGTV videos depending on the selected tab.
- **Options Menu**: An options menu (three horizontal lines at the top right) provides access to settings, archived posts, and other features

The profile page is a personalised space where users can manage and view their content, and access settings. It's structured to provide a clear overview of the user's activity and content on Instagram.



4. Camera Interface:

- **Access**: Accessed via the plus icon in the center of the navigation bar or by swiping right on the home page.
- **Capture Modes**: Offers various capture modes like Photo, Video, Story, and Reel.
- **Filters & Effects**: Provides a range of filters and effects that can be applied in real-time.
- **Editing Tools**: Post-capture editing tools for cropping, adjusting brightness/contrast, and adding text or stickers.
- **Preview & Post Options**: Before posting, users can preview the capture, tag people, add location, and configure sharing settings.

The camera interface is intuitive, offering a plethora of tools and options for users to create and customize their content effortlessly. Instagram's camera section is so comprehensive with a rich feature set that it is almost a camera app in itself.

Overall, Instagram's UI is one of the most well defined, intuitive and visually engaging UIs, ensuring users can easily navigate through the app, engage with content, and discover new accounts or posts.

APP CRITIQUE: LINKEDIN

USER SEGMENTATION:

Career Professionals:
Description: They seek networking for career advancement, job searching and switching. They need targeted job recommendations and professional development content.

Entrepreneurs and business owners:
Description: They use the app for talent acquisition, business insights, and brand visibility.

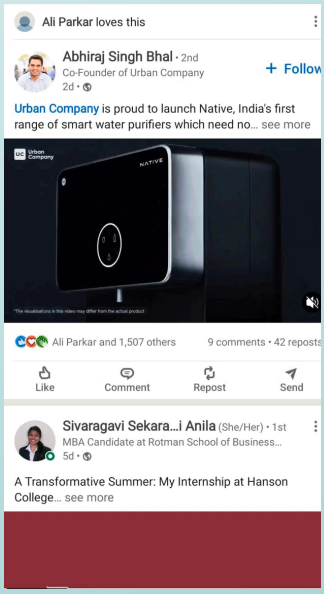
Freelancers and Consultants:
Description: These are independent professionals offering services. They need the app for client acquisition, project collaboration and staying updated on industry trends.

Influencers:
Description: They seek an audience for their content and also use the app to promote third party content and advertisements to monetise their account.

Among these user segments, career professionals form the largest cluster. For them, the primary task is seeking for a newer job opportunities and thus this becomes the most impactful use case of the LinkedIn. The user journey for this looks as -

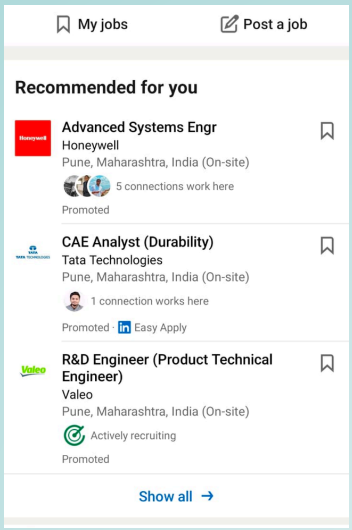
1. Profile creation
Analysis: This section is well guided by an assistant. Job recommendation algorithms work well if the profiles are not incomplete. Although LinkedIn guides the user with next steps, it doesn't incentivize them and hence most profiles are left incomplete
Suggestions: Implement interactive prompts and tooltips to guide users in creating comprehensive profiles, emphasizing the importance of details.

2. Networking and research:
Analysis: The first page(Home tab) is a feed of posts which most likely serves the use case of staying updated with industry and trends.



3. Job search

Analysis: The 'Jobs' section is at the last which is not an issue because it is necessary to network and stay updated before jumping into a job search. The design of the Jobs tab is pretty straightforward.



Under the jobs tab, the 'Search' changes to 'Search jobs' which is unlike most of the other apps. Such features are something that I would like to be gone under A/B testing because users are not used to such changes. Next, the jobs coming at the top are recommended ones which are based on the profile of the user (which might not always be updated or the user might be looking for a different domain altogether). Users may find the job search filters complex or overwhelming, hindering their ability to efficiently find relevant opportunities.

Suggestions: A bar could be added below the search jobs section giving users the flexibility to choose if they want to go for the recommended section or choose from the other available options (which would include a list of domains). Further, conduct usability tests to optimize the filter interface, ensuring it is user-friendly and aligns with the needs of different job seekers.

4. Selection and application process:

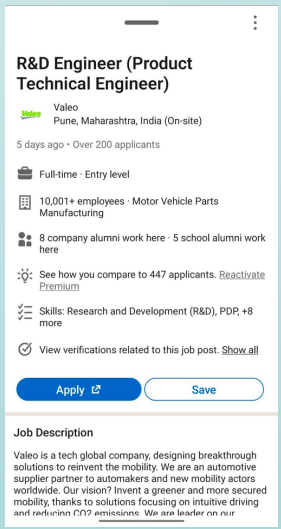
Analysis: After clicking on the specific job, details page opens up with an 'Apply' tab which directs you out of the app. Moreover, the screen contains lot if information and features which might not be equally relevant.

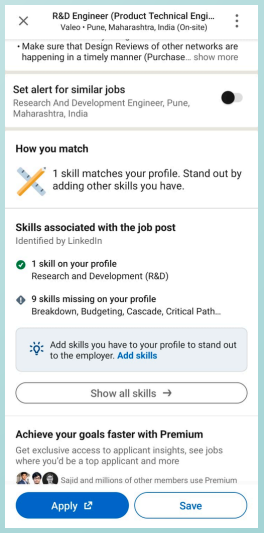
Suggestions: A Product Manager would ideally want to conduct a study here to find out the most useful features and then prioritise and optimise the page design accordingly. Additionally, Job-Person fit is another feature which might come handy to users to click on the jobs efficiently saving their time in reading every job description.

In dissecting the LinkedIn app through each step of the consumer journey, it becomes evident that while the platform excels in connecting professionals worldwide, there is still a scope for improvement.

The intricacies of the application process, the quality of job recommendations, and the balance of visibility settings all play pivotal roles in shaping the user's LinkedIn experience.

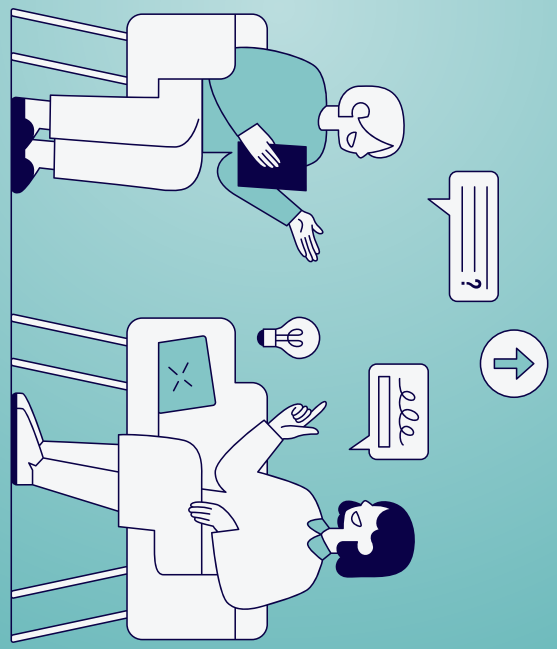
A similar analysis can be carried out for other use cases and other customer segment journeys.





SECTION 4

PM INTERVIEW CASES



GUESSTIMATE TIPS

INTRODUCTION

Guesstimates are used as a tool to test a candidate's problem-solving and reasoning skills. They also give the interviewer a sense of the candidate's ability to think creatively and outside the box.

BASIC TIPS

- Think of the problem from various perspectives (supply side, demand side), and choose the one that appears more suitable (simpler assumptions possible).
- Stick to basics. Try sticking to round numbers to the best extent possible.
- Some key filtering criteria are given in the table on the next page as a reference. These are - gender, rural/urban, age and income splits. The numbers are only for reference, and candidates can use alternate assumptions as well.
- Candidates are free to make their own assumptions wherever needed. However, always ensure to confirm your assumption with the interviewer before proceeding.
- Keep the calculations clean and simple.
- Ask as many relevant clarifying questions as possible to ensure you clearly understand the problem's scope and requirements.

SOME COMMONLY USED FIGURES:

India Specifics*		
Total Population		1.4 Billion
Total Area		3.3 Mn sq. km
Avg. people/household		4.5
Gender Split		
Male	50%	700 Million
Female	50%	700 Million
Rural Urban Split		
Rural	70%	980 Million
Urban	30%	420 Million
Age split		
0-20	40%	560 Million
21-40	30%	420 Million
41-60	20%	280 Million
60+	10%	140 Million
Income Split		
Low	20%	280 Million
Middle	40%	560 Million
Upper Middle	30%	420 Million
High	10%	140 Million
Major Cities		
Delhi	2%	~30 Million
Mumbai	1.5%	~20 Million
Bangalore	1%	~15 Million
Kolkatta	1%	~15 Million
Hyderabad	0.7%	~10 Million

GUESSTIMATE 1

PROBLEM

Interviewer: Estimate the number of Instagram uploads occurring in India.

ASK CLARIFYING QUESTIONS

Candidate: I'd like to ask some clarifying questions. Do you want me to estimate the number of uploads per day or per year?

Interviewer: Uploads per day.

Candidate: Do uploads consist of posts or stories or both?

Interviewer: Consider only posts.

Candidate: Are we considering personal or business accounts?

Interviewer: Consider only personal accounts for now.

Comments: The clarifying questions showcase the candidate's understanding of the different types of uploads that can be made on Instagram (posts, stories), and the user profiles (personal, business). It portrays a sense of comprehensiveness.

STRATEGY FORMULATION

Candidate: I will approach this problem by estimating the number of personal users on Instagram with the average number of uploads they make each day. First, I shall estimate the number of personal users. Then I will come up with the average number of uploads figures by splitting these users into categories based on frequency of use.

Interviewer: Alright, go ahead.

GUESSTIMATE

(Candidate writes down the following on a sheet of paper)

General equation:
uploads = [# of users on Instagram] * [avg. number of uploads per person each day]

Candidate: To estimate the number of personal users of Instagram, I will multiply the following parameters: (Candidate writes down the following on a sheet of paper)

[# of users on Instagram] = [population of India] * [% population with smartphone and access to internet] * [% of people on Instagram] * [% of monthly active users]

Candidate: The population of India is 1.4 billion. Due to an all-time high penetration of mobile services, I estimate that 40% users have a smartphone with access to internet. Instagram is the third most popular app on the Play Store with over a billion downloads, so I assume that 50% of these users have Instagram. About 30% of these accounts would be fake or inactive, leaving us with 70% monthly active users.

Interviewer: Sounds reasonable.

(Candidate writes down the following on a sheet of paper)
[# of users on Instagram] = 1.4 billion * 40% * 50% * 70% = 196 million

Candidate: I'll round that off to 200 million. Now, I'll estimate the average number of uploads per person per day. For this, I'll divide the monthly active users into three categories, with rough estimates for the percentage of users in each category from my own experience:
1.Light users: 1 post every month on an average - 30% = 60 million
2.Moderate users: 1 post every 10 days on an average - 60% = 120 million
3.Heavy users: 1 post every 2 days on an average -10% = 20 million

Therefore, number of posts per day = 60 * 1/30+ 120 * 1/10 + 20 * 1/2 ≈ 24 million posts

Additional ways of segmenting which could be done:

- Age/gender wise segmentation for posting habits
- Users could be Indians or tourists making posts in India

Interviewer: Sounds good, thank you.

COMMENTS

The candidate provides rationale for every estimate they make. They also simplify their calculations by rounding off. This is a good trick, as numbers can start getting complicated when segmentation is done. The Candidate also provides alternate ways of segmenting, which shows that they understand that trends can vary across user demographics.

GUESSTIMATE 2

Interviewer: Estimate the revenue that YouTube makes in a day?

Interviewee: Interesting! To estimate the total revenue, I will have to break down the same into revenue streams. As per my knowledge, YouTube has multiple revenue streams - Ads, YouTube music, YouTube TV, affiliate marketing, etc. Ad revenues form the majority of this. Am I missing any revenue stream? Do you want me to go into a particular revenue stream or across all of them?

Interviewer: Since you mentioned that ad revenues form a bulk of it, let's deep dive into that.

Interviewee: Sure, do you want me to consider the ad revenues of YouTube globally or in a specific geography?

Interviewer: Good that you asked that! Lets look at estimating the revenues specifically in the US.

Interviewee: Alright so the approach I'm going to take is as follows -

Youtube ad revenues = Total number of ad views x avg charge per ad view

Youtube ad revenues = Videos watched per day x Avg ads/video x avg charge per ad view

For estimating the total number of ad views, there are two ways we can go about it - estimating from the demand side as to how many videos do people watch on YouTube on a daily basis. The second way is from the supply side as to how many video views does YouTube have the capacity to serve. Since, in this case capacity will not be a problem as YouTube can easily scale up its infra to serve the demand, we should go via demand estimation. Does this sound fair?

Interviewer: Alright, that does sound like a plan. Lets go ahead.

Interviewee: For estimating the total number of ad views in a day, I will start with the population of the US, segment it into age intervals and then assume a percentage of them that watches YouTube videos on a daily basis. This will give me average total daily Youtube viewers in the US. Then, I will assume the number of videos watched by each of these age segments on a daily basis to get to the total number of Youtube video views in a day. Assuming a percentage of this has ads on it, we can get to the total ad views per day in the US.

I will start with the total US population as 300 m with an average life expectancy of 80 years and assume a uniform age distribution.

Estimating the avg number of YouTube users per day -

- 0 - 5 yrs : doesn't watch => 0
- 5 - 10 yrs : watches on 50% of the days => (5/80)*300*50% = ~10m
- 10 - 20 yrs: watch daily => (10/80)*300*100% = ~40m
- 20 - 40 yrs: watch daily => (20/80)*300*100% = 75m
- 40 - 60 yrs: watch on 75% of the days => (20/80)*300*75% = ~60m
- 60 - 80 yrs: watch on 50% days => (20/80)*300*50% = ~40m

So now we have the daily viewers across age segments we can go ahead with calculating the daily views -

- 0 - 5 yrs : 0 daily views
- 5 - 10 yrs : watch 5 videos per day => 5*10 = 50m views
- 10 - 20 yrs: watch 20 videos per day => 20*40 = 800m views
- 20 - 40 yrs: watch 10 videos per day => 10*75 = 750m views
- 40 - 60 yrs: watch 5 videos per day => 5*60 = 300m views
- 60 - 80 yrs: watch 5 videos per day => 5*40 = 200m views

Hence, we get the total number of views as = 50+800+750+300+200 = 2.1 billion views

We will assume that of these 2.1 billion views, 50% of them have ads showing up on them => This gives us a total of ~1bn ad views per day.

Total estimate the ad revenues, I'll assume a per $10 charge per 1000 views to the advertisers.

Total ad revenues = 10 * billion/1000 = 1 $10 million per day

Interviewer: Alright anything else that you would want to see in this?

Interviewee: I would like to have a sanity check on this answer. I'm getting a per annum Youtube revenues of = $10 * 365 = $3.65 billion

I know that Google makes ad revenues of ~$100 billion on an annual basis of which YouTube contributes around 10-15%. This brings global youtube revenues to around $10-$15 billion.

$3.65 billion forms a big chunk of this revenue coming from the US alone. It sounds sane given the higher internet penetration in the country as compared to other geographies.

Interviewer: Alright, sounds fair, we can end the interview here.

GUESSTIMATE 3

Interviewer: Estimate the market size for driverless cars.

Interviewee: Interesting! I've a few questions regarding the problem statement before I get into structuring a solution for the same. So, is there a particular geographic area that we are considering for the estimation?

Interviewer: Yes! I would like you to estimate the market size for the same in the US.

Interviewee: Sure. Also, do you want me to estimate market size in terms of unit volumes or in terms of total value of cars sold?

Interviewer: That's a good question. Let's look at it in terms of the unit volumes.

Interviewee: Great, we can always get to the total value by multiplying with an average unit price. So, that makes sense. Lastly, the market size will include both B2B and B2C sales. Should I consider both?

Interviewer: Lets only go for the B2C or purchases made by US households.

Interviewee: Understood! So here's how I'll structure this problem as - I'll formulate the market size of driverless cars as a percentage of the total new car purchases in the US -

Driverless cars unit sales = New car purchases in the US x Market share of driverless carsNext, I would estimate the total car purchases in the US in an year by considering the replacement cycle of the car -

New car purchases in an year = Total number of cars in the US/Car replacement cycleTo estimate the total number of cars in a country - I will start with the population of the country and estimate the number of households in the same by assuming an average household size. Further, I will then multiply the same with an average car per household assumption -

Total number of cars in the US = Total number of households x Avg. cars per household

Total number of households = Total population/Average household size

So, does this sound like a fair approach?

Interviewer: Yes, that sounds fair! Lets go ahead.

Interviewee: Great! So, let's start with the US population of 300 million. I will assume an average household size of 4 in the US to get the total number of households in the country as - 300/4 = 75 million households.

Further, given that the US is a high income country combined with a much higher percentage of women in the workforce, we can safely assume that on an average, the households in the US will have a car each for both the elder male and female members. Hence, I will continue by assuming the average number of cars per household as 2.

Hence, we can formulate the total number of cars in the US as = 75 * 2 = 150 million

Now, I'll estimate the total car sales in the US on an annual basis. For the same, using my prior knowledge, I'll assume a car replacement cycle of 10 years. This means that each car would be replaced by a new one after 10 years of usage. This gives us the total number of cars sold in an year in the US as - 150 million/10 years = 15 million cars per year

Interviewer: Great, so now that we have reached the total number of car sales in a year, how would you estimate the driverless cars market size?

Interviewee: I know that in the US, Tesla is the largest producer of driverless cars. Further, with the model 3 being its most affordable car in the driverless segment, Tesla had sold about 300k cars which is around 1.66% of the total market size. I'll assume this market share to further increase marginally to around 2% to get total driverless cars sale as -

15 million x 2% = 300,000 cars

Hence, we reach a market size of 300k driverless cars being sold in the US per year.

Interviewer: Interesting! Alright, we can close this year.

GUESSTIMATE 4

PROBLEM

Interviewer: Estimate Zoom's daily server usage during the lockdown period

ASK CLARIFYING QUESTIONS

Candidate: I'd like to ask some clarifying questions. My understanding is that Zoom's server usage would comprise of the following:
1.Storing information about its users and their call logs
2.Streaming the data (audio, video and text) to and from participants in a zoom call
3.Miscellaneous processing tasks (backend)

Among these, I believe that the second point would take up the majority share of the server usage. Shall I proceed with this assumption?

Interviewer: Yes, that sounds right.

Candidate: Do you want me to account for global usage or constrained to particular geography, such as India?

Interviewer: Consider the usage for India.

Candidate: Are we considering personal, student, or business accounts?

Interviewer: You can consider business accounts for now.

Candidates: Among business accounts, should I account for both event streaming or just professional calls between employees?

Interview: You may proceed with the latter.

Comments:
The clarifying questions showcase the candidate's understanding of the different types of data usage that Zoom may have and the types of users it caters to.

STRATEGY FORMULATION

Candidate: I will approach this problem by estimating the number of employed personnel in India working white collar jobs. My assumption is that most of these people would be using some video conferencing app for their work day. Zoom has two main competitors that I can think of- Microsoft Teams and Google Meet. However, Zoom has the major share of video-conferencing app users, so I will assume that 60% of the white- collar employees use Zoom.

Interviewer: Alright, go ahead.

Candidate: I will use the following general equation:
Daily server usage = # of Zoom users * data usage per user
=60% of white collar employees * data usage per hour * # hours per user

Candidate: First, I will start with the number of zoom users. As there are a large number of jobs in India that are in the manufacturing and agriculture sector, I will assume that 10% of the workforce is employed in white collar jobs. And to estimate the workforce, I will assume that 80% of adults in the age range of 20-60 in India are employed. I will assume that the population is uniformly distributed between the ages of 0-80, so 50% of the population is in the 20-60 range.# of white collar employees= 10%* 80% * 50%* population of India = 56 million. 60 % of this is roughly 35 million.

Among the white collar employees too, there would be a gradation in usage based on how critical collaborative tasks are to their job or how much time of their work day is spent in a meeting. So for estimating data usage, I will consider three categories of users:

Heavy users: >50% of the work day spent in a meeting => >4 hours
Medium users: 1-2 hours
Light users: <1 hour

Candidate: I will make a simplifying assumption that users are distributed among these categories roughly normally, so 20% of the users are heavy users, 60% are medium and 20% are light users and that the usage for the categories is exactly 4, 1.5 hours and 0.5 hours respectively.So, average usage per user = 20%*4 + 60% *1.5 + 20% *0.5 =1.8 hours, which is roughly 2 hours.

Candidate: Finally, to get the data usage per hour, drawing from my personal experience, a zoom call functions best when I have access to at least 1 Mbps of data. So I will assume that a zoom call uses 1 Mbps of data in a call. This equates to 360 Mb of data in a n hour.

So, finally we have:
35 million zoom users * 2 hours * 360 Mb/hour = 25,200 million Mb = 25.2 Tb/ day

Interviewer: Sounds good, thank you.

Comments:
The candidate provides rationale for every estimate they make. They also simplify their calculations by rounding off. This is a good trick, as numbers can start getting complicated when segmentation is done.

GUESSTIMATE 5

Interviewer: Estimate the number of WhatsApp chats occurring in a day.

ASK CLARIFYING QUESTIONS

Candidate: Sure. I'd like to understand the context a little better. Are we trying to gauge this for a particular geography?
Interviewer: Yes, let's focus on messages in India.
Candidate: Okay. I'd further like to know which user types should we focus on: WhatsApp for business users or casual users?
Interviewer: Good point. Let's discuss about the casual users.

DESCRIBE YOUR APPROACH

Candidate: Thanks. To approach this problem, I'd like to first estimate the number of Whatsapp users in India. Then I'd like to segment this population and find out the average number of chats per segment to arrive at our larger question. Does this approach sound good?
Interviewer: Yes! Please go ahead.

BREAK THE PROBLEM AND COMPUTE

Candidate: Firstly, the number of WhatsApp users in India can be broken down as,
# of WA users = Population of India * internet penetration % * % having smartphones * % having WhatsApp installed.
Interviewer: Sounds right, please go ahead.

Candidate: Great! So I'll make a quick estimate of our multipliers. India's population is roughly 1.4 Billion people. About 50% of India has internet penetration with regular access. The majority of people use internet on smartphones these days, and hence, the percentage of people with internet access who have smartphones would be close to 70%. Finally, as WhatsApp is one of the core apps on smartphones, I will assume that 80% of people have it and use it regularly.
This leads to number of WhatsApp users in India as 1.4 Billion*0.5*0.7*0.8, or ~550 Million.

Interviewer: Perfect! That's a very good estimate. How will you go about the number of messages in a day?
Candidate: Please spare me a moment to segment our users.
I'd like to firstly classify our users on their age into 15-29, 31-49 and 50+. The usage and affinity varies across these age groups. Then for each of the age group, I would break the users into heavy users, medium users and light users. Does this approach sound good?
Interviewer: Yes, I think this is fair.

Candidate: Okay. I would distribute 60% of WhatsApp users in the first age category of 15-29, 30% in 31-49 and 10% in 50+.
Further, for the category 15-29, 60% would be heavy users with ~80 chats sent a day, 20% medium users with ~ 40 chats a day and the final 20% of light users with ~10 chats a day. This would lead to an average chats per day for the 15-29 age group as 60%*80 + 20%*40 + 20%*10, or ~60 chats per day.
I would similarly compute average number of chats for the other age groups. Should I continue with this approach and arrive at our final answer?
Ensure to check-in with the interviewer in middle of large computations.

Interviewer: The approach looks good to me. I have some data for you, the average number of chats for the middle and old age group are 25 chats and 8 chats a day.
Candidate: Thank you! With all the data bits now ready, I can build-up to our final solution.
# of chats per day in WA = Total WA Users* [(% of user 15-29 * avg. chats per day of 15-29) + (% of user 30-49 * avg. chats per day of 30-49) + (% of users older than 50 * avg. chats per day of users older than 50)]
Candidate (contd.):
This computes as,
# of chats per day in WA = 550 Million * [(60% * 60)+(30% * 25)+(10% * 8)], or, ~24 Billion.

Hence, we have approximated the number of WhatsApp chats per day in India is close to 24 Billion.

Interviewer: Thank you! We can close this interview.

PRODUCT DESIGN: DO'S & DON'TS

DO'S

Laying down the structure: Giving the interviewer a broad overview of how you plan to approach finding solutions to the given problem can be beneficial. It helps the interviewer follow your answer better and helps them understand the rationale behind each step in the approach.

Naming personas based on attributes: Giving your persona a proper name, e.g. Rahul, is common as it makes referring to the persona easier. However, a better approach would be to name the persona by their defining attribute (which is key to your solution), e.g. Sports Enthusiast.

Map the customer journey: A simple way to approach brainstorming user needs is to first think about the series of actions the intended customer would perform in the context of the question. For example, if the question is based on e-commerce, write down the list of actions a user performs to buy a product on an e-commerce platform. The needs and problems at each step can then be listed down.

Reporting needs as stories: User stories are a concise way of defining a user's needs.

Format: As a *(user persona)*, I want to *(need)*,so that *(rationale)*.

DON'TS

Jumping to Solutions/Features: You will be judged more on the approach to the solution and the logic behind the various decisions you take to find the solution. Hence, it is critical to walk the interviewers through your approach.

Spelling out the framework: Frameworks are meant to structure your approach. Avoid mentioning the name of the framework or being too obvious in your application of it. A good practice would be to add some variations to the framework.

Reporting only the unique solution: One always feels the pressure to come up with a unique solution with advanced application of AI/ML/AR/VR, leaving out the most obvious solutions. While it is important to come up with creative solutions, it is also necessary to show that you can identify the obvious solutions that satisfy important customer needs.

Being married to your first solution: It is recommended to come up with multiple features to provide alternative strategies.

NEW PRODUCT DESIGN 1

Design a mobile application for an office space sharing business for Airbnb.

Comment: The first step of the candidate is to clarify any ambiguities in the problem statement.

Candidate: Before moving further, I would like to understand a bit more about the problem at hand. What is the exact business model of this office sharing app? For whom would we be designing the app, the owner or the renter? And lastly, what is the objective of this app.

Interviewer: Good questions. The whole business idea is that people with extra office space can rent out to start-ups or freelancers, Airbnb will take the empty space and remodel it, but the owner will take care of the operations. We need to design the app for both owner and the renter. As of now, the objective is to get as many customers as possible on board.

Candidate: That's helpful. I am assuming that the person who is renting the space is the owner, just to reduce the complexity of the design. Also, I believe we are not focusing on monetization for the business as of now, later for sure, please correct me if I am wrong. However, I will address the payment issues between the owner and the person renting it. Now that I understand the business, I will look for user segments, then find their needs, build user stories, search for solutions and create an MVP to fit these solutions.

Comment: After understanding the problem statement, the candidate clears all his/her assumptions beforehand. Also, he/she gives an overall flow of how he/she is attacking the problem. This helps to keep the interviewer intact with the broad view, and aligned with the candidate's thought. It's always good to first give a high-level view and then go on to the detailing.

Candidate: There are 3 users for this application
1.Owner
2.The person renting it
3.Airbnb
Which one should I focus on?

Interviewer: You may decide that.

Candidate: For any multi-sided platform, it is important to solve the chicken-egg problem. Taking the example of Uber who first focused on on-boarding the drivers, I will also focus on owners. Since we are focusing on the owner, we must first define their persona. Mostly, the owner will be a person or a company who have extra space in their building which can be monetized. They will be comfortable in sharing their space to a stranger. If it is a company, this will be an additional source of money.

Comment: The candidate chose a particular user to focus, but he/she also gave a proper rationale for that. He/she backed it up a similar business model like of Uber to rightly choose the user. It would have been better if the candidate would be gone deeper about the user(the owner) like their demographics, their size, their location etc.

Candidate: Few of the needs of the owners would be
1.Finding reliable people to rent it to, also the ability to approve or
decline requests
2.Furnish and maintain the space
with office supplies
3.Estimated rent and information on
rental periods and owner's rules
4.Payment and customer support
5.Security of the place

The user needs can be analyzed by navigating through the steps the owner would need to go through the whole process.

So the needs of the user should be identified at each stage of the process.

Comment: It's always good to break down the whole process into intermediary stages and figure out the customer in these stages, this helps in getting the exhaustive list of user needs, and avoids missing of any important part.

Interviewer: Fair enough. But how will you prioritize among the different needs of the user?

Candidate: Since we are focusing on MVP, I will look at the most needed features which are 2, 3, and 4. So my user stories as an owner are : Keep a check on the office supplies and other furniture, replenish them whenever necessaryI want to know the rent I will get so that I can decide whether it is meaningful for me or notA way to interact with the system to confirm/cancel bookings and settlements

Comment: Prioritizing the requirements is one of the key roles of a product manager. Since its a boundary spanning role, product managers are bombarded with multiple tasks of the product, from the business team, from the engineering team, from the marketing team etc, but the product managers have to make hard calls on prioritizing the requirements. Some of them use the effort-impact matrix to do this, however, there are multiple other frameworks as well.

Candidate: For an MVP, I will address all the stories individually first

Solution to story 1:

- Display the list of all the supplies and furniture available in the location.
- Provide the capability to the place orders through the app itself,
- Airbnb can provide support.
- An ability to display to the renters what all furniture and equipment is present in the office.

Solution to story 2:

- Option for the owner to send the rent to the person renting it
- Rent estimation based on Space occupied
- No. of days Special requests/Customisation
- Number of people/employees

Solution to Story 3:

A tab to manage request for renting. Option to confirm/cancel or chat with an interested person. Have a payment gateway to pay full rent in advance. Provide a weekly settlement into the owner's account. Cancellations/Refunds will be handled by Airbnb in between settlements

Interviewer: Would you go ahead with all of these?

Candidate: No, I will evaluate the solutions in each story based on customer impact and its ease of implementation. After the evaluation, I will select 1,2 in S1, 2 in S2, 1,2 in S3 for the MVP. Do you like me to go in detail about these?

Interviewer: No, this is good. Thanks for your time and all the best.

Comment: Candidate could have explained here a bit more about how selected the features for the MVP, however, since the interviewer confirmed then it's fine. It's always better to back up your arguments with data and logic, interviewers are looking for critical thinking of the ability of the candidate.

NEW PRODUCT DESIGN 2

Perugia's voting turnouts have stagnated at 55% in the last few years. The Government of Perugia has hired you to design an application to improve voter turnout. What are some of the features you would recommend for such an application?

Interviewee: Okay. Can I start with a few clarifying questions about the problem?

Interviewer: Sure, go ahead.

Interviewee: What has been attributed to as the reason behind the stagnation of the turnouts? Are the people skipping the vote from a specific demographic? What are the current methods of voting in Perugia? What is the specific aim of the government with the application, and is it supposed to serve as a method of voting at all?

Interviewer: The voters who have skipped have cited reasons of lack of knowledge of the candidates, lack of interest in the voting process, confusion about how the registration and voting process works, the perception that their vote doesn't count for much and that they were busy on the day of voting.

The government runs awareness campaigns, but they have reached and affect only a few of the voters. The people skipping are primarily the new voters from the ages of 18-30 years of age. The current voting method is just a ballot paper system, and the government doesn't aim to change that. With the application, the government wishes to attract more voters to the process and engage them during the election season to get them interested and vote more. Does that help?

Interviewee: Yes, it does. From what I could gather, the government would like the application to make the voters more engaged with the voting process and show up to vote. As the most affected user base is the new voters, I'll keep the application focused on solving their needs. Is that okay?

Interviewer: Yes, that is the government's target segment as well.

Interviewee: Okay. From what you mentioned, it seems the new voters suffer from both the absence of information on multiple levels of the voting process and the lack of interest in the voting process.

This also shows that the currently accessible mediums are unable to solve this problem, be it conventional media or otherwise. Looking at the voting process itself, it could include information on the performance of the current/previous candidate, the profiles and activities of all contesting candidates, their manifestos, the current problems faced by the majority of the citizens, how to enter the voting process and where to vote. If these look fine, can I take a minute to think over the solution?

Interviewer: Sure, go ahead.

Interviewee: Alright. Keeping the needs in mind, I have come up with the following features in the app:

- Addressing lack of information: Candidate Scorecard: This would be curated using the information from their previous work, their previous election performance, their headlines in the news, criminal and other charges against them. Since we are working with the government, some of this information can be directly sourced from the candidates, and the rest can be gathered from external sources.
- Application-based registration: By digitizing the voter registration through the app, the voter would be able to register from anywhere using the app, reducing the time and money taken to set up and use an extensive offline registration system.
- News section: Capture the headlines to inform the user of the current issues in the news, which would be localized to their voting region.

Timeline of the voting process What's my booth and navigation to the voting booth.

Addressing lack of interest:

- Gamification of the voters: Running an app-based daily/weekly quiz to find the most informed voters. There may be perks associated with that, like preferential voting etc., depending on the rules and laws of the land.
- Notification centre: Using notifications specifically to keep the voter engaged about the happenings of the process and the critical dates.
- Connect with the candidates: The candidates can leverage the app to communicate directly with their voters specific to the region. On the other side, the voters could ask questions and upvote the questions that they find relevant. It would be in the candidates' interest to answer more of the questions asked since it would only increase their reach, and the voters would be more engaged.
- Non-transferrable badges for voters: Perks like badges, QR to your exclusive photo at the booth etc, could be incorporated into the app to encourage voting. Do these look fine, or should I explore more features?

Interviewer: No, these should do. Could you elaborate on one of the user workflows that you see the user having on the application?

Interviewee: Sure. Talking about a scenario a week before the election when the app has the most information. The user would open the app and see their profile details with the gamified score (could be a

percentile of how much more informed they are than the others on the app), the next quiz they should attempt, and the latest news on their voting area. The user takes the quiz and moves to the candidate profile tab. The user can select the current/previous elected member who is re-contesting. The app gives the user a candidate scorecard, their recent work in the office, the candidate's latest headlines, and the recent legal charges against them (if any). The user selects the work tab, which then elaborates on their recent participation in the parliamentary debate, the bills they presented etc. The user reads the information and closes the app. That would be a flow where the candidate engages with the app and consumes some information. Does that work?

Interviewer: Yes, that is good enough. Thank you.

NEW PRODUCT DESIGN 3

You are a PM at Disney. You have been asked to design an application to improve the customer experience at Disney theme parks. What would you do?

Interviewee: I would like to ask a few questions before diving into the solution. I want to verify whether my understanding of the offerings of the theme park is correct. The parks generally offer amusement rides and shows for entertainment. Theme parks typically have eateries for customers to eat. Some theme parks have an in-premise hotel for customers to stay.

Interviewer: You can assume that the theme park you are developing for has all the offerings you mentioned except hotels.

Interviewee: I will start with identifying the primary user of the application and then listing down the actions they take while visiting the theme park.
The most common visitors of theme parks would be families where the adults are accompanying the children. The decision taker would be one of the parents, so they would potentially be the primary user of the application.

The users perform the following activities: -
Book the entry ticket to the park
Decide on the attraction to engage with
Navigate to the selected attraction Make the payment if the attraction has an associated fee
Visit eateries for meals or refreshments

Based on the activities I listed down, the users will have the following needs:-

- Ticket Booking: I would want to explore the different tiers of tickets available and their offerings to find the tier which best suits my budget and interests.
- Picking the amusement ride/show: During this phase, the customers would have the following needs: -
- I would like to understand the experience of the attraction I am planning to visit to ensure that the attraction would be interesting to me.
- I would also like to understand the restrictions (e.g., age, height) of the ride/show of my choice to validate that I am eligible to visit it.

I would be visiting the park for a fixed amount of time and want to make the best use of my time.

I want to visit an attraction when it is not too crowded so that I don't have to wait in line. Navigate to selected attractions: I need help figuring out the way to the ride or the show I want to visit.

Eateries: I want to pick an eatery that would provide me with the food I like.

Interviewer: The pain point assessment seems fair. Assume that you had the budget to solve only one of these problems. Which one would you select, and what feature would you build for it?

Interviewee: I would prioritize the solutions based on the following parameters: -

1.Effectiveness of Existing Solutions

2.Value customer would derive from the problem being solved

Note: Analysis give in the table below

Based on the selected parameters, I would choose 'Ride Picking' as the pain point to be solved.

Solution:

The goal of the features to be built would be to help the customers optimize the time spent in the park.

- Attraction Catalogue: The feature would help customers explore the rides and shows available in the park by filtering them based on their requirements.

The rides can be filtered based on the following parameters: -

- Age, Weight, Height – To ensure eligibility
- Attraction Type – Audio-Visual Shows, Rollercoasters, Games etc.

The feature would give the following information about the attraction:-

- Rules & Regulations
- Video of the ride
- Timings with available slots (If it's a scheduled show)
- Expected waiting time Customer Reviews
- The feature would have an option to pre-book scheduled shows/rides.

	Pain Point	Existing Solutions	Effectiveness Existing Solution	Value derived by the customer
1	**Ticket Booking**	Third Party Booking platforms	High	Medium
2	**Ride Picking**	Going through the list of attractions on the park website or on the park brochure	Low – existing solution is tedious	High
3	**Navigation**	Google Maps, Physical maps & direction boards in the park	Medium	Medium
4	**Eateries**	Park website, Brochures	Medium	Low

- Itinerary Optimizer: The feature would assist the customers in building the itinerary for a visit.
- The optimizer would be a chatbot that would take the following
- information: -Duration and Timings of the visit
- Number of visitors
- Preferences of each visitor (type of rides/show they want to see)
- The feature's output would be an itinerary that allows the customers to visit the most attractions, which are to their liking, in the duration they are present in the park. The feature would enable the customer to make changes to the itinerary manually.

NEW PRODUCT DESIGN 4

India's start-up ecosystem has grown immensely in the past decade. As a product manager at LinkedIn, suggest features to leverage the growth in the start-up ecosystem.

Interviewee: LinkedIn's mission is to "connect the world's professionals to make them more productive and successful." Is it fair to assume that LinkedIn has successfully attracted the various stakeholders of start-ups into registering to their platform (as shown by the steady growth in their user base)?

Interviewer: Yeah, that's true.

Interviewee: The goal of the additional features of the platform should then be to increase customer engagement. The goal would be achieved by banking on LinkedIn's strengths - its large user base, presence of experts from all domains, data analytics capabilities, and services like LinkedIn learning.

Moving ahead, I would figure out the stakeholders in the start-up ecosystem and list down their needs. There are two primary stakeholders in the start-up ecosystem: -

1.Founders: A person or group of people who come up with the fundamental idea that forms the
core business of the start-up
2.Investors: A high net-worth individual/group that provides capital to start-ups in exchange for ownership equity in the
company

Interviewer: Those are the two key stakeholders. Before moving to the needs, could you tell me which stakeholder you would prioritize?

Interviewee: As the business goal is to increase customer engagement, I would prioritize the two stakeholders based on the potential for increased customer engagement. Investors are generally backed by a professional team or institution and have the capital to get personalized services for their needs. Thus, they have limited requirements for additional services and limited engagement potential. On the other hand, many founders do not have the monetary backing or human capital to meet their business needs, thus creating multiple potential avenues to engage with them.

Interviewer: Assume that you have the resources to build features only for your priority stakeholder.

Interviewee: A typical journey of a start-up from the founder's point of view is the following:
1.Ideating and Conceptualization: The founder develops a business idea and conceptualizes a product
or service offering.
The needs during the phase are -
a.Market Research
b.Guidance for conceptualization
c.Idea Validation
2.Commitment: The founder moves from a concept to a company, putting their research into practice.

The needs during the phase are –

- Prototyping/Simulating idea
- Building a team
- Training the team
- Building the Minimum Viable Product

3.Traction & Scaling: The founder takes steps to grow the company which is still in its infancy. The
needs during the phase are –

- Raise funding
- Marketing product/service
- Soliciting customer feedback
- Refining product/service

The features are prioritized based on the compatibility with LinkedIn's mission and strengths, value to the customer and lack of existing solutions fulfilling the needs. Based on the above criteria, need for guidance, training the team and raise funding are prioritized.

I would like to suggest the following features to satisfy the prioritized needs:
-

1.Mentors connect: Industry experts can be designated as mentors (like
a blue tick). Start-up founders can search for mentors based on
industry and domain of expertise. 2.VC Marketplace: A dedicated
platform on LinkedIn where start- ups can post their elevator pitch and funding requirements would
be established. Investors can review the pitch, connect to the
start-up's team, and negotiate.
3.LinkedIn learning skills gap analyzer: LinkedIn learning would
be integrated into the training and feedback system of the company. The feature would analyze the team's existing skills and required skills and suggest training courses
accordingly.

NEW PRODUCT DESIGN 5

Ola management has recognized travel needs of school kids as a potential business opportunity. Design the experience on Ola to meet their needs

COMPREHENDING THE SITUATION

Interviewee: Would the service be integrated into the existing Ola application or a new application? What would be the region of operation?

Interviewer: This service needs to be integrated into the existing Ola environment. The region of operation is India.

Interviewee: The target audience for this service would primarily be children falling in the age group of 4- 17 years. As per my understanding, the goals for the service would be:-
User acquisition

SETTING UP A NEW REVENUE STREAM

Before moving on to the features, I would like to identify the key stakeholders, their needs and the problems they face with the existing services.

Interviewee: Sure, go ahead .

IDENTIFYING STAKEHOLDERS

The key stakeholders would be the service users (school kids – consumers, their parents - customers) and the drivers. We will be designing the service from the point of view of service users because the new service will be drastically different compared to their usual services from the service user's point of view but similar from the driver's point of view.

Users will be segmented based on:

- Distance between school and residence – A student residing very close to the school can walk to the school, use a bicycle, or be dropped off by the parent and will probably not avail a bus/van service. On the other hand, a student who resides far away from school would look for a more comfortable mode of transport as he/she will have to travel for a longer time.
- Affordability (parent) – The choice of service also depends on how much the parents can afford.

Target user segment: Our service should target the middle-income segment. Higher-income segments

prefer personal vehicles and school bus services and are less likely to look for cheaper alternatives. The lower-income segment is not an ideal segment to target at launch. The service could later be expanded at a future stage to cover all the remaining segments.

Interviewer: That sounds fair. What other factors will you consider while building the service.

Interviewee: Another decision- making point would be the kind of service Ola would be planning to offer. The standard services available are listed below:-

- Bus – Cheaper, more secure, more timely and disciplined service
- Auto – Unsafe, informal, multiple vendors, more options to choose from
- Van – Flexible timings, multiple vendors, more options to choose from, safer option
- Cab – Expensive, target higher-income segments
- Bicycle – Ideal option fortravelling small distances

Target Service Segment: As the current Ola service already includes travel via autos, cars and bicycles, we can target all the service segments excluding buses, as including buses could add huge establishment cost.

Interviewer: That sound alright.

Interviewee:
Step 1: Driver/Vehicle Selection
1.Vehicle condition isn't good
2.Lack of driver/vehicle options
3.Prices are same irrespective of quality of service
4.Prices are same irrespective of quality of service

Step 2: Commute
1.Lack of discipline, very untimely
2.Drivers take uninformed/irregular leaves
3.Information asymmetry regarding driving skills
4.Many drivers don't have insurance/required paperwork
5.Overcrowding in the vehicle
6.Not very secure

Step 3: Payments
1.Irregular price hikes
2.One to one dealing with the drivers
3.No receipts or official paperwork

All the above-given problems are must-solves to engender trust in the parent's mind and ensure the safety and comfort of children. The incremental effort of adding features to solve the above pain points is also low. Therefore the MVP should solve all the listed problems.

Interviewer: Okay Lets move on to the solution.

SOLUTIONS

Interviewee: The solution is an end- to-end school drop and pick up service with the following features

Ola verified drivers: The drivers on the platform will be tested and verified by Ola (driving tests and background checks) to allow parents to be sure of the driving standards of the driver. The driving quality will also be kept in check using GPS tracking on the vehicles at all times.

Free trial rides: To onboard new customers, a free trial ride will be offered so that the parents can experience the product features live,

meet the driver and validate the vehicle quality before subscribing to the service. If the parents do not like the free trial ride, they can report the same on the app and provide feedback. A customer service rep will get in touch if needed.

Real-time location updates: Notification in the app for relevant locations – driver starts the ride, driver 10 mins away from your location, driver at your place, driver at the school, etc.

Backup arrangements: Ola will offer to arrange for a backup at no extra charges if the regular vehicle breaks down or the driver is unavailable.

Workflow of the application: -

Step 1: Signing In
Create a profile with name ,age, photograph
Select school and home address

Step 2: Select Service Type & Driver

- Select service type - van/auto etc. Look at price estimates for each
- service
- Select driver
- Look at driver profile & vehicle condition information
- Schedule test ride with the driver
- Finalize driver

Step 3: Payment and Service Agreement
Select Subscription Model - Monthly/Yearly
Sign Online Agreement

Step 4: Avail the service
Confirm Schedule for the week. (Driver to be notified for minor tweaks in schedule)
See the live location of driver
Select backup vehicle in case of driver/vehicle unavailability

NEW PRODUCT DESIGN 6

You are a PM for Microsoft Surface and Delta asked you to design the perfect airline entertainment system. You decide the features, scenarios, and users. What would you do?

Interviewee: I have some clarifying questions to ask before we proceed.

Interviewer: Sure, go ahead

Interviewee: Since airline entertainment systems already exists, do I need to make improvements to the existing system or design one from scratch? What are their objectives with this new application? What is the problem with the current product that you are planning to solve for the user?

Interviewer: You would need to make improvements to the existing software system. The hardware, however, can be changed with the new corporate relationship between MS and Delta. MS wishes to become the leading provider of the entertainment system for airlines, and Delta wishes to improve the user experience. The current usage of the system remains low, especially on shorter flights.

Interviewee: Yes, are there any constraints?

Interviewer: Yes, we are looking at a 6-8 months timeline of making the changes, and the budget is not a constraint.

Interviewee: Got it. Based on our discussion so far and my understanding, the needs of the users could look like this:
Casual Entertainment
Information on the flight
Other Information
Communication
Does this sound okay?

On the user persona side, I think there would be the following user personas that we can look at:
The busy business traveler already has internet access
The bored economy traveler
The international leisure traveler
Would you like me to focus on any specific persona?

Interviewer: We would ideally like to serve all of them, but let's look at the domestic bored economy traveler for the problem.

Interviewee: Sure. So, let's dig deeper into the user persona. These are travellers who are flying for personal use and are usually at their own expense.

They are on a short flight and can't usually afford the in-flight Wi-Fi. The current system gives them minimal support unless they spend money to access the live tv, shows etc. and is virtually unusable.

The user is left to their own devices to spend the time of the flight. Addressing their requirements, with the needs in the background, I would like to propose a few features:

- Interface with OTT platforms: Users can utilize their existing subscriptions during transit and continue their watchlist.
- Home screen with options based on topics and areas of interest: This would better assist the users in understanding the scope of the content. Currently, the system shows the type of entertainment options like movies, tv-series, radio, music etc.
- Incorporate MS surface abilities like drawing, MS 365 ecosystem apps: The surface ability gives the user the apps to create drawings, documents, notes, etc. The user could also email their work to themselves once done and wouldn't need to log in to secure the user better.
- The MS 365 ecosystem with a login option would give additional access to MS Teams and the user's past work for messaging and collaborating.
- MSN available freely on the system: The user would have access to information and stay up to date with current happenings. Incorporating MSN as a free service would serve both companies better.

Detachable systems for drawing or reading mode

Making free and premium sections: to encourage users to rely on the entertainment system for the flight time rather than look for their own devices.

Quick Navigation to all important options

Would you like me to prioritize some of these features?

Interviewer: Yes. As stated, let think of an MVP that can be built within the stipulated timeline.

Interviewee: Sure. Since time is our only constraint. I will focus on the features with the highest impact and the least effort not only in development but also in the implementation. I've eliminated the physical changes to devices as they require a change in both the software and hardware. Therefore, the features we can focus on for the MVP could be:

Incorporate MS 365 ecosystem apps like OneNote, MS office without the login

Interface with the OTT platforms

MSN access on the system

Making free and premium sections

These features enable the user with more entertainment options than currently available and make the experience more convenient to use. Moreover, it strengthens the corporate relationship between Delta and Microsoft, increasing their stickiness.

Assumptions:

The current system can take a software update

Revenue generation has not been a focus since we looked at only improving the user experience

- Changes in content were also put out of the current scope of the problem

Metrics:

- Increase in the usage time of the entertainment system overall
- Increase in the number of users accessing the entertainment system more a minimum threshold time
- Increase in the customer satisfaction of the entertainment system showing a better perception of the system in the user's mind

Would you like me to elaborate further on any part?

Interviewer: No, this should do. Thank you.

NEW PRODUCT DESIGN 7

Design an Alexa skill for artists/designers that can be monetized.

COMPREHEND THE SITUATION

Candidate: Let me just clarify the problem statement. I need to design an Alexa skill for artists or designers. I am assuming that this means that the skill I am designing shall have a voice-interface?

Interviewer: Yes. You may feel free to add other interfaces too.

Candidate: Okay. Are we targeting a particular segment of artists? Sculpters/painters/graphic artists, etc.?

Interviewer: You can consider any artists or designers.

Interviewer: You can consider any artists or designers.

IDENTIFY THE CUSTOMERS & REPORT THEIR NEEDS

Candidate: I will first try to identify certain user personas that the Alexa skill can cater to. Among artists, the most attractive market that has a demonstrable monetization potential is that of graphic artists and designers. They use products like Adobe Illustrator, GIMP, etc. to create designs ranging from posters and flexes to logos for clients. This is one market segment that I can think of targeting. Their needs include searching for appropriate graphics to be used in their designs and being able to create the designs in a time- efficient manner.

The second segment I can think of are UI designers who create mock ups of ads, apps, and websites. These customers can be catered to with the value proposition of having voice- assisted designing.

Interviewer: Sounds good.

Comment: This problem statement requires you to create a revenue generating product, so it is good to identify use cases up front that have demonstrable monetizing potential.

CUT, THROUGH PRIORITIZATION

Candidate: As the business goal is to have a monetizable skill, I will prioritize among the two personas on monetization potential. I am aware that Adobe products are provided for a high subscription fees. Thus, I think that professional graphic designers

are much more willing to pay for a product that they can use, whereas for UI designers, prototyping is a secondary activity. So, I would choose graphic designers as my target users.

List Solutions

Interviewer: In what way would this Alexa skill cater to graphic designers?

Candidate: The typical user journey for a graphic designer is: ideation- >planning->searching for graphics/vector clipart to be used- >designing->iterating. One of the issues that I used to face when I would create posters for my college fest was finding appropriate graphics to be used on the posters. That is one pain point that can be addressed among the users: search for graphics to be used in their designs, as an add on tool for some graphic design software.

Interviewer: So, how would this work?

Candidate: The user can call, "Alexa, find me animal clip-arts". Apart from asking for clip-arts from a premium library that can be built by partnering with firms like ShutterStock, the users can also use Alexa to make use of voice assistance to rapidly prototype designs and storyboards. They can ask for the right elements and with a touch interface and the voice interface reposition elements on their canvas to generate quick designs.

Interviewer: What are the possible revenue streams from this product?

Candidate: The primary revenue stream is from the sales of the Alexa- based add-on (which would have to be done in conjunction with the graphic design software). The second stream is from commission on sale of premium designs from shutterstock.

Evaluate Trade-offs

Interviewer: As a PM, would you work on this product to bring it to market?

Candidate: No. My reasons for not pursuing this product to market would be:
1.The size of this market is very small, and developing such a skill would require a lot of product development effort. It is not easy
to seamlessly integrate voice plugins with sophisticated
software like Illustrator.
2.Amazon focuses on creating value for users and creating a complete ecosystem. As a skill that would be
a secondary add on to a graphic software, that would be opted by a
few graphic designers, and that shows limited potential outside
this market, I would not recommend developing this
product.

Comment: The candidate draws on personal experience. This gives more credibility to their assumptions and prioritization criteria. They also show that it is okay to answer the 'Would you build this product?' question negatively, if you can give a logical and/or business reason for the same.

NEW PRODUCT DESIGN 8

Design a product for social media apps to fight Covid-19

COMPREHEND THE SITUATION

Interviewee: Is the main purpose of this product to access information easily and reduce misinformation going around and be a reliable source for data?

Interviewer: Yes, Along with this we would also use this product to build strong support communities in this difficult times

Interviewee: Sure, We can also share safe practices to be followed.

Interviewer: Sure, Go ahead

CUSTOMER IDENTIFICATION AND REPORTING THEIR NEEDS

Interviewee: I would like to segregate users based on the emergency and their impact on the situation. So, the main segments would be

Healthcare workers
Covid -19 patients
Government
Others (Friends, family and everyone else)

I would like to first define the user needs of these segments and pick the most critical one to solve. Is that okay?

Interviewer: That sounds okay

Interviewee: Defining user needs for each of the categories

Healthcare workers:

- Need to communicate with other hospitals and government officials to confirm protocols to be followed
- Availability of kits, medicines etc. As this is a heavy stress period, concentration should be there on the mental health of healthcare workers.
- Express concerns and update the status of patients etc.

Covid - 19 Patients:

- Confirmed patients: Steps to be followed next and actions to be taken
- Non - confirmed patients: Starting from testing procedure, what are the steps to be followed

Government:

- Send alert messages when new cases are detected
- General information and rules to be followed - in terms of conveying them to citizens or checking if they are being followed etc.,
- Track confirmed patients and returnees from abroad

Act as single channel to address all the frequent concerns and questions that public might have Act as a medium to let public know of the new schemes by government

All the other users
Act as single trusted source for all the information
Protocols to be followed as advised by WHO and other governments

For the purpose of this interview I would like to address the issues faced by healthcare workers as they are the ones most impacted and have to spend long hours at hospital, away from family and are really important for us to catch a hold of the situation.

To define these personas a little more, they are mostly doctors, nurses, hospital staff and other staff related to healthcare. The assumption here will be that most of these people are well connected with technology and are often 25 years or older.

POSSIBLE SOLUTIONS - HEALTHCARE WORKERS ESPECIALLY DOCTORS AND NURSES

In the problems mentioned above for the healthcare workers I would suggest the following solutions

- Online counselling app to healthcare workers working directly with Covid patients: This will be available for both healthcare professionals and their families to help them get through this stressful situation
- Information portal : This will act as a source where doctors from different hospitals will be able to update different requirements they have (like blood, plasma, PPE kits etc.,) and update multiple things like number of patients etc., and because large amount of people already use different platforms this will be helpful

One of the other objectives in these times is morale building. If we enable users on facebook to appreciate the real world events happening around them, this would act as a motivator for health care workers in these tough times. This can probably include sending appreciation stories to their profile, or give some badges or ratings on different platforms.

While suggesting these you may face some questions like by whom, who will support the platform and funding etc., be careful to not quote something totally unrealistic.

CUT THROUGH PRIORITIZATION

To come up with the prioritization list let's consider engineering costs and the number of people this will impact.

So, in my opinion the Information portal has the highest ROI amongst the suggested options because in these tough times, it is hard to get correct data and harder to fulfill the requirements needed. This might act as one of the ways in bridging that gap.

This can be followed by Online counselling app as keeping doctors mentally healthy is also critical.

At the end we can have facebook stories where others can appreciate the real word happenings to increase morale of doctors and nurses.

SUCCESS METRICS

- #people using this service (in terms of regions, professions in a time frame like 1 months, 3 months etc.,) - helps us understand the attractiveness of our features

- #of requests answered by government officials in terms of supplying kits, filling up vacancies, addressing concerns (converted) - on daily basis

SUMMARY

To summarise, data has a crucial role to play to understand the spread of disease, capacity of workforce we have, necessity of medical equipment etc., so the process of data sharing should be seamless, for this reason the idea of an information portal is prioritised.

NEW PRODUCT DESIGN 9

Design a new bicycle renting app to be used in a college's campus.

Interviewee: So we have to design a bicycle renting application for the people staying in large college campuses. I want to confirm that the goal of this product will be to provide an easy mode of commute for the college's residents? Also, will these bicycles be allowed to be driven outside campus?

Interviewer: Yes, this is the goal. The bicycle can be driven only inside the campus.

Interviewee: Then this can be a mobile application, thus, giving users the flexibility to plan the commute from anywhere. Moving forward, I will first identify the customers for the product, then list down and prioritize their needs. Then, I will brainstorm the solutions and prioritize among them for the MVP. Lastly, I will define the success metrics for the product. Does this plan look good?

Interviewer: Yes, please continue.

Interviewee: There can be following type of users:

Residents:

- College students: They would be young (generally, 18 yrs to 30 yrs) and tech savvy. However, they will have lower willingness to pay. They might use the bicycles for faster commuting between hostels and academic complexes, and for leisure campus rides.
- Faculty members: The younger section of faculty members might be interested in utilizing the bicycles for commuting within campus. Their willingness to pay would be higher than that of students.
- College staff: They would comprise guards, mess workers and workers in food outlets. Fewer of these will be tech savvy and have higher willingness to pay than students. Their main goal would be to travel from one location to another faster and carry goods.
- Visitors: They may visit the campus during fests, conferences and competitions. Their willingness to pay might be higher when compared to resident students. Their goal would be to travel from dorms to event locations faster and sightseeing.

Whom shall I focus on for this interview?

Interviewer: Lets focus on students.

Interviewee: Sure. I will first list down the needs of a student as per their user journey.
1. A student would want to find the nearest available bicycle so that she is aware of where the bicycle can be picked from. Furthermore, she would block the bicycle for some time so that no one else takes it by the time she reaches there.

2. She might need directions to reach the nearest bicycle stand.

3. After reaching the location, she would have to identify her bicycle and unlock it to start the ride.

4. New students might want to be navigated to the destination. Students will also require an emergency contact number whom they can reach out in case the bicycle breaks in the route.

5. She will end the ride after reaching the destination so that she can stop the meter from charging her anymore.

6. She would want to make the payment through the apps she commonly uses so that she doesn't have to maintain a new wallet.

7. She would want to generate recurring bicycle booking requests so that she can have a mode to commute for each of her classes.

8. She might want to synchronize her academic calendar with bicycle booking so that a bicycle can be tentatively blocked for her.

For MVP, we would initially focus on fulfilling the following needs so that students rent the bicycle for easy commuting: 1, 2, 3, 5, 6.

Brainstorming few features for MVP of mobile application and also measuring their impact:

	Features	User Impact	Cost of building impact
1	GPS on college campus map for finding the nearest pick up point with available bicycle	High	Low
2	Maintaining state of bicycle: available, not available, soon available	High	Low
3	Blocking the bicycle and making it unavailable for other users for a time duration	High	Low
4	Measuring estimated time to reach destination and making it soon available at that point	High	High
5	If a bicycle is not currently available, the user should get to know an estimated availability time	High	High
6	Whenever users leave the hostel, provide him with a push notification if a bicycle is available near them	Low	High
7	The user should be able to navigate to the available bicycle	Low	Low

8	The user can scan a barcode on bicycle through app to unlock it and start the ride	High	Medium
9	The user enter the destination and be navigated to the nearest drop point from there	Medium	Low
10	The user should be able to end his ride through the mobile app	High	Low
11	When the user ends the ride, the bicycle should be locked automatically	High	Medium
12	The user should not be allowed to end a ride if he is not at a drop location	High	Low
13	The user should be able to link UPI and wallets with the app for ease of payments	High	Medium
14	The user should able to see the ride duration and distance travelled for past rides	Low	Medium
15	Gamification can be implemented by maintaining a leaderboard on the distance travelled and also providing users with badges for motivation	Medium	High
16	An alert should be shown if user takes the bicycle outside the campus	High	Low

Based on this, I will prioritize the following features: 1, 2, 3, 8, 10, 11, 12, 13, 16.

To measure the success of the product, I would track the following metrics:

- No. of app downloads per month
- No. of new registrations per month
- No. of rides per month
- Average duration of a ride
- Average distance of a ride
- Percentage of pickups from each point
- Percentage of drops from each points

Would you like me to do something more?

Interviewer: This is good. Thank you!

NEW PRODUCT DESIGN 10

YOU ARE PM OF SPOTIFY. YOU WERE ASKED TO COME WITH SOLUTIONS TO IMPROVE SPOTIFY AS A PODCAST APP

Interviewee: Spotify is a music and podcast streaming platform. It follows freemium model. It gives listeners access to millions of songs and other content from creators all over the world.

Interviewer: It's correct.

Interviewee: Before delving deep into the problem deep into the problem, I need to understand for whom, are we trying to improve the product for? Listeners or creators. are we have targeting any particular demography or are we have having budget or timeline constraints?

Interviewer: Thanks for asking that. We are focusing only on the customers at the moment. For the simplicity of the case we can assume there are no demography, budget or timeline constraints.

Interviewee: Multiple stages of user journey involve acquisition, activation, engagement, retention, and referral. For the given problem statement, engagement and retention stages should be focused. I'll prioritize Engagement since a good engagement increases chances of good retention, if that's fine with you?

Interviewer: Sure, Go ahead.

Interviewee: Based on my understanding, there are three user personas for Spotify podcast users.

1. College Students
2. Working Professionals
3. Retired Citizens

College students are primarily interested in music and podcasts. However, they struggle to find the right topics for podcasts, often losing interest and leaving halfway. Working professionals, who listen to both music and podcasts, are more focused on podcasts related to their professional work. Despite having a higher attention span, they also get distracted while listening to long podcasts. Retired professionals, who have more spare time, also struggle to find the right recommendations due to the various types of recommendations on the app's home page.

I would like to concentrate on working professionals, as those are people who tend to use podcasts more and expect a learning out of the it.

Interviewer: That's a good understanding. You can go ahead with your line of thought.

Interviewee: Thanks. Going forward, I see that the current pain points for working professionals are:

1. The user finds it hard to capture learnings from a podcast
2. The user needs to put in efforts to consolidate all their learnings in a single place & make it easy to refer back

High-priority issues and poor present satisfaction levels will be given priority. I'll give priority to the first issue, which is that consumers find it difficult to take notes on podcasts while listening to them, as I believe that the first problem is more important to power users and that user satisfaction is currently low.

Interviewer: That sounds fair. What product interventions or features you would like to introduce to mitigate the pain points?

Interviewee: In order to diversify the content usage by the customers, we can introduce a function equivalent to "Podcast Snippets". With this feature, viewers may scroll through and view brief excerpts from different podcasts. Clicking on a clip may take a user to the original podcast channel, where they may listen to the entire episode, if they find it fascinating and would like to investigate more. This strategy could pique users' interest in podcast genres they never would have considered.
To increase engagement among the users, we can have a community feed for podcasts after it attains specified number of followers. Where the creators can share read notes and conduct Q&A sessions.

Interviewer: Those are some interesting suggestions. Can you also list down some success metrics for your suggestions?

Interviewee: Sure.
Success Metrics:
(a) increase in the number of people that subscribed to a podcast via the feed and listened to a new episode each month
(b) an increase in the quantity of users monthly users who discovered a podcast through the feed and went on to listen to several episodes
(c) No. of users saving snippets, Avg. snippets saved per person

INTRO TO GO TO MARKET STRATEGIES

A Go-to-Market (GTM) strategy is a detailed action plan designed to launch a product or service in either an existing or a new market. It focuses on reaching the right set of customers at the optimal time through effective channels. The ultimate objective of a GTM strategy is to develop a comprehensive plan that outlines how a business will attract, acquire, and retain customers. This involves clearly communicating the value proposition to potential customers, ensuring they understand the unique benefits and advantages of the product or service.

A GTM problem can be thought under the below steps:

Define your market

This includes the type of market segment you are trying to enter including B2B, B2C etc, the size of the market segment, details about the industry your product will be placed in et al. Knowing a market is essential to determine the product market fit.

Define your buyer persona

Understanding the customer is extremely important to any product management interview question. The candidate needs to identify the target customer for the product and understand their needs and sentiments. Persona may also include age range, location and other demographic data as well as Income level, Education level etc.

Develop a clear value proposition

Understand the product and the value it adds to the customer's life. A unique value proposition can also help the company differentiate its products from the competitors. A popular formula is:

"For (target customer) who (statement of need or opportunity), our (product name) is a (product category) that (key benefit, compelling reason to buy). Unlike (primary competitive alternative), our product (key differentiator)."

Identify your competitors

Another area of prime importance is knowing the competitors the product is up against. The candidate can list down the competitors and try to determine what makes their products unique. This exercise can also help a product manager highlight the value proposition of their product to pitch why a customer must buy their product over competitors.

Choose a pricing model and Strategy

There are three approaches a candidate can choose to price a product:

Value-Based Pricing: This is a powerful way to price a product which is operating in a niche market. It works by usually charging a percentage (10% to 50%) over amount the consumer is paying to solve the problem through a alternate product or the amount of value that the product can add to the customers life.

Competitor-Based Pricing: This is more appropriate in competitive markets. A firm can try to match the average price of similar products and then adjust the structure according to their offer over and above the competitor's offerings.
Cost-Based Pricing: This should not be preferred in most situations. This technique involves adding a premium over the costs the company incurs in production.

Plan marketing and sales tactics

Next, the candidate must discuss a robust marketing plan through which the word about the products reaches the target consumer. The below marketing funnel can be considered while thinking about marketing strategies for products:
Awareness: Paid Ads, content marketing, social media marketing, etc.
Interest: Email Campaigns, Free Trials etc.
Decision: Lead Nurturing & Qualification, Sales Enablement Tools (CRM).
Adoption: Onboarding Tutorials, Support Services etc.
A candidate can also try to map out the customer's journey to determine various touchpoints and decide on how to increase awareness about their products.

Launch Strategy

After deciding the pricing and marketing techniques for the products, the candidate must also try to determine the optimal mix of channels through which the product can be launched. There should be a healthy mix of options through which the product can reach the consumer. Further, entering at the right time may also be important factor in a product's success.

Post Launch Measure and Adjust

The candidate must also define the metrics which the firm can use to determine the success of the product. Some examples can be - Customer Acquisition Costs, Rate of Acquisition, Daily Active Users etc.

Possible Risks

The candidate can highlight any risks associated with the product launch to cover all areas of the strategy comprehensively. Some of identified risks may include - Government Regulations, Distributors, Exit Barriers, etc.

GO TO MARKET STRATEGY 1

Interviewer: Our company wants to launch a social fitness product. How would you develop its go-to-market strategy?

Candidate:
(Ask clarifying questions about the product, what it does, what problem it solves and how, target market and the goal of the launch)
What kind of a product is this? Is it a tangible product like smartwatches or a smartphone app?

Interviewer: It is a smartphone app where users can connect with other fitness enthusiasts. The users can share updates about their workouts, diet plans, progress etc. with each other. They can work together through video conferencing on the app itself and compete against other users in groups or individually in regular challenges hosted on the app. The aim is to encourage users about health and fitness through peer motivation.

Candidate: Okay. Are you partnering with other third-party apps?

Interviewer: Yes, users can also share their updates on other social networking accounts like Facebook and Instagram. It can also connect to smartwatches and wearables to enable automatic updates.

Candidate: So, who are the target users for this app? I assume it would be fitness enthusiasts who are also tech-savvy.

Interviewer: Yes, you are right. We are targeting people in the age group of 20-60 years, living in metropolitans and urban areas of India.

Candidate: Is this a new product or an existing one being launched now in India? And, what is the goal of this launch?

Interviewer: This is a new product, and India is the first country where it would be launched. The MVP launch and beta testing were successful and now we want to launch the full version in the market. The goal, therefore, is to build awareness, drive adoption and generate revenue.

Candidate: Okay, thank you for answering my questions.

(Analyse the market situation – take company, product, customer, competitor, and market landscape into consideration)

The company is a sports equipment company that manufactures footwear, sports and casual apparel. They are looking to diversify their offerings by

entering the smartphone app space. There are some synergies on both sides of the business. Major competitors would be other fitness apps currently in the market like MyFitnessPal, cult.fit, HealthifyMe, FTTR etc. Today, especially in the covid era, physical activities and social interactions have reduced. This product covers both of these aspects. Moreover, people are becoming aware of the importance of good health and the fitness industry is in the growing phase in India.
Now that we have analysed different components, we can go to the launching strategy. I will divide the activity into 3 stages: **pre-launch, launch and post-launch**.

Pre-Launch:
Pricing – The product could be a free app with ad based revenue or follow a freemium model or could be a mixture of both (depends on the discussion with the interviewer)

Create buzz – through marketing activities like releasing teasers on social media channels and promotion by fitness influencers

Launch:
Place – Since this is an app, it will be released via app stores(android and iOS)

Promotion - start marketing the product actively by employing digital and social media marketing tactics. We should also advertise the product in gyms and wellness centres. And we can also use SEO to help gain traction for the application on the web as well as on the app stores.

Post -Launch:
Now, we will measure the performance by tracking relevant metrics. Some examples could be:

Acquisition - conversion rate, cost of customer acquisition, # of downloads Retention - churn rate

Engagement – avg. session time, what features are being used, # of updates/posts posted

Customer satisfaction – feedback in the form of ratings and reviews, NPS scores

Revenue and market share – are we meeting our revenue and market share goals

I will use the feedback information and engagement information to plan for v2 potentially.

GO TO MARKET STRATEGY 2

MARKET ENTRY STRATEGY FOR THE HAND- HELD GAMING DEVICE FOR PC GAMES AT AFFORDABLE RATES.

Marketing Goals:
(*Ask clarifying questions)*

1.Position the product into a niche segment entailing the go-to PC gaming with seamless PC connectivity.
2.Create space into already competitive hand-held gaming space.

Overview:
(Enquire about the company's details, the prevailing situation, and the value proposition the company seems to fulfil)

Valve Corporation has developed a hand-held gaming device that purports to maintain steam PC library games into a hand-held device. The hardware offers an opportunity to experience high-end gaming experience on a hand-held device. It aims to leverage an already established position in the PC gaming market by enabling a user to play any game in the user's library. It is more than a hand-held console and leans towards a portable PC with various storage and CPU options. The software is optimised towards a smooth gaming experience which may need an expensive setup in a traditional PC.

Product:
(What are the product, its specifications, and the current market entry strategy envisioned by the company?)

A hand-held gaming console powered by AMD options ranges from 64GB mme to 512 GB nvme (There is a large gap between mme and nvme. Why is the mme option even considered?). The specs seem enough to run a typical game in 720p resolution. Current go-to market entry is based on pre-booking with a delivery date a year ahead of the launch date. Comment on an ideal go-to-market strategy for such a product and any improvements that may be incorporated into the steam deck's launch.

Competitors:

1.Nintendo Switch: Leader in the segment offering two modes of playing viz. plugging the switch to dock or playing as a stand-alone device. An extensive gaming library is available, entailing numerous Nintendo titles.

2.Gaming Mobile devices: High-end gaming devices like Asus ROG, One plus pro, and Ipad pro cater to multiple

users' needs, viz gaming and productivity. The increasing library of mobile games has pushed users against stand-alone consoles. However, the gap between mobile games and PC games persists.

Customers:
(*What does a typical customer look like? What are the primary requirements of a typical customer? Any niche customer segment which may be looked at?)*

1.**The gaming enthusiasts** looking for a more convenient setup for gaming: These users are already familiar with the steam interface offerings and would like the product as an extension for the primary setup.

2.**On the move enthusiasts** who may be gaming enthusiasts but do not use services offered by the steam. These customers need to be made aware of the value proposition of Steam Deck. It may be assumed that these customers are looking at the steam deck to be their primary device.

3.**Heavy mobile gamers** looking for better alternatives for mobile gaming: These customers want a better experience than mobile gaming. Again these are looking for a device that may be their primary gaming device irrespective of the PC platform.

Context:

The portability of the device has become an important consideration in recent times. Cloud services may enable a better experience with lower hardware specifications and costs in future. However, a gap persists in the "on the go" gaming market.

Proposed Initiatives:

1.Marketing via direct channels to the already subscribed users of the Steam PC platform. A discount should be offered in pre-booking stage to bring the most loyal
customers on board.

2.The mobile gamers may be targeted via advertisements in the standard games available on both
platforms.

3.Top players of the famous games should be roped in. The streaming
of the games played on the portable device should be incentivized to remove inhibitions in the minds of prospective buyers.

4.Partnerships with the top- grossing gaming companies should
be initiated, which should incorporate interoperability between the platforms. The dedicated hardware and software
of the device shall bring in more
users.

Summarize:
(*Indicate how the objectives are met with the initiatives.)*

By addressing the high-performance need of the current mobile users and the portability need of frequently mobile users, steam may carve out unique space for its product. The targeted message towards the existing users about interoperability, synchronisation shall help the company to induct them.

GO TO MARKET STRATEGY 3

Interviewer: Swiggy has decided to enter the table reservation category in restaurants. Formulate a launch strategy for the same.

Candidate:
(Ask clarifying questions about the product, what it does, what problem it solves and how, target market and the goal of the launch)

Can you tell me more about the product, like if you want to create a new app for this requirement, or would this be a new feature in the existing app?

Interviewer: So the product is still in the idea conceptualisation phase. I was hoping you could help me with that.

Candidate: All right, for table reservation, there would be three major stakeholders, the customers, restaurants and Swiggy. I would suggest creating a new feature in the app itself as Swiggy already has a large customer base, therefore creating awareness and growing customer adoption about this product would be easier. Next, I want to look at the pain points of the stakeholders involved to come up with the product features.

1.**Customers** – Customers usually aren't aware of the different restaurants in their cities. They would like to search for restaurants offering dining according to their cuisine and budget preferences. Under cuisine, they would like to know the variety of dishes served. Customers do not like waiting, so they would want to know the waiting time beforehand. They would also like to know about the availability of parking space, timings etc. And most importantly, they would like to reserve tables so that they don't have to join a queue when they reach the restaurant.

2.**Restaurants** – Restaurants are aware that if they cannot handle demand effectively in their restaurants, they would lose out on customers who had to wait too long. On the other hand, if a restaurant reserves a table for a customer, then they can't give it to another customer. They must keep the table vacant until the designated customer arrives. Due to this, there are chances of them losing out on prospective customers.

Interviewer: But don't you think Zomato already provides these features?

Candidate: Yes, so one differentiating feature of this product would be that customers can also pre-order their dishes while reserving a table. This way, they wouldn't have to wait for their orders to be prepared in the restaurant. Swiggy can provide estimated arrival times of customers to the restaurants so that they can start preparing their orders accordingly.

Interviewer: Okay, I have one more question. If the customer has booked a table for a particular time and is late, what happens then? Should it continue to be blocked at the cost of losing another customer?

Candidate: We can provide a buffer time beyond a customer's reservation time till which the restaurant will reserve the table for them. If the customer arrives later than the allotted buffer time then he/she will have to wait if the restaurant is fully occupied.

Interviewer: Okay. Considering this to be what the application does, how would you go about launching it?

Candidate: I would like to divide the marketing activities into three buckets. **Pre-launch, Launch, and Post-launch.** I would like to go through this one by one.

Pre-launch
Swiggy can release this product in big cities like Mumbai, Bangalore, Delhi etc. first, where the dining culture is big to observe customer adoption. They can create hype around the new feature through marketing activities like releasing teasers on social media channels, putting up billboards.

Launch
We can start marketing and promotional activities to build customer awareness: traditional and modern channels .

Post Launch
Now, we will measure the performance by tracking relevant metrics. Some examples could be:

Acquisition - conversion rate, cost of customer acquisition.

Engagement – # of reservations, bounce rate, % of reservations where customer visited the restaurant out of total reservations, % of reservations where customer arrived before the end of buffer period.

Customer satisfaction – feedback in the form of ratings and reviews, NPS scores.

Revenue and market share – Are we meeting our revenue and market share goals?

I will use the feedback information and engagement information to expand it to other cities potentially.

GO TO MARKET STRATEGY 4

MARKET ENTRY STRATEGY FOR AN ALL-ENCOMPASSING MENTAL HEALTH APP/PRODUCT

Marketing Goals:

(*Ask clarifying questions)*

1.Position the product as a one-stop solution for mental health needs. The offerings should range from detection, self-help, preliminary diagnosis, expert help, and regular motivation.

2.Address users' privacy concerns about user data sharing and the confidentiality of users' information and diagnosis.

3.Increasing the uptake by other stakeholders in the program viz.. regulatory bodies, certified practitioners, academics and associations dealing with the matter.

Overview:

(*Enquire about the prevailing situation and value proposition the company sims to fulfil)*

Mental health is being recognised as an important element of overall wellbeing. However, it is continued to be ignored. The mobile and other electronic devices present an immense opportunity to engage users in this space. It shall include all aspects of mental health, from awareness to diagnosis. The increased awareness about mental health should prompt a user into using the App. Thus, an element of the offering should entail awareness about mental health and removing its stigma. Various organisations, including many non-profits and governmental ones, are engaged in the sector. The company aims to be a one-stop solution for such needs. The current App focuses on the specific aspects of mental health, whether mindfulness, anxiety or limiting digital time.

Product:

(*What is the product, and what are the privacy concerns that may perturb a user?)*

The product should focus on prevention, preliminary diagnosis, awareness, and expert help. The product aims to track user data across various platforms, and thus, certain inhibitions may arise about data sensitivity and its sharing with others. Additionally, mental problems still carry a social stigma. The user should not have any concern about a breach of private information in any regard. Consequently, the company has adopted no data sharing policy.

The product shall have various features like a meditation session, mindfulness session, gratitude journal. The relevant options are to be suggested to a user without overwhelming them. The primary option to take user input includes filling in keywords about their day, rating it, searches, and App usage data. The motivation game feature is an innovation of the product which presents a relevant historical anecdote to uplift the mood by asking the user to make the right choice.

The recommendations and advice are to be based on expert help and guidance. In this regard, experimental features shall be available to limited users. The necessary regulatory obligations regarding privacy concerns and expert qualifications are given due importance and highlighted in the product.

COMPETITORS

There exist numerous products targeting specific aspects of mental health. Digital well being is now being offered in the OS of most devices. Some enable users to demand therapy sessions at any time. Others enable users to have a gratitude journal or takes them through simple exercises to uplift their mood. Others take a spiritual take towards mental health and address common concerns. On a different note, many products enable users to develop a healthy routine, habits, and dietary plan, essential for mental wellbeing.

The problem with such offerings is that a person must be self-aware about the problem they face and must look for the right application, which may not be true.

Customers:
(*What does a typical customer look like? What are the primary requirements of a typical customer? Any niche customer segment which may be looked at?*)

1.An overworked individual facing difficulty in balancing personal life with professional life. The resultant anxiety has affected both aspects of their life.

2.A typical individual stuck at home in times of COVID. The sheer amount of change faced by such individuals has caused numerous changes in their psyche. The unknown nature of experience has caused them not to be able to decipher their situation correctly.
3.The silent majority are placed in various conditions which may be suffering from some mental problem yet remain unaware. The diversity of the segment prevents drawing sweeping generalisations. It may range from a student struggling through their course work, a homemaker who recently found themself in an entirely new environment or even a worker faced with severe workplace hazards every day.

Proposed Initiatives:

1.Awareness via direct channels on social media sites. The non-profit and government bodies should be roped in to advertise about the prevalence and harmful effects of mental problems. It is along with term initiative, which should be scaled up in due time.

2.Marketing via direct channels to the already subscribed users of similar products. The dedicated user therapist should be leveraged.

GO TO MARKET STRATEGY 5

Interviewer: You are PM at the Facebook building for India. You're tasked to come up with a Go to Market strategy for a dating application that Facebook has come up with.

Candidate: Before I move on, I want to clarify the marketing goal. It is to launch the application in India?

Interviewer: Yes, you need to come up with a strategy to launch this application in India.

Candidate: Sure. I would like to start off by analyzing the current dating application market. Dating is among the top-grossing categories on the Google play store in India. People are willing to pay for such products. The top applications which in can recall are Tinder, Bumble. They are a global product. There are quite a few local products as well. But all these are dating applications first. None of them have the capabilities that Facebook has, in terms of Social graph, user preferences, etc. Most of the users are between the ages of 18 - 27. They are usually in college or early stages of their professional careers. Most of the dating platforms follow a freemium model. They also use ads to generate revenue. One of the main reasons for a freemium model is to get the users to use the product and realize the value before asking them to pay. There is also the aspect of network effects. Getting more users on the platform is very important for its success.

Interviewer: So how does Facebook fit in this market? What can Facebook do differently?

Candidate: Talking about Facebook, I think the goal of Facebook is to bring the world close together and help build communities. A dating platform that enables building new connections fits right into its vision. The key strength of Facebook is its data. It can leverage this data to find an almost perfect match. However, dating experiences vary from culture to culture. Facebook has to keep in mind that dating expectations are completely different in India compared to western countries. So the data on users and their networks can be leveraged to build a product that people can trust. Dating platforms in the market struggle to keep the platform clean of bots and fake accounts. This is a huge problem, which is hard to solve without

taxing the actual users to give out personal information. However, the current platforms have the advantage of understanding the users' behavior. They also have brand equity. This would take quite some time to build. Facebook can have features like fraud detection and a better recommendation system to win over the customers' trust. So Facebook should focus on providing a platform for building lasting relationships as they have enough data about the users and their behavior.

Interviewer: Okay, this is good. Considering this to be what the application does, how would you go about launching it?

Candidate: Sure. I would like to divide the marketing activities into three buckets. Pre-launch, Launch, and Post-launch. I would like to go through this one by one.

Interviewer: Sounds great. Go on

Candidate: Okay.

Pre-launch - Facebook should ideally start by testing it in a few regions within India. Probably start with metro cities like Mumbai, Bangalore where the dating culture is more progressive. Facebook should create hype around the product. They could release teasers, small clips showing what the product could do.

Launch - They should launch the product by creating a virtual/ live event. All the competitors in the market are kind of established, so Facebook should seek to make an impression on its users.

Post-launch - Facebook should actively market the product and educate the users about how this platform is different from its competitors. They can use existing products like Facebook and Messenger to market this product.

Now, talking about distribution; Facebook can launch this as a separate application. Facebook has access to 300 million Indian users and they can use this user base to market the product to.

Interviewer: Now let's move on to the retention of these users.

Candidate: Sure. When it comes to dating platforms, retention is dependent on the value people derive in terms of connections made. Facebook should not focus on whether the said match goes on a date or meet up in real life as there are a lot of other factors beyond Facebook control. The primary focus should be on how happy a user is with the kind of people they meet on the platform. We can measure this by seeing how long they interact with each other on the platform and also if they add each other on Facebook as friends. We can also ask them for feedback a week after they start interacting with each other. We need to message the feedback in such a way that the users realize why the feedback is being taken.

Interviewer: How would you go about pricing this product?

Candidate: The product should be free for all users. New experiences can be added later which can be made paid. We can also tap into live experiences with virtual parties, events, etc. They can also create a pro version for the user which would unlock several features to find better matches or even find matches quicker.

Interviewer: This is good. Could you please summarize the whole plan?

Candidate: Sure. The goal for Facebook would be to stand out and differentiate itself from the competition. The focus should be on building long term relationships, be it friendship or dating. Moving on to marketing, Facebook should focus on creating hype towards the product. This will also set the right expectations for the users. They should back that up with a launch that involves its users. However, it would be a product that would determine success. Most platforms don't have enough data to find the perfect match for a person. Therefore, the value derived from the platform is not maximum. Also, these platforms struggle to identify bots or fake accounts on their platform. Facebook can differentiate itself by solving these very problems.

GO TO MARKET STRATEGY 6

Interviewer - The company “Grocers” wants to launch a product which does grocery delivery service. Give us a Go-to-market strategy for the company.

Candidate - What are the goals of the company? What are the company’s strengths and weaknesses?

Interviewer - The company wants to capture market share in Tier-2 cities and towns in the country. The product has vernacular language support, which adds further strength to the company’s goal of entering Tier-2 cities and towns.

Candidate - What does the product do and what is the value proposition for the customer? What are the strengths and weaknesses of the product?

Interviewer - The product delivers goods listed on the app/website to your doorstep, with a minimal delivery charge. With the stress on social distancing, the product gives the freedom of shopping from home.

Strengths: The product captures data and the same can be used to predict demand, improve product features, introduce new services, make the logistics cheaper. The product has features like scheduling of delivery, taking limited credit, vernacular language support.

Weaknesses: Security of customer data, Technical glitches in the product, Mismatch or errors by the human agents in the system like delivery agents.

Candidate - How do competitors position their products? What are their strengths and weaknesses?

Interviewer - Given the current changing scenario, we can divide the competitors based on the delivery channels as Online Delivery platforms and physical stores.

Within Online Platforms, we have Hyperlocal services (like Bigbasket, Grofers, Swiggy Genie etc) and E-Commerce platforms (like Amazon, Flipkart etc). The hyperlocal delivery services now have a quicker delivery time when compared to the E-commerce platforms. E-Commerce platforms are venturing into grocery delivery now lately. The advantage that the online platforms have are the aggregation of orders and that

successively helps in planning the deliveries and procurement better. Also, tech products give abundant user data, from which insights can be mined and used.

Within the traditional physical stores, we have Organised and Unorganised retail. In organised retail, we have big players (like Reliance Fresh, Big Bazaar, D-Mart etc) and small players (like local supermarkets). Big players have the advantage of big basket size and enjoy economies of scale Under unorganized retail, we have the local kirana stores. The unorganized retail usually works with low volume and high margin sales.

Since our target market is Tier-2 cities and towns, we do not have Big Organized retailers and Hyperlocal delivery services to a large extent. The major competitors still would be physical stores only.

Candidate - Who are our customers? What do customers want to use the product for? How do they pay for the product?

Interviewer - With the pandemic situation, working professionals in metros have moved to WFH. They are either working from their homes in the same city or have moved to their homes in Tier-2 cities and towns. They now live with their parents, who are aged and have a higher risk due to the pandemic.

They would take sufficient precautions to ensure that the aged parents are safe and would be inclined to use all the means available to ensure that. The customers can use our product to do their usual grocery shopping in a safer manner.

We offer different payment options for the customers like Cards, Wallets, UPI and also take credit with a limit based on the customer profile. There is subscription-based payment for certain products such as milk, which can be delivered daily. Our revenue stream includes the delivery fee and the margins that we get on each order.

Candidate - What are the trends? What are the legal and regional issues?

Interviewer - The Online e- commerce industry is currently growing very fast due to the Covid'19 scare. Many new players are coming in, like Amazon, Flipkart who are getting into the grocery category to meet the increased demand amidst Covid'19.

Candidate - I want to segregate the marketing activities in 3 headers, pre-launch, launch, and post-launch. I'll cover different aspects like Product, Distribution, Promotion, Partnerships, Retention strategy, Pricing that are associated with the launch.

- Speaking about the features that the product must have, While moving to Tier-3 cities and towns, it is necessary to focus on regional features like vernacular languages, buying habits of consumers, traditional customs etc.

- Regarding distribution , the product can be available as an App on the Playstore, Website and Ordering via Phone. Ordering via phone is an important channel to consider because our goal is to capture tier-2 cities and towns.

- About Promotions, An outbound marketing strategy can be used to cover the audience in masses, again due to the nature of consumers in Tier-2 cities. While a significant number of people in these places would have ordered online, our outbound marketing strategy should be focused on making this a habit.

.

- Partnerships with wholesalers and suppliers. Partnerships with local farmers to buy fresh produce, at low prices.

- To keep users actively using the product, retention , strategies like Promotions and discounts based on a Loyalty program.

- Our Pricing strategy can be offensive or defensive. The offensive could be price leadership, penetration pricing, price/performance pricing, or promotional discounts. Defensive could be value-based, maintaining the highest price, price skimming, or bait-and-switch pricing. We could go with an Offensive pricing with promotional discounts to capture market share from Local supermarkets and Kirana stores.

Interviewer - Can you summarize the launch strategy for the company and wrap it up?

Candidate - The company is aiming to capture a market share in the grocery space in Tier-2 cities and towns in India, through Online delivery service.

To target the Tier-2 consumers effectively, the product has vernacular language support, Ordering via Calls, Outbound marketing strategy to educate and create awareness among the people.

Partnerships with farmers and mandis would enable procuring fresh supplies in a lower cost. Loyalty programs to ensure customer retention, accumulated data would enable better logistics planning and hence cheaper delivery costs.

The product pitches in the enable the convenience of shopping from home, amidst and post Covid'19 where physical distancing is very important. The competitors here are Local supermarkets and Kirana stores where the consumers have to interact physically. Whereas our product enables consumers to follow physical distancing.

GO TO MARKET STRATEGY 7

Interviewer: Lyft wants to enter Indian market, what should be its go to market strategy?

Candidate: Lyft has both ride sharing and Food delivery. Is there anything specific that I should be focusing on?

Interviewer: Let's focus on ride sharing.

Candidate: In ride sharing should we focus on only cars or should we also look at Autos and bikes?

Interviewer: We'll just keep it to cars for now.

For this lets first analyze the situation in the market.

Candidate:

Company
Lyft is a ride sharing app like OLA and Uber. It connects riders and drivers and allows an easier mode of transportation. Lyft has many verticals in other countries like Uber does in India (Autos, Shared, normal etc.,). We need to decide in which segments we are going to launch initially and the future plan.

Lets now explore the opportunity on the segments. one of the main things to keep in mind is the traffic in India, along with this we also need to keep in mind demand and supply for these segments. My suggestion in here is to take use of high number of cars existing already and give good benefits to drivers so they use lyft more than other competitors

Coming to our customers most of them might not know about the brand itself and most of them might be regular customers to other competitors we have. We need to give them a compelling reason to try out our app, can be discounts, more safety etc.

As we know we have 2 major competitors currently and many upcoming ones. Uber expanded into this market with a lot of competition and was successful, if we have a compelling differentiating factor we can also be successful as customers look for best options for them. We can probably tie up with local car dealers to create localisation feelings among customers which would probably encourage drivers to join us. We should also invest heavily into

marketing initially and probably be a loss making firm like Amazon initially to get customers to try our product. In addition to this our pricing strategy is unique and will help us in remaining competitive.

Market Conditions

If we divide India into rural, suburban and Metropolitan we need to decide on where to launch first and what would make sense to achieve our goal. Because major customers are there in Metropolitans that would be the first place to launch according to me.

Regulatory Issues

Along with these we also need to consider regulatory issues and rules that the country has so that we won't get into trouble later which might spoil the brand reputation.

Now that we have analysed different components we should go to the launching strategies. Let's divide the activities into 3 stages: Pre launch, Launch itself and Post launch.

Pre-launch

- Marketing Activities
- Billboard ads, news paper ads, pamphlets, experience videos from other countries etc.
- Enough supply according to the demand in the area
- Offer monetary benefits to drivers better than competitors
- Referral programs
- Tie ups with local dealers
- Training and adhering to rules
- Compulsory training for drivers to improve customer experience as they are the face of our brand

Launch

- Promotions and discounts for riders, customers
- Advertisements in local news papers, pamphlets Tie ups with corporates
- Blogs with influencers and celebrities to create positive images etc., can be done.

Post Launch

This is where we measure the metrics. I think these are the metrics which we make more sense in this case.

For Acquisition

- #customers in time frames like 1 month, 3 months and 6 months
- #drivers in time frames like 1 month, 3 months and 6 months
- #app downloads in time frames like 1 month, 3 months and 6 months

For Revenue

- Expenditure/ride - 3 months Vs 6 months
- Revenue/ride - 3 months Vs 6 months

For Customer Experience

- # of repeat customers in a week
- Wait time (average) in a day
- #cancellations in a day
- Ratings given by customers on average

In addition to all this we should proactively reach out to customers with complaints to resolve them soon for better experience and also reach out to people for feedback for continuous improvement.

users who have already recognised the problem.

3. The partnership with various organisations must be leveraged to bring in users. The reach may be enhanced by partnering with content creation agencies working in similar spaces, even on an ad-hoc basis.

4. The differentiators and one-stop solution aspects should be highlighted in relevant discussion forums across the internet. The populace engaged here determines the behaviour of numerous users.

5. The partnership with academics and other aspects of innovation must be highlighted across the channels. Yet still, it should not glamorise mental issues and ensure reservations in some aspects.

6. The motivation games may be popularised by embedding them into the social media feeds and other relevant platforms

Summarise:

(Indicate how the objectives are met with the initiatives)

The first step towards resolution is taken by recognising the diverse problems and informing the users about the same. Suggestions for specific features shall enable appropriate resolution. Certifications and no sharing policy shall address the privacy concerns. Bringing stakeholders onboard shall be a slow process that must be done brick by brick.

GO TO MARKET STRATEGY 8

Interviewer: The tata group wants to incentivize their customers and want to increase their loyalty towards the brand. We need your help to prepare a loyalty program for the conglomerate.

(Comprehend the situation)

Candidate: To make sure I understand the situation correctly, Tata group wants to design a conglomerate wide incentive program which would cover all the brands falling under the Tata brand including Taj, Titan, Tanishq, automobile etc.

Interviewer: Yes, that is correct.

Candidate: As I understand, many brands under Tata group already has their standalone loyalty program. May I please know what will happen to these programs.

Interviewer: You can assume that all the existing loyalty programs will be replaced by the new loyalty program that you would suggest.

Candidate: I am not sure that I understand the problem statement. Do the Tata want to make the existing benefits (reward points) that customers have earned null and void?

I believe that would make the customers angry.

Interviewer: Good that you mentioned it. In your solution, ensure that there is a smooth transition for the existing reward system.

Candidate: Okay. One last question before I lay out my solution, will the brands have a say in incentivizing their customers or the complete control will be with the of Tata group?

Interviewer: We would like you to think how this Loyalty program should be structured. Also, comment on its technical feasibility.

Candidate: Okay, can I take 2 minutes to structure my thoughts and provide a solution?

Interviewer: Sure.

Candidate: So according to me, different stakeholders like the Tata Group, brand owners and store owners should have the option to award reward points to the customers. Also, the system should have the capability to award different points to consumers using different brands. For example, different reward

should be awarded to customer of Starbucks and customer of Titan.

Interviewer: Makes sense, carry on.

Candidate: So, the structure should be as follows:

- Tata Group points: Under this, the Tata group can award points to all the customers irrespective of the brands.
- Brand points: Under this, any independent brand can award points to their customers. They can define rules for the loyalty program.
- Store owners: Every store owner should have the ability to award points to the local customers.

Another thing I need to add is that a customer can be eligible for any points, depending on the rules defined. A customer can get Tata group points, brand promotions and even points by the store owner.

Interviewer: Usually when the campaign runs, the points are provided in a particular city. How will you ensure that?

Candidate: Fair enough. I believe if all the store owners are asked to update the offers, it might be quite ineffective. We should probably give this functionality in brand points and Tata group points. Generally, the configuration should be done on store id which acts as an identifier for the store.

Interviewer: That seems like a good plan. Can you explain the implementation plan?

Candidate: Okay. I know that we need a unique identifier to identify each person uniquely. It is hard to provide the unique number to the customer as they would tend to forget. I suggest we use the mobile number of the consumer as a unique identifier. However, with this, we will have one limitation. The identity of the person will be saved at the Tata group level and not the brand level. To counter this, we need to create a tag where the customer registers.

Interviewer: Can you suggest other things to consider for a good loyalty program?

Canadiate: Sure, please let me think about it for a minute.

The whole idea of accruing the points is to have the ability to redeem the points. Usually, the points can be redeemed at the same brand where they were accrued. I believe Tata group can look to allow some cross- brand redemptions. For example, the points accrued at Titans can be used to redeem at Starbucks.

Interviewer: Interesting, are you done with your solution?

Candidate: One last point which we discussed in the beginning about the current reward points of the users. I believe if we are going with the new Loyalty program, all the existing points should be mapped to the new customer identifier so that customers can use those points in future. I believe the personal data of all the existing customer is also important to import. Therefore, there needs to be a one-time data migration activity to get the old data in the new system. That will be all, sir.

Interviewer: Thank you for the insights.

Suggestion:

Here, the candidate could have talked about the currency value to be used across the loyalty program. Ideally, the value of one point gained at Titan would be same as the value of one point at Starbucks. The number of points to be awarded should be same across the brand, although the number of loyalty points earned can be different for a fixed amount of bill at different stores. For example, Titan may give 2 points per Rs 1000 spent and Starbucks may give 5 points per Rs 1000 spent in that brand.

GO TO MARKET STRATEGY 9

Interviewer - WinZO has seen a great success in the Indian market, and now is looking to expand to other geographical areas. After thorough research, we have found that Brazil would be a suitable market to expand. Can you chalk out a Go-to-market strategy for the same?

Candidate:
(Ask clarifying questions about the product, what it does, what problem it solves and how, target market and the goal of the launch)

Can you tell me more about the product and its features? Also, do you want to create a new app with new games for this requirement, or would this just be a new version of the existing app?

Interviewer - WinZO is a social gaming platform with over 100 games. Currently it has over 20 Cr registered players who compete against each other for real money. About the second part of the question - I was hoping you would help me with the same.

Candidate - Alright. Before I move on, I want to understand two things:

1. You have described WinZO as a gaming "platform". Does it mean that not all games are owned by WinZO?
2. Are all the games paid on the app? And how are the games priced?

Interviewer - You are right - the games on the app are a mixture of owned and hosted games. WinZO collaborates with game developers to host their games on the app either on a fixed fee or a revenue sharing model.

Most of the games on the app are paid. Users can use the practice mode for free if they want to. Pricing of the games is different across different formats of the games. There are single player games, two-player live games, tournaments, etc.

However, the ticket value for each game is small (ranging from INR 2, 5, 10, 20, 50 to 100), considering our target audience which is typically from Tier-2,3,4 cities.

Candidate - Thank you for providing the details. I would first analyse the market situation - taking the company, product, customer, collaborator and competitors into consideration. Then, I shall explore the pricing and launch strategy.

Interviewer - In the interest of time, I would like you to focus on the product, customers and collaborators. We can assume that the pricing strategy would be similar to that in India for different formats.

Candidate - Alright. When it comes to the product and customers, I feel that there is a need to look at the games, which are the key value proposition offered to the customers.

There will be a significant difference between games liked by people in Brazil. WinZO has games like Ludo and Snakes & Ladders in India which are traditionally relevant. These may not be performing in Brazil as much as they do in India.

For the other casual games, I feel Brazil as a country is more inclined towards games related to football compared to India which has more cricket fans.

Interviewer - That's right. So, how do you go about choosing which games to add in the platform? What all factors would you consider?

Candidate - For choosing the games, I would be looking at these factors:

- *Player preferences:* Looking at the general trends, popular game genres like sports, FPS, MOBA, etc.
- *Cultural Context:* Games that align with cultural elements or themes of the country might have a greater appeal (Like Ludo in India).
- *Existing Platforms:* Looking at the other competitors, the kind of games they offer and top grossing games.
- *Collaborators:* Looking at possible partnerships with games and game developers to host them onto the platform, along with top grossing games in Apple Store, Google Play Store, etc.
- *Tech & Infra:* Device preferences of the users (mobile vs tabs vs console), internet stability etc. will help decide to host cloud-based lightweight games, or client-based games.
- *Integration:* When we consider possible partnerships, we should also look at technological feasibility of integration (Unity, Android, etc.)

This should be giving a good starting point to decide on the games. After the launch, we can also take inputs from the users about the games they would like to see on the platform.

Interviewer - This looks good. What all metrics do you think would be important to track when we decide to release the application?

Candidate - We should definitely be tracking the progress of the app after the release. Few useful metrics could be:

Acquisition - Installs, Conversion Rate, Cost of Customer Acquisition.

Engagement – # of games played, time spent in the app, DAU, MAU

Customer satisfaction – feedback in the form of ratings and reviews, NPS scores.

Revenue – Amount of real money spent, Amount of money withdrawn, CLTV, etc.

I will use the feedback information and engagement information to strengthen the product further.

Interviewer - Sounds good. Thank you for your time.

INTRO: PRODUCT METRICS

As a Product Manager, your role revolves around metrics in one way or another. Metrics are the numbers that describe what is going on with your products. They are also called KPIs, i.e., Key Performance Indicators. However, metrics, KPIs, and OKRs (Objective and Key Results) differ from each other.

Metric: Metrics are the quantitative measurement used for assessment of any product's performance
KPI: A specific type of metric that lets you know quantitatively how successful a process is at achieving the desired outcome.
OKR: An OKR is a metric that defines a big-picture objective for the team or the company as a whole, along with a limited set of critical results used to measure that objective's success. It is similar to a KPI but differs in that OKRs are typically more aggressive and represent larger-scale initiatives.

While prioritizing the metrics, we should be aware of the vanity metrics that boost the team's motivation. They can even serve as very indirect indicators that your products have an increased potential for success. For example, the number of views on a Youtube Donation campaign video is good to follow. However, there can be millions of views but 0 donations. Hence, the number of views is the vanity metric.

The most important metric to focus on is the *Product North Star metric*. A North Star metric must fulfil three criteria: *lead to revenue, reflect customer value, and measure progress*. The north star of some of the famous companies are Spotify: Time spent listening, Amazon: Number of purchases per month, etc. Several frameworks can be referred to define the requisite metrics for a product. Candidates can choose to focus on any of the frameworks discussed in the *PM Concepts* section, like AARRR, HEART, etc.

Now, Let's dive into the success metrics for various types of products:

1. **General Product Metrics**: These are not specific to any product, but the common metrics we should be familiar with such as:
a.*Net Promoter Score*: We must have seen the question, "How likely are you to recommend the product/service to others, on a scale of 1-10 ?". This is a measure of customer loyalty and satisfaction. High NPS means healthy customer relations. NPS ranges between -100 to 100. NPS= %Promoters - % Detractors.

b.*Churn Rate:* The Number of customers who stop using the product in a particular duration. This is important for subscription- based products

c.*Cost per Acquisition*: The cost of acquiring one customer. It is generally compared with the customer lifetime value

d.*Life-Time Value*: the value of a customer over their lifetime. All customer-facing products should measure this to understand the value derived over the lifetime of a customer

2. **Mobile-based applications**:

a.*Downloads*, Activations, and Uninstalls: Measure the Number of downloads, activations, or logins and the uninstalls per month. Activations depend on the product

b.*Daily/Monthly Active Users*: Number of users using the app daily or monthly.

c.*Session length and interval*: Session is the duration between app open and close. These give the measure of engagements.

d.*Ratings:* The star rating of the app on AppStore or play store.

e.*In-app purchases*: Revenue metric

3. **API Products**:

a.*API calls and actions:* The Number of times the API is called (request or response) in a month. Actions are the specific task done through the API call, such as booking, purchase, etc. It gives the effectiveness of the API

b.*Failure Rate*: Number of times the API was called, but no response was sent. This can be due to missing arguments in the calls or any internal issue

c.*Uptime and Downtime*: Duration for which the API was working and not working.

4**. E-commerce websites:**

a.*Order and Average Order Value*: Number of orders and average monetary value of the order. This is helpful for future projections and updating the investors.

b.*Shopping Cart & Checkout Abandonment:* Number of customers who add items to the basket and do not checkout, and Number of customers who do not complete the purchase but reach the checkout. If these are high, experiments can be run to reduce the value.

c.*Visit Sale conversion*: This is the % of all visitors who make a sale. This can be related to the shopping cart and checkout abandonment metrics.

5**.Content-based websites:**

a.*Click-Through Rate, CTR:* It measures the effectiveness of advertisements. It is calculated as: Number of clicks on the advertisements / Number of times the ad was shown.

b.*Cost per Thousand Impressions (CPM)*: It is the price of 1000 advertisement impressions on a web page, also known as cost per mile.

c.*Bounce Rate*: It is the % of users landing on a webpage, clicking on nothing, and leaving. It helps in measuring the performance of specific landing pages

d.*Email Subscriber Rate*: It is the % of total visitors who subscribe to email newsletters. It indicates the effectiveness and quality of the content

e.*Percentage of shares*: This is the percentage of visitors who share at least one article vs. the total visitors. It indicates the viral growth of the pages.

Again, metrics can be divided based on their usability.

1. Product Usage Metrics
2. Revenue Metrics
3. Engagement Metrics
4. Conversion Metrics
5. Operational Metrics

PRODUCT USAGE METRICS:

1. **Daily Active Users (DAU)**
2. **Monthly Active Users (MAU)**
3. **User Retention Rate**
4. **Churn Rate**
5. **Session Length:** The amount of time a user spends on your product in a single session. This metric indicates how long users are engaged with your product each time they use it. Longer sessions often mean that users are finding your product useful or entertaining.
6. **Time to First Use:** The time it takes for a new user to start using your product after signing up or downloading it. This metric shows how quickly new users can get started with your product. Shorter times are better, as they indicate an easy and intuitive onboarding process.
7. **Feature Adoption Rate:** The percentage of users who use a specific feature of your product within a certain period. This metric helps you understand how popular or useful a new feature is. A high adoption rate means users are finding the feature valuable and are incorporating it into their regular use of the product.

REVENUE METRICS:

1. **Customer Acquisition Cost (CAC)**
2. **Customer Lifetime Value (CLTV)**
3. **Average Revenue Per User (ARPU):** The average amount of money earned from each user over a specific period, on an average.
3. **Monthly Recurring Revenue (MRR):** The predictable revenue your company expects to earn every month from subscriptions or recurring payments. This metric shows the steady income you get from your subscribers each month. It's crucial for subscription-based businesses.
4. **Annual Recurring Revenue (ARR):** The predictable revenue your company expects to earn every year from subscriptions or recurring payments. This metric is similar to MRR but on a yearly basis. It helps in long-term financial planning and forecasting.

ENGAGEMENT METRICS:

1. **Net Promoter Score (NPS)**
2. **Customer Satisfaction Score (CSAT):** A metric that measures how satisfied customers are with a specific interaction or overall service, usually on a scale from 1 to 5. Higher CSAT indicates higher satisfaction.
3. **Product Stickiness (DAU/MAU Ratio):** This metric shows how often users come back to your product within a month. Higher ratios indicate that users find your product essential in their daily lives.
4. **Feature Usage Frequency:** The rate at which users interact with a specific feature within your product. Higher frequencies indicate that the feature is popular and useful to users.

CONVERSION METRICS:

1. **Conversion Rate (from trial to paid):** The percentage of users who move from a free trial to a paid subscription. This metric shows how effective your free trial is at convincing users to pay for your product.
2. **Cart Abandonment Rate**
3. **Lead-to-Customer Conversion Rate:** The percentage of leads (potential customers) who become actual paying customers. This metric shows how effective your sales process is at converting interest into actual sales.

OPERATIONAL METRICS:

1. **Time to Market:** The amount of time it takes to develop a product from the initial idea to its release in the market. This metric shows how quickly your team can bring a new product to customers. Shorter times mean faster development & quicker delivery of new features.
2. **Number of Bugs/Defects:** The total count of errors or issues found in the product. This metric indicates the quality of your product. Fewer bugs or defects mean a more stable and reliable product.
3. **Sprint Velocity:** The amount of work a team completes in a single sprint, usually measured in story points or tasks. This metric shows how productive your development team is during each sprint. Higher velocities indicate a more efficient team.
4. **Cycle Time:** The amount of time it takes to complete one work item, from start to finish. This metric shows how long it takes to move a task from the beginning to the end. Shorter cycle times indicate faster completion of work items.

TIPS TO ANSWER METRICS QUESTIONS

1. Understand the Context

- **Clarify the Question:** Make sure you understand what is being asked. Is the focus on a specific feature, the overall product, or a new initiative?
- **Identify Stakeholders:** Determine which stakeholders are relevant (e.g., users, the company, partners).

2. Define the Goals

- **North Star Metric (NSM):** Identify the core metric that represents the primary value your product or feature delivers. This should align with the long-term success of the product.
- **Success Metrics:** Define additional metrics that measure specific aspects of success related to the goal. These should be SMART (Specific, Measurable, Attainable, Relevant, Time-bound).

3. Consider the Customer Funnel/Journey

- **Awareness:** Metrics related to how users find out about the product (e.g., brand awareness, reach).
- **Acquisition:** Metrics that track how users start using the product (e.g., sign-ups, downloads).
- **Activation:** Metrics that measure initial user interactions (e.g., first use, onboarding completion).
- **Retention:** Metrics that show how well users are staying engaged over time (e.g., DAU, MAU, user retention rate).
- **Revenue:** Metrics that track monetization (e.g., ARPU, MRR, conversion rate).
- **Referral:** Metrics that measure how users promote the product (e.g., NPS, referral rate).

4. Select and Explain Metrics

- **Choose Relevant Metrics:** Based on the goals and customer journey, select the most relevant metrics to track.
- **Explain Each Metric:** Describe what each metric measures and why it is important for evaluating the product's success.

5. Provide Examples

- **Use Real or Hypothetical Data:** Illustrate how these metrics can be tracked and analyzed using real or hypothetical data.
- **Analyze Trends and Insights:** Explain how you would interpret the data to gain insights. What trends would indicate success or potential issues?

6. Discuss Actionable Steps

- **Optimization:** Discuss how you would use the metrics to optimize the product. What actions would you take based on the data?
- **Iteration**: Explain how you would iterate on the product or feature to improve the metrics over time.

7. Highlight Tools and Techniques

- **Mention Tools:** Discuss any tools or techniques you would use to gather and analyze the metrics (e.g., Google Analytics, Mixpanel, SQL).
- **Data Collection Methods:** Explain how you would ensure accurate and reliable data collection.

8. Conclude with Impact

- **Link to Business Objectives:** Tie your analysis back to broader business objectives. How do these metrics support the overall goals of the company?
- **Long-term Vision:** Discuss how tracking and improving these metrics aligns with the long-term vision for the product or feature.

After the explanation of customer funnel and every metric, you can choose to discuss the evaluation and prioritization of metrics if time permits. It is suggested to ask the interviewer regarding this before explaining these steps. But these prioritisation hints more clarity to the interviewer.

METRICS 1

Interviewer: You are a product manager at Instagram. How will you measure the success of Instagram shops?

Candidate: I would like to start by shedding light on my understanding of Instagram shops to check whether it's accurate. Please correct any inaccuracies that I might state. Instagram shops is a new shopping experience feature which allows businesses to setup online shops with product listings on Instagram. The buyers can visit the profiles of these businesses which include brands or stores and view the products on sale. They can even place the orders and the purchase the product through Instagram if the business allows it. Since Instagram is one of the highest used social media platforms, Instagram shops allows businesses to showcase their products without requiring buyers to visit their website for product discovery.

Interviewer: The description is accurate. Please proceed ahead with the rest of the solution.

Candidate: So, in summary Instagram shops serves a platform for buyers and sellers to interact. Buyers can now directly view the product offerings of their favourite brands and stores on Instagram itself. I observe that Instagram Shops is a two-way platform and both buyers & sellers are stakeholders here.

For defining the metrics, I would like to proceed with defining the journey of both the stakeholders on the app.

Interviewer: That should be fine. Please go with the journey

Candidate: The seller follows the below steps to set up shop on Instagram:

- Checking the eligibility: It checks whether- the Business is in a supported market, Product is eligible to be sold, Business is compliant will all terms & conditions and Business has a website.
- Creating a Business or Creator account: It would involve adding important info like website, email, phone number, address.
- Linking to a Facebook page: It is helpful in setting up the shop on Facebook marketplace as well
- Importing a product catalogue: The product offerings need to be listed on Instagram and the items need to be added to the catalogue
- Activating shopping: If the seller wants the entire checkout process to go through on Instagram shops, they can activate this feature. Otherwise, the buyer needs to select the product and checkout on the seller's website to complete the purchase.
- Regularly updating the catalogue items, prices, and quantity available

For the buyer, the steps involved would be:

- Searching for the brand or store on Instagram or viewing an ad of their brand or store
- Visiting the page of the brand or store and clicking on 'View Shop
- Going through all the listed offerings
- Selecting the required product and adding it to cart
- Once all the products are added, begin the checkout process by adding personal details, address, and the payment information.
- The checkout is complete once the payment is successful, and the order is placed

Interviewer: That is correct. You have correctly identified the user journeys. You can proceed with defining the success metrics.

Candidate: I'll start by defining the metrics for the sellers at each stage of their user journey.

- Awareness - The seller needs to be aware about the existence of Instagram shops. E.g., Number of signups on Instagram shops every month and Percentage of businesses who have viewed marketing mails and ads on Instagram shops
- Engagement: This step gauges if the seller can use the feature and is able to setup the shops successfully. E.g., Percentage of sellers who setup shop after becoming aware of Instagram shops, Percentage of sellers going through each stage of the seller journey i.e., finding leakage at each step whether the setup process is left in between, and Percentage of sellers update their product catalogue every week, every month
- Conversion: This step sees if the seller successfully manages to sell products on Instagram shops • Percentage of sellers using Instagram for checkout. (Sellers can check out through Instagram or redirect buyers to their website)

Instagram shops is currently free to use. However, there are plans to monetize it based on a commission of every executed order.

The revenue can be calculated as Revenue = No. of shops * Avg order value * Avg. no. of orders per shop

The revenue metrics can be:

• No. of shops on Instagram
• Average order value of every order
• Average no. of orders per shop

Interviewer: That looks good. Please proceed with defining the success metrics for the buyers

Candidate: I'll follow a similar approach for buyers by following the steps of their user journey

- Awareness: This step checks whether the buyer is aware about Instagram stores, E.g., Percentage of users visiting the shop after visiting the Instagram profile of a business, Percentage of users clicking on ads about the Instagram shop of a brand.
- Engagement: Average time spent on viewing the products in a shop, Average no. of products checked out in detail in the shop.
- Conversion: Percentage of buyers adding products to the cart among all shop visitors, Percentage of buyers completing the order process at each stage i.e., personal information, address & payments (Leakage in the checkout process)

I believe this covers important metrics of Instagram shops for both buyers and sellers.

METRICS 2

Interviewer: You are a product manager in the Google Pay team. You have been given the task of defining the metrics for tie-in promotions on Google Pay. How will you go about it?

Candidate: I believe I can start by describing how tie-in promotions work on Google Pay to get my context right. Please correct if my description is inaccurate at any point. Google Pay is used for making payments. People use it for transferring money to their friends or making online payments mostly via UPI or cards. Once the user makes a payment, they get a scratch card as a reward. On scratching the card, users get rewards which can be cashback or coupons of various companies. I believe these coupons of various companies like Lenskart, Mamaearth, Zomato & others are tie-in promotions.

Interviewer: Your description of tie-in promotions is accurate. Essentially tie-in promotions involve a company partnership with another company for promotions. You can proceed further with your analysis.

Candidate: The metrics need to be aligned with the business value. The tie-in promotions are an important factor for users to repeat payments using Google Pay. Users will keep using Google Pay if they get relevant offers on some occasions. Similarly, promoters will partner with Google Pay if they benefit out of the partnership by gaining more customers & revenue.

Interviewer: The business goals seem accurate. Please proceed with the metrics.

Candidate: I will cover the metrics from aspects of both users and promotional partners.

From the user's perspective, the metrics will be covering the frequency of promotions, its utilisation, and its impact on user engagement on the platform. We would ideally expect the promotions to be useful for the users.

No. of successful uses of promotions = No. of payments * Percentage of tie-in promotion scratch cards * Percentage of tie-in promotions used.

The metrics are:

- No. of payments/ Payment rates every month
- Percentage of scratch cards which are tie-in promotions (Scratch cards be cashback, promotions, or better luck next time)
- Percentage of promotions getting expired
- Average of no. of different brands/types of promotions given to the user
- Percentage of repeat promotions of a specific type or brand.

From a promotion partner's perspective, the metrics will cover how Google Pay and promotion partners can benefit from the exercise.

No. of successful use of a specific promotion = No. of payments * Percentage of promotional coupons for a given partner * Percentage of a specific coupon * Percentage of users successfully using the promotional coupon.

The metrics are:

- No. of tie-in promotion partners
- Retention rate of promotion partners
- Churn rate of promotion partners
- Average number of promotional coupons shown for every partner

Percentage of promotions which get expired without use, both overall and for a given partner Average number of recipients of a given promotional coupon
Percentage of users successfully using a given promotional coupon (this needs to be shared by the partner)

Overall, these metrics will help track the success of tie-in promotions and help identify some issues. For example, if a promotional coupon by Lenskart has very low conversion rates, it can be discontinued or be modified to increase the usage.

Interviewer: I think this covers most of the important metrics for tie-in promotions.

METRICS 3

Interviewer: You are the PM for Facebook Marketplace. What metrics would you track to measure the health and success of the feature?

Interviewee: I would first like to clarify my understanding of the feature and the stakeholders involved. Facebook marketplace is a platform within Facebook to discover, buy and sell items with people in your community. The functionality caters to two stakeholders – the buyer and the seller. The feature allows sellers to list both new and secondhand items they want to sell that would then show up on the marketplace tab. Buyers can then explore the listed products in both the marketplace tab and groups in which product details are shared.

I'll assume that Facebook's goal with the feature is to increase customer engagement on the platform.

Interviewer: You are right. Please go ahead.

Interviewee: Before I list down the metrics, I would like to map out the journey of both the stakeholders to understand the goal of each action the feature asks the user to take.

Seller's Journey:
1.Visit the marketplace tab
2.Select the option to list an item
3.Add item description and price
4.Receive an offer to buy from the buyer
5.Accept or reject the offer
6.Receive the payment
7.Handover the item
The last two steps happen outside Facebook's ecosystem

Buyer's Journey
1.Visit the marketplace
2.Search for a product
3.Contact the seller
4.Make the payment
5.Receive the product
An alternate customer journey would start with buyers discovering a product they like in groups

Interviewer: Your understanding of the journey of the stakeholders is fair. Let's move ahead to the metrics

Interviewee: I would divide the metrics into Awareness, Activation, Conversion and Retention metrics.

1.Awareness: The goal would be to measure whether users can discover Facebook Marketplace. #Users who have visited Facebook Marketplace, #Clicks on Facebook Marketplace tab per day

2.Activation: The goal of the phase is to measure the level of active engagement with the feature. #Products listed per week, #Bounce Rate. #Clicks on product listings per week

3.Conversion: The goal of this phase is to measure how effective is the feature in getting users to perform the primary action, i.e. buying a product. #Unique Interactions per week, #Purchases made per week

4.Retention: Retention indicates the users' propensity to come back to the platform and use the feature again. #Repeat Purchase Rate, #Repeat Product Listing Rate

The Activation and Conversion metrics should be tracked for individual categories to understand differences in buying and selling process for different product categories. With the goal of increasing customer engagement in mind, the North Star Metric would be - #Unique Interactions per week. #Purchases per week would be used as the counter metric to ensure that product quality on the platform is not compromised for increasing user engagement

METRICS 4

Interviewer: There is a decline of about 10% in cart additions (Add to Cart) in the last 3 days in Myntra. Can you help us investigate the problem?

COMPREHEND THE SITUATION

Candidate: Sure, to make sure we are on the same page, I would clarify the problem statement. The cart addition refers to adding an item through the 'Add to Bag' button on Myntra and I have to find the root cause behind the decline in the cart additions.

Interviewer: Yes, that is correct.

Candidate: Myntra has both mobile application (android and IOS) and website. Are we looking at a generic 15% decline on all platforms or is it specific to any one platform i.e. app or website?

Interviewer: Good question, we are noticing the decline in the Myntra android app.

Candidate: Is the decline seen across all categories of products or in any specific category of product?

Interviewer: The decline is seen throughout all the categories.

Candidate: Okay. I will take just 2 mins to prepare the root cause analysis framework and explain it further.

Interviewer: Sure, take your time.

Framework

Candidate: Okay, so I am assuming that all the analytical tools are working fine and we do not need to check the tool that are providing us with the data.

Interviewer: Yes, you can assume this. We have checked the logs multiple times.

Candidate: Thank You. I will divide the framework in 2 parts: External factors and internal factors.

1. External factors: What competitors have done in the last 3 days. This can include any new product launch or new announcement by the competitor. This can take away some of the existing customers of the Myntra or their purchase decision can be delayed. We can also look at macro-economic changes: is it the seasonality impact or has been there any particular news in the market?

2. Internal factors: Is there any new announcement or campaigns run by the Myntra? Has Myntra made any design changes in the app or released any updates in the android app?

Apart from these factors, we can look at the customer journey to see whether any step before the cart addition has seen any glitch or bugs. If none of this is the reason, then probably we can look at some of the factors that impact the purchase decision of the consumer.

Does the approach look good or do you want me to make any changes?

Interviewer: Yes, that looks great. Please continue.

Candidate: Has there been any sale or product announcement by the online of offline competitors of Myntra?

Interviewer: No, nothing unusual. To the best of our knowledge, they are running their regular campaigns.
Candidate: Is it safe to assume that there have been no macroeconomic changes that can influence customer behavior? Personally, I did not come across any such news.
Interviewer: Yes, it is safe to assume.
Candidate: Okay, so I will move on to the internal factors. Firstly, has there been any new announcement or campaigns run by the Myntra? The information can help me identify whether customers are waiting for any special event to shop or has there been decline in any one product category and increase in another.
Interviewer: The decline is spread over all the categories and not specific to any product or gender. Also, we are running regular campaigns that we run throughout the year.
Candidate: Interesting. My next question would be whether Myntra app has released any new updates?
Interviewer: Yes, we have released few updates to improve the user experience of the customer.
Candidate: Are these updates related to the customer journey of the user and does it involve changes in Add to Bag feature? **Interviewer**: Yes, it involves design changes related to 'Add to Bag' feature. We want you to analyze this thing further. What other factors do you think can be there that can lead to decline?
Candidate: Okay. The first thing I can think of is whether this update released for the android app was a force update or not. If it was not a force update, then some consumers might be using the older version and the backend services might not support the older version. Second, we can look at the success rate of the 'Add to Bag' API in the last 3 days to analyze if the API is working fine or not. Thirdly, I hope consumers are aware of the new changes in the app and they might not consider the changes fishy and abort the transaction. Interviewer: Actually, when the consumer clicks on the 'Add to bag' button, that action is not being captured by the API.
Looks like there was a mistake in the deployment. Thank You for resolving the issue.
Candidate: Do you want me to analyze it further by going through the customer journey?

Interviewer: No, I think that will be all.

METRICS 5

Interviewer: You are the Digital Product Manager of Big Basket. What would be the metrics you would like to follow?
Interviewee: Okay, first I want to confirm that my understanding of Big Basket is correct.
Big Basket is an Indian food and grocery store eretail store. A user can order edible products and get them delivered at their doorstep. They have also cataloged few other products like aluminium foil and mugs. Moreover, the users can use both a mobile app and website for placing an order.
Interviewer: Yes, you may move forward.
Interviewee: Could you please share the business goal which we would like to report through the metrics.
Interviewer: Assume that the goal of the business is to increase the value of orders placed.
Interviewee: I will create a user journey map, identify which actions indicate success and failure towards placing order and then define the actionable lagging indicators. Should I do it for both, website and mobile app?
Interviewer: You can define the metrics for only the website for now.
Interviewee: Sure. The value of orders can be formulated as follows:
Total Value = # of order * average value per order # of order = # of visitors * % of Viewed/added items to cart * % bought the items # of visitors = # of new visitors + # of retained visitors.

- The user would go through the following journey
- The user either lands the home page through url or the product page through a search engine.
- The user then browses through the website either through navigation or through the search box.
- The user adds the products he wants to purchase to the basket.
- The user can also save the product. For saving, Big Basket will ask the user to login or register.
- The user can create custom lists and add the items to the list for repeated purchase. For ordering the products from the basket also, Big Basket will ask the user to login or register.
- The user can review the items we want to buy and then click on checkout.
- Following this, the user is navigated to the checkout screen. Here, the user can select or add the delivery address.
- He/She can also select an appropriate delivery slot and apply for a voucher.
- Users can also opt to buy the BBstar membership if she/he plans to order frequently with high value carts.

The user is then navigated to his preferred payment gateway for placing the order. After placing the order, the user is navigated to the confirmation screen. Confirmatory email and message is also delivered. The user can connect with the customer care any time to make a change in order or delivery time slot or requesting cancellation.

The user receives the confirmation after the order is delivered at the doorstep.

In this journey, the user goes through 5 stages: **Awareness**: The users can visit Big Basket either through url or search engine. Through search engines also, the user can navigate through organic search results or inorganic paid results. Hence we can have following metrics:

- No. of visits per month
- Percentage of visits through organic results per month
- Percentage of visits through inorganic results per month
- Rank on search engine results

Engagement: The user browses through the website to find products. The following can be measured to improve user engagement:

- Bounce rate per product or category page per month
- Average no. of pages visited per session per month
- Average time of a session per month
- Average quantity per product added to
- the basket per month
- Number of saved lists per month
- Percentage of users using referrals per month
- Percentage of logged in users per month
- Percentage of new registrations per month
- No. of complaints per month
- No. of orders per user per month
- Value of returns per month
- Daily active users
- Monthly active users

Conversion: After adding the products to the basket, the user reviews the basket and checks out. The following can be measured to improve conversion:

- Average number of items in the basket per month
- Average value of items in the basket per month
- Basket abandonment rate per month
- Order cancellation rate per month
- Average order value per user per month
- Percentage of users who select similar products during checkout per month
- Percentage of users applying vouchers per month
- Value of vouchers applied per month

Retention: The user should again opt for Big Basket for further purchases within 2 weeks. The following can be measured to improve retention:

- Percentage of users opting for BBstar membership per month
- Percentage of users buying again per month

Referral: The users can also be a marketer for our product through word of mouth and sharing referral code.

- Percentage of users sharing referrals per month

Interviewer: This is great. Thank you!

METRICS 6

Interviewer: How will you measure the success of WhatsApp's in-app video playing feature? Describe the feature & its goal

Candidate: First, I would like to describe the feature and my understanding of its objective.

WhatsApp has introduced an in-app video playing feature, whereby users can play YouTube videos from within the chat window itself. I think the objective of the feature is to keep the users within the app context, so that the time spent on WhatsApp can be increased. There are three types of interactions that can be done when this feature is in use:

1.Expand the video to full screen

2.Close the video

3.Watch the video on YouTube by clicking on the YouTube icon

Interviewer: That sounds good. Please proceed.

WALK THROUGH THE CUSTOMER JOURNEY

Candidate: The user's journey for using this feature would look like this:

- Activation: The user receives a link to a YouTube video on WhatsApp
- Engagement: They click on the link, and the video starts playing within the app
- Engagement: They can expand the video to full screen, or stop viewing by clicking on the close button.
- Alternately, they can click on the YouTube button to be taken to the YouTube app. If they chose this option, they can either continue browsing on YouTube (Leakage), or return to WhatsApp.

QUANTIFY BEHAVIOUR IN THE CUSTOMER JOURNEY

CANDIDATE: SINCE THIS FEATURE WOULD BE AVAILABLE ONLY ON PHONES WITH AN ANDROID VERSION THAT SUPPORTS IN-APP WINDOWS, AND AN UPDATE VERSION OF THE APP, ALL THE METRICS SHOULD ACCOUNT FOR THAT.

Activation

- % of users that play a link they are sent
- Relative activation rate: % of users with this feature available that play a video v/s % of users without the feature who play a video(which opens the YouTube app instead)

Engagement:

- % of users who played the link and watched the video (either partly or wholly)
- Relative return rate: % of users who stayed in WhatsApp after they stopped watching the video v/s of users without the feature who return to WhatsApp after watching the video
- Time spent on WhatsApp by users with this feature v/s without this feature

Leakage:

- % of users with the feature opting to view the video on YouTube
- % of users who opted to view the video on YouTube that returned to WhatsApp

EVALUATE THE METRICS

Candidate: The engagement metrics will give a picture of whether the feature has boosted time spent on WhatsApp and whether it enables users to engage with YouTube content more frequently. Ease of measurement: The metrics can be easily tracked by logging actions of the users.

WRAP UP

Candidate: To summarize, the in-app video playing feature is an effort by WhatsApp to boost time spent by users within WhatsApp as well as enable more media consumption for users. Primarily, the relative activation and retention rates will help evaluate these value propositions of the feature.

METRICS 7

Interviewer: How do you measure the success of the Instagram story highlights features?

UNDERSTAND THE FEATURE AND ITS OBJECTIVE

Comment: The first step for deciding product metrics is to understand the product objectives well.

Candidate: I would like to clarify my understanding of the Instagram story highlight feature and its objective. So, basically, it is a feature to describe a particular account in crisp selected stories. The intention of the highlights could be multiple. For instance, if we talk of a photography page, the highlight might include its best photo works to indicate the level of quality to its clients. If the Instagram page is about a tech product, its highlight may include how to use the product in stepwise, or maybe the answer to the FAQs about the product. So, the objective of the feature is to reduce viewer's time to understand/know about the Instagram profile/product in a short span of time and thus increase the conversion rate for these accounts. Conversion might mean different things for different accounts, for instance, for the personal account it could mean increased followers/likes, for business accounts, it could mean increased product sales etc. Also, Instagram has introduced a story archive feature which is the backbone of highlights feature. So, even after 24 hours of posting a story, you can get those stories in the archive.

Interviewer: Bang on. You are right, please go ahead.

Candidate: Thanks. So, users can make any old stories a highlight, which will be present on the Instagram account until and unless it is removed manually by the account owner. It just acts like key highlighted posts for the account. These highlights are like small circular icons, similar to story icons present on top of all of the posts on the account page. So, there are two stakeholders in this system:

- **Account owner:** The account owner follows the below steps:
  - Click on the new button on account page
  - Select the Instagram stories to be highlighted
  - Confirm the stories to make them the highlight
- **Account visitor**: Once a user visited the Instagram account, they can see all the highlighted story in circular icons on the account page. Then they can click on the highlight of their choice and they will be able to get the story like any other Instagram story.

Interviewer: Yes correct, you have got the product correctly. Now tell me, how will we measure its success.

Candidate: Let's go through the customer journey in phases. I have already explained the detailed steps for both the account owner and visitor. Now, let's get into the customer journey stages.

Awareness: The account owner should be made aware that such a feature exists. It is currently being done at two places, on the account page and on the story page.

- Average # of times highlight button clicks per story on the story page
- Average # of times highlight button clicks on the account page per account page visit per user

Engagement: The engagement here indicates whether the account owner is able to engage with this feature or not, basically the highlights they created. For account visitors, the engagement depends on various other factors like time spent by visitors on other's account seeing these highlights and so on.

- Percentage of number highlights created to the number of stories posted per user per month.
- Percentage of number highlights created, from account page to the number of stories posted per user per month.
- Percentage of number highlights created, from story page to the number of stories posted per user per month.
- Average time spent on watching other account's
- highlights per user per month.

Conversion: The whole purpose of the feature is to improve the conversion for users. So, once a user has engaged with other's highlights, he/she should be able to quickly understand about the other's account and make decisions whether to follow or not, whether to buy their product or not etc.

- Average time spent on another account's page per user per month.
- Percentage of highlight viewers following other's page per user per month.
- Percentage of highlight viewers ordering services from other's account per user per month
- Average time spend on other's account before following them per user per month
- Average time spent on other's account before placing the order for their products/services per user per month.

Comment: Always try to quantify the metrics like what is the time range we are talking about, is it per month, is it per day etc. Also, each stage of the product has different objectives and thus the metrics should be aligned with that. Thus, we must be first clear what we expect from a particular stage, for instance, what is expected out of Awareness stage, or conversion stage and then align your metrics to their objectives.

INTRO TO PRICING

The pricing strategy for a product would be one of the following three strategies:

1.**Cost Based**: Cost based pricing is simply charging the customer the direct and indirect product cost along with a certain markup. This approach is generally used when the product is a commodity and can't be differentiated easily. Even in cases when other strategies are more relevant, it is good to do a sanity check if the costs are getting covered and the profit margin is in line with expectations.

2.**Competitor Based**: This technique involves considering competitors' price as reference, and adding/subtracting a premium based on additional features (if any) and willingness to pay for them. Examples of this are online airline prices and prices charged by food delivery platforms.

3.**Value Based**: Unlike the previous two approaches, this involves pricing a product basis the value it generates for the customer. The value is often the differentiation it offers over its competitors, or the opportunity cost of not having the product. This value can be calculated by focusing on a single customer segment, comparing the next best alternative, understanding our product differentiation, and quantifying the value of this differentiation in monetary terms.

Before you delve into the pricing strategies, it is important to define the objective of the case. Objective sets the way forward towards deciding the correct approach to arrive at the right price. The objective can be, for example, attaining a 10% market share within 3 years of product launch or combatting the entry of a new competitor in the market. In an interview, the objective would be either told by the interviewer upfront (rare) or more often, arrived at by discussion between the interviewer and the interviewee.

Once the objective has been decided, we need to analyze the following, keeping in mind our target audience:

Product offerings: We must lay out the key offerings our product has which would be of interest to our customer segment.

Competition Landscape: We need to identify the major competitors which exist in our market. Competition is any player who tries to fulfill the same customer needs as our product.

- Customer's Willingness to Pay (WTP): For our target audience, we need to identify their willingness to pay for our product. This can be decided considering the nature of business, opportunity cost of having no such product (this could be value of time saved by using the product or current spending of customers on close alternatives), or urgency to fulfill the needs catered by our product.

- Position on Product Life Cycle (PLC) Curve: We must also consider where our product lies on the PLC curve. A product at the introductory stage usually attracts users who are open to experimentation and usually have a higher WTP. This is also the stage where the price reference for the future is established. In growth and maturity, the competition starts increasing and the customer WTP starts declining. At this stage, our product needs to take either a low cost-based position or a premium product offering-based position, and each would have a separate target audience. At the decline stage, a prudent approach is to simply extract whatever revenue is possible and shut down the business or move into a new arena.

In majority of cases, using one of these three techniques (Cost, Competitor, or Value-based) directly should be sufficient in an interview. But sometimes, the interviewer expects a more comprehensive approach for arriving at the pricing technique. You could use the models and techniques shown next:

- Upfront Pricing (Perpetual license): In this case, the customer pays upfront for the product and gets lifetime access to the same. While the license is perpetual, product updates and new feature releases might not always be offered after a certain cut-off date. This pricing technique is more common in B2C segment than the B2B segment, as the latter generally values continuous product support, security patches, and product updates more than the B2C segment.

- Subscription Based: In subscription-based pricing, the customer pays a fixed amount basis their usage, usage being generally defined by time duration (annual/monthly), computation, or storage (followed by cloud service providers). More and more firms are migrating from upfront to subscription-based pricing models due to its clear advantages like recurrent revenue stream and reduced upfront cost for customers.

- Gain Sharing: This is still an evolving pricing strategy and might be considered in B2B space. In this strategy, the revenue is directly linked to the gains made by the customer due to our product. This raises the customer's confidence in our product and reduces their financial risk in case the product doesn't work. On the flip side, it is often difficult to accurately define the exact gains for the customer due to our product alone. This can often be a source of disagreement and is a major challenge towards the adaptation of this pricing technique.

Quite often, the pricing is a combination of some upfront price and then a recurrent subscription cost. The upfront price can be justified if the product needs customer-based customizations, installation cost etc. The recurrent cost can be attributed towards product maintenance and support.

The upfront and subscription-based pricing techniques can be further classified into:

Cost/Competitor/Value Based approach (already covered)

Fremium: In this case, the customer gets access to some basic set of features for free, and pays for complete access. The free part provides the user an opportunity to try out the product and decide whether it is suitable to their needs. Zoom, Grammarly, Spotify are all examples of fremium pricing model.

Flat Rate: In flat rate pricing, the entire product is priced at a flat price and offers all the features. The only variation introduced can be by offering monthly/annual subscription options. This is a simple to implement pricing technique but fails to extract varying customer surplus of different segments.

Usage Based: This is the pay-as- you-go approach, where the customer is billed basis their product usage, usage being defined in units of data/computations etc. This is used generally by cloud computing and storage providers, customer support services etc.

Feature Bundling: In this pricing approach, multiple product bundles are introduced, each priced differently and catering to different customer segments. The features and functionalities offered are customised for the target segment. Linkedin subscription is an example of the same. Some example of different bundling tiers is:

Free, Individual, Team
Starter, Professional, Company

While no pricing approach is incorrect, some approaches would be more appropriate than others in the contextual setting we are analysing. The rationale for the chosen approach needs to be justified using the analysed factors.

PRICING 1

Interviewer: JioSaavn is a music streaming service that wants to launch a family music plan for its users. How would you go about it?

Interviewee: Sure, from what I understand JioSaavn wants to launch a new subscription plan aimed at families. So I have to come up with a pricing strategy for this plan?

Interviewer: Yes, go ahead.

Interviewee: Okay, and I assume the goal of the pricing strategy is to attract more users. Especially, the users who are not earning or those who are taken care of by others in the family.

Interviewer: Fair enough. How do you go about pricing such a plan?

Interviewee: From what I know music streaming is a very competitive market in India and I would like to concentrate on competitor based pricing. However various players target different customer segments. To identify the most appropriate competitors to benchmark against, I would look at the customer segment that we would be targeting with this plan and identify the competitors targeting the same customer base. I assume JioSaavn's main user base is those listening to regional songs and this plan would be targeted towards the same user segment. (One can use a matrix to better represent the market)

Interviewer: Sure, go ahead.

Interviewee: I think Gaana and Wynk are the appropriate competitors. I would look at their pricing strategies and the offerings they provide to their customers.

Interviewer: Here is the pricing chart.

Streaming Service	JioSaavn	Gaana	Wynk
Pricing	Rs 99 for 1 month; Rs 999 for 1 year	Rs 99 for 1 month; Rs 999 for 1 year	Rs 99 for 1 month
Free features	Listen to tracks with ads, Lyrics	Listen to tracks with ads, Lyrics	Listen to tracks with ads
Paid features	Unlimited downloads for offline listening; 320kbps audio; No ads;	Unlimited ad-free tracks, Download songs offline, HD Audio;	Unlimited ad-free tracks, Download songs offline
Family plan	-	499 (upto 5 people)	-

Interviewee: Looking at the chart, I see that there is very little differentiation wrt to the offering from each streaming service. I think the appropriate pricing would be Rs 499 for 5 users or Rs 399 for 4 users. We can probably run experiments to see which of these offers would be more attractive to the users. There is another data point I would look at to ensure that we are not missing out on any opportunities. I would look at the number of subscribers to each of these streaming services, especially the number of paid users. If we are significantly lagging behind the competitors, I would consider pricing it more aggressively. If not, I would stick to the competitors' pricing range. The market is very competitive, and being aggressive about pricing might lead to a price war.

To summarize, I would price it the same as the competition but maybe run a few experiments to find the right value proposition. I would also take into consideration the market and JioSaavns position in the market. I would make sure the pricing strategy is aligned with the overall strategy of the company.

Interviewer: Great! This sounds good.

PRICING 2

Interviewer - With the launch of the iPhone 12 a few months away and keeping in consideration Apple's focus on growing their market share across the world. How do you think Apple should go about pricing the iPhone 12 lineup?

Candidate - Is there any particular manner would you like me to go ahead with, for example:
Competitor based pricing
Value based pricing
Cost based pricing

Interviewer - You are free to go ahead anyway you would like.

Candidate - Sure, in that case I would prefer going with the Competitor based pricing. It would be great if you could further explain the product portfolio for the iPhone 12 lineup?

Interviewer - Sure, there are going to be four models, namely- iPhone 12, iPhone 12 Max, iPhone 12 Pro and iPhone 12 Pro Max.

Candidate - Can you tell me the key differentiators between the models?

Interviewer - The screen sizes are going to be 5.4", 6.1" and 6.7" respectively with 120HZ XDR OLED display for the Pro models. The Pro models will have a triple camera setup whereas the non Pro models will only have dual cameras. Apart from this you're free to make any other assumptions.

Candidate - Who are the main competitors for the iPhone 12 and iPhone 12 Pro models?

Interviewer - You can consider the OnePlus 8/Huawei P30 Pro and Samsung S20+ to be the competitors respectively.

Candidate - How much do these retail for?

Interviewer - OnePlus 8 retails for $699, Huawei P30 Pro retails for $599 and Samsung S20+ retails for $899.

Candidate - Do I need to consider different storage variants or just a single base storage variant for all the models?

Interviewer - Just consider the base models with 128GB storage.

Candidate - Thanks for the background information, there are three different ways we can look at pricing

Customer's willingness to pay
Competitive pricing
Cost-based pricing

First I'll start looking at the product from the customer's perspective. The customer will look for the best alternative to our product or find a substitute product. Negotiators call it BATNA or the best alternative to a negotiated agreement. Here I would call it the customer's willingness to pay.

In the case of the iPhone 12, the next best alternatives are prices at $599 and $699. This gap of $100 is something Apple can think about playing around in. The perception for an OnePlus device is that it's a good product but the cost doesn't justify the high premium they are asking for their brand. When it comes to Huawei P30 Pro the brand image is not that of a premium smartphone but that of a Chinese manufacturer. If Apple prices the iPhone at a cost lower than that of the Huawei P30 Pro then that would give out the perception that the iPhone 12 is just a cheaper version of the superior Pro models. That would be detrimental to the iPhone 12 as a standalone phone. Hence it is logical to cost the iPhone 12 more than $599.Apple also need to consider the screen size of the iPhone 12. At 5.4" the iPhone 12 would be significantly smaller than the OnePlus 8. In mass consumer markets like India, the perception is that bigger the smartphone screen the more premium and desirable it is. Hence pricing the iPhone 12 at a cost higher than OnePlus 8 would hit the sales severely in such markets. Hence it is suggested that the iPhone 12 be priced between $599 and $699.

As for the iPhone 12 Pro model, its competitor the Samsung S20+ retails for $899. We also need to consider the bigger iPhone 12 Pro Max model. It would be better to have a Pro model iPhone which costs higher than Samsung's flagship model. There is always the bigger iPhone 12 Pro Max model to be the best that Apple offers which can command a premium over other competitors' flagship devices. Pricing Apple's flagship iPhone 12 Pro Max higher that all the other competitors will help in two ways:

Help maintain Apple's brand image of being the most premium smartphone and it's always a statement to carry an iPhone.
It would help skimming from the top and tap into the super-premium/luxury smartphone market.

Interviewer - So what are your recommendations?

Candidate - For the iPhone 12 model, I would suggest to cost it at $649. This would help undercut the mass popular OnePlus 8 while preserving the iPhone's brand name by not becoming a cheap phone or being compared to the Huawei P30 Pro. This can help in selling more quantity of iPhones and the revenue would be driven by the surge in demand for the apple accessories, app store licensing fees and also from other services like apple music and iCloud storage that iPhone provides.

The larger of the non Pro models, the iPhone 12 Max can be priced at $749 as a better alternative to the iPhone 12 and would also seem superior to the $699 priced OnePlus 8.

Coming to the Pro series of phones, here I would suggest that the iPhone needs to preserve its premium brand image. Even if Samsung comes up with a more expensive series of phones before the next cycle of Apple iPhones, the cost of iPhone 12 Pro should be at least on par with the flagship phones from other manufacturers. Hence I would suggest to price the iPhone 12 Pro model $100 over the price of the competitor. There is always a cheaper non Pro series for a customer with a smaller budget. The iPhone 12 Pro Max, being the flagship product offered by Apple, commands a premium for the larger screen over the standard pro model. Hence I would suggest to price it at $1099.

Interviewer - Can you summarize the pricing for me?

Candidate - The iPhone 12 should be priced at $649, the iPhone 12 Max should be priced at $749. The Pro models would charge a significant premium, partly because of the technology used in them and partly because of it being the flagship offering by Apple. Hence, the iPhone 12 Pro should be priced at $999 and the iPhone 12 Pro Max should be priced at $1099.

PRICING 3

Interviewer: Recently, Disney has partnered with Hotstar to provide its Disney+ streaming services in India. This poses a major threat to the user base of Netflix. As a product manager at Netflix, you need to formulate a competitive response in terms of the pricing and product strategy of Netflix in India.

Interviewee: So, I understand that the objective here is to preserve and grow the user base (market share) of Netflix in India. Is that right?

Interviewer: Yes, that sounds fair. Go ahead.

Interviewee: Alright, we can look at the pricing policy of Netflix through three lenses –

- Competitor based pricing
- Value based pricing
- Cost based pricing

I would look at Hotstar and Primevideo as the only two major competitors of Netflix. Can you tell me more about the current pricing policy of these platforms?

Interviewer: Following are the pricing models of the three players.

Hotstar			
Plan	**Monthly price**	**Annual subscription price**	**Benefits**
VIP	Not Available	399	Disney+ movies, sports, hotstar specials
Premium	299	1499	All of above + American shows + Disney originals

Prime Video			
Plan	**Monthly price**	**Annual subscription price**	**Benefits**
Amazon Prime	129	999	Free early amazon delivery, video, music subscription

Netflix			
Plan	**Monthly price**	**Annual subscription price**	**Benefits**
1 screen	500	Not available	-
2 screens	650	Not available	HD streaming
4 screens	800	Not available	UHD streaming

Interviewee: It seems that from a competitive angle, a Netflix subscription becomes economical only when shared among 4 different users. Further, Netflix doesn't provide annual subscriptions. This can increase churn rates due to higher monthly costs for the users. Further, Netflix shows a clear instance of decoy pricing wherein its cheapest pricing subscription of 500 is a decoy to stimulate users to opt for the marginally higher priced options which in turn cost users less on an individual basis.

Looking at the pricing strategy of competition, I would suggest Netflix to decrease its lowest price to a competitive level close to 300. This aligns with increasing the user base within the highly price conscious customers in India. It would be economical for viewers to buy a single screen plan while eliminating the market friction involved in finding a co-subscriber for the "2 screens" version to reduce cost.

Moreover, it would not negatively impact the top-line, as the change in product mix is likely to be compensated by the higher number of users buying the low-price subscription.

Further, I also recommend a second-degree price discrimination by providing annual subscriptions. This would attract the users with a lower price point perceived on a monthly basis and would also help reduce user churn that happens on a monthly basis due to higher monthly subscription charges.

Interviewer: Alright, sounds fair. What else you would be looking at?

Interviewee: We also need to analyze the value proposition of the three platforms and how they stimulate trial and stimulate renewal. Hence, to do so, I would be looking at the following parameters across the platforms.

	Hotstar	Primevideo	Netflix
Trial simulation policy	Provides some tv, sports and news content for free + 30-day free trial	30-day free trial	30-day free trial
User segments	- TV series and movie watchers - News watchers - Sports watchers	- TV series and movie watchers	- TV series and movie watchers
Competitive Advantage	Exclusive content from Disney and Marvel. Strives to get the most popular content such as Game of Thrones on the platform.	Primevideo is offered along with other Amazon prime incentives. The price of the complete bundle is the lowest. Partners with Bollywood producers to get the movies on the platform at the earliest.	Has the highest amount of Foreign video content. Unlike its competitors, it emphasizes the most on high quality originally produced exclusive content.
Emphasis on regional content	All Disney content dubbed in major regional languages; High amount of regional content available	High amount of regional content available, Major emphasis on Bollywood	Relatively low regional content available. Majority of content not dubbed in regional languages

The value that Netflix provides lies in its originally produced content such as House of Cards, etc. However, this competitive advantage is weathering off due to popular Disney and MCU content coming up on Hotstar. This entails that from a value perspective, it's difficult for Netflix to justify a price premium based just on its exclusive content.

Netflix should look at increasing its trial. An offensive strategy for the same can be to provide some content for free coupled with its existing 30- day free trial policy, something that Hotstar has been doing since its inception. To increase conversion of these prospects to paid-users, Netflix can adopt a strategy of providing the pilot episodes of some of its most popular web series for free. This would psychologically increase the tendency of users to buy a subscription.

I would also recommend Netflix to inculcate third-degree price discrimination by providing separate content options to users at different price points. In case of Netflix, these product options can be

- Non-original content – American TV shows, movies, etc.
- Netflix originals content + Non-original content.

Decoy pricing should be used in these product options to stimulate users to buy the complete bundle of Netflix originals and non-original content.

Further, Netflix should lay emphasis on dubbing at least the most popular content to regional languages to enhance its reach to people from across the country.

Moving on to cost based pricing, it doesn't make sense to analyze it from a cost perspective as the variable cost involved in serving every additional subscriber is close to null. This industry comprises primarily of high fixed costs of obtaining content licenses, content production, IT infrastructure costs, etc. and hence shouldn't be looked at through a cost-plus pricing angle.

Interviewer: Looks good. Can you summarize your recommendations for the pricing and product strategy?

Interviewee: Alright! So, we saw that from a competitive viewpoint, Netflix is priced much higher. Further, due to its decoy pricing, its lowest price point of 500 comes at a much higher level than its competitors. Hence, I recommend reducing this lower price point to a competitive level of 325 and shift the decoy pricing towards the 2 product-based options of non- original content and non-original + Netflix original content.

Further, they should also implement annual subscriptions to provide for lower monthly price points for this price-conscious market, at the same time reducing the subscription churn rate.

It can also look at increasing the trial by providing select content for free and adopting a partially freemium model. Also, given the heterogenous market, Netflix should focus on regional content to provide value to the customers and justify its premium pricing.

Interviewer: Alright, sounds good. We can close the case here.

PRICING 4

Excerpt: While most pricing questions cater to setting the appropriate price for new products or services, negotiating vendor contracts is another kind of pricing problem that a manager might encounter

Interviewer: Hi, You are a product manager at a large gaming company. It has only one application in the market - a multiplayer first-person shooter game. The server for this game is hosted with a third-party vendor who is dissatisfied with the pricing contract that is currently in place and is asking for the payment of overage charges. How would you proceed?

Candidate: That's an interesting question. To begin with I would like to clarify some aspects of the problem.

Interviewer: Go ahead

Candidate: I would like to understand what kind of pricing structure is currently in place, and what is the reason for which the vendor is asking for the renegotiation?

Interviewer: Currently we have a pay-per-use payment model. Here the amount to be paid per user depends on the band in which the lump sum requirements of the server capacity lies. During the times of COVID, there has been a surge in the server capacity used by our application, which has motivated the vendor to demand an overage.

Candidate: Alright. So this surge in usage could be both due to an increase in the usage per active user, or a spike in the number of active users itself. Which one has been the case with us?

Interviewer: Good question. So we have had a spike in the number of users and in addition, the average time for which a user engages with the application has increased too.

Candidate: Also, to understand the business requirements better, I would like to know more about the business model in place. Typically such applications would either generate revenue based on adverts or based on registration fees, or a mix of the two. If the model is ad-based, we would be in a marginally better position, because the increase in engagement time would translate to

higher ad revenue per user. In case of registration fees, the existing fee would have been decided considering the break-even point with the ongoing server costs - hence the loss of profits would be higher.

Interviewer: Fair analysis. Assume for simplicity that there are no advert based revenues, we are simply using a registration fee-based model.

Candidate: Great so we can approach the issue in the following way. There are two counter forces to shape the response to the situation - Server costing & Product pricing.

Servers can either be our own or from a third-party. The third-party server additionally can be existing or new. The existing one can have the ongoing costing model, or it can be renegotiated.

Since the ongoing costing model is clearly loss-making, we would exclude that from our analysis.

On the pricing end. We can either continue with the current pricing strategy or shift to a new one. The new strategy can, in turn, be registration-based or usage-based. Is there something else I should consider?

Interviewer: No this looks perfect. What parameters would you evaluate these options on?

Candidate: We could evaluate the options on the revenues (a proxy of pricing success), cost (a proxy of costing success), and risk (a sanity check on whether the option makes sense for the business).

Interviewer: Great. Now you can present your analysis and recommendations

Candidate:
H=High, M=Medium, L=Low

S.No.	Alternative	Rev	Cost	Risk	Evaluation
1	Own Server	L	H	H	Although this option will reduce the reliance on third parties in the future, it has potential risks of downtime during migration. It would involve a fairly high cost and effort. Further, the customers might become inactive once the COVID situation improves, leading to unused capacity. Since the process is long, the revenue realization might be lower than the potential.

2	Third-party server - new	M	L	M	While the costs are low, there would be a risk due to migration, albeit lesser than that posed on migration to own server because the timeframe would be lower and expert assistance would be available from the new vendor. The cost for the transition would be lower than that of option 1 and revenue realization would be higher due to a shorter and smoother transition.
3	Third-party server - existing, renegotiation	H	L	M	The risks related to migration would be the least, however, due to the dependency on one firm, there might be a loss of bargaining power - hence an overall moderate risk. The revenue realization would be maximum given no transition losses. The cost will be less because no capital or fixed payments would have to be done.
4	Current pricing	L	L	M	Maintaining the current pricing would lead to no effect on cost as such, medium risk because the business might suffer losses as a whole and low revenue generation because the usage spike won't be monetized in the best way.
5	Increment registration fees	M	L	M	If the registration fees are incremented, it would lead to an increase in revenue, however, the increase might not be sufficient to recoup all the costs, hence moderate. The cost might not be affected again. In terms of risk, it is moderate because, on one hand, some customers might leave because of increased costs, while on the other, there is still a risk due to only partial recoupment of costs.
6	Usage-based pricing	H	H	H	Usage-based pricing would lead to total monetization of the additional costs from the new users, hence the revenues would be high. The costs would be high as well because having metering of usage would require changes in the application. The risk would be high as well because old customers - who would ideally be using the application more - might exit due to higher costs.

Based on this I would like to recommend options 3 and 5, due to low risks and high returns.

Interviewer: That'd be all, thanks!

PRICING 5

Interviewer: Assume you are the new product manager in our Amazon Prime business and are deciding pricing. The vice president would like to lower the price from $79.99 per year to $69.99 per year. Making your own assumptions, develop the financial projections for this decision.

Interviewee: With lower prices, customer demand will increase. The new Prime customers might also purchase more things. However, increased demand will reduce membership revenues from existing Prime customers and increase shipping costs for existing non-Prime customers. Let's assume there are 5 million Prime customers.

1.Existing Prime customers
With a $10 price reduction that reduces annual revenues by $50 million.

2.New Prime customers
We need to estimate how many new Prime customers will be gained from the changes. Finally, let's consider additional revenue from the membership, along with incremental sales. Let's say Prime membership jumps 20 percent due to the price cut. That's an additional 1 million Prime customers. The incremental membership revenue is roughly $70 million. Average shipping costs per customer is around $40 per year. Let's say that moving these customers into Prime increases shipping costs by $20 per year per new Prime customer. That's an additional $20MM cost. Lastly, we can anticipate more frequent purchases due to free 2-day shipping. Let's say there's an additional $40 worth of purchases per new Prime customer per year. Amazon's gross margins are 25%, so that's an extra $10 per year per new Prime customer.

To recap on the net annual impact:
Existing Prime customers
-$50 million net (loss)
New Prime customers
+$60 million net = +$70 million member revenue -$20 million incremental shipping costs + $10 million incremental gross margin

Interviewer: So what's the overall change in revenue, and what's your recommendation?

Interviewee: Overall, we're looking at a $10 million annual increase from reducing price by $10. I recommend we reduce price.

INTRO TO RCA

FRAMEWORK

1. Ask clarifying Questions:

- Reiterate/Clarify what the product does
- Clarify the time period, geography, product sub-segment, type of customer (persona, new, existing), platform where the metric change is observed or is there any seasonality involved in the change
- Is the metric change sudden or gradual over the time duration
- Clarify what the metric actually means and how is it calculated
- Check if there has been any recent change in how the metric is calculated
- Check whether the BI tool that reports the metric is working fine.

2. Internal factors:

- Traverse the user journey of the customer on the app and check at which point is the change detected
- Check if there have been any tech-related issues like recent major UI releases on the application, leading to either technical glitches, non-intuitive UX, or performance issues
- Any changes in the offerings on the app, with respect to the pricing(demand-side) or internal salaries, service quality, or cannibalizing product offerings (supply-side)
- Any trends with respect to the user reviews of the products

3. External factors:

- Any new Govt. regulation or law affecting product performance in any way
- Any environmental effect like a natural calamity or pandemic
- Competitor Analysis in terms of new entrants or existing ones offering more lucrative options or sale or any other cheaper alternative that has appeared on the scene
- Any recent PR issues or negative sentiment on social media or News
- Any change in consumer behaviours or psychology

Tips

- Clearly call out any assumptions you make due to lack of information.
- Always keep the interviewer informed and ask for clarification when making assumptions or hypotheses.
- Do not apply or mention the framework directly
- Allow the interviewer to provide guidance and steer the discussion in the right direction.
- Understand how RCA applies at different stages of the product lifecycle (e.g., design, development, launch, post-launch)
- Interviewers may present you with hypothetical scenarios where a product issue arises and ask for a solution as well. Practice thinking on your feet and applying RCA techniques to solve these hypothetical problems.

RCA 1

Interviewer: Instagram recently launched a new feature allowing users to purchase merchandise of clothes shown in Reels. However, the initial adoption rate for this feature is very low. Can you walk me through how you would start investigating this issue?

Interviewee: Let me know if I understood correctly about the product in question. Instagram is a popular social media platform that revolves around visual content sharing. Its core value proposition lies in enabling users to express themselves through photos, videos, short form content, connect with a global community, and discover inspiring content.

Interviewer: Yes, you are right about the product.

Interviewee: Firstly, I will narrow down the scope of the problem. Is this specific to a geography? Or Is it specific to app users? Or any such specific demographic.

Interviewer: The low adoption rate has be been low for all geographies.

Interviewee: Is this a recent issue, or has it been gradually declining since the feature was launched?

Interviewer: It's a recent issue. Candidate: Have there been any technical issues or performance problems reported with this feature?

Interviewer: No technical issues have been reported.

Interviewee: Is the low adoption seen across all types of Reels or concentrated in some categories?

Interviewer: It has been observed across the content.

Interviewee: **T**hank you for the information. Now, I'd like to analyze both internal and external factors.

Have there been any external factors such as changes in the social media landscape or competitors launching similar features or government regulations?

Interviewer: No, there haven't been any significant external changes.

Interviewee: Okay, let's move on to analyzing the internal factors. Let me chart out the user journey of user interacting with reels.Instagram users open the app's homepage. Then, they open a reel on their timeline. They, then, do vertical scrolling to watch new recommended reels. They can like, comment or share a reel in the feed when they are watching it or they can follow the content creator. Users would open the profile page of content creator to check old content of the content creator. Let me know where the new feature got introduced and has users been dropping at any part of the above user flow.

Interviewer: Good analysis of user flow. The new feature was introduced on the profile page of content creator. Where user gets option to visit the page to buy merchandise of content creator by clicking on the profile image of content creator. No, user metrics have not changed in the old user flow.

Interviewee: Based on this new information, the user flow would be modified as follows. Instagram users open the app's homepage. Then, they open a reel on their timeline. They, then, do vertical scrolling to watch new recommended reels. They can like, comment or share a reel in the feed when they are watching it. If they like the reel and content creator, they open the profile page of the content creator and click on the profile image of the content creator to buy their merchandise.

Interviewer: That's right. That would be the new user flow.

Interviewee: Since, problem is observed across the content and user metrics have not changed at any part of the old user flow. What has been the rate of users clicking on the profile image and clicking on the merchandise link

Interviewer: There was no change in rate of users clicking on the profile image. However, rate at which users click on the merchandise link has been minimal to say the least.

Interviewee: Since, the whole user flow metrics have been normal till the last point. The problem could be awareness about the feature.

Interviewer: Yes, you are right. Users did not know that the feature existed. Only users who read or heard about the feature release interacted with it. How can we improve awareness about the feature?

Interviewee: Following could be done so as to increase adoption rate. Send notifications to all users or targeted user segments when they open the Instagram app informing them about the new shopping capability. Have a "Shop from Reels" spotlight as the first post users see in their feed for a period of time to announce the feature. Show reels that are ads to users on how to use the feature. This way interventions can be made on current the user flow to nudge user behaviour.

Interviewer: Thank you. We can close the interview.

RCA 2

Interviewer: Let's say you're a Product Manager at YouTube, and you've noticed a 10% drop in overall watch time on the platform this week. How would you go about analyzing this issue?

Candidate: Let me know if I understood correctly about the product in question. YouTube is a leading video-sharing platform that allows users to upload, share, and view a wide variety of videos. With its vast and diverse content library, including user-generated content, official music videos, and educational resources, YouTube offers a one-stop destination for entertainment, information, and creative expression.

Interviewer: Yes, you are right about the product.

Candidate: Firstly, I would like to seek more context about the problem. Could you provide insights into whether the drop is consistent across all content categories, or if specific types are more affected?

Interviewer: The drop seems to be consistent across most content types.

Candidate: Understood. Next, I'd like to understand if this drop is observed on all devices or if it's more pronounced on mobile or desktop.

Interviewer: It appears both have the same drop.

Candidate: Great. I'd also like to know if specific regions are experiencing a more significant drop in watch time compared to others.

Interviewer: It seems to be a global issue without any significant regional variations.

Candidate: Lastly, have there been any recent changes in how watch time is calculated or measured on the platform?

Interviewer: No, there haven't been any changes in the calculation method recently.

Candidate: When did this drop in watch time start, and has it been consistent over the past week?

Interviewer: It has been consistent.

Have there been any external factors such as changes in the social media landscape or competitors launching similar features or government regulations?

Interviewer: No, there haven't been any significant external changes.

Candidate: With that context, I would proceed to internal factors analysis. Do we have any indications that data accuracy might be an issue? Are there discrepancies in our metrics or reports?

Interviewer: The data appears to be accurate and consistent with other data sources.

Candidate: Have there been technical problems, such as increased downtime, latency, or errors in the mobile app? Could these technical issues be discouraging users from engaging with the platform?

Interviewer: No, technical issues observed.

Candidate: Let's understand the user journey and see where the problem is occurring for users. Users open the YouTube app. Then, they search for a video based on keywords or click on a video recommended to them. They watch the video. They click on search for a new video or a new video gets auto played after the video is played.

Interviewer: This user journey is appropriate and relevant.

Candidate: Let's pinpoint the point in user journey where the watchtime is dropping. are fewer people launching the YouTube app?

Interviewer: No.

Candidate: are people finding less content that intrigues them to watch? Or not receiving relevant search results? or are fewer titles and thumbnails enticing users to click videos open?

Interviewer: Relevant keyword results are shown. Recommendation algorithm is working well.

Candidate: are people watching videos for shorter durations once they start?

Interviewer: The watch time per video has been normal.

Candidate: Has there been change in drop offs after watching the video? If yes, it could be due to disabling of default auto-play because search and recommendation are working normally

Interviewer: Yes. This was the reason for drop off of watching times. Thank You.

RCA 3

Interviewer: Thank you for joining us today. We are facing a significant issue with our Uber app. There has been a sudden spike in cancellation rates, and our customers are experiencing poor service as a result. I'd like you to walk me through how you would approach identifying the root cause of this issue.

Candidate: Thank you for the opportunity. To start, can you clarify what the Uber app does in this context and which specific services are being impacted by these cancellations?

Interviewer: The Uber app connects riders with drivers for transportation services. The issue is primarily with ride cancellations.

Candidate: Understood. Could you specify the time period during which this increase in cancellations has been observed? Is it a recent development, or has it been a gradual increase over time?

Interviewer: It's a recent development, primarily observed in the last couple of weeks.

Candidate: That's helpful to know. Is there a particular geographic region or city where this problem is more pronounced, or is it widespread across all areas where Uber operates?

Interviewer: The issue seems to be more prevalent in metropolitan areas, but it's not limited to a specific city.

Candidate: Interesting. What kind of customers are being affected the most by these cancellations? Are they new users, existing users, or a mix of both?

Interviewer: Both new and existing users are affected, but we've noticed more complaints from our frequent users.

Candidate: Thank you. On which platform are these cancellations happening? Is it more on iOS, Android, or both equally?

Interviewer: The cancellations are happening across both platforms, so it doesn't seem to be platform-specific.

Candidate: Got it. Have there been any recent changes to how the cancellation metric is calculated or reported? Any updates to the BI tool that tracks these metrics?

Interviewer: No, the way we calculate and report cancellations hasn't changed recently.

Candidate: Understood. Now, let's look at the customer journey on the app. Typically, a customer would open the app, enter their destination, request a ride, and wait for a driver to accept the ride.

The customer is then notified of the driver's details, price and ETA.

Interviewer: Yes, that's correct.

Candidate: That's very helpful. Moving on to internal factors, has there been any recent update or major release to the Uber app, particularly anything that might affect the user experience or the functionality of the driver's side of the app?

Interviewer: Yes, we recently rolled out an update that allows drivers to see the drop-off location before they accept a ride.

Candidate: Interesting. Let's go back to the user journey once. Can you specify at which point in this journey the cancellations are occurring?

Interviewer: The drivers are usually cancelling the rides after accepting them but before reaching the user's pickup location.

Candidate: That provides valuable insight. Have there been any technical issues or glitches reported since this new feature was released?

Interviewer: No technical glitches have been reported with this feature.

Candidate: Aside from the new feature, have there been any other recent changes in the offerings on the app, such as pricing, internal salaries, service quality, or any new product offerings that might influence driver behaviour?

Interviewer: No, the main change has been the new feature allowing drivers to see the drop-off location.

Candidate: Let's consider external factors as well. Are there any new regulations or laws affecting our service? Any environmental factors like recent natural disasters?

Interviewer: No new regulations or environmental factors that we're aware of.

Candidate: How about competitor actions? Have there been any significant changes in competitor offerings or promotions that could be impacting driver behaviour?

Interviewer: Not that we've noticed. Our competitors seem to be operating as usual.

Candidate: Have there been any PR issues or negative sentiment on social media that could be influencing driver or customer behaviour?

Interviewer: We haven't noticed any significant PR issues or social media backlash recently.

Candidate: Based on our discussion, it seems the root cause might be related to the recent update allowing drivers to see the drop-off location. Drivers could be cancelling more frequently if they find the drop-off location undesirable. This matches the feedback you've received from the drivers. Addressing this feature, perhaps by implementing incentives for completing rides to less desirable locations or refining the feature to balance driver preferences with customer needs, might help reduce the cancellation rates.

Interviewer: That makes sense. Thank you for walking through the analysis so thoroughly.

RCA 4

Interviewer: Great to have you here today. Let's dive into the case. Netflix subscriptions have reduced on a month-to-month basis by 8%. How would you approach identifying the root cause of this decline ?

Candidate: Thanks for the opportunity. To get a comprehensive understanding, I'll follow the RCA framework. Let's start by asking some clarifying questions. Could you confirm what exactly the product in question does?

Interviewer: Sure, Netflix is a streaming service that offers a wide variety of TV shows, movies, documentaries, and more, available to subscribers for a monthly fee.

Candidate: Got it. Now, regarding the time period, is the 8% decline observed just in the last month, or has this been a trend over several months?

Interviewer: The decline has been observed over the last month.

Candidate: Understood. Is this decline observed only in India, or is it a global phenomenon?

Interviewer: It's specific to India.

Candidate: Okay. Are we seeing this decline across all customer segments, or is it more pronounced in a particular segment, such as new subscribers or long-term users?

Interviewer: It seems to affect both new and existing subscribers.

Candidate: Noted. Is this change in subscription rates sudden, or has it been gradual over the month?

Interviewer: It's been quite sudden.

Candidate: That's helpful to know. Could you clarify how the subscription metric is calculated? Is it the number of active subscriptions, revenue, or something else?

Interviewer: It's based on the number of active subscriptions.

Candidate: Thanks. Has there been any recent change in how this metric is calculated or reported?

Interviewer: No, the calculation method has remained consistent.

Candidate: Lastly, is the BI tool that reports this metric functioning correctly?

Interviewer: It seems to be working fine.

Candidate: Great, thank you. Now, let's look into internal factors. To do this, I'll map out the user journey on the Netflix app: users typically log in, navigate to the payment section to adjust any payment preferences and pay, set preferences of language and genre and then proceed to search for or watch recommended content. Is that correct?

Interviewer: Yes, it sounds perfect.

Candidate: Moving on, have there been any recent tech-related issues or updates or major UI releases that could have contributed to this decline? Any glitches, non-intuitive UX changes, or performance issues?

Interviewer: Yes. There was an update related to instant messaging service to customers reminding them of scheduled auto debit subscription setting. Furthermore, a new readily available and accessible auto debit cancellation button has been introduced in the payment section of the websites and app.

Candidate: Okay, that's really insightful. Have there been any changes in the offerings on the app, such as pricing adjustments, changes in service quality, or new product features that might be cannibalizing existing offerings?

Interviewer: There haven't been any significant changes in offerings.

Candidate: Understood. Are there any notable trends in user reviews that could give us insight into their satisfaction or dissatisfaction?

Interviewer: We had noticed some negative reviews mentioning the ease of cancellation due to the new regulations.

Candidate: That's insightful. Now, let's consider external factors. Has there been any new government regulation or law, other than the one you've mentioned, that might affect Netflix's performance?

Interviewer: Yes, the Government of India has recently introduced other policy changes aimed at increasing consumer protection. These include stricter data privacy laws and more stringent rules around auto-renewal of subscriptions across various digital services.

Candidate: That's important to know. Can you confirm if the mentioned updates were due to these factors ?

Interviewer: Yes, that's correct.

Candidate: How about competitor analysis? Have any new entrants or existing competitors launched more attractive options or promotions recently?

Interviewer: Some competitors have been offering discounts, but nothing drastically different from usual.

Candidate: Noted. Have there been any recent PR issues or negative sentiment on social media or in the news about Netflix?

Interviewer: No major PR issues, but the new regulation has been a topic of discussion.

Candidate: Finally, are there any observable changes in consumer behaviour or psychology that could be influencing this decline?

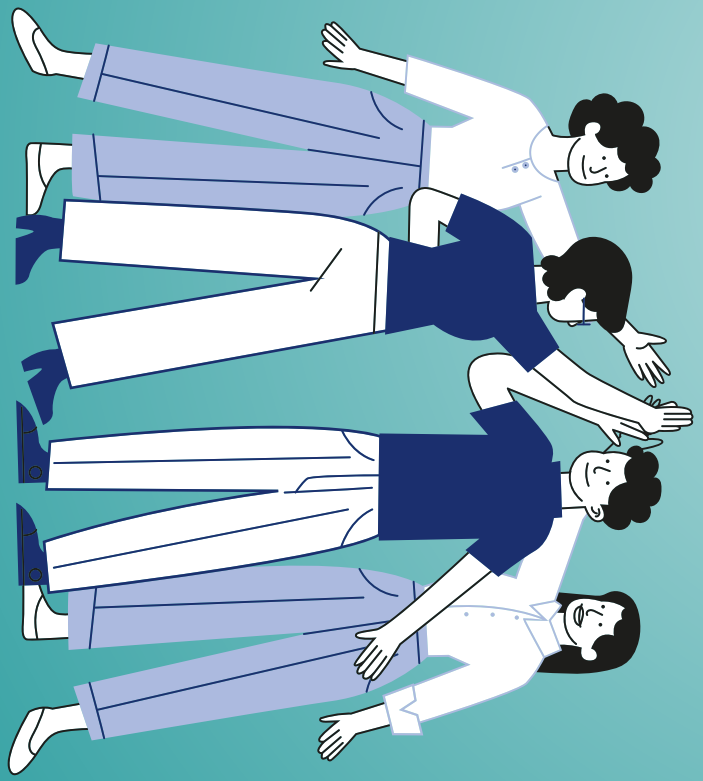
Interviewer: The users have been opting for cancellation of auto debit feature in significant numbers.

Candidate: That makes sense. The government regulation requiring text reminders and an easily accessible cancellation button seems to have made users more aware and cautious about automatic payments, leading to the observed decline in subscriptions. Addressing this might involve enhancing the value proposition to retain users and perhaps offering reminders or loyalty programs of the benefits of staying subscribed.

Interviewer: Sounds good, thank you!

It's time to...

MEET THE TEAM



COLLABORATORS

Revamped Edition: 2024-25

Manish Nath
B.Tech Mechanical Engineering, IIT Bhubaneswar
Previous Company: Amazon

Dhananjay Patil
B.Tech Mechanical Engineering, College of Engineering Pune
Previous Company: Siemens

Sai Prashanth Vankayala
B.Tech Electrical & Electronics Engineering, NIT Warangal
Previous Company: Oracle

Sundhar V
B.E. Computer Science and Engineering, Anna University
Previous Company: Zoho, Licious

Kislaya Kumar
B.E. Electrical & Electronics, BITS Pilani
Previous Company: Nvidia

Akriti Agrawal
B.Tech, Information Technology, JC Bose of Science and Technology, Faridbad
Previous Company: KPMG

Atul Chaurasia
B.Tech Computer Science & Engineering, SRM Chennai
Previous Company: EY

Sujit Kumar Mallik
B.Tech Mechanical Engineering, NIT Rourkela
Previous Company: Caterpillar Inc.

Abhishek Mathur
B.E Computer Science & Engineering, Chandigarh University
Previous Company: Gist Impact

Saurabh Wani
B.Tech, ICE, NIT Jalandhar
Previous Company: Cadence

Aayush Bhansali
B.Tech IT, College of Engineering Pune
Previous Company: Schlumberger

COLLABORATORS

AY: 2023-24

Shaswat Dixit

B.TECH Computer Science, Manipal Institute of Technology
Previous Company: Baxter

Guruprakash A

B. Tech Production Engineering, NIT Trichy
PGP'24, IIM Ahmedabad
Previous Company: Novartis Healthcare

Chanakya Yadav

Dual Degree, Metallurgical & Materials Engineering, IIT Madras
Previous Company: Zensar

Mandeep Verma

B.TECH Computer Science, Vellore Institute of Technology
Previous Company: Société Générale

Shreyas Likhar

B.TECH Mechanical Engineering, IIT Bombay
Previous Company: Bajaj Auto

Obada Mohammad Anwar

B.Tech, Computer Engineering, DCE
Previous Company: Citi

Kritish Puri

B.TECH Software Engineering,, DCE
Previous Company: Citi

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AY 2021-22

ANAND PATIL

B.TECH Electronics Engg,
IIT (BHU) Varanasi
Previous Company:
Microsoft

ANNU BANSAL

B.E Electronics & Comm.
Engineering, BIT Mesra,
Previous Company: Amadeus
Labs

ASHUTOSH CHANDRA

B.TECH Chemical
Engineering, IIT Delhi
Previous Company:
Innovacer

ASITAVA SARKAR

B.TECH Electrical
Engineering, IIT Kanpur
Previous Company:
Gartner

JAY BHAVIN SHETH

B.TECH Electrical
Engineering, IIT Jodhpur
Previous Company: Intel

MANPREET SINGH

Dual Degree Mechanical
Engineering IIT Kanpur
Previous Company: Loylty
Rewardz

MRIDUL KOTHARI

B.TECH Metallurgical & Materials
Engg, IIT Kharagpur
Previous Company: Morgan
Stanley

NANDIT JAIN

B.TECH. Chemical
Engineering IIT Kanpur
Previous Company: EXL,
Thales

ROHAN PATNAIK

B.E. (Hons) Computer Science
BITS Pilani, Goa Campus
Previous Company: Morgan
Stanley

ROHIT JOSHI

B.TECH Production and
Industrial Engg, IIT Roorkee
Previous Company: American
Express

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B.TECH Electrical Engineering, NIT Calicut

PGP'22, IIM Ahmedabad

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B.TECH Mechanical Engineering, IIT Guwahati

PGP, IIM Ahmedabad

Previous Company: Capgemini

SHOBHIT SINGH

B.TECH Mechanical Engg, MNIT Jaipur

PGP, IIM Ahmedabad

Previous Company: Dassault Systems

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B.TECH Production and Industrial Engineering, IIT Roorkee

PGP, IIM Ahmedabad

Previous Company: Green Affiliates Pvt Ltd

SOORYA RAMESH

B.TECH Computer Science & Engineering, SASTRA University

PGP'22, IIM Ahmedabad

Previous Company: Amazon

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B.TECH Information Technology, DTU PGP, IIM Ahmedabad

Previous Company: Visa Inc.

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B.E Electronics & Communication College of Engineering Guindy

PGP'22, IIM Ahmedabad

Previous Company: Genesys

COLLABORATORS

AY 2020-21

ABHISHEK KIKANI

B.E. MS Operations Research, Columbia University

PGPX, IIM Ahmedabad

Previous Company: Charles River Development

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B.TECH Chemical(TXT), Institute of Chemical Technology

PGP, IIM Ahmedabad

Previous Company: Pivy (Founder)

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B.TECH Mechanical Engineering, IIT Bombay

PGP, IIM Ahmedabad

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B.TECH Mechanical Engineering NITK Surathkal

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Previous Company: Oracle

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B. TECH Mechanical Engineering, MNIT Jaipur

PGP, IIM Ahmedabad

Previous Company: NTPC

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B.TECH CivilEngineering, IIT Kanpur

PGP, IIM Ahmedabad

Previous Company: Capgemini

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B.TECH ElectricalEngineering, IIT Roorkee

PGP, IIM Ahmedabad

Previous Company: ZS Associates

DHAVALA V S ADITYA

B.TECH MechanicalEngineering, IIT BHU

PGP, IIM Ahmedabad

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B.TECH ComputerEngineering, College of Engineering Pune

PGP, IIM Ahmedabad

Previous Company: Citicorp Services

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B.TECH Electronics&Communication IIT Kharagpur

PGP, IIM Ahmedabad

Previous Company: Texas Instruments

COLLABORATORS

AY 2020-21

JAYANT JAIN

Bachelors in Design
IIT Guwahati

PGP, IIM Ahmedabad

Previous Company:
D.E.Shaw

KUNAL JAIN

B.TECH
ElectricalEngineering
IIT(BHU) Varanasi

PGP, IIM Ahmedabad

Previous Company: EXL
Services

MAYANK CHAUHAN

BTech, Chemical
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Engineering NITK
Surathkal

PGP, IIM Ahmedabad

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BE
ComputerScience,
BITS Goa

PGP, IIM Ahmedabad

Previous Company:
HSBC Technology

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B.TECH Computer
Engineering, DTU

PGP, IIM Ahmedabad

Previous Company:
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Previous Company:
Qualcomm

MAURICE PATEL

Int. MSc. AppliedMathematics,
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PGP, IIM Ahmedabad

Previous Company: Accenture
Digital

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B.TECH Electronics &
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Engineering,
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PGP, IIM Ahmedabad

PREMNANDHAKUMAR

B.E Mechanical
Engineering, PSG
College of Technology

PGP, IIM Ahmedabad

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B.TECH ChemicalEngineering, IIT Madras

PGP, IIM Ahmedabad

Previous Company:

Kabbage

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B.TECH Electronics&Communication, NIT Surat

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Previous Company: Medibuddy

SEEMA

BE Computer Science, Manipal Institute of Technology

PGPX, IIM Ahmedabad

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B.TECH Electronics & Communication, IIT Roorkee

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Previous Company: Citrix Systems R&D

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PGPX, IIM Ahmedabad

Previous Company: Sopra Banking Software

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PGP, IIM Ahmedabad

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BTECH + MTECH Computer Science, IIT Madras

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Bachelors in Design, NIFT Bangalore

PGP, IIM Ahmedabad

Previous Company: Future Group

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AY 2020-21

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PGP, IIM Ahmedabad

Previous Company: Oracle

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B.TECH Computer Science, SGGSIE&T, Nanded

PGP, IIM Ahmedabad

Previous Company: Tata Consultancy Services

YUVRAJ SINGH BAWA

B.TECH MechanicalEngineering NITK Surathkal

PGP, IIM Ahmedabad

Previous Company: Microsoft

ABOUT THE BOOK

This is a book for the aspiring Product Manager, new age tech product enthusiast and truly anyone interested in understanding what makes the technology around us so engaging and effective.

The book aims to bring the average reader up to speed on today's most relevant technologies, including artificial intelligence and machine learning, and methods of problem solving, such as design thinking. It also cuts straight to learning by doing, critiquing apps that every reader has used, ranging across Uber, Amazon and Zomato.

Depending on the reader's level of knowledge and understanding of Product Management, they may find tremendous value in each chapter of this book, or crisp insights by leafing through specific sections.

Why is this book needed?
In today's age of information, it is easy enough for aspiring Product Managers to find resources which layout the basics of the PM interview, as well as information on what to expect from the industry. As Product Management continues to grow as a career of choice, and become increasingly competitive, a need was felt for a resource which builds not just a theoretical understanding, but a practical experience and intelligence.

This book therefore was born out of industry relevant discussions and sessions conducted by the ProdMan Club of IIM Ahmedabad & the ProdX Club of IIM Ahmedabad. In each chapter, the learnings and perspectives of the youngest generation of students entering the industry are used to provide the reader with the advantage of cutting edge industry knowledge. These chapters are further refined with insights, springing from a healthy discussion and dissection of concepts, by a group of budding Product Managers. Finally, ideas are presented through simple user journeys and relatable app critiques.

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